

Cattle Production

(Choose one of the following domestic animals to rear: Poultry, Goats, Fish or Rabbits)

Competency: You should be able to select desirable cattle for production and breeding and to carry out all processes involved in their rearing



KEY WORDS

- Breed
- Breeding
- Culling
- Dissect
- Dual purpose
- Exotic
- Feeding requirement
- Feeding standards
- Fodder
- Indigenous
- Notifiable disease
- Parasite
- Quarantine
- Ration
- Roughages
- Ruminant
- Rumination
- Selection

LEARNING OUTCOMES:

By the end of this chapter, you should be able to:

1. distinguish different breeds of cattle
2. understand the physical characteristics of cattle and show skills in selecting the best breed for rearing
3. understand the internal organs, particularly the digestive system and related physiological processes
4. understand female and male reproductive systems
5. understand and outline the different breeding and selection systems used in cattle
6. apply artificial insemination in rearing of cattle
7. understand the significance of good nutrition in rearing of cattle, and be able to prepare a balanced ration from locally available foodstuff
8. efficiently use high-quality roughage in rearing cattle
9. evaluate different cattle management systems
10. show skills in control of diseases and parasites
11. be able to detect milk abnormalities using a strip cup
12. keep records of breeding, management and production
13. be able to carry out cattle management practices
14. understand the importance of milking techniques and clean milk/meat production
15. show skills in handling cattle products and bi-products

Cattle Breeds and their Characteristics

A breed is a family of cattle with special characteristics that gives it a distinct class. There are many different breeds of cattle in Uganda but they are grouped broadly into two categories; the indigenous breeds (tropical origin) *Bos indicus* and the exotic breeds (temperate origin) *Bos taurus*.

Indigenous Cattle (*Bos indicus*)

These are cattle native to the tropics and they include the **Zebu** e.g. Boran, Nandi, Nkedi and Masai cattle, (ii) **Sanga** e.g. Ankole cattle, and (iii) crosses of the two above (Nganda, Nkedi, Nyolo etc).

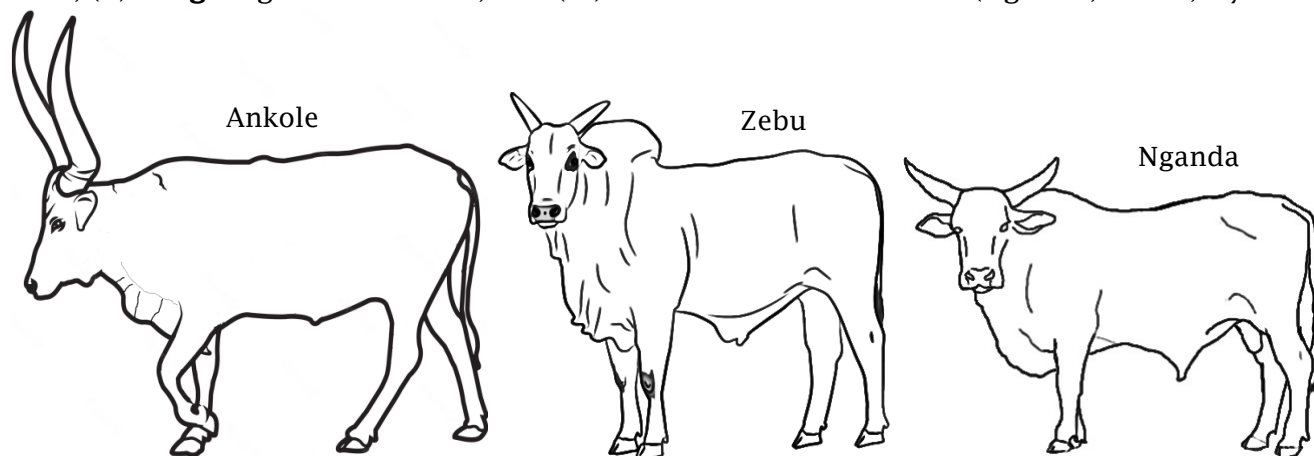


Fig. 1.2 Indigenous cattle

Characteristics of indigenous cattle breeds		
Ankole cattle (<i>Bos taurus indicus</i>)	Zebu (<i>Bos indicus</i>)	Nganda
Large in size; bulls 450-730kg and cows 410-550kg White large thick long horns Coat usually red or brown sometimes with shades and patterns Pendulous sheath and dewlap Area reared; mainly in the western and southwestern regions	Smaller in size; bulls 400-500kg and cows 230-280kg Short horns Coat usually red or grey sometimes with shades. Large humps, large dewlap and sometimes drooping ears Area reared; mainly in the eastern and northern regions	Small body size Short horns colour varies but often red, brown or black or spotted; white with brown/red spots or white with black spots Area reared; found mainly in Buganda and Bunyoro regions

General characteristics of Indigenous cattle

- They have well developed large dewlap and navel.
- They have prominent hump.
- They are generally horned.
- They are relatively small in size as compare to the exotic breeds.
- They are tolerant to tick bone diseases.
- They have good adaptation to high environmental temperature.
- Usually aggressive with poor dairy temperament.
- Have long legs and long face
- They have thin loose skin with short smooth shiny coat.
- They have a slow maturity rate,
- They produce less amounts of meat and milk compared to exotic breeds.
- Produce milk is of high butterfat content.

Exotic Cattle (*Bos taurus*)

These are cattle of temperate origin. They are fast growing and high yielding for milk or beef and have inferior adaptation to tropical conditions. These cattle breeds are usually used for cross breeding and upgrading purpose to improve indigenous cattle.

General characteristics of Exotic cattle

- i. They Grow faster
- ii. The size and length of the body is large
- iii. Their back is straight without a hump
- iv. Dewlap is less developed
- v. Have thick skin with long hair
- vi. They are hornless or with small horns
- vii. Have large udder
- viii. Have short stout legs
- ix. Less tolerant to parasites and diseases
- x. Produce more milk and carcass weight
- xi. Less tolerant to high temperature
- xii. Have short calving interval
- xiii. Milk has relatively low butterfat content.

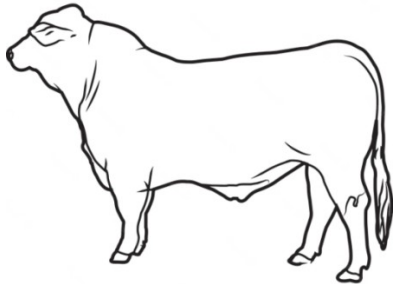


Fig. 1.3 Beef cattle characteristics

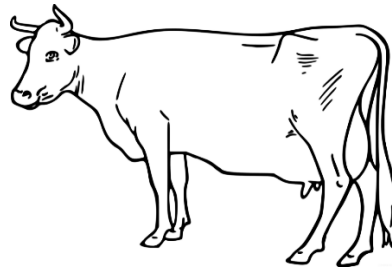


Fig. 1.4 Dairy cattle characteristics

Table 1.1 Beef breeds.

Examples of beef breeds	
Aberdeen Angus Origin: North East Scotland Colour: Usually Black Shape: Cylindrical, compact and polled. Size: Mature bulls weigh 900kg. Mature cows weigh 720kg.	
The Hereford Origin: England. Colour: Deep red and white-faced. Shape: Cylindrical, compact and small horns. Size: Average weight of bulls is 817kg.	
Shorthorns Origin: England. Colour: Red, roan or white. Size: Bulls 700-900kg, cows 545- 630kg.	
Galloway Origin: Scotland Colour: Black with a white belt. Size: Bulls 750-950kg, cows 450- 680kg.	

Table 1.2 Dairy breeds.

Examples of dairy breeds	
Friesian Holstein (largest dairy breed) Origin: Holland. Colour: Black and white. Size: Cow weighs 600kg; Bull weighs 950kg. Highest milk producers of all dairy breeds, but with least butterfat content; 3% -4%.	

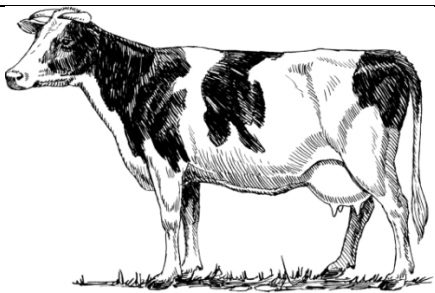
Ayrshire Origin: Scotland. Colour: White with brown markings. Size: Cows 360-590kg; Bull: 500-720kg; Conformation: Straight top lines, horns are long and face upwards. Milk production is second to Friesian with butterfat content of about 4%.	
Guernsey Origin: Guernsey Island off the coast of France. Colour: Yellowish brown to red with white legs, switch and girth. Size: Bull 780kg, cow 500kg. Conformation: Udders are less symmetrical. Average milk production 6100kg with butterfat contents of 5 % hence the yellow colour of milk.	
Jersey (smallest of all the dairy breeds) Origin: England. Colour: Yellow brown with black muzzle and switch. Size: Bull 700kg, cow 400kg. Conformation: Dished forehead, have straight top-line and level rumps with sharp withers. Average milk 4270kg, butterfat content 6% - 7%.	

Table 1.3 Differences between beef and dairy cattle

Beef Breeds; These are breeds that are kept specifically for beef production. They have the following characteristics: - They grow faster and mature early. - Have good conversion of feed to meat. - They produce high quality beef. - They have blocky body shape. - Have high percentage of carcass weight (60% or more). - Deep chest and girth and short legs. - Long-broad back and straight top and lower lines. - They have large heads with thick necks - Larger in size than dairy cattle	Dairy Breeds; These are kept specifically for milk production. Dairy breeds have the following characteristics: - They are wedge or triangular in shape. - Have large stomach. - Have good feed conversion to milk - They are docile with mild temperament. - Large, well-developed udders and teats. - Have lean bodies; Lean and smooth neck - Have large and long milk wells and veins - Wide and well set hindquarters to accommodate the large udder. - Have smaller heads and slim necks - Have large bellies - Smaller in size than beef cattle
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Dual Purpose Breeds

These are cattle breeds that produce high quantities of milk as well as quality carcass at the end of their milking life. These breeds have well developed body and put on flesh readily when fattened at the end of their milking life.

Examples of dual purpose breeds

Sahiwal

Origin: India and Pakistan.

Size: Bull 650kg, cow 400kg.

Colour: Reddish brown.

Conformation: It is a hornless, well developed hump, with deep body, sloping rump, broad forehead, distinct dewlap and resistant to tropical diseases and high ambient temperatures.

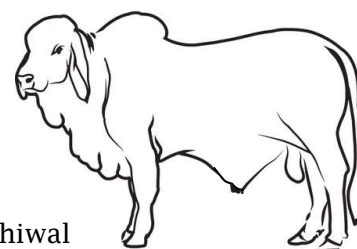


Fig.1.5 Sahiwal

Redpoll

Origin: England

Size: Bull 810kg, cow 550kg.

Colour: Deep red with a white nose.

Conformation: Polled-deep girth and short legs.

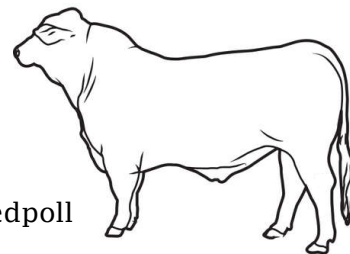


Fig.1.6 Redpoll

Simmental

Origin: Switzerland

Size: Bull 1300kg, cow 680kg.

Colour: Light red and white patches on head.

Conformation: It has broad and straight back, with well-sprung ribs and deep girth. It is well fleshed at rear quarters, has well suspended udders and large teats.



Fig.1.7 Simmental



Activity 1.1

Identifying cattle breeds and their characteristics

You will need

Notebook, pen, internet, observation list and protective gears

Key question: What are the physical characteristics of different breeds of cattle?

In groups:

1. Write a letter to the farm manager of a nearby cattle farm asking for permission to visit the farm
2. Prepare a check list of how you are going to engage the farm manager.
3. Put on protective gears. What will they include?
4. Visit the farm and obtain information as per the check list
5. What cattle breeds are reared on the farm and their characteristics.
6. Find out from the farmer the average productivity of each breed.
7. Suggest the adaptations that have allowed the cattle breeds to survive on the farm.
8. Further classify the cattle as beef or dairy using their productivity qualities.
9. Individuals record their findings in their note books.

Selecting the Breed of Cattle to Rear on the Farm

Selection involves choosing a breed with good and desirable qualities to rear. Whether to rear local or exotic cattle breeds or to rear beef or dairy cattle, the choice on which type of cattle to rear will depend on the following factors; (1) The adaptability of the breed to the local environmental conditions (high temperature and drought), (2) the availability of the breed, (3) ease of rearing (skills needed to manage the breed), (4) level of output (milk or meat) from the breed, (5) tolerance to common parasites and diseases, (6) marketability of the breed or its products, (7) availability of pasture (quality and quantity), (8) amount of capital needed, (9) growth rate and weight gain and (10) ease of breeding the animal.

Table 1.3 Advantages and disadvantage of rearing exotic and indigenous cattle

Advantages of rearing indigenous cattle	Disadvantages of rearing local cattle
• Have good resistance to most parasites and diseases • They have good tolerance to high temperature. • They can easily survive on poor dry pasture without losing much weight. • They are tolerant to drought conditions. • Their meat is highly priced in the market. • They have good mothering ability and with few reproduction problems. • Produce good quality milk of higher butter fat content. • Can survive in a wide range of ecological conditions. • Have good ability to walk long distances in search of pasture and water • They have low cost of maintenance.	• They have low body weight thus fetching less profit on the market. • They have low dressing weight. • They produce less milk and meat. • They take long to reach maturity. • Have long calving intervals. • Have poor dairy temperament. • Harbor most diseases which they can spread to susceptible animals and man.

Advantages of rearing exotic cattle	Disadvantages of rearing exotic cattle
• Have short calving intervals • Grow and mature faster • Easy to fatten for marketing • Produce more milk and meat • Fetch good price in the market • Have tender meat • Have relatively good cool temperament • Have good feed utilisation i.e., high feeding efficiency	• Costly to acquire • Require high level of management • The cost of maintenance is usually high. • Less tolerant to high temperature and drought. • Highly susceptible to most parasite and diseases. • Produce low quality milk • Low demand for their products in the market. • Can't survive well on poor pasture Require good quality pasture



Activity 1.2

Skills in selecting cattle breeds to rearing

You will need

A notebook, pen, internet, reference books and observation list

Key question: what are the factors to consider in selecting cattle breeds to rear on the farm?

In groups:

1. Write a letter to a nearby cattle farmer asking for permission to visit the farm.
2. Prepare an observation list of the breeds and their characteristics.
3. Put on your protective gears and ensure safety precautions.
4. Carry out a farm visit, observe, identify and record cattle breeds reared. Which cattle breed is dominant on the farm. Ask the farmer for reasons for the dominance of the breed identified.
5. Brainstorm and list the desirable characteristics considered when selecting good breeds to rear
6. Discuss the advantages and disadvantages of rearing; (i) Indigenous and exotic cattle (ii) Dairy and Beef cattle.
7. Share your findings with the rest of the class.
8. Record your findings in your notebook.

Reasons why indigenous cattle are able to survive in the tropical conditions

- i. They have good adaptability to high tropical temperatures; the skin is thin and extended by means of a dewlap this provides large surface area of the skin and numerous sweat glands which ensures efficient heat dispensation from the body. The skin coat also bears short shiny hairs which reflect much of the sun heat.
- ii. Tropical conditions favour development of pathogens and prevalence of animal parasites. Most indigenous cattle have good resistance to most tropical parasites and diseases.
- iii. Tropical are characterized by long dry conditions. These conditions affect the growth of pastures and availability of water. Indigenous cattle have good ability to withstand water and pasture shortage during dry spells. They have low water turnover and good forage utilization
- iv. Indigenous cattle are light and with well-developed long legs allowing them to move distances along the savanna in search of water and pastures without much loss in the body conditions.
- v. Indigenous cattle are small in size as compared to exotic cattle breeds. The small body sizes reduce the water and feed requirements. This allows indigenous cattle to cope up with shortfalls in water and food supplies in the tropics.

Reasons for the low productivity of indigenous cattle in Uganda

- i. Most indigenous cattle breeds are of poor quality with inferior genes in terms of milk and meat production which leads to low productivity.
- ii. Poor feeding of indigenous cattle; more often indigenous cattle depend on poor natural pastures with low nutritional values. This lowers their productivity.
- iii. High prevalence of parasites and diseases in most parts of the country affect the animals' genetic potential, and lower productivity in terms of milk and meat production.
- iv. Poor climatic conditions characterized by long drought periods which lead to drop in supply of pasture and water to cattle.
- v. Poor quality feeds and drugs available in the market which worsen the problem of poor nutrition and health in cattle which in turn leads to low productivity.
- vi. Poor methods of cattle rearing; most farmers still use the traditional methods of cattle production such as nomadic herding, inbreeding, and late detection of heat and diseases.

- vii. Inadequate supporting infrastructure such as improved abattoirs, all season roads, cold stores, milk coolers and markets.
- viii. Inadequate extension services to the farmers that would otherwise provide training and build capacity of farmers in forage conservation, breeding, and disease control.

How to improve the productivity of indigenous cattle

- i. Upgrading of genetic potential of indigenous cattle through selection and controlled breeding with improved cattle breeds.
- ii. Improving the feeding and nutrition of indigenous cattle through use of high quality preserved forage and feed supplements.
- iii. Providing effective parasite and disease control through routine vaccination and pest control.
- iv. Practicing dry season livestock feeding through pasture preservation.
- v. Training farmers on appropriate, safe and cost-effective husbandry practices to enhance production and increase quality of animal products.
- vi. Develop and expand a nationwide market information system that is accessible to all farmers.
- vii. Construction of water conservation system to provide water during drought.



Activity 1.3

Finding out the reasons for the survivability of indigenous cattle, and ways of improving their productivity

You will need

A notebook, pen, internet, reference books

Key question: what are the reasons for the survivability of indigenous cattle to tropical conditions and how to improving their productivity?

In groups:

1. Using the internet or agriculture reference book and journals, research and brainstorm and,
2. Identify the reasons why indigenous cattle are able to survive the harsh tropical conditions.
3. Outline the factors that contribute to the low productivity of indigenous cattle
4. Suggest the ways of improving the productivity of indigenous cattle.
5. Share your findings with the rest of the class.
6. Record your findings in your notebook.

Activity of Integration	Heifer International Uganda has decided to donate heifers to the youth in your community as part of its objectives of improving youth livelihood. To make the project successful, the donors have given the beneficiaries a chance to choose the breed of cattle they would prefer. Most youths lack farming skills and cannot make informed decision when selecting the breeds of cattle to rear.	
		
Task	Using the skills so far acquired in cattle breed selection, prepare a document that you will present during the youth meeting in your community. The document should indicate the factors that they should consider in selecting a suitable cattle breed, adaptable exotic and indigenous breeds and the merits and demerits of rearing indigenous and exotic breeds	

Exercise 1.1	Self-Check Questions
	In Uganda, farmers rear different breeds of cattle. However, indigenous cattle breeds are the most widely reared. As an expert in cattle production, 1. (a) Give five reasons for the popularity of indigenous cattle in Uganda. (b) Explain why indigenous cattle have been very successful in the tropical conditions. 2. (a) What factors should a farmer consider when selecting a suitable breed of cattle to rear on the farm? (b) Give four advantages and disadvantages of rearing exotic cattle. (c) Compare the different breeds of cattle in terms of appearance

Cattle Management Systems

The systems of cattle management can be grouped into 3 categories; the extensive, Semi-extensive and intensive management system. The systems of cattle production are based on the climatic conditions of the area, pattern of agriculture, the availability of land and the land use pattern, level of advancement in technology and cultural norms.

The intensive cattle production system

The intensive cattle production system is one in which cattle are fed in confinement with limited or no access to grazing, and where grazing is allowed, limited land is provided and high carrying capacity occurs. Examples of intensive production system is ranching in perimeter fencing and paddock grazing, stall feeding where cattle are confined and fed on cut pasture and concentrates. In Uganda, intensive production on grazing land is not common perhaps due to poor pastures, droughts and incidence of animal parasites and diseases.



Fig. 1.9 Paddock grazing



Fig.20 Stall feeding/Zero grazing

Advantages of intensive production system

- Parasite infestation is considerably reduced.
- It permits the introduction of labour-saving mechanical devices such as self-feeders.
- Cattle are protected from bad weather such as sun heat, storm, cold and wind.
- Feed and water intake can effectively be controlled.
- Animals save much energy leading to quick weight gain
- Manure can easily be collected for use in crop gardens or generation of biogas.
- It requires less space.

Disadvantages of intensive production system

- It requires a lot of capital investments, to construct the cattle stalls and paying labour.
- Feeding costs are also higher than in the extensive system.
- Cattle don't get enough exercise and as such they become weak.
- The system requires a lot of labour in feeding and cleaning of the houses.

The extensive management system

In the extensive system, cattle graze over large areas of land. The area is usually marginal land or land unsuited for crop production. The system involves low carrying capacities, uses small inputs of capital, supplementary feeds and labour and little management is practiced except driving the animals to the field and bringing them back home to the sleeping yards.

An example is the nomadic herding and transhumance, and the traditional free range, mostly practiced in the northern and eastern Uganda and on marginal lands around game reserves.



Fig.1.8 Nomadic grazing of cattle.

Advantages of extensive management system

- It requires very little capital.
- There is a lot of saving in feed costs since pigs scavenge for their own feeds.
- Labour requirements are very low.
- Manure is evenly deposited in the field.
- Animals get plenty of exercise.
- Little damage to the local environment and soils usually caused by overuse of chemicals in intensive farming.

Disadvantages of extensive management system

- Requires more land for grazing.
- It leads to underutilisation of land
- Predators can attack cattle as they graze
- Animals lose a lot of energy searching for quality pasture.
- Poor quality pasture is available for grazing leading to low output.
- Less yield is obtained from the animals
- Droughts can lead to devastating losses.

Semi-intensive

The method allows varying degree of compromise between extensive and intensive production. There is limited grazing and stall-feeding of cut grass, crop residues and concentrates or tethering where the animal is tied by a rope to a peg or tree allowing it to access the grazing area provided.

**Activity 1.4 Identifying cattle management systems**

You will need A notebook, pen, internet and camera

Key question: what systems are used in management of cattle?

In groups:

1. Write a letter to a nearby cattle farmer asking for permission to visit the farm.
2. Prepare an observation list of the cattle management systems.
3. Put on your protective gears and ensure safety precautions.
4. Carry out a farm visit, observe, identify and record cattle management system practiced.
5. Ask the farmer for reasons for the cattle production system observed on the farm.
6. Discuss with the farmer on the limitations related to the cattle management system practiced.
7. Brainstorm and mention other alternative cattle management systems which could be used in production of cattle.
8. Present your findings to the class and record conclusions in your notebooks.



In Agro-pastoral production systems, cattle graze in private or public pastures and provided with additional feed material, particularly crop residues.



Using the internet and reference books research about nomadic pastoralism and transhumance and how they affect cattle production and the environment.

Activity of Integration

In most parts of Uganda, cattle farmers still practice nomadic pastoralism. The change in land use policy, drought, animal diseases and parasites have caused significant economic losses, and negatively impacts on livelihoods of pastoral communities. Besides, the practice is destructive to the environment. Thus, there is a need to change the system of cattle management.

**Task**

Prepare a poster you will use to sensitise the pastoralists in your community on the better systems of cattle management.

Exercise 1.2**Self-Check Questions**

There are different cattle management systems practiced in different parts of Uganda today. As a senior three student, use the knowledge so far acquired to

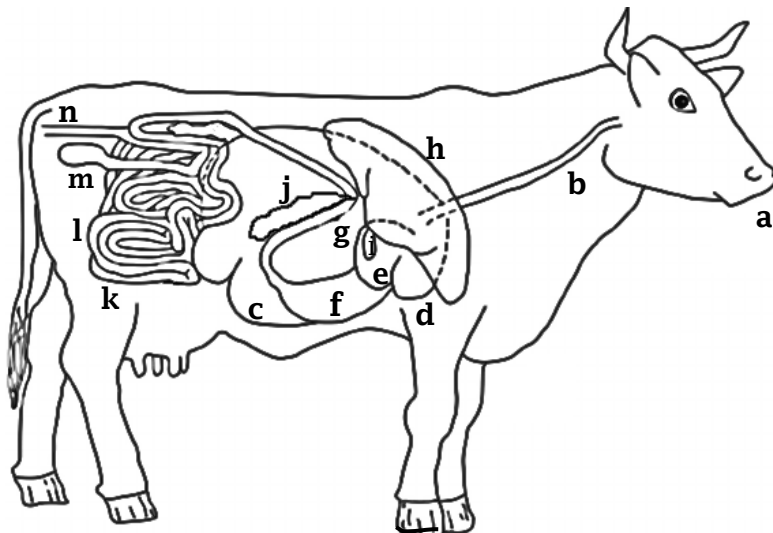
- (a) Outline the systems of cattle management practiced in Uganda.
- (b) Outline the effects of pastoralism on the environment.
- (c) Give the advantages and disadvantages of zero grazing.
- (b) Give your advice on how nomadic pastoralism can be improved.

Internal Organs of Cattle

Cattle, just like other farm animals, have many internal organs and systems which help the animal's body to perform several physiological roles such as digestion and reproduction.

The Digestive System of Cattle

We looked at the digestive system of a ruminant while studying domestic animal rearing in senior two. Cattle are ruminants and their digestive system is similar to that of goats and sheep. At this level, we shall make a more detailed study of the digestive system by dissecting a ruminant and making a study on it. Before we do that let's do the activity below to remind ourselves about the structure and functions of the digestive system.



- | | |
|------------------|------------|
| a - Mouth | l - Ileum |
| b - Oesophagus | m - Caecum |
| c - Rumen | n - Rectum |
| d - Reticulum | o - Anus |
| e - Omasum | |
| f - Abomasum | |
| g - Duodenum | |
| h - Liver | |
| i - Gall bladder | |
| j - Pancreas | |
| k - Jejunum | |

Fig.1.21 Digestive system of a cow (in situ)



Activity 1.5

Understanding the digestive system of cattle

You will need

A notebook, pen, internet, reference books

Key question: What organs make up the digestive system of cattle and their functions?

In groups:

1. Using the internet or any agriculture reference book, research and draw the digestive system of a cow, showing the following parts; mouth, oesophagus, rumen, reticulum, omasum, abomasum, small intestines, caecum, large intestines, rectum and anus.
2. Discuss the function of each part labeled.
3. Using a manila paper and an appropriate ICT software, prepare a presentation of your findings to the rest of the class and record conclusions in your notebook

As earlier seen, ruminants have digestive system made of the stomach which has four partitions (rumen, reticulum, omasum and abomasum), intestines divided into three parts (the duodenum, jejunum and the ileum), the caecum and the colon.

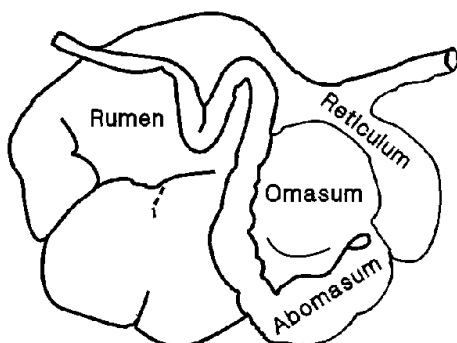


Fig.1.22 Ruminant stomach

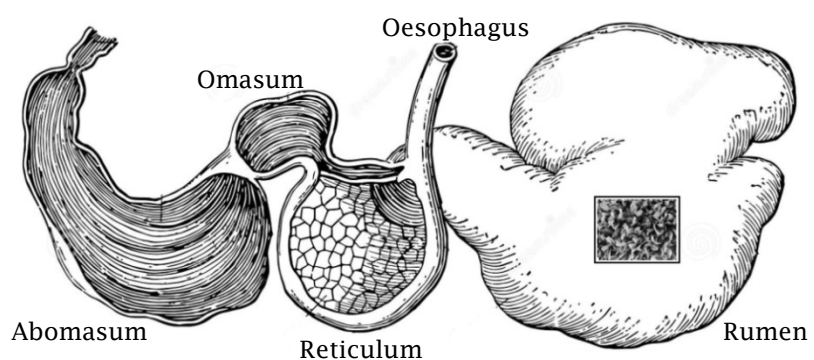


Fig.1.23 Dissected compartments of the ruminant stomach

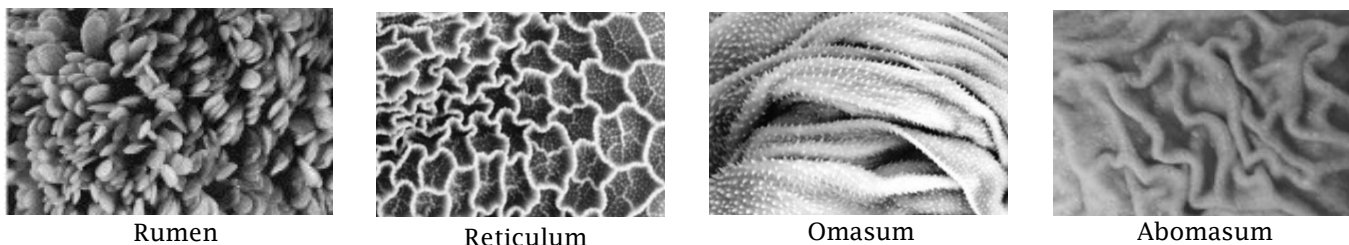


Fig.1.24 The structural appearance of the inside of the four compartments of the ruminant stomach

Adaptation of each compartment to its function;

The rumen; it is large to accommodate or store more food for the animal. It also has many papillae to increase the surface area for the absorption of volatile fatty acids in the rumen, and it has numerous microbes that help in the fermentation of food and the microbes produce cellulase enzyme that digests cellulose in the food the animal eats.

The reticulum; it has numerous intersecting laminae of different height that allow separation of fine food particles from large food particles.

The omasum; it has numerous parallel laminae. The laminae have numerous short papillae which increase the surface area for the absorption of water from the food and also allow the grinding of food particles to very fine texture.

The abomasum; it has a glandular epithelium that is highly folded to increase the surface area for the secretion of gastric juices containing pepsinogen and acids. Acids activate Pepsinogen to pepsin, which digests protein to peptides and polypeptides. The acids also kill germs in the food.



Because digestion of food involves progressive breakdown of food to finer particles, the size of the particles gradually reduces in size as food moves along the digestive tract. Thus food particles in the rumen are coarse than those in the omasum

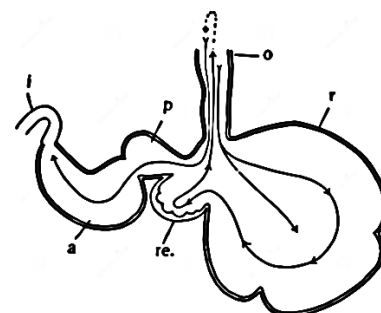
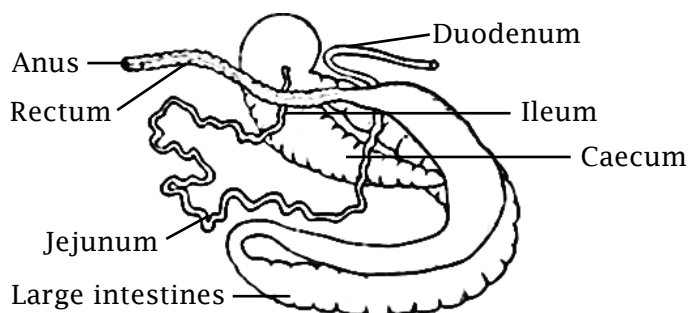


Fig.1.25 A section of the ruminant digestive system

Fig.1.26 Movement of food in the stomachs

Adaptations of each part

Intestines; a long coiled tube to provide more surface area for absorption of nutrients in the food.

Caecum; large sac having microbes that produce cellulase enzyme which break cellulose in the food.

Large intestines; large tube that collects undigested food from the ileum



Farm animals such as horses, rabbits and donkey are able to digest grass and yet they possess single stomachs. How is this possible?



Activity 1.6 Understanding the physiology of the digestive system of cattle

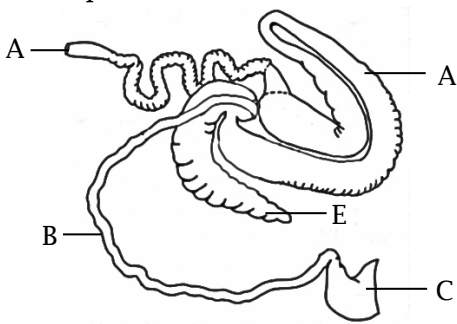
You will need A notebook, pen, reference books, digestive system, dissection kit

Key question: How does the digestive system of a cow function?

In groups

1. Put on protective gears. What will they include?
2. View a dissection to identify and state the functions of main parts of the digestive system.
3. Cut open each compartment of the stomach and describe the appearance of each.
4. Remove the food content from each compartment and describe the appearance of each.
5. Explain how each compartment is adapted to perform its function.

Activity of Integration	Farmers in your community have a challenge of highly polluted environemnt and more often, their cattle pick and swallow pieces of plastics and metals during grazing, leading to fialure of digestive system and sometimes death of their animals. This causes heavy economic losses to farmers which discourages farming.	
		
Task	You have been invited by the parish chief to sensitise the local people on how to dispose farm wastes safely to prevent pollution of the environment and to cattle from ingesting metals and plastics and to explain to them how the ingested plastics and metal move and affect the digestive system.	

Exercise 1.3	Self-Check Questions
	Farm animals must eat, digest and utilise the nutrients in the food. As a learner who has studied digestive systems of farm animals, (a) Make an annotated drawing of the digestive system of a ruminant (b) Mention the roles played by the parts of the digestive system of a cow. (c) Describe the nature of the food materials found in the rumen, reticulum, omasum and abomasum (d) The drawing below is of a section from the ruminant digestive system. Label the part marked A - E.  A B C D E

The Reproductive Systems of Cattle

The reproductive system of cattle is somewhat similar to that of other farm animals but with slight differences in Sows and Does. These have an extended uterine horn and are multiple ovulators, producing many young ones (litter).

The Reproductive System of a Cow

This is made up of the two ovaries, two oviducts (fallopian tubes), two uterine horns, uterus, cervix and vulva. The bladder is connected to the reproductive tract by the urethra.

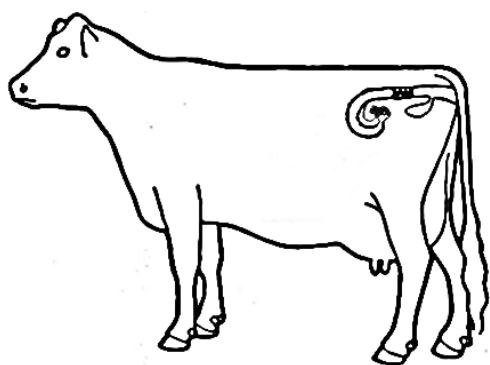


Fig.1.27 Position of the reproductive system of a cow

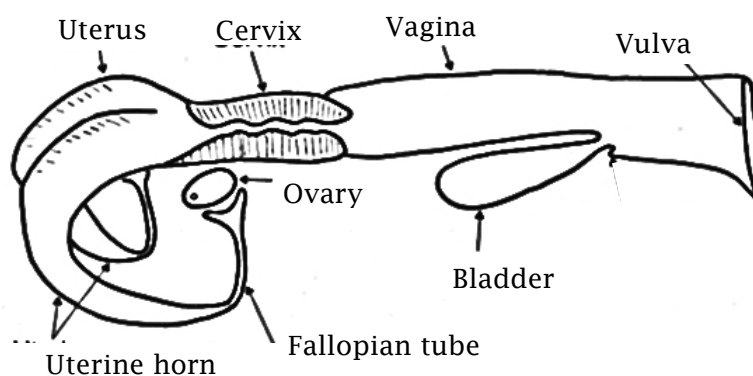


Fig.1.28 Drawing of the reproductive system of a cow



From the knowledge obtained about reproduction in goats while in senior two, give the functions of parts of the reproductive system of a cow



Activity 1.7 Understanding the reproductive system of a cow

You will need

Notebook, pen, display of reproductive system of a cow with the parts labelled; A ovary, B uterine horn, C cervix, D vagina and E vulva

Key question: what is the structure of a reproductive system of a cow?

In groups:

1. Put on protective gears.
2. Make a study on the display of reproductive system of a cow and identify the parts labeled.
3. Make annotated drawing of the display and discuss the function of each part.
4. Share your findings with the rest of the class and record conclusions in your notebook

The Reproductive System of a Bull

The reproductive system of a bull consists of two testicles, secondary sex organs (epididymis, sperm ducts and penis) and the three accessory glands (seminal vesicles, prostate and bulbourethral or Cowper's gland). These organs work together in the formation, maturation, storage, transport and deposition of the spermatozoa in the female reproductive tract.

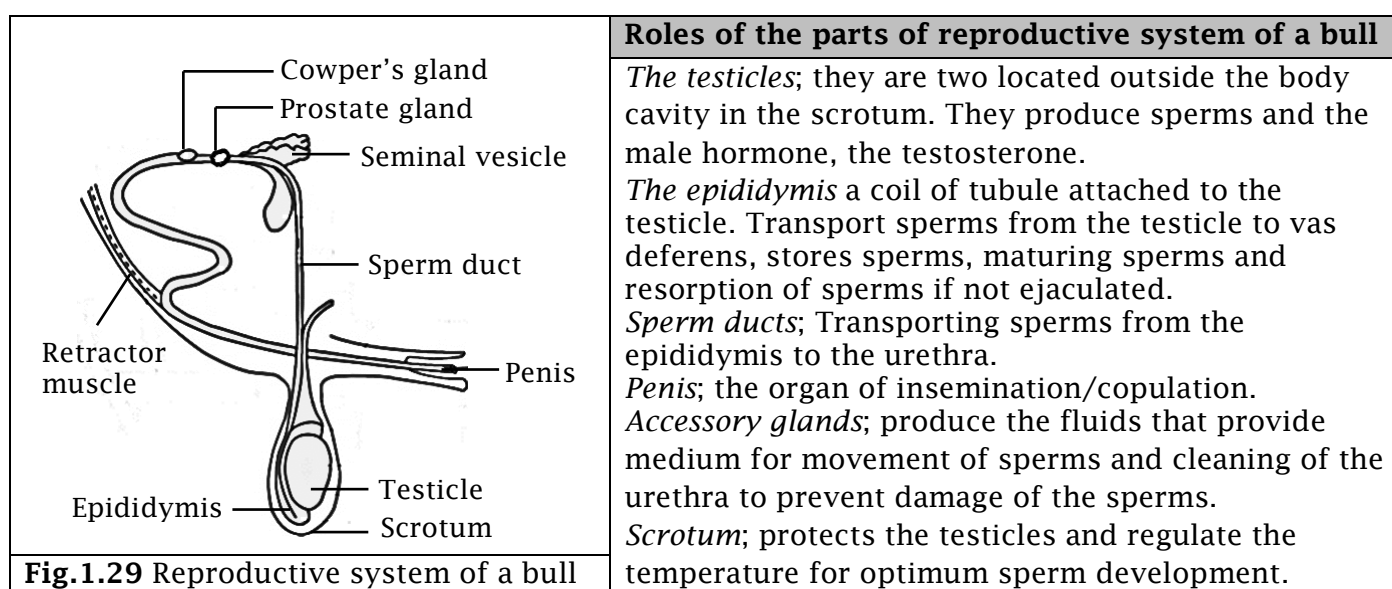


Fig.1.29 Reproductive system of a bull



The location of the testicles outside the body cavity is essential for normal sperm formation, which occurs only at 4 degrees to 5 degrees below body temperature. One or both testicles occasionally fail to descend into the scrotum and are retained in the body cavity. This condition is known as cryptorchidism.



Critically think about the mechanism how the scrotum regulates the temperature of the testicles.



Activity 1.8 Understanding the reproductive system of a cow

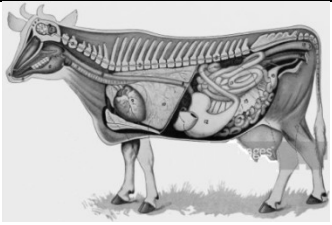
You will need

Notebook, pen, manila papers, markers, display of reproductive system of a bull

Key question: what is the physiology of the reproductive system of a bull?

In groups:

1. Put on protective gears.
2. Make a study on the display provided and identify the parts labeled
3. Make annotated drawing of the display and discuss the function of each part.
4. Share your findings with the rest of the class and record conclusions in your notebook

Activity of Integration 	The study of internal organs of cattle deals with the anatomy and physiology of mainly the digestive and reproductive systems. Many of your colleagues in the class donot understand this concept.	
Task	Write a speech to be presented to fellow learners about the structure and functioning of the reproductive system of a cow and that of a bull.	

Exercise 1.3	Self-Check Questions
	Fill in the blanks. Select the suitable answers from the word bank Connects vagina to uterus, Source of both the reproductive hormone and ova, Produce sperms and hormones, Regulates the temperature of testicles, Insemination of sperm into the female, Produce the fluids that render sperms motile Word bank: ovary, testicles, sex glands, cervix, scrotum, penis

Selecting and Breeding of Cattle

Selecting of Cattle for Breeding

Selection is the choosing cattle with the best characters from the herd for use in breeding, so that they pass these good characters to their progeny. **Culling** on the other hand, means removing of unproductive animals with undesirable characters from the hard or farm. **Classing** is the grouping of selected animals for culling into grades to achieve market specifications.

Advantages of selection

- It ensures that only the best cattle are selected for breeding.
- Concentrate the desirable characters in the herd.
- It reduces the spread of diseases in the herd.
- It ensures that animal with undesired characters are removed from the herd.

Table 1.4 Selection qualities for cattle

Dairy cattle	Beef cattle
- Wedge shaped appearance of the body - Lean neck - Large well-developed udder and teats - Cool dairy temperament - High longevity-productive life - High fertility - High milk and fat production - Low calving difficulty - High disease resistance - High feed efficiency-high food conversion - Good adaptability to local environment	- Blocky body conformation - Cool temperament - Disease and parasite resistance - Good adaptability to the local environmental conditions - High carcass percentage - High food conversion ability - Good mothering ability - Ease of calving - First growth rate and weight gain - Large body size

Methods of Selection of breeding stock

- Individual or mass selection:** an animal is selected based on its phenotypic traits.
- Pedigree selection:** pedigree is simply a record of ancestry. The ancestry records are used to predict those animals (descendants) that would transmit good traits to the future generations. Ancestry records include records of great grandparents, grandparents and present parents.
- Progeny testing:** this is the selection of the males using production records of its female offspring in certain traits that are sex-limited, for example milk production.
- Performance of other relatives (family selection):** the breeder selects an animal for breeding basing on the performance records of its relatives. This is because animals from the same family carry the same blood-lines or genetic traits.



Activity 1.9

Finding out the different selection systems used in cattle production

You will need Notebook, a pen, internet, agriculture text books

Key question: what are the different selection systems used in cattle production?

In groups:

- Using the internet or any other agriculture reference material, research and note down;
 - The meaning of selection as used in animal production.
 - The selection qualities for different types of cattle.
 - The methods used in selecting cattle on the farm.
- Using a slide presentation (power point presentation), present and discuss your findings with the rest of the class.

Breeding of Cattle

Breeding is the process of mating selected animals to produce offspring which have the desired qualities. **Inheritance:** is the genetic transmission of traits (or genes) from the parents to the offspring. These traits (genes) are carried in the male and female gametes.

Reasons for Breeding cattle

- To produce cattle that are resistant to parasites and diseases
- To produce cattle that have high food conversion ability
- To produce cattle that have high growth rate
- To produce animals that have large body size
- To produce animals that have high fecundity
- To produce animals that have high productivity (milk and meat)
- To produce animals that have high adaptability to local environment
- To produce animals that produce high quality milk and meat
- To produce cattle that have cool temperament for ease of handling
- To produce cattle that have appropriate body conformation, for example, wedge shape in dairy cattle and rectangular shape in beef cattle
- To produce cattle that have high good mothering ability



Polledness is a desirable condition in both dairy and beef cattle. How is this condition important to cattle farmers?



Selection, controlled breeding and culling lead to the improvement of animal's genetic potential. Other method of improving farm animals include; the introduction of better animals from other areas on the farm and improving the environment the animal lives and proper animal management (proper feeding, disease control, good housing).

Breeding Systems

Inbreeding; this is the mating of animals that are more closely related to each other i.e. mating animals that have one or more ancestors in common. Inbreeding simply reduces genes that are heterozygous and increases genes that are homozygous. It concentrates gene in the population that are of merit.

Advantages of Inbreeding

- It establishes uniform families in a population that are distinct from each other.
- Conserves the good genes in a population that can be used to upgrade other animals.
- It increases homozygosity in the herd.
- It is used to develop pure lines that are used in cross breeding to produce hybrids
- It increases the ability of the animal to produce offspring like itself.

Disadvantages of inbreeding

- It leads to loss of hybrid vigour.
- Can lead to emergence of recessive genetic defects originally hidden in the population.
- It reduces fertility in animals.
- Leads to high rate of prenatal mortality due to recessive genes.
- Poor genes can be concentrated leading to bad characters in the herd

Cross breeding; the mating of animals of different breeds. The result from crossbreeding is a **hybrid**. The superior performance inherited by a hybrid is called **hybrid vigour**.

Examples of hybrid vigour include; Good temperament, High adaptability to local environment, High food conversion ability, High growth rate, High productivity (milk and meat), High quality products, High resistance to parasites and diseases and Large body size	Advantages of cross breeding • Leads to hybrid vigour • Can be used to improve inferior breeds • Reduces the spread of inheritable diseases
--	--

Back crossing; this is where a crossbreed offspring is bred back to one of its parents. Back crossing aims at increasing the characters of one of the parents in the offspring.

Outcrossing; this is the mating of animals of the same breeds but of different origins. The aim is to bring in some new genetic variations often called 'new blood' into the herd and this is usually done by bringing in a new sire on the farm.

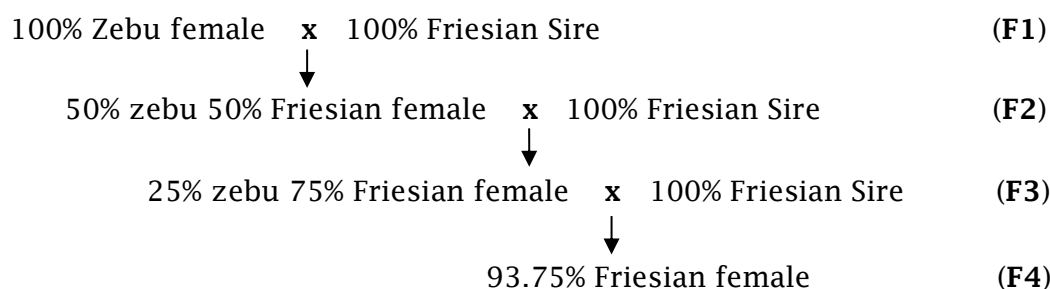
Close breeding; this is the breeding of very closely related animals. i.e. sib mating- between brothers and sisters and parent-sib mating between parents and offspring.

Line breeding; this is the mating of distantly related animals that share a common ancestor, e.g. cousins and cousins, granddaughters versus grandsires etc. the system aims at preserving good qualities of superior ancestors.

Grading up

This is also called top crossing. It usually involves crossing of a poor breed (but with some desired characters) with an improved breed (but with some lacking characters). A four generation cross is usually accepted as a purebred. Example of a grading-up programme involving two breeds: a Zebu cow and a Friesian bull. Remember Friesian has good traits for milk production but very low disease and pest resistance and very low environment adaptability. Zebu has poor milk traits but, has good environment adaptability and highly resistant especially to tropical pests and diseases.

A grading up programme for Zebu will therefore work this:



The F4 generation animal that is 93.75% Friesian and 6.25% zebu is acceptable as pure Friesian. Such pure breed should be with good adaptability to the environment, and resistant to pests and diseases and with good dairy traits.



Activity 1.10 Understanding the importance of cattle breeding

You will need a pen, notebook, reference textbooks and internet

Key question: what is the importance of controlled breeding of cattle?

In groups:

1. Research about the meaning of animal breeding and explain how selective breeding could improve the performance of a herd.
2. Discuss the different breeding methods used on the farm and the advantages and disadvantages of each.
3. Give the meaning of hybrid vigour and give examples of hybrid vigour in farm animals.
4. Each group should prepare a presentation to the rest of the class.

Methods of Mating (Breeding) Cattle

Natural Mating

This involves the use of a healthy proven bull with desired traits to directly serve the cow.

Advantages of natural mating

- It is less laborious.
- It increases chances of conception since the male detects the female on heat more easily than a farmer.
- It is an ideal method for animals that don't show clear signs of heat.
- It eliminates insemination costs.
- It increases farmer's certainty of the sire of the offspring.

Disadvantages of natural mating

- There is high chances of uncontrolled breeding.
- Aggressive and heavy bulls can injure a cow.
- One bull can only mate a limited number of cows within a given period of time.
- Semen is wasted since a lot of it is released by the bull into the female and yet only one sperm is required to fertilise the ovum.
- There is a risk of spread of diseases since there is physical contact and exchange of body fluids.
- It is expensive to keep a bull.

Artificial Insemination (AI)

This is a breeding method in which an instrument such as inseminating tube or syringe or pipette is used to introduce semen into the reproductive tract of a cow. The semen so used should be got from healthy and qualifying males in terms of meeting the breeding objectives. AI is one of the Assisted Reproductive technologies used in Uganda. Other technologies in this category include multi ovulation and embryo transfer and the use of sexed semen.

Semen collection techniques

1. *Use of artificial vagina:* in this method a bull is allowed to mount a teaser cow and the penis is held and directed into the artificial vagina where the bull ejaculates and deposits semen. The semen is then collected, examined, diluted and stored.

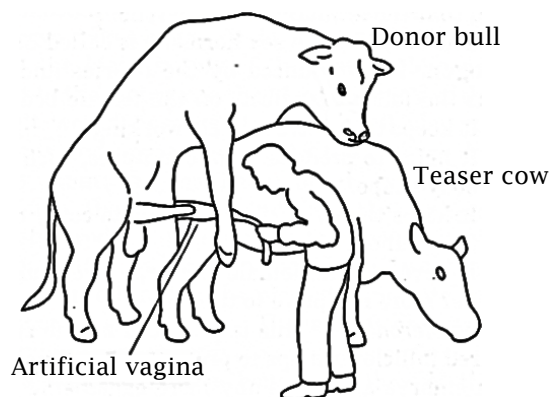


Fig 1.30 Collecting semen using AV.

2. *Massage of the ampullae:* gentle caressing of the genital parts will stimulate the sire to erect and ejaculate. The released semen is then collected and used to inseminate animals on the farm.
3. *Use of a breeder bag:* this resembles condoms used in human beings. The bag is attached to the penis or inserted in the vagina where the sire deposits semen.
4. *Recovery from the vagina:* the bull is allowed to mate a cow and deposited semen is recovered from the vagina using a gloved hand or vaginal spoon.
5. *Electro-ejaculator:* low voltage electrode is inserted in the bull's rectum to stimulate the nerves of the reproductive system causing the bull to ejaculate.

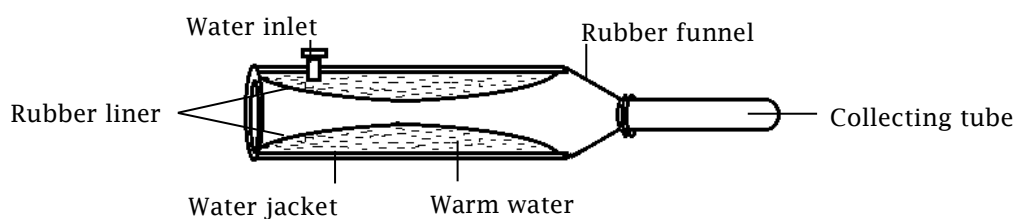


Fig 1.31 Artificial vagina used in collecting semen from bulls

Preparation of AV for collection of semen

- i. Cleaning the AV with clean water
- ii. Lubricating the inner rubber liner with non-spermicidal lubricant
- iii. Filling the water jacket with warm water 45°C to 50°C

Handling of semen after collection

After semen has been collected, it is taken to the laboratory for examination for quality; viscosity, motility, concentration, percentage of live spermatozoa, and freedom from deformities.

Then the semen is diluted with suitable diluent, semen is strawed and then frozen to -196° C

Limitations of artificial insemination

- i. Lack of storage facilities for the semen especially in rural areas.
- ii. Most local cattle do not show obvious signs of heat. This makes AI difficult.

- iii. Poor skill among many farmers in detecting heat and managing animals on heat.
- iv. Lack of enough good quality semen available in the semen bank.
- v. Conservativeness/unwillingness of farmers to castrate inferior bulls and use AI.
- vi. The scattered nature of most farmers makes it difficult as the inseminator cannot make quick contact with them when cows are on heat.
- vii. Inadequate extension staff to sensitize the farmers about the benefits of AI..
- viii. Lack of training facilities for locally-based inseminators.
- ix. Lack of demonstration centres for AI technology in most parts of the country.
- x. Poor transport network making it hard for inseminators to be in contact with farmers.
- xi. Communal grazing which encourages random mating hence limiting the use of AI.

Procedure of inseminating a cow

Insemination of the cow should be done within 6 to 12 hours following visible of signs of heat.

1. Restrain the animal on heat to be inseminated. Animal should be in a standing heat.
2. Wash and disinfect the hands
3. Put on gloves on one hand.
4. Lubricate the gloves with soapy water.
5. Thaw the semen; remove the semen straw from the refrigerator and put it in warm water at room temperature to activate the sperms.
6. Insert the semen straw into the inseminating instrument (a syringe or a catheter).
7. Insert one gloved hand into the rectum and locate the cervix. Hold the cervix through the elastic rectal wall.
8. Use the other hand to insert the inseminating instrument through the vagina and the cervix
9. Release the semen in the lower part of the uterus just in front of the cervix
10. Gently remove the inseminating instrument and then release the cervix and withdraw the gloved hand from the rectum.
11. Release the animal and isolate it from the males on the farm.

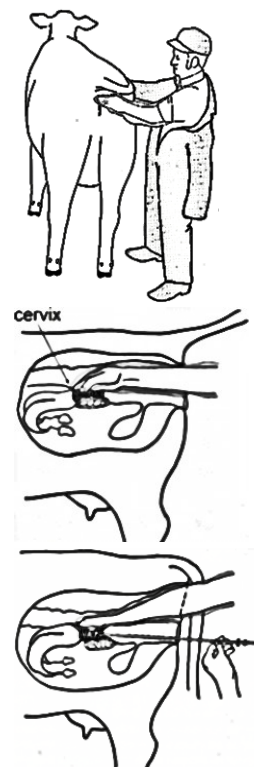


Fig 1.32 inseminating a cow

Note: cows coming on heat in the morning should be inseminated in the afternoon of the same day and those that come on heat in the afternoon should be inseminated in the morning of the next day.

Advantages of using artificial insemination to serve farm animals

- i. It increases chances of fertilisation since sperms are deposited at the lower part of the uterus where they can easily swim to the oviducts to fertilise the ovum.
- ii. It is easy to control inbreeding since no bull is kept for breeding on the farm.
- iii. It controls the transmission of breeding diseases such as contagious abortus.
- iv. Since bulls do not come in contact with cows, it eliminates chances of injuries to cows.
- v. It Eliminates expenses of keeping and maintaining a bull.
- vi. Farmers who may not afford to buy a superior bull can use artificial insemination to impart good qualities in the off-spring of their cows at an affordable cost.
- vii. Semen from one superior bull can be used to serve many cows.
- viii. Semen can be stored for a long time even after the death of a bull.
- ix. Many animals can be served from a single ejaculation.
- x. Risky and hostile bulls which may cause injuries to the farmer are excluded on the farm.
- xi. Semen of a good breed can be made available in remote areas.
- xii. Infertility or sterility can be easily detected. For example, should the animal come on heat many times without conceiving, it shows that either the cow or the bull is infertile.
- xiii. Heat synchronization can be used to plan or decide when the young ones should be born

Disadvantages of using artificial insemination to serve farm animals

- Checking for signs of heat and transferring the females from the farm to the station where insemination is done is tiresome and time consuming.
- Skilled labour is required in carrying out the insemination.
- Semen may die or its quality lowered due to storage and transportation problems.
- If the donor bull had harmful traits, they can be spread quickly to many offspring.
- There are low chances of conception if heat signs are not monitored accurately.
- Little will be achieved if inferior semen is used.
- It may not be successful if the females show silent heat.
- Lack of quality sperms and inseminators in rural farming areas.

Embryo Transfer

In this method eggs (ova) are harvested from a high quality female, fertilized in tubes and the embryos that develop are transferred into foster mothers where they grow until birth.

Advantages of embryo transfer

- Possible to implant embryo from a high quality female to a less superior female and obtain a high performing offspring.
- Increases reproduction rates of females i.e. increases the number of offspring per female.
- Allows the breeder to quickly multiply the genes of the top quality cows in the herd
- It is easier to transport embryos in test tubes than the whole animal.
- Embryos can be stored for long period awaiting availability of a recipient female.

Disadvantages of embryo transfer

- The technique is expensive; the high cost of embryo production and transfer
- Requires trained personnel to handle
- Requires special equipment for fertilization and storage of the embryos.



The embryo transfer process begins with injecting cows with a hormone to produce more ova at ovulation. The cow is then artificially inseminated, and after seven days, the embryos are then removed from the uterus using a catheter.



Activity 1.11 Finding out how artificial insemination is carried out in cattle

You will need

a pen, notebook, a video on collection of semen and artificial insemination, projector or TV, agriculture reference books and internet

Key question: how is semen collected from a bull and inseminated in a cow?

In groups:

- Watch a video on semen collection and note how the process is carried out and how the collected semen is prepared for use in artificial insemination.
- Note the steps being carried in artificial insemination and the restraints needed for a cow.
- Research about the advantages and disadvantages of AI over natural mating.
- Discuss the limitations of embryo transfer and artificial insemination.

Each group should prepare a presentation for the class.

Breeding Efficiency of a Herd

Breeding efficiency of a herd refers to the ability with which the herd is able to reproduce and multiply. Or, the number of offspring per breeding animal in a given unit of time.

Infertility is a temporally inability of the animal to reproduce.

Sterility is a permanent inability of the animal to reproduce.

Breeding efficiency is calculated using the formula; $B.E. \% = \frac{(n-1)365}{D} \times 100$

Where; B.E. % = Breeding efficiency in percentage

n = Number of total calvings

D = Total number of days from first to last calving

Calving interval refers to the period from one calving to the next calving. In cattle, the calving interval is one year.

Calving interval is calculated using the formula; $CI = \frac{\text{Number of months from previous to current calving}}{\text{Number of cows in calving}}$

Causes of low breeding efficiency in cattle

- i. Poor feeding of the animal affecting proper development of the uterus and ova and the growth of the foetus
- ii. Infection of the uterus preventing ovulation and implantation of embryo.
- iii. Serving the cows at very low or high age
- iv. Poor insemination techniques used
- v. Very low or very high environmental temperature which reduce sexual activities
- vi. Physiologic dysfunctions such as imbalance in the reproductive hormones, low number of functional ova, poor development of the uterus and irregular heat in cows.

How to maintain high breeding efficiency in the herd

- i. Good feeding of breeding animals on balanced diet and mineral supplements.
- ii. Giving breeding animals a rest period after calving to allow the reproductive system to repair
- iii. Serving the cows at the proper age i.e., not when too young or too old.
- iv. Serving animals at the right time; if AI is being used, cows should be served towards the middle and later stage of oestrus, as ovulation occurs 14 hour after oestrus.
- v. Control of diseases and providing good health care to breeding animals.
- vi. Using of well trained and skilled inseminator to carry out artificial insemination.
- vii. Keeping proper breeding records of all animals on the farm as it helps to know when each animal will go on heat and thus prepare for artificial insemination services in time.
- viii. Carrying out pregnancy diagnosis and giving good care to the pregnant animals.

Causes of infertility in farm animals

- i. The age of the animal: fertility increase with increase in the animal's age and later diminishes as the animal overages. This is due to the reproductive defects associated with old age.
- ii. Poor feeding of the animal: unbalanced diet results into poor growth of the reproductive system i.e. poorly developed ovary or defective uterus that makes reproduction difficult.
- iii. Hormonal imbalances within the animal's body.
- iv. White heifer disease i.e. presence of persistent hymen that prevent sperms from moving to the uterus to fertilise the eggs.
- v. Lethal genes within the dam and the bull.
- vi. Cystic ovaries that prevent ovulation to take place.
- vii. Cryptorchidism; a bull calf is born with both testes retained in the abdominal cavity.

Factors that may cause an animal to fail to become pregnant after being served

1. Serving an animal too early or too late during oestrus.
2. reproductive diseases
3. Cystic ovaries that prevent ovulation to take place.
4. White heifer disease i.e. persistent hymen prevents sperms from fertilising the egg.
5. Poor feeding of the animal prior to mating.
6. Using poor AI techniques that may lead to depositing semen in wrong position.
7. Presence of lethal genes that destroy the zygote soon after fertilisation.
8. Using unviable sperms during artificial insemination.
9. Presence of antibodies in the female that kill sperms.
10. Hormonal imbalance within the cow's body that prevent implantation of the zygote.



Activity 1.12

Finding out the meaning breeding efficiency and the causes of low breeding efficiency in cattle

You will need

a pen, notebook, reference textbooks and internet

Key question: what factors affect the breeding efficiency in cattle?

In groups:

1. Use the internet or any other agriculture reference book to research, discuss and write about the meaning of breeding efficiency, and the causes of low breeding efficiency in cattle.
2. Discuss the difference between infertility and sterility in cattle.
3. Suggest reasons why a cow may fail to become pregnant after being served.

Each group should prepare a presentation for the class.

Pregnancy in Cows

Pregnancy refers to a condition where the cow has developing embryo in her uterus. The period between conception and birth is called *gestation period*. The average gestation period of cows is 282 days or about 9 months.

Signs of pregnancy in a cow

- i. Absence of oestrus (the cow does not go again on heat after being served).
- ii. Increase in the size of the belly.
- iii. Development of the udder tissue especially during the late pregnancy. This is visible in heifers or cow becoming pregnant for the first time.
- iv. High levels of progesterone hormone in the blood, body fluids and milk.
- v. Enlargement of the uterine horn.
- vi. Enlargement of uterine arteries.
- vii. Presence of the foetus in the abdomen as indicated by movements and feel.
- viii. Slight increase in the body temperature and pulse rate.
- ix. Increase in the body weight.
- x. Presence of the sticky and thick mucus sealing the cervix.

Pregnancy Diagnosis

This involves checking for pregnancy in the cow after being served. The knowledge of whether the cow is pregnant or not is important to the farmer in the following ways:

- i. It helps in feeding program; pregnant cows require extra feeding to compensate for the increased demands by the foetus.
- ii. It allows the farmer to reorganize for the next servicing of the cow especially if the cow did not become pregnant in the first service.
- iii. It allows the farmer to study and monitor the progress in the developing embryo, and in case of a problem to seek veterinary help.
- iv. It facilitates record keeping and preparation for calving and post calving care.
- v. Helps when selecting animals for culling. Cows which do not get pregnant after consecutive servicing are regarded as infertile and such cows are uneconomical to keep on the farm.
- vi. Helps in selection of more fertile animals.

Methods used in pregnancy diagnosis

1. **Using the visual signs of pregnancy** such as enlargement of the belly, failure of the animal to go again on heat after being served etc.
2. **Palpation per rectum:** the hand is inserted into the rectum to feel the uterine horns. Sign indicating pregnancy may include: enlargement of the uterine horn, enlargement uterine arteries and presence of the foetus.
3. **Use of x-ray studies:** X-ray picture is used to effectively establish pregnancy especially after the foetal bones have begun to form and calcify.
4. **Biologic tests:** biologic tests are based on the presence of the progesterone. The increase in the level of progesterone levels in the urine or blood indicates pregnancy.



Activity 1.13

Finding out the signs of pregnancy and its diagnosis in cows

You will need

a pen, notebook, reference textbooks and internet

Key question: what are the signs of pregnancy and the methods of diagnosis of pregnancy in cows?

In groups:

1. Use the internet or any other agriculture reference materials to research about the signs of pregnancy in cows.
2. Discuss the methods used to diagnose pregnancy in cows and give reasons why it is important for a farmer to know the pregnancy status of his animals on the farm.

Each group should prepare a presentation for the class.

Caring and Management of Pregnant Cows

- i. The farmer should first carry out pregnancy test to establish whether the cow is pregnant.
- ii. Proper feeding of the animal on balanced diet to cater for the increased nutrient needs for the mother and the foetus.
- iii. Vaccination and protecting cows from infections and immediate treatment of sick cows.
- iv. Isolation of the animal from the males as these may continue chasing the pregnant animal resulting into stress and abortion.
- v. Regular parasite controlling by spraying and controlled dipping and deworming.
- vi. Keep the animal away from severe weather such as strong rainfall and sunshine.
- vii. Give the animal enough rest and avoid overworking of the animal.
- viii. Give the animal plenty of clean drinking water regularly.
- ix. At 7th month of pregnancy, it should be dried off (stop being milked if it was milking) and its feeding improved. This allows more nutrients to go to the rapidly developing embryo.
- x. Towards the end of pregnancy the animal should be steamed up by giving it extra-nutritious feeds to provide more nutrients for the developing embryo.
- xi. Towards the end of pregnancy the animal should be isolated and placed in the delivery paddock or pen, laid on floor with soft absorbent dry plant material.
- xii. For heifers, at 7 months of gestation, they should be introduced to the milking practice.

Drying off and Steaming up of pregnant animals

Drying off

This is a practice of stopping a lactating animal from producing milk in preparation for the next birth and lactation. It is mostly done when the animal is left with about two months to give birth. Drying off in cows starts at 7 months and in goats at 3 to 3½ months of pregnancy.

Reasons for drying off pregnant lactating animals

- i. To allow the animal to replace lost nutrients in milk before the next lactation.
- ii. To allow the udder tissues to repair and rebuild before the next lactation.
- iii. To allow the animal to gain some weight before the next lactation.
- iv. To allow the cow to replenish mineral reserves mainly calcium, phosphorus and magnesium.
- v. To allow more nutrients to be directed to the fast growing and more demanding embryo.
- vi. To increase milk yield in the next lactation.
- vii. To give a chance for treatment of the udder infections such as mastitis and sores.

Methods of drying off

Drying-off occurs as the milk pressure in the udder equals blood pressure, thus stopping the production of milk. There are several systems of drying off farm animals. In all systems, it is helpful to reduce the feed and water intake during the drying-off process. It is also helpful to treat for mastitis immediately after the last milking.

Dry cow therapy: this involves use of milk depressants to stop milk synthesis and secretion, and antibiotics to stop subsequent infections and mastitis.

Incomplete milking: in this method, the animal is milked halfway for two to four days after which milking is stopped completely.

Intermittent milking: the animal is milked on alternate days for 3 to 5 days after which, milking is stopped completely.

Secession: in this method, milking is stopped completely when time for drying off is reached.

Care of a dry cow

- i. Immediately after drying-off the cow, move it to the dry-cow housing area.
- ii. Reduce on the amount of feeds and water given to the cow for about one week
- iii. Give mastitis treatment to the dry cow.
- iv. Protect the dry-cow from bad weather.
- v. Observe the cow for signs of mastitis, indicated by udder quarter becoming swollen or “hot”.
- vi. Feed the dry cow on extra and nutritious feeds. a practice known as “steaming up”
- vii. Treat the dry-cow against worms.
- viii. Ensure good sanitation around the cow.
- ix. As the time for calving approaches, the cow should be moved to maternity pen, stall, or area.

Steaming Up

This is the provision of extra feeds of high nutritive value to an animal during the last weeks of gestation. The animal is provided with good pasture or forage and given more concentrates as supplement feeds. If the animal is lactating, it should first be dried off first to avoid draining of the nutrient in milk.

Advantages of steaming up

- It provides the extra nutrients needed by the developing embryo.
- It ensures that the animal builds up body reserves of nutrients needed for milk production.
- Prevents nutritional disorders associated with lactation e.g. milk fever, grass tetany, ketosis.
- Stimulates the development of the udder tissues leading to high milk production.
- Since concentrates are intentionally provided in the milking parlour, it helps to accustom the heifers to the milking parlour to stop disturbance during milking.
- It reduces the possibility of pregnancy toxaemia and twin lamb disease in sheep.
- It makes the animal fit physiologically for the next milk secretion.
- In poorly grown heifers, it provides extra nutrients for their full growth and development.
- To ensure birth of healthy, vigorous young ones
- To promote good health of mother
- It helps the animal to build reserves for energy used during birth.



Activity 1.14

Understanding the care for the pregnant cows on the farm

You will need

a pen, notebook, reference textbooks and internet

Key question: what care should be given to a pregnant cow?

In groups:

- Research about the care that should be given to pregnant cows on the farm.
- Discuss the meaning of the following terms; drying off and steaming up, and describe the methods used to dry off lactating-pregnant cows on the farm.
- Give the benefits of drying off and steaming up of incalf cows

Each group should prepare a presentation for the class.

Activity of Integration 	You are a concerned citizen. There is unplanned roundom mating of cattle in your community. You are worried that the practice could lead to a generation of less productive cattle in the area.	
Task	Write a letter to be read at the village council meeting explaining the dangers of unplanned mating of cattle, the criteria they can use to select breeding cattle and the benefit associated with using artificial insemination.	

Exercise 1.4 	Self-Check Questions As an animals scientist with good knowledge of animal breeding, (a) Define selection as used in livestock breeding (b) Give four example of hybrid vigour in dairy cattle (a) Describe the different systems of breeding cattle on the farm (b) Give two advantage and disadvantage of cross breeding (a) Outline the steps involved in handling semen after collection (b) Describe how you can inseminate a cow using frozen semen (a) Define the term breeding efficiency and outline the causes of low breeding efficiency in cattle (b) Suggest measures that can be put into place to improve the breeding efficiency of the herd. (a) Outline the signs of pregnancy in cows. (b) Explain why a cow may fail to become pregnant after being served (c) What care should be given to pregnant cows on the farm?
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Nutritional Requirements of Cattle

Cattle, like other farm animals, need food to grow, maintain the body, reproduce and to produce milk. The dietary needs of cattle vary with age, breed, stage of production, performance level, and weight. Health and weather, also impact on the nutritional requirements. Cattle need water, protein, carbohydrates, fats, minerals, and vitamins. water is needed in the greatest amounts daily and is served to cattle ad libtum (provided all the time).

Nutritional requirements of cattle at different stages

Cattle have distinctly different nutritional needs depending on the following; level of production, age, body size and gestation cycle. Thus, the four primary stages that affect nutritional requirement are;

Growth; this depends on the rate of increase in body mass. At this stage, protein requirement is higher because the young cattle are gaining more muscle than fat. When cattle need to gain weight to increase their body condition score, this is also considered growth.

Maintenance; this is basic energy the animal needs to do things like breathe, digest feed and maintain body weight. The larger the animal, the greater it's maintenance requirement, especially energy and protein.

Reproduction; early on in a pregnancy, a cow's nutritional needs are low. As the pregnancy advances, so does its nutritional needs. Thus, the nutritional needs will depend on the expected calf birth weight and the stage of gestation, especially the last three months of pregnancy when the foetus is growing rapidly.

Lactation; this is the time between birth and weaning, when the cow is producing milk. Nutrient requirement are based on the amount of milk at peak lactation and the composition of the milk. Thus cows producing more milk with more fat and protein, will have more nutrient requirements.

Table 1.6 Feeding requirements for cattle

Age	1 day - 4 weeks	4 - 8 weeks	8 - 13 weeks	4 - 15 months	Lactating cows
Fluid intake	Colostrum 2-2.5L daily for 3 days Then, 2.5L milk Water ad lib	2.0L milk Water ad lib	Water ad lib	Water ad lib	Water ad lib
Concentrates	50g calf starter Calf pellets	100-250g calf starter Calf pellets	500-750g calf starter	1.500-2.500g Grain mixtures	2.500-3000g Grain mixtures
Forage	Small quantity of hay or green grass	Small quantity of hay or green grass	2-5 kg green grass, 1-2kg hay	5-20kg green grass, 6-7kg hay	20-30 kg green grass, 4-5kg hay
Salt lick	Not fed	Not fed	Provide ad lib	Provide ad lib	Provide ad lib

Balanced Ration

A balanced ration is the amount of feed that will supply the proper amounts and proportions of nutrients needed for an animal to perform a specific purpose such as growth, maintenance, lactation or gestation. A balanced ration should provide protein, energy, minerals and vitamins from dry fodders, green fodder, concentrates, mineral and vitamin supplements in appropriate quantities to enable the animal to perform optimally and remain healthy.

Locally available feedstuffs for feeding cattle

Cattle have well developed digestive system allowing them to effectively utilise low quality forages and feed stuffs. A wide range of locally-available foodstuffs are available for feeding cattle.

1. Concentrates are classified as protein, energy (basal) and mineral concentrates.
 - *Protein concentrates* e.g. fish meal, soybean meal, blood meal, cotton seed cake, sunflower seed meal
 - *Energy concentrates* e.g. maize bran, wheat bran, rice bran, cassava chips, sorghum meal, dried distillers grain
 - *Mineral concentrates* e.g. crushed oyster shells, sea shells, crushed bones, crushed rock phosphate and limestone.

These concentrates can be mixed in varying proportions to make a balance ration for cattle.

2. Roughages are feedstuffs with high crude fibre and low energy and protein content. They are mainly of plant origin. They include; hay, silage, fodder, pasture and crop residues.

Roughages contain indigestible dietary fibres which stimulate the churning of food in the rumen and aid the passage of food and waste products through the gut. It also absorbs water as it moves through the colon allowing the faecal matter to be released easily.



Roughages contain less nutrients but add bulk to animal feed to provide satiety. This reduces the amount of concentrates the animal eats thus reducing bloating.



Activity 1.15 Understanding the importance of good nutrition in rearing of cattle

You will need a pen, notebook, reference textbooks and internet

Key question: why is good nutrition important in the rearing of cattle?

In groups:

1. Using the internet or agriculture reference materials, research about the food sources and the role of the following nutrients in the diet of cattle
(i) proteins (ii) calcium (iii) water (iv) vitamins
2. Explain the following terms as used in animals nutrition; balanced ration, maintenance ration, production ration and steaming up
3. Discuss the nutritional requirements of cattle at growing, lactation and maintenance stages.
4. Prepare a balanced ration required by dairy cattle from locally available foodstuff.

Present your findings to the rest of the class.

Making Efficient Use of High-Quality Roughage in Rearing Cattle

Pasture growth and surply to cattle is usually high during rainy season. During this period, the pasture is in plenty and canot all be grazed by cattle. The excess pasture can be harvested and conserved by drying (hay) or fermentation (silage). The forage can also be processed into pellets and cubes and used for feeding cattle during dry periods. It is important that the forage be utilized at the proper stage of growth i.e, at flowering stages. During this stage, the nutrient content of the plants is high, dry matter content and digestibility are fairly high.

Crop residues are palatable and provide high quality roughages on the farm. Maize stubbles, potato vines, cereal stalks, banana psuedostems and leaves can be chopped and used for feeding cattle. Roughages are important in the diet of cattle as it provides a suitable environment for survival and action of the rumen microbes, helps in proper functioning of the rumen by providing regurgitating material and prevent bloating and provides bulk to the feed and satiety to the animal.



Activity 1.16 Understanding how high-quality roughage is prepared

You will need a pen, notebook, Napier or Guatemala grass, panga or chaff cutter, rock salt, protective gears, and internet, wheel burrow and a feeding trough

Key question: how to prepare high-quality roughage for feeding cattle?

In groups:

1. Assemble all the equipment and tools necessary
2. Put on protective gears
3. Select the grass and legume just at flowering stage. Why should forage be at flowering stage?
4. Harvest the forage (50% grass and 50% legume) and remove any non palatable plants.
5. Chop the forage into small pieces of about 5 to 10cm. Why is this important?
6. Dissolve rock salt in water and sprinkle the salt water on the forage. Why is this important?
7. Serve the chopped fodder in a clean feed trough to the cattle or you can conserve it for future use. What are the methods used to conserve forage?
8. Clean the tools and equipment, remove the protective gears and wash up.

Make a typed document indicating how to prepare a good quality roughage for feeding cattle.

Activity of Integration 	It is reported in the local news papers that more than 5000 heads of cattle have died due to draught that has hit Ntoroko district for the last few months. The district receives plenty of rainfall during the season and there is usually plenty of pasture. Pastoralism is the main economic activity providing livelihood to most households	
Task	As an agriculture student, write a document to be read at the district council meeting explaining how the situation can be prevented from happening again	

Exercise 1.5 Self-Check Questions	
	As an animal nutritionist, suggest (a) Four most important nutrients for cattle. (b) Four factors that determine the nutritional requirement of cattle (c) The value of roughages in the diety of ruminants. (d) Two examples of feedstuff with high roughage content.

Metabolic Disorders Associated with Feeding in Cattle

Disorders in livestock caused by feeding practices in which the normal metabolic balance and functioning is disrupted include;

Milk fever (Hypocalcaemia)

Milk fever is as a result of lack of calcium in lactating animals. It occurs in high yielding dairy cows one or two days before or after calving. The disease is common in older cows and high yielding cows.

Cause of Milk fever; inadequate supply of calcium, and poor calcium-phosphorus ratio in the animal's body.

Symptoms of milk fever in a cow

- The first sign is loss of appetite and severe depression.
- There is slight drop in body temperature (below 38°C).
- The cow is unable to stand and lies down, often with the head resting on the shoulder.
- There is hyper excitability and muscular tremours.
- The animal becomes paralysed with the hind legs stretched out behind.
- The rumen stops functioning and bloat sets in.
- If untreated the cow becomes comatose and dies within 1 day after the first signs.

Treatment; slow intravenous infusion or sub-cutaneous injection of 600-800 ml of 20% calcium borogluconate to raise blood calcium level.

Prevention

- Feeding the cow on a diet containing high supply of calcium and phosphorus in the ratio of 1 : 1 during steaming up. This ensures concurrent absorption of Ca and P, and high calcium reserves and supply in the animal's body before and after calving.
- Drying-off the cow two months before calving to reduce calcium depletion through milk and to allow more calcium reserve in the animal's body.
- Delayed or incomplete milking in the early stages of lactation to prevent calcium depletion from the animal's body.
- Feeding the cow on feeds rich in vitamin D to increase calcium absorption in the gastrointestinal tract.
- Proper selection of cows which are less susceptible to milk fever. This is because some breeds and some animals within breeds are less susceptible to milk fever.

Grass Tetany (Hypomagnesaemia)

This is also known as **grass staggers** or **milk tetany** in calves. It occurs most commonly in high yielding dairy cows at peak lactation that are grazing on lush grass.

Cause of grass tetany; deficiency of magnesium in the circulatory system.

Symptoms of grass tetany in a cow

- The cow is unsteady and sways as it walks.
- The cow becomes nervous and muscle tremours around the head, shoulders and flanks.
- The animal fall over and paddle its legs frantically.
- Over salivation and frothing of the mouth.
- Increased and strong heart beat and respiration.
- If left untreated, the cow dies within 2 to 3 days.

Treatment; Sub-cutaneous injection of 200 ml of 50% magnesium sulphate, with calcium borogluconate included in the treatment. Treatment must be administered as quietly as possible as the handling and injecting procedure can aggravate the symptoms.

Prevention

- Feeding the cow on diet rich in magnesium.
- Delayed or incomplete milking in the early stages of lactation to prevent magnesium depletion from the circulation system.
- Controlled use of nitrate and potash fertilizers in the pasture as too much supply of potassium inhibits magnesium absorption by pasture plants.
- Avoid grazing cows on lush grass or if animals are grazing on fast-growing lush grass, provide feed supplements rich in magnesium.

Ketosis (Pregnancy Toxaemia or Acetonaemia)

This occurs in high yielding cows within a few weeks of calving, usually in housed cows fed mainly on conserved fodder.

Cause; Ketosis is caused by low glucose supply in the circulating blood.

Symptoms of ketosis in a cow	Treatment;	Control
- Sudden drop in appetite of the cow. - Rapid loss of body condition (emaciation). - Drop in milk yield. - The animal's breath has a sickly sweet smell of acetone. - Milk tests positive to acetone. - Nervous signs such as salivation, chewing, teeth grinding, staggering, walking in circles, blindness and aggression. - Mild muscle twitching. - Faeces become hard covered in mucus. - Coma and death follow within a few days if treatment is not provided in time.	Intravenous infusion of 500 ml of 40% glucose, plus twice daily dosing of 150 ml of propylene glycol (a glucose precursor) for 4 days. Give corticosteroids to reduce ketone production and to increase blood glucose.	- Adequate feeding with feeds rich in roughage and carbohydrates. - Providing housed animals with fresh green fodder regularly. - Providing the cow with adequate exercise to stimulate digestion and metabolic processes.

Bloat (Tymphanitis or Hoven)

This is a distention of the rumen with gases produced by fermentation of food by microbes in the rumen. In ruminants, the left side of the abdomen behind the ribs becomes swollen.

Cause; during fermentation of food in the rumen, large quantities of gases, namely CO₂ and methane are produced. These gases are normally removed by eructation (belching). If for any reason this gas doesn't escape, the rumen swells up with gases and causes bloat.

There are two types of bloat; gassy bloat and frothy bloat.

Gassy bloat: occurs due to obstruction of the oesophagus that prevents escape of gases from the rumen by eructation (belching). The oesophagus may be choked by a hard lump of food or rumen liquid may cover the end of the oesophagus preventing escape of gases.

Frothy bloat: this occurs due to foaming in the rumen and gases are trapped in tiny bubbles which cannot escape in the normal way.

Predisposing factors

- Sudden change of animals on diet of fresh young pastures. These are rapidly fermented, producing large quantities of gases and froth in the rumen.

- ii. Feeding animal on much succulent feeds e.g. sweet potatoes, cassava peelings and fruits; these tend to ferment rapidly and produce froth during churning of food in the rumen.
- iii. Feeding animals on a sole ration of grains or concentrates: these tend to ferment rapidly in the rumen, producing large quantities of gases.
- iv. Infection with diseases: some diseases such as rabies and tetanus cause failure of the animal to eructate hence causing gassy bloat.
- v. Blockage of the oesophagus by lumps of food or other objects swallowed by the animal.

Symptoms of bloat

- i. The left side of the abdomen behind the ribs becomes swollen.
- ii. Lack of appetite due to distress and fullness of the rumen.
- iii. Difficulty in breathing due to pressure exerted by the swollen rumen on the diaphragm and lungs.
- iv. The animal stands with its legs spread wide apart and its head extended forward.
- v. A cow lies down and fails to stand up due to distress.
- vi. If left untreated, death occurs in a short time due to excessive pressure on the diaphragm and the heart.

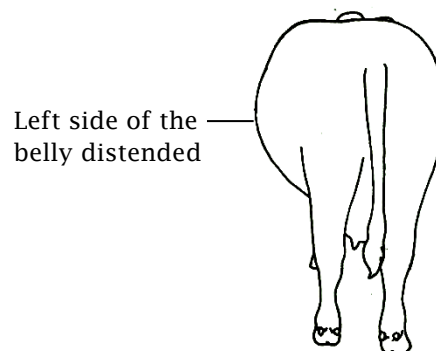


Fig 1.33 Bloated animal

Treatment

- i. Affected animal should be prevented access to the food causing bloat and drenched with an antifoaming agent such as vegetable oil to break down the froth in the rumen.
- ii. Use of stomach tubing; the tube is passed through the mouth into the rumen to allow the gas to escape from the rumen. The tube can also help to push any obstruction in the oesophagus into the rumen so that eructation is possible. Also antifoaming agents can be poured directly into the rumen via the tube.
- iii. Use of trocar and cannula; used to puncture the rumen to allow gases to escape from the rumen. Also antifoaming agents can be poured directly into the rumen via the cannula.
- iv. Use of broom stick method; in this method a wooden rod is inserted across the mouth to keep it open so as to allow eructation and easy escape of gas via the mouth.
- v. Exercising the animal to stimulate eructation; the animal is haltered and moved around the farm to agitate the rumen contents and to allow escape of entrapped gas.

Prevention

- i. Providing daily feeding of hay to livestock grazing on lush grass or grain concentrates.
- ii. Susceptible animals should be regularly drenched with ant-bloat drugs.
- iii. If animals are to be changed to a diet of fresh green vegetation, e.g. young improved pasture or fresh legume, it should be made gradually over several days.
- iv. Avoid feeding animals on young pasture or highly succulent feeds.
- v. Avoid giving animals excess grains and protein-rich concentrates.

Procedure of using the trocar and cannula to treat bloat

1. Prepare all the required equipment and drugs; restraints, disinfectants, antibiotics and trocar and cannula. The cannula should be cleaned and disinfected.
2. Restrain the animal.
3. Lift the front feet to relieve pressure on the diaphragm and the heart.
4. Disinfect and then insert the trocar into the cannula
5. Disinfect the site to be punctured
6. Place the sharp end of trocar on the left side of the belly near the last rib.
7. Strike the trocar sharply on the belly to puncture the stomach
8. Hold the cannula in place and withdraw the trocar to permit gas to escape
9. Then remove the cannula
10. Treat the wound with antibiotics and release the animal. Monitor to ensure the animal is ok.

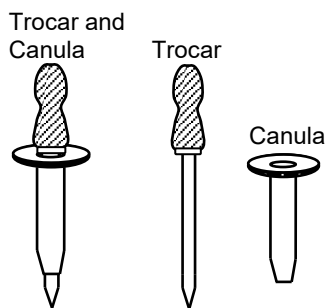


Fig 1.34 Trocar and cannula

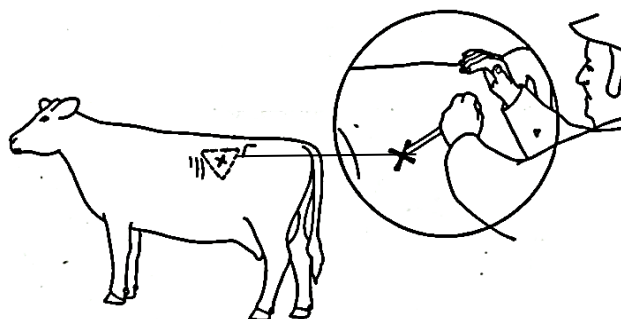


Fig 1.35 using trocar and cannula.

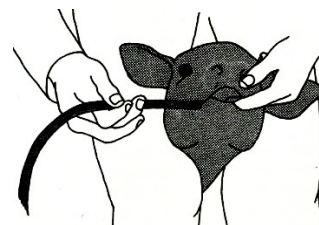


Fig 1.36 Stomach tubing



Activity 1.17 Finding out the metabolic disorders associated with cattle feeding

You will need a pen, notebook, reference textbooks and internet

Key question: what are the metabolic disorders associated with feeding cattle?

In groups:

Use the internet or any agriculture reference materials to research the on the metabolic disorders associated with cattle feeding and in each, discuss the causes, symptoms and control measures. Type your findings and present them to the rest of the class and record conclusions.



Diseases and Parasites Control in Cattle

Parasites and diseases negatively affect the health and welfare of cattle, and they pose a heavy burden on communities that rely on cattle production for their livelihood.

Disease Control and Prevention in Cattle

Cattle, just like other farm animals are affected by many diseases. Some of these diseases are caused by pathogens such as bacteria, viruses, protozoa and fungi. Others are as a result of poor nutrition and others as a result of distress from the environment the animal lives in.

Effects of diseases on the farm

- i. Some are transmissible between cattle and man and can lead to death of farm workers.
- ii. They lead to death of animals on the farm there by causing losses to the farmer.
- iii. They lead to loss of weight of the animal's body. This lowers the sale value of cattle.
- iv. Leads to decrease in milk production in dairy cattle.
- v. Leads to fall in quality of cattle products. This results into rejection of milk and meat and their products in the market.
- vi. Some diseases may cause reduction in the fertility of cattle.
- vii. Diseases lead to reduction in the growth rate of cattle.
- viii. Diseases increase the cost of production especially buying drugs and hiring vet doctors.
- ix. Some infectious diseases such as foot and mouth or anthrax may lead to instituting of quarantine on the farm or even closure of the entire farm by authorities.
- x. Diseases reduce the efficiency of feeding cattle (feeding more for less).

Importance of keeping cattle healthy

The following are the reasons why cattle should be kept healthy:

- i. Healthy cattle grow faster and reach maturity early.
- ii. Healthy cattle give high-quality milk and meat
- iii. It helps to keep the cost of production low by saving on the cost of treating diseases.
- vi. It controls the spread of diseases to other livestock and human beings.
- vii. It enables cattle to live longer and live more productive lives.
- viii. Healthy animals produce more yields.
- ix. Good health enables livestock to breed more regularly.

- x. Health cattle produce vigorous and healthy calves.
- xi. Healthy cattle are attractive to buyers and fetch more prices in the market.

Table 1.6 Signs of good and ill-health in cattle.

Signs of good health in cattle	Signs of ill-health in cattle
- The cow responds readily to stimuli - The cow moves freely in a normal posture without limping - The cow has normal body temperature - The cow has good appetite and eats and drinks freely - The cow produces normal yield of milk - The animal has normal pulse rate - The animal produces faeces of normal consistence; not too hard or too watery. - Eyes are bright and alert and no discharges - The ears are erect and active against flies - The rumination process is regular and the animal regularly chews the cud - No discharges from the nose, anus and vulva. - The skin coat is clean, smooth and shiny - The cow breathes smoothly and regularly at rest.	- Lack of appetite; the cow fails to eat/drink - High or low body temperature. - Irregular breathing, coughing - Severe decline in milk production. - Abnormal discharges of fluids from the natural openings on the body such as the eyes, ears, anus and mouth. - Loss of weight or emaciation. - Increased or reduced pulse rate. - Inconsistency in the texture, colour and smell of an animal's faeces and urine. - Rough coat, loss of hair and abnormal outgrowths or swelling on the skin. - Bright red, pale mucus membranes. - Dullness, listlessness and restlessness. - Aggression or making of abnormal sounds. - Lameness or difficulty in standing/walking - Diarrhoea/constipation/blood stained dung

Reasons why cattle diseases are common in most parts of Uganda

- i. Ideal climate that favours existence and multiplication of diseases and vectors.
- ii. Increased uncontrolled movement of cattle from one area to another.
- iii. Poor farming methods such nomadic herding and communal grazing which encourage the spread of diseases.
- iv. Prolonged draughts which stress cattle reducing their resistance to infections.
- v. Lack of effective drugs in the market for treating animal diseases effectively.
- vi. Inadequate extension services to advise farmers on effective disease control measures.
- vii. Existence of many wild animals which are carriers and reservoir for most cattle diseases.
- viii. Weak laws on the importation and movement of cattle and cattle products in the region. Thus cattle and cattle products carrying diseases are easily introduced into the region.

Predisposing factors of cattle to diseases

Predisposing factors are conditions inside or outside the body of an animal that make the animal vulnerable to disease or injury. They include the following:

Genetic factors or breed of the cow: East Coast Fever for instance, causes severe symptoms or even death to exotic cattle and mild symptoms to indigenous cattle.

Age of cattle: some diseases attack cattle at a certain age. Calf scours, for example, attacks calves while mastitis attacks older lactating cows.

Level of production: high-milking cows, for example, are prone to milk fever.

Sex of the animal: some diseases are sex-linked. For example, contagious abortus, and mastitis which attack cows and orchitis which attacks bulls.

Hygiene of the environment the animal lives in: unhygienic conditions are a source of infections such as coccidiosis and calf scours.

Health of the animal: sick animal have weak immunity and can easily get other infections.

Injuries: An injury on a teat of a cow, for example, can lead to mastitis.

Level of nutrition: good feeding of cattle increases the animal's resistance to infections. However, poor feeding reduces immunity and predisposes livestock to deficiency diseases.

Physiological condition of the animals e.g. pregnant or lactating.

Prevalence of disease vectors around the animal,

Overcrowding of animals leading to easy spread of infectious diseases,



Activity 1.13 Finding out the signs of ill-health in cattle

You will need a pen, notebook, reference textbooks and internet

Key question: what are the signs of ill-health in cattle?

In groups:

1. Use the internet or any agriculture reference materials to research and note the following
 - (a) the signs of good health and ill-health in cattle.
 - (b) the reasons why it is important to keep cattle healthy on the farm.
2. Why are cattle diseases so common in Uganda.
3. What factors predispose farm animals to infection?

Present your finding to the rest of the class and record conclusions.

Descriptions of Animal Diseases

Communicable diseases: these are diseases which spread from one animal to another by direct or indirect contact. e.g. foot and mouth disease, bird flu.

Notifiable diseases: these are highly infectious and contagious diseases that in case of outbreak of such diseases on the farm, they should be reported to the authorities such as veterinary officers or police, for appropriate and effective control measure.

Zoonotic diseases: these are disease which can be transmitted from animal to man or from man to animals. e.g. tuberculosis, anthrax, brucellosis.

Acute diseases: these are diseases with sudden onset, severe symptoms and can result into death in very short time. e.g. blue tongue, foot and mouth disease, anthrax.

Epidemic (also called epizootic) diseases: these are diseases which spread very fast in a large area and affect large populations of animals. e.g. rinderpest, foot and mouth disease.

Indicators of disease outbreak on the farm

The following brief descriptions outline the signs of diseases outbreak in a herd;

- i. Abnormal levels of deaths occurring in herd on the farm.
- ii. Sudden drop in the level of milk production.
- iii. Progressive emaciation of most animals on the farm.
- iv. Failure of the animal to eat or reduced feed intake.
- v. Skin lesions (lumps/nodules on or under the skin), scabs, oedema and skin thickening.
- vi. Discharges from the mucosal orifices of the head (eyes, nostrils and mouth).
- vii. Severe/persistent diarrhoea with or without blood stains.
- viii. Infertility and abortion in adult females.
- ix. Change in behaviour of animals due to nervous disorders.

Ways through which diseases spread on the farm

- Introduction of diseased animals from other farms
- By indirect contact, through feeding troughs.
- By direct contact with the sick animal
- Through contaminated food fed to health animals
- Through vectors such as ticks, tsetse flies, vampire bats and lice.
- Through contact with infected dead animals.
- Rodents and birds moving from one farm to another or from one animal to another.
- Visitors from infected areas visiting the farm
- Through contaminated litter brought on the farm.
- Through mating; a bull carrying a disease can spread it to cows during mating.
- Through use of contaminated equipment such as injection needles and surgical blades.
- Vertical transmission from the mother to the calf during parturition.
- Through the wind especially for air borne diseases such as tuberculosis.

Management of sick animals on the farm

- Isolate the animal from the herd; this ensures special treatment for the animal and prevents the spread of the disease to other animals in the herd.
- Provide shelter to protect the animal from bad weather such as storm and sunshine.
- Provide adequate treatment to the animal. This may require hiring a veterinarian.
- Provide good feeding to the animal depending on its nutritional requirements.
- Provide adequate clean water to the animal
- Provide vitamin supplements to boost the animal's immune system and to speed up the healing process.
- Provide good hygienic conditions around the animal to stop secondary infections.
- Keep proper records of the sick animal in relation to the type of the disease, type of treatment given, dosage, and progress



Disease-causing organisms can enter an animal's body through the mouth, nose, eyes, anus, ears, vagina, penis, umbilical cord, teat canal, wounds and vector bite



Activity 1.14

Describing the way cattle diseases spread on the farm and the indicators of disease outbreak on the farm

You will need

a pen, notebook, reference textbooks and internet

Key question: how do cattle diseases spread on the farm?

In groups:

1. Using the internet or agriculture textbooks, research on and discuss the mechanisms through which animal diseases spread on the farm. List the indicators of disease outbreak in a herd
2. Give the meaning of the following terms; zoonoses and a Notifiable disease?
3. Suggest the care that should be given to sick animal on the farm.

Present your finding to the rest of the class and record conclusions.

Common Diseases of Cattle

Mastitis

This is the inflammation of the mammary glands (udder) due to infection with bacteria especially *Staphylococcus aureus*, *Streptococcus agalactiae* and *Corynebacterium pyogenes*.

Predisposing factors of animals to mastitis

- i. Poor hygiene on the farm: poor hygiene increases the risk of contamination of the teat cups, or the hands from the environment or from the milk of other milking cows.
- ii. Cut or sores on the teats and the udder: these makes it easy for the microorganisms to invade the teats and the udder.
- iii. Improper use of milking machines; too high vacuum may damage the udder tissues thus making it prone to infection with microorganisms.
- iv. Faulty milking machines: cracked teat cups and faulty vacuum regulator may damage the teats and udder tissues making the cow prone to infection with microorganisms.
- v. Poor hand milking techniques: pulling of the teats during milking may lead to bruising and internal injuries to the teats thus making the udder prone to infection.
- vi. Genetic or species factor: some animals and some animal breeds are more prone to mastitis than others. This is due to natural immunity in different animals.
- vii. Level of production: high milk producers are more prone to mastitis than low producers.
- viii. Pendulous udder: due to swinging action of the udder in some animals, teats and the udder become more prone to damage from objects, twigs and grass in the pasture, and in the process increase the chances of teat and the udder infection by microorganisms.
- ix. Grazing in bushy/overgrown pastures: this can cause bruising of the teats leading to infection

Symptoms of mastitis

- i. The udder becomes swollen causing heat and pain.
- ii. When palpated, the udder feels abnormally hard and warm.
- iii. When the udder is touched or palpated, the animal experiences pain and kicks.
- iv. The milk from the udder is blood stained, contains clots, flakes and pus.
- v. Reduced milk yield due to fibrous scar tissue in the udder, causing cessation to milk secretion by alveoli.
- vi. Milk becomes abnormally thick or watery.
- vii. Milk tastes salty.

Detection of mastitis

Detection of mastitis can be done using one of the following techniques;

1. **Using indicator paper:** this has four yellow spots one for each teat. Milk from each quarter is squirted on to the spot. Milk from the affected quarter changes the yellow spot to green or blue-green while milk from normal quarter does not change the colour of the spot.

2. **Using a strip cup:** this is made up of a cup fitted with a black pad. Milk is drawn from each quarter onto the pad. Any changes in colour or consistency are easily detected.

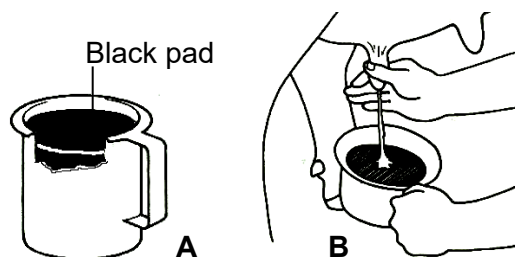


Fig 1.34 A Strip cup and B using a strip cup.

3. **Rapid mastitis test Using California Mastitis Test (CMT):** this indicates the somatic cell counts of milk. Milk is mixed with a reagent and swirled. The thickness of the gel indicate the severity of mastitis.



Fig 1.35 California mastitis test (CMT).

Treatment of mastitis

Infusion of antibiotic ointment into the affected quarter after milking and cleaning the teat.
Giving a combination of intramammary and intramuscular injection with antibiotic

Control of mastitis

- Ensuring good hygiene on the farm to reduce contamination of the teats and teat cups.
- Disinfection of the teat cups before use to minimize the spread of the disease from one animal to another.
- Use of disposable towels to prevent the spread of the disease from one cow to another.
- Stripping of the udder completely after milking as the milk remaining in the udder make the udder prone to infection.
- Regular servicing and repair of the milking machines to ensure proper functioning and to avoid possible injury to the udder which predispose to mastitis.
- Use of dry cow therapy at drying off. This help to prevent and destroy bacteria responsible for mastitis development.
- Dipping the teats in disinfectant solutions (teat dips) every after milking to prevent infection of the teats.
- Suspected or cows on mastitis treatment should be milked last and the milk discarded.
- Control flies by regular spraying or good sanitation. Flies may spread mastitis from one animal to another animal.
- Culling cows with chronic mastitis to prevent the spread of the infection to other cows.

Anthrax

This is a bacterial disease of all warm-blooded animals and man caused by *Bacillus anthracis*. The disease is highly infectious and infected animals discharge large numbers of *B. anthracis* in blood, faeces and other excretions to the environment where they form spores on exposure to the air. Spores can survive for several years in the environment.

Predisposing factors

- Drought that forces the cattle to graze close to the ground thus ingesting contaminated soil.
- Windy conditions that carry spores from the ground into the air, inhaled by animals.
- Feeding animals on feedstuff containing infected animal products.
- Communal grazing and watering of animal in water holes used by different animals.
- Bite from biting flies can also transmit infections mechanically to animals.
- Contact with sick animals or its discharges.
- Wounds on the animal body which act as entry point for pathogens.

Symptoms

- High fever of up to 42°C in cattle.
- Sudden death of animal within short time.
- Dead animals discharge thick, dark blood from the external openings of the body (mouth, nostrils, anus and vulva).

- iv. Visible mucous membranes are congested and haemorrhagic.
- v. Oedematous swelling of the throat causing difficulty in breathing.
- vi. Abortion in pregnant animals.
- vii. Reduction in milk production.
- viii. Milk produced may be bloodstained or yellow in colour.
- ix. Colic or dysentery due to infection of alimentary canal.
- x. Depression, dullness and loss of appetite by the animal.
- xi. Lack of rigor mortis (stiffening of muscles after death) in dead animals.
- xii. Excessive swelling of the stomach after death.

Treatment; Because of the rapid course of the disease, animals are usually found dead or too sick for effective treatment. However, early treatment with antibiotics such as oxytetracycline and penicillin can be successful.

Control

- i. Strict legal regulations (quarantine) to prevent and minimize the spread of the disease.
- ii. Proper disposal of infected carcasses by burying deep into the soil, or burning to ash to prevent contamination of the environment.
- iii. Regular vaccination of all susceptible animals on the farm.
- iv. Immediate isolation and treatment of suspected animals with appropriate antibiotics.

Contagious Bovine Pleuro Pneumonia (CBPP)

This is an infectious disease of the lungs caused by *Mycoplasma mycoides* in cattle. Animals are infected by the inhalation of infected droplets expelled by infected animals.

Predisposing factors	Symptoms
• Congestion of the animals in a house and during transit. • Mixing of animals of different ages. • Poor ventilation of the animal houses • Introduction of carrier animals on the farm.	• Rise in body temperature (fever). • The animal breathes rapidly. • Coughing. • Difficult and painful breathing. • Sick animals stand with their head extended and are reluctant to move. • Death in a few weeks if the animal is not treated.

Treatment; Intramuscular injection with sulfadimidine and tylosin every 12 hrs for 3 days.

Control

- i. Regular vaccination with live Mycoplasma-based vaccines.
- ii. Culling of affected and carrier animals (lungers).
- iii. All new cattle stocks should first be quarantined and tested to detect infected animals which should be treated or culled.
- iv. Restrict animal movement and avoid communal grazing.

Foot Rot

This is a disease of the feet of mainly sheep and goats caused by infection with two bacteria, *Fusobacterium necrophorum* and *Bacteroides nodosus*.

Predisposing factors

- i. Keeping livestock in damp place most of the time.
- ii. Over growth of hooves.
- iii. Keeping animals on contaminated pasture.
- iv. Congregation of large numbers of animals together.

Symptoms

- i. Reddening of the skin between the digits, often called scald.
- ii. In severe cases there is separation and under-running of the hoof.
- iii. Abnormal nodding of the head as the animal is moving.
- iv. Lameness of varying degree subject on the severity of lesions and the number of feet affected
- v. The animal becomes reluctant to walk and tends to lie down.

- vi. The animal tends to crouch and graze in a kneeling position.
- vii. The animal holds one of the foot off the ground.

Treatment; trimming the affected hoof and then stand treating it with appropriate antibiotic (e.g. 10% zinc sulphate or 5% formalin and 5% copper sulphate). Or spray the affected foot with an antibiotic aerosol or injection with appropriate antibiotic.

Control

- i. New animal brought on the farm should be quarantined first and examined for the disease. Any affected animal should be rejected or treated.
- ii. Isolation of affected animals and treating them before they are allowed to join the flock.
- iii. Regular walking of animals through the foot bath containing suitable antibiotic.
- iv. Regular vaccination of animals to protect them from the infection.

Foot and Mouth Disease (FMD)

This is an infectious viral disease of cloven hoofed animals and occasionally man. Infected animals excrete virus in saliva, milk, semen, faeces, urine and exhaled air.

Symptoms	Control
i. Rise in body temperature to about 42°C. ii. Sudden drop in milk production in lactating animals. iii. Depression and anorexia. iv. Small vesicles on the feet and in the mouth (tongue, gums, lips and dental pad). v. Excess salivation and reluctance to eat due to lesions in the mouth. vi. Shedding of the hooves due to secondary bacterial infection of the lesion on the feet. vii. Lameness and inability to walk due to lesions and wounds on the feet. viii. Inability to work or reduction in power output in draught animals.	• Regular vaccination of all animals on the farm. • Destruction of bedding and leftover feeds of infected animals to prevent the spread. • Restriction of movement of livestock and people from suspected areas (quarantine). • Disinfection of animal premises and equipment where the disease has occurred. • Slaughtering of all infected animals and proper disposal of carcasses by burying deep into the soil or burning to ash. • Reporting suspected cases to authorities such as veterinary doctors or government officials.

East Coast Fever (ECF)

Also called Theileriosis or Corridor disease. It is a tick-borne protozoal infection of cattle caused by *Theileria parva*. The protozoan is spread by the brown ear tick (*Rhipicephalus appendiculatus*).

Symptoms	Control
i. Rise in body temperature of up to 41-42°C. ii. Enlarged lymph nodes due to localised multiplication of parasites in the lymph nodes. iii. Difficulty in breathing and soft cough due to fluid in the lungs. iv. Blood stained diarrhoea. v. White or yellow discolouration of the eyes and gums. vi. Anaemia and jaundice due to invasion of the red blood cells by the parasite. vii. Emaciation due to muscle wasting. viii. Loss of appetite. ix. Death within 2-3 days in exotic cattle if not treated on time.	• Regular spraying or dipping of cattle with recommended acaricides. • Controlled grazing of animals by fencing to prevent animals from straying and picking ticks from the environment. • Fencing to control stray animals carrying infected ticks from coming into the farm. • Ley farming to control ticks in the pasture by ploughing and burying ticks in the ground during cultivation of crops. • Controlled burning of pastures to kill ticks on the ground and ticks questing. • Isolation and treatment of infected animals • Vaccination of all animals on the farm. • Selection and rearing of resistant breeds.

Treatment; early injection of affected animal with parvaquone and buparvaquone.

Trypanosomiasis

This is a protozoal infection of all farm animals and man caused by Trypanosome parasites e.g., *T. congolense* and *T. brucei*. It is known as Nagana in cattle and sleeping sickness in humans.

Symptoms	Control
i. Anaemia and loss of body condition. ii. Recurrent fever for up to three months. The fever comes every after 12 days and lasting 2 to 4 days. iii. Discharge from the eyes. iv. Oedematous swelling under the skin. v. Paralysis of the hind quarters of the animal. vi. Death of the animal after long period of clinical signs usually within 3-12 months.	• Control of tsetse flies through clearing of bushes, spraying tsetse habitat with insecticides, and use of tsetse traps. • Isolation of infected animals to prevent the spread of the infection. • Treating sick animals with trypanocidal drugs • Rearing animals in tsetse fly-free zones. • Dress farm animals with tsetse fly repellants to prevent tsetse fly bites.

Anaplasmosis (Gall sickness)

This is a rickettsial infection of red blood cells of domestic ruminants with *Anaplasma marginale*. The parasites are transmitted by Boophilus ticks and biting flies.

Symptoms

- i. High body temperature (fever).
- ii. Anaemia due to infection of the red blood cells.
- iii. Mucous membranes, e.g. gums and eyes are very pale or yellowish in colour (jaundice).
- iv. Loss of appetite and eventual drop in milk production.
- v. Incoordination, difficult breathing and abortion in pregnant cows.
- vi. Death in few days after clinical signs.
- vii. Post mortem examination shows yellow discoloration of tissues, enlargement of the spleen and gall bladder with thick brownish-green bile, hence the name 'gall sickness'

Control

- i. Vaccination of all animals on the farm.
- ii. Tick control measures such as double fencing, rotational grazing, controlled burning of infested pasture and regular dipping and spraying with appropriate acaricides.

Heart Water (Cowdriosis)

A rickettsial infection of ruminants caused by *Cowdria ruminantium*. Infection is spread by a three-host Amblyomma ticks.

Symptoms

- i. High body temperature.
- ii. Incoordination characterized by circling (the animal turns around) and with high stepping gait
- iii. Nervous signs of anxiety and uncontrolled movement of jaws, lips and the eyes.
- iv. Oedema of the brain due to parasite damage of cells lining blood vessels.
- v. Death within one or two days if not effective treatment given.
- vi. Post mortem examination shows a yellowish brown fluid in the abdominal and chest cavities and in the sac surrounding the heart, hence the name 'heart water'.

Control

- i. Treating sick animals with tetracycline administered early before nervous signs develop.
- ii. Vaccination with a live vaccine.
- iii. Regular dipping and spraying of animals with appropriate acaricides to prevent tick bites.
- iv. Suspected animals should be isolated and treated promptly.



Activity 1.15 Understanding the different types of diseases affecting cattle

You will need A pen, notebook, reference textbooks and internet

Key question: what are the examples of cattle diseases, their symptoms and control measures?

In groups: using the internet or agriculture reference books,

1. Discuss the causes, symptoms and control of the following diseases in cattle; east coast fever, heart water, foot and mouth disease, anthrax, and mastitis.

2. Describe how a strip cup is used to test for mastitis in a milking cow.
3. Outline the current legal regulations put up by the government to control notifiable diseases in the country (*use the internet*)

Record your findings and share with the rest of the class.

General methods used to control livestock diseases on the farm

The following are some of the measures used to control diseases in livestock:

Use of prophylactic drugs: animals are given drugs routinely to avoid infections. Examples of prophylactic drugs are coccidiostats which are given to calves to control coccidiosis and sulphonamides which are used to control trypanosomiasis.

Vaccination: vaccination provides resistance or immunity against certain diseases.

Isolation of infected animals to prevent the spread of infection to other animals.

Proper nutrition: feeding animals on well-balanced ration helps to prevent deficiency diseases such as milk fever and grass tetany. It also gives the animal the ability to resist infections.

Control of internal parasites by regular deworming to keep the immune system strong.

Treatment of sick animals to prevent the spread of such diseases to other animals.

Control of vectors: vectors such as ticks and tsetse flies transmit diseases. Controlling such organisms using appropriate methods also helps in controlling the diseases they transmit.

Quarantine: during an outbreak of infectious disease such as foot-and-mouth disease, movement of cattle and their products is restricted to avoid the spread of the disease to other areas.

Proper housing: providing of proper housing helps to avoid exposing the animals to predisposing factors for diseases such as pneumonia, foot rot and mastitis.

Hygiene: some diseases can be controlled by observing proper hygiene, for example cleaning the house and equipment using disinfectants and proper carcass disposal helps to control diseases.

Use of healthy breeding stock to prevent breeding diseases such as brucellosis and trichomoniasis. Selection of disease-resistant animals also helps in controlling diseases.

Restricting visitors on the farm; visitors may carry the disease to the farm. They should be restricted or screened or disinfected before entering the farm.

Double fencing: this protects animals from those of poorly managed neighbouring herd.

Buy feeds from certified suppliers: infected feeds can introduce diseases on the farm.

Buy animals from reliable farms to avoid introducing diseases on the farm from other farms

Proper disposal of dead animals e.g. by burying them deep into the soil or burning them to ash. This helps to destroy pathogens and prevent the spread of the disease.



In Uganda, the government has put in place legal interventions to control animal diseases by enacting laws on the movement of animals and animal products in the country, and regulating importation and export of animals and animal products.



Activity 1.16 Finding out how cattle diseases are controlled and treated on the farm

You will need a pen, notebook, veterinary officer and protective gears

Key question: What steps are followed in control and treatment of cattle diseases?

In groups:

1. Write a letter to the cattle farm manager seeking for permission to visit the farm and participate in treating cattle.
2. Put on protective gears and move to the farm.
3. Which disease prevention and control measure do you find at the farm gate?
4. Move to the crush where several animals have been restrained. Why is restraining animals vital?
5. Observe and participate with the guidance of the vet in the spraying, temperature taking, pulse taking, hoof trimming and injection of animals. which tools are being used to control and treat cattle diseases? What other methods are being used to control diseases on the farm?
6. Clean your hands and tools you have used and store them properly

Record your findings and share with the rest of the class.

Activity of Integration 	While attending a community meeting about “health living” you heard the medical doctor say sharing houses with livestock is the major cause of persistent human diseases in the community. This statement left many people wondering how this is possible.	
Task	Design a poster to make the people understand the doctor’s statement	

Exercise 1.7	Self-Check Questions
	As a student who has who has studied animal diseases and their control, 1. (a) What do you understand by the term notifiable disease, give two example of notifiable diseases in Uganda. (b) Describe two appropriate measures that can be used to control notifiable diseases on the farm. 2. (a) Explain the factors that predispose cattle to disease infections. (b) List the general methods that are used to control animal diseases.

Parasite Control and Prevention in Cattle

Parasites are organisms which depend on cattle partly or wholly for their existence and in the process cause discomfort and ill-health to the host. Parasites are classified as **external** (or ecto-parasite) and **internal** (or endo-parasite);

External Parasites

External parasite: also called ecto-parasites, are parasites which live on the outside of the animal’s body mainly on the skin of the animals. Examples include; ticks, mites, flies, lice, keds, and fleas.

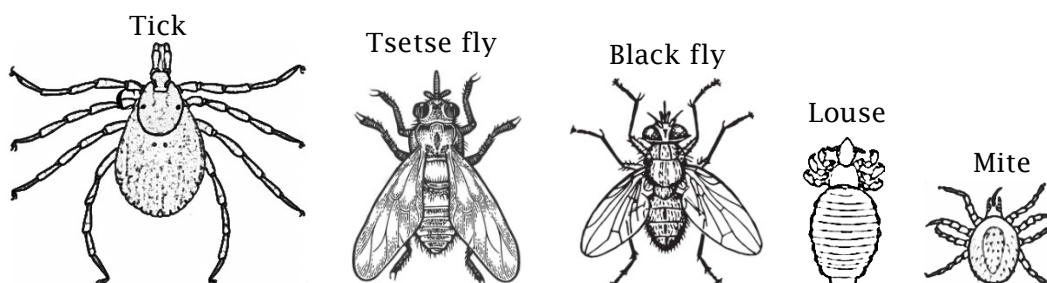


Fig 1.35 example of some ecto-parasites.

Ticks

These spend most of their life time on the skin of the animal. There are two types of ticks soft ticks and hard ticks. Hard ticks, on dorsal surface have a hard protective shield (scutum). Whereas soft ticks don’t have hard protective shield on the surface.

Table 1.7 Tick species of economic importance

Tick species	Site of attachment	Parasite spread	Disease caused
Brown ear tick <i>Rhipicephalus appendiculatus</i> No of hosts 3	in and outside the ears, eyes lid, base of horns, tail brush, under the tail	<i>Theileria parva</i> <i>Theileria Lawrencei</i> <i>Theileria mutans</i>	East coast fever corridor disease Benign theileriosis
Red-legged tick <i>Rhipicephalus evertsi</i> No of hosts 2	deep inside the ears, under the tail, around the anus	<i>Babesia bigemina</i> <i>Theileria parva</i> <i>Theileria mutans</i>	Red water East coast fever Benign theileriosis
Blue tick <i>Boophilus decoloratus</i> No of hosts 1	face, neck, dewlap, flanks	<i>Anaplasma marginale</i> <i>Babesia bigemina</i>	Anaplasmosis (gall sickness) Red water
Bont tick <i>Amblyomma variegatum</i> No of hosts 3	on ears, dew lap, belly, udder, flanks, scrotum and tail brush	<i>Cowdria ruminantium</i>	Heart water
Bont-legged tick <i>Hyalomma</i> spp. No of hosts 2 or 3	udder, scrotum, tail brush between the hooves.	<i>Theileria annulata</i> <i>Ehrlichia bovis</i>	Tropical theileriosis Equine babesiosis

Effects of ticks on the host animals

- Vector of diseases such as East coast fever, Red water (babesiosis) and heart water.
- Suck blood from animal leading to anaemia.
- Cause wounds on the skin paving way for infections from bacteria and flies.
- They cause damage to the skin/hide thus reducing its quality.
- Cause irritation to the animal making it uncomfortable and reducing feed intake.
- Cause tick-worry to the animal leading to discomfort and reduced feed intake.
- Increase the cost of management; buying acaricides and hiring labour for spraying.



Use the internet and search for different types of hard and soft ticks. Observe the appearance in terms of body shape, colour, and mouth parts. Record your findings.

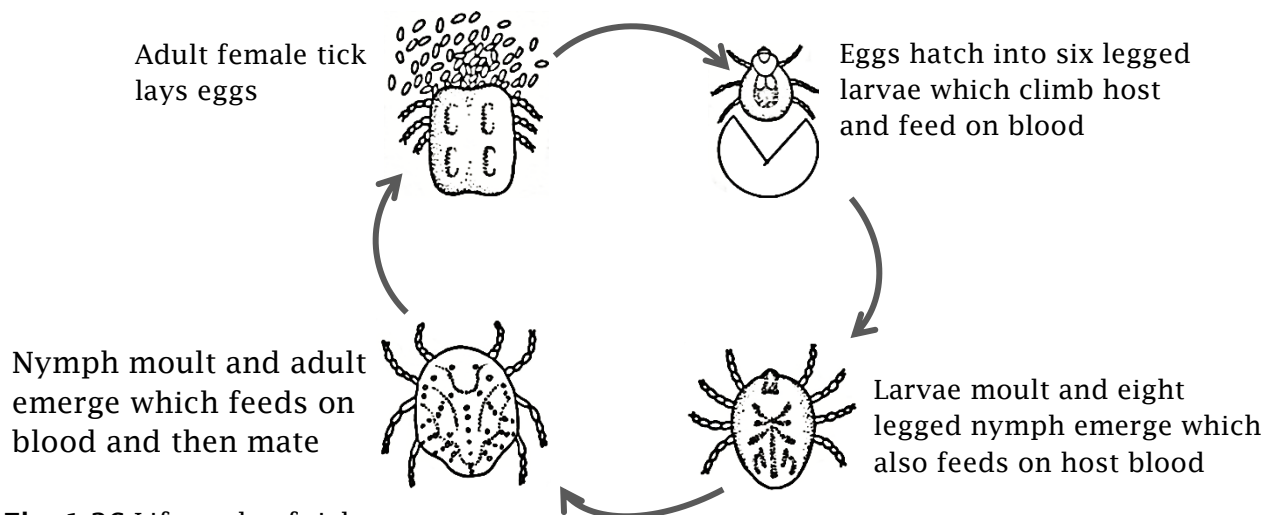


Fig. 1.36 Life cycle of ticks

The One-host tick

This is a tick that completes its life cycle on one host. Examples of one-host ticks include the following: Blue tick (*Boophilus decoloratus*), Cattle tick (*Boophilus microplus*) and Texas fever tick (*Boophilus annulatus*)

Life cycle of a One-host tick

This requires one host to complete its life cycle

- Adult female tick lays eggs on the ground which hatch into larvae,
- The larvae climb onto the first host, feed on blood and become engorged.
- Larvae moult into nymphs while still on the host, feed on blood and become engorged.
- The nymphs moult into adults, feed on the same host and mate.
- The engorged female adults drop to the ground to lay eggs and start the cycle again.

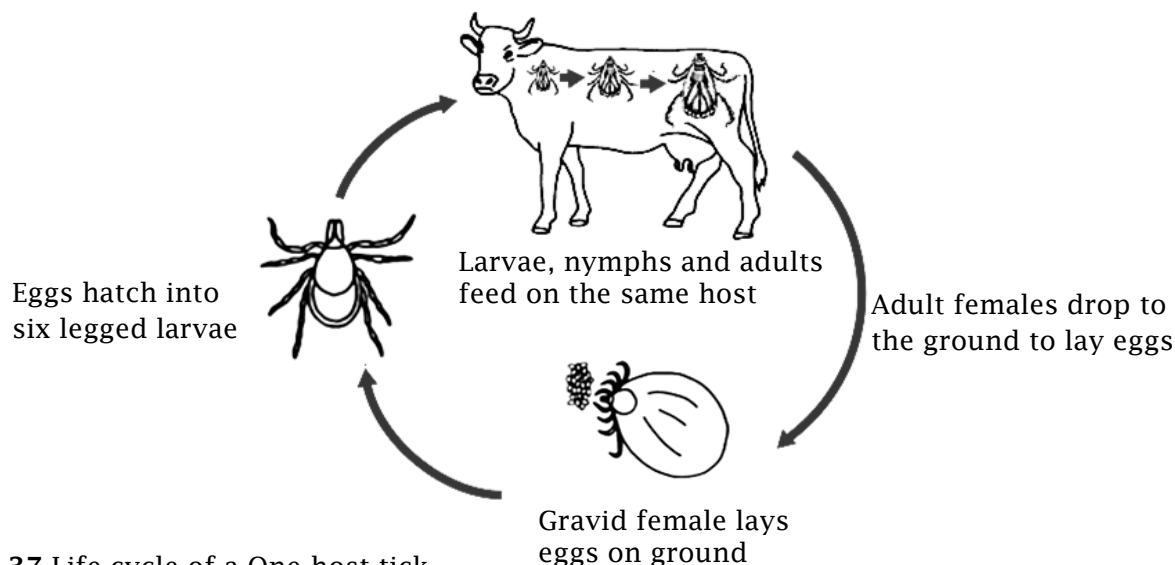


Fig 1.37 Life cycle of a One-host tick.

The Two-host tick

This tick completes its life cycle on two hosts. Examples of two-host ticks include: Red-legged tick (*Rhipicephalus evertsii*), Brown tick (*Rhipicephalus bursa*), Bont-legged tick (*Amblyomma* spp.)

Life cycle of a two-host tick

This requires two hosts to complete its life cycle.

1. The adult female tick lays eggs on the ground. Eggs on the ground hatch into larvae,
2. The larvae climb onto the first host. Larvae feed on blood and become engorged.
3. Larvae moult into nymphs and feed on the same host.
4. Engorged nymphs drop to the ground and moult into adults.
5. The adults climb onto a second host, feed on blood and mate.
6. Engorged females drop to the ground to lay eggs and restart the cycle again.

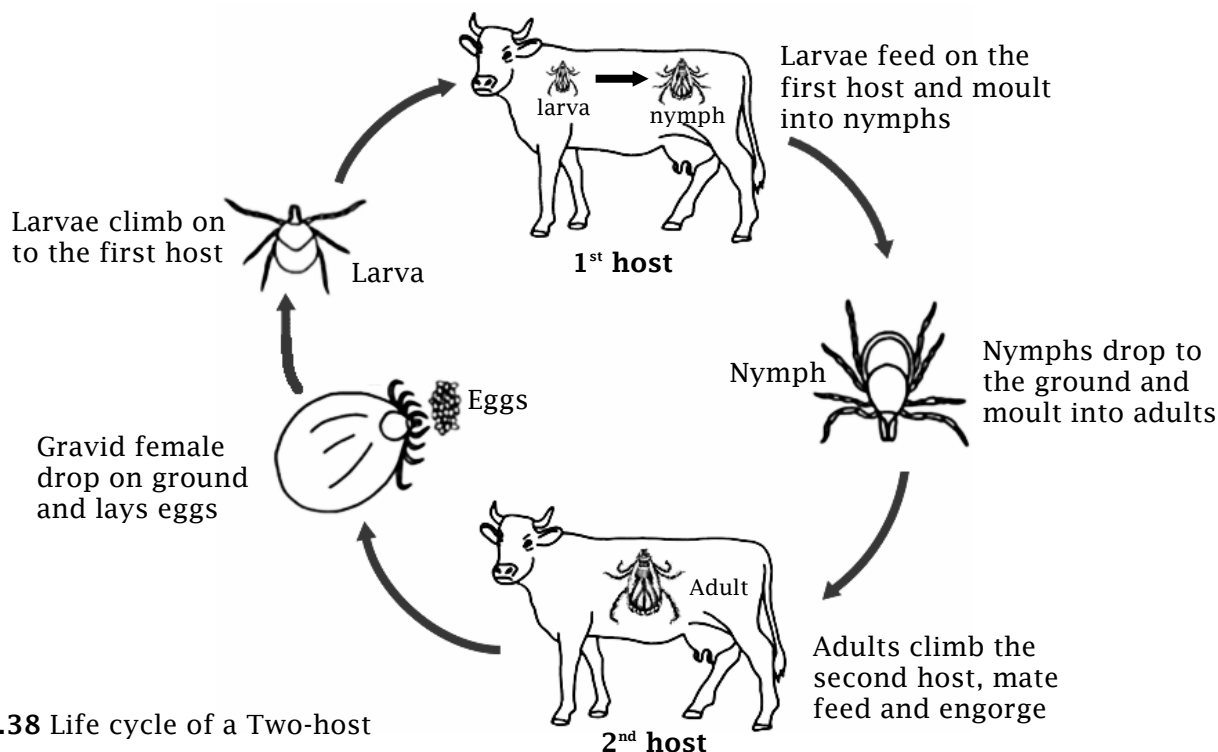


Fig 1.38 Life cycle of a Two-host

The Three-host tick

This is a tick that completes its life cycle on three hosts. Examples: Brown ear tick (*Rhipicephalus appendiculatus*), Bont tick (*Amblyomma variegatum*) and

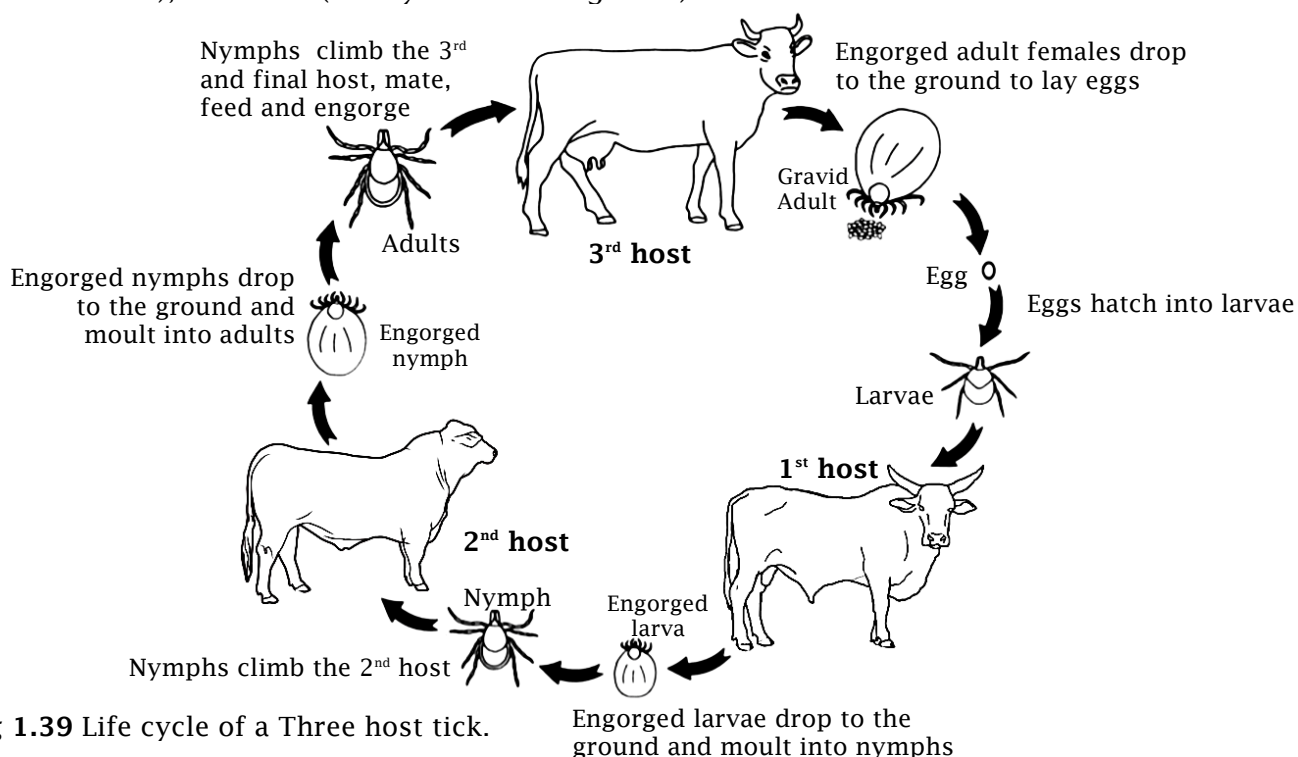


Fig 1.39 Life cycle of a Three host tick.

Life cycle of a Three-host tick

This requires three hosts to complete its life cycle.

1. The adult female tick lays eggs on the ground.
2. Eggs on the ground hatch into larvae.
3. The larvae climb onto first host, feed on it, drop to the ground and moult into nymphs.
4. The nymphs climb the second host, feed and drop to the ground to moult into adults.
5. The adults climb onto a third host, feed on it and mate.
6. Engorged females drop to the ground and lay eggs to start the cycle again.

Adaptation of ticks for their parasitic mode of life

- i. Possess well developed mouthparts to help them attach on the host and to suck blood.
- ii. They possess legs (4 pairs) in adults or three pairs in larvae. Legs help in movement to search for the host or to look for a suitable part for attachment.
- iii. The brown/ dull colour helps them to camouflage on host animals.
- iv. Possess cuticle around the body to protect them against dehydration during hot weather
- v. Possess claws on their legs for clinging on to the skin coat during movement.



Activity 1.16 Finding out the different types of ticks on the farm

You will need Protective gears, a pen, notebook, observation list, internet

Key question: what is the effect of tick on cattle?

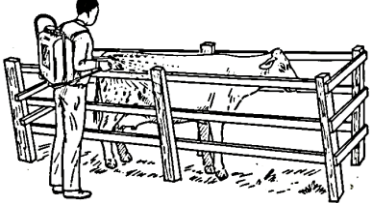
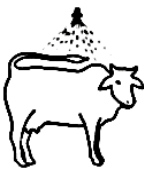
In groups:

1. Put on your protective gears
2. Visit a nearby livestock farm, observe and Identify the different types of ticks found on the animals, giving the sites of attachments of ticks on the animals.
3. From the observations made on the different ticks, how are they adapted to live as animal parasites? Discuss the effects of ticks on farm.
4. What advise do you give to the farmer to control ticks on his/her farm.

Present your finding to class and record conclusions in your note books

Control of ticks on the farm

- i. Regular spraying/dipping animals in suitable acaricides
- ii. Hand picking and killing of picked ticks by burning or crushing
- iii. Controlled grazing of animals by having perimeter fence around the farm to prevent introduction of ticks from stray animals on the farm.
- iv. Controlled burning of the pasture to kill ticks on the ground.
- v. Ploughing the pasture land to destroy ticks on the ground by either burying them or exposing them to harsh conditions and predators.
- vi. Proper sanitation in the animal premises and regular disinfection of the premises.

Steps in hand spraying of cattle to control ticks	Step 1	Step 2	Step 3	Step 4	Step 5
					
Using BBRH procedure; The Back, Brisket, Belly, Rear and the Head	spray the backline; from the shoulders to the tailhead	spray the belly, udder or testicles	spray the belly, udder or testicles	spray the anus, vulva and base of tail up to switch	spray the head, face, neck, base of horns and ears



Mites, flies and lice are parasites of economic important in livestock. Explain the effect of each and how they can be controlled on the farm.



Activity 1.17 Skills in hand spraying of cattle to control ticks

You will need Protective gears, hand sprayer, acaricide, clean water and a crush

Key question: what procedure is followed during hand spraying of cattle to control ticks?

In groups: On the school farm, demonstrate how you can hand spray cattle to control ecto parasites.

1. Prepare all the equipment to use; the sprayer should be in proper working condition. Mix the acaricide in the sprayer according to the manufacturer's instructions.
2. Restrain the animal properly in a crush.
3. Start by spraying the entire backline of the animal from the shoulders to the tail head.
4. Spray the sides in a zigzag manner to trap and retain the spray wash from the backline.
5. Spray the belly and spray with the nozzle facing upwards. Give more attention to the flanks, the udder or the scrotum.
6. Spray the rear and spray the tail head, around the anus, the vulva, and then the tail switch to ensure complete wetting.
7. Spray the head and spray the face, the base of horns, the ears, the neck, the front legs, flanks and the heels. Give more attention to the inside of the ears.
8. Leave the animal for about 2 minutes in the crush to allow the acaricide to drip off from the animal's body.
9. Release the animal.
10. Clean all equipment used with plenty of water.

From the above state why

- (a) It is important to restrain the animal before you start to spray it.
- (b) Proper mixing of acaricide is important
- (c) It is important to follow proper procedure during hand spraying of the animal
- (d) It is vital to hold the animal for some time before you allow it to move to the pasture.

Present your finding to class and record conclusions in your note books

Internal Parasites

These are parasites that live inside the host. Internal parasites that affect livestock include roundworms, tapeworms and flukes.

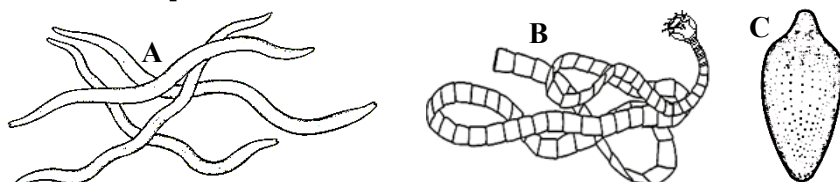


Fig 1.40 Internal parasites; A roundworms, B tapeworm, and C liver fluke.

Symptoms of attack of internal parasites

- General emaciation or loss of weight.
- The eyes are dull and pale.
- Distended belly.
- Dullness and depression.
- Rough coat.
- Persistent diarrhoea. Blood stained dung
- Swelling under the jaws.
- Excessive appetite or loss of appetite.
- Presence of eggs, parasite in the dung.
- Reduced growth in young animals.
- Reduced milk production.

Effects of internal parasites on cattle

- Retarded growth and loss of weight.
- Reduced production.
- Damage to internal organs such as the liver.
- Blockage of internal organs such as the bile ducts.
- They cause irritation.
- They lead to loss of appetite in animals.
- They lead to death, in severe cases.
- Lowering the quality of meat from the cattle



Activity 1.18 Finding out the symptoms and effects of worm infection in cattle

You will need a pen, notebook, reference textbooks and internet

Key question: what are the symptoms and effects of internal parasites in cattle?

In groups: using the internet or any other agriculture reference books;

1. Give examples of internal parasites that affect cattle
2. Give the signs that indicate worm infection in cattle .
3. Discuss and brainstorm the effects of endo parasites in cattle management.

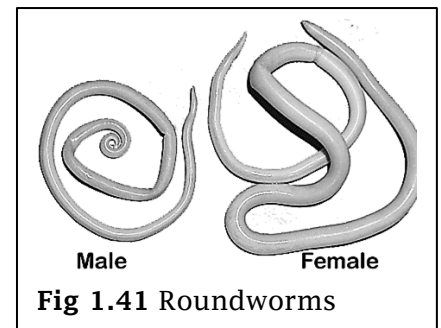
Present your finding to the rest of the class and record conclusions.

Roundworms (*Ascaris* spp.)

Roundworms inhabit the alimentary canal of cattle.

Life cycle of roundworms (*Ascaris* spp.)

1. The adult lays eggs in the alimentary canal of the host animal.
2. The eggs are passed out with dung.
3. The eggs hatch into larvae.
4. The larvae climb onto vegetation and moult through a process called ecdysis and become encysted.
5. The host animal swallows the larvae as it feeds.
6. The larvae shed their cyst in the small intestine of the host to form juvenile worms.
7. The juvenile worms penetrate through the intestinal walls into the bloodstream.
8. Through blood, they migrate to the heart, and then to the liver and lungs.
9. During migration, worms change from one level to another within their juvenile stages.
10. In the lungs, the larvae move up the trachea where they are coughed out and swallowed with sputum into the intestines.
11. In the intestines, the larvae grow and develop into the adult stage.
12. The male and female worms mate and eggs are laid to restart the cycle.



Liver flukes (*Fasciola hepatica*, *Fasciola gigantica*)

They commonly infect cattle and sheep. They are found in the liver.

Life cycle of liver flukes (*Fasciola gigantica*)

1. Adult flukes in the liver produce fertilized eggs that enter into the alimentary canal and are passed out with dung.
2. Under favourable conditions of warm temperatures and right moisture, eggs hatch into the first larvae stage called miracidium, which is ciliated.
3. The miracidium swim around and on coming into contact with their intermediate host, a water snail, penetrates the snail.
4. Once in the snail, the miracidium hatch into different larvae forms called sporocysts
5. The sporocysts develop further to a stage called redia.
6. The rediae develop into another larval stage called cercaria, which is tailed.
7. At this time, the cercariae leave the snail and swim away to attach on grass.
8. Here, cercariae lose their tails and form cysted and infective stage called metacercaria.
9. Cattle (primary host) swallow metacercariae during grazing or drinking infected water.
10. After being swallowed, the cyst coat dissolves to release young flukes.
11. The young flukes bore through the walls of the intestine and find their way to the liver, where they mature into adults.
12. In the bile duct, the adults produce eggs to start the cycle all over again.

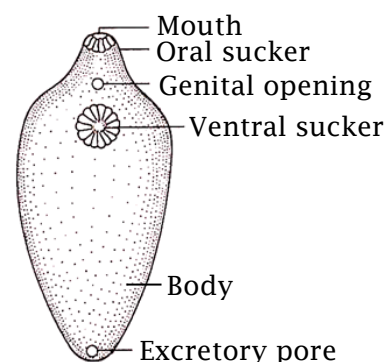


Fig 1.42 Parts of a liver fluke

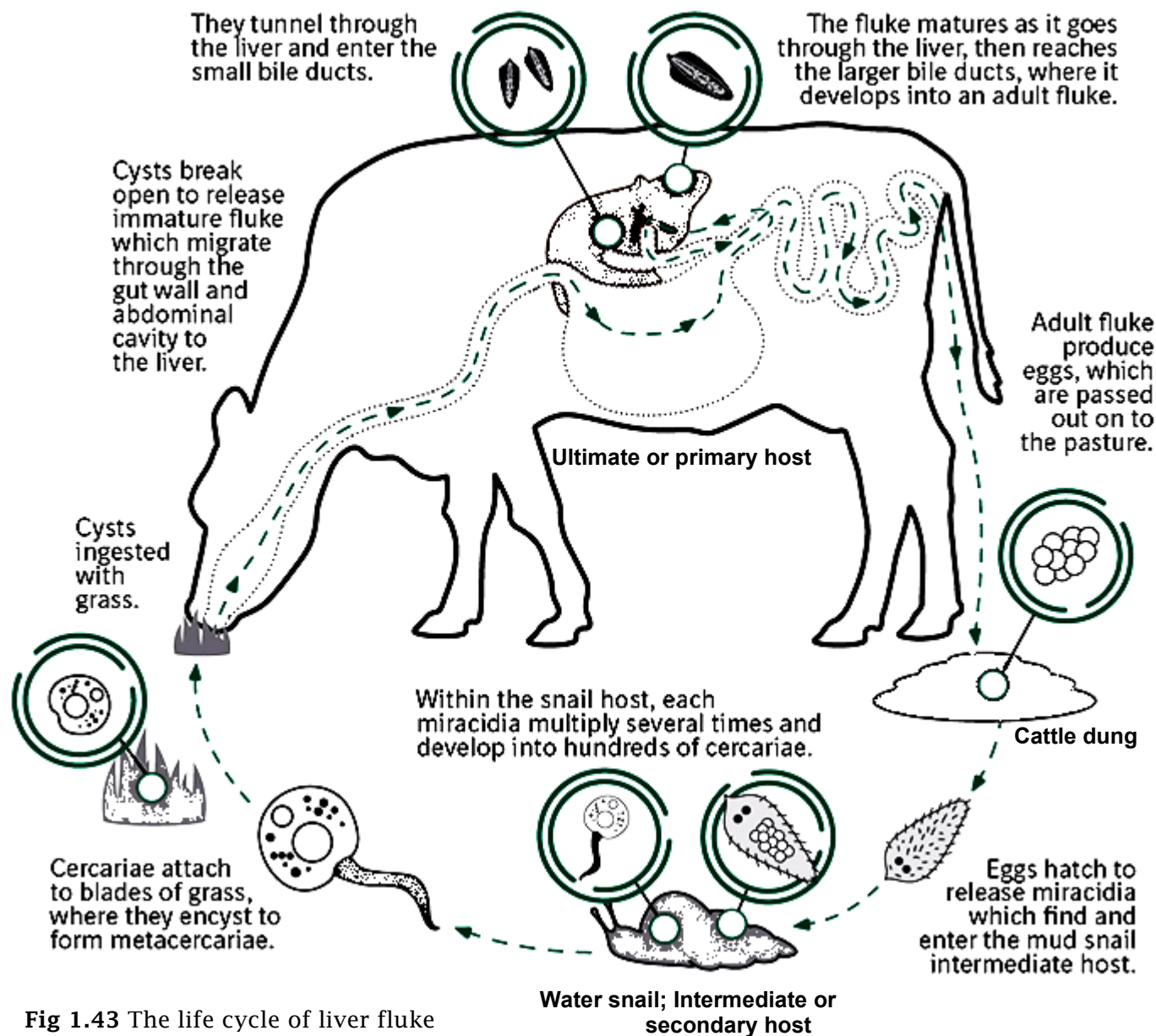


Fig 1.43 The life cycle of liver fluke

Tapeworms (*Taenia* spp.)

Tapeworms are long thin, flat worms which attach by suckers/hooks in the intestines of the host. .

Life cycle of tapeworms (*Taenia* spp.)

1. The last mature segment or proglottide of the adult tapeworm breaks off into the digestive system of human beings (primary host).
2. Humans pass out faeces with proglottides, which have embryonated eggs,
3. The eggs are ingested by cattle (intermediate host) during feeding.
4. Once in the intestines of the intermediate host, the eggs hatch into six-hooked embryos.
5. The embryos penetrate the intestinal walls and enter the bloodstream.
6. They are then moved to the heart from where they are distributed to body muscles. In the muscles, the embryos form cysts called bladder worms or *Cysticercus* larvae.
7. At this stage, the bladder worms develop suckers, hooks and rostellum.
8. Under cooked beef is eaten by man, the cysts dissolve in the intestine and the parasites become active and develop to mature stage.
9. The adults release proglottides to start the cycle again.



In the life cycle of a beef tapeworm, man is the primary host and cattle the secondary host. But on many occasions, mature tapeworms have been found in cattle. How is this possible?

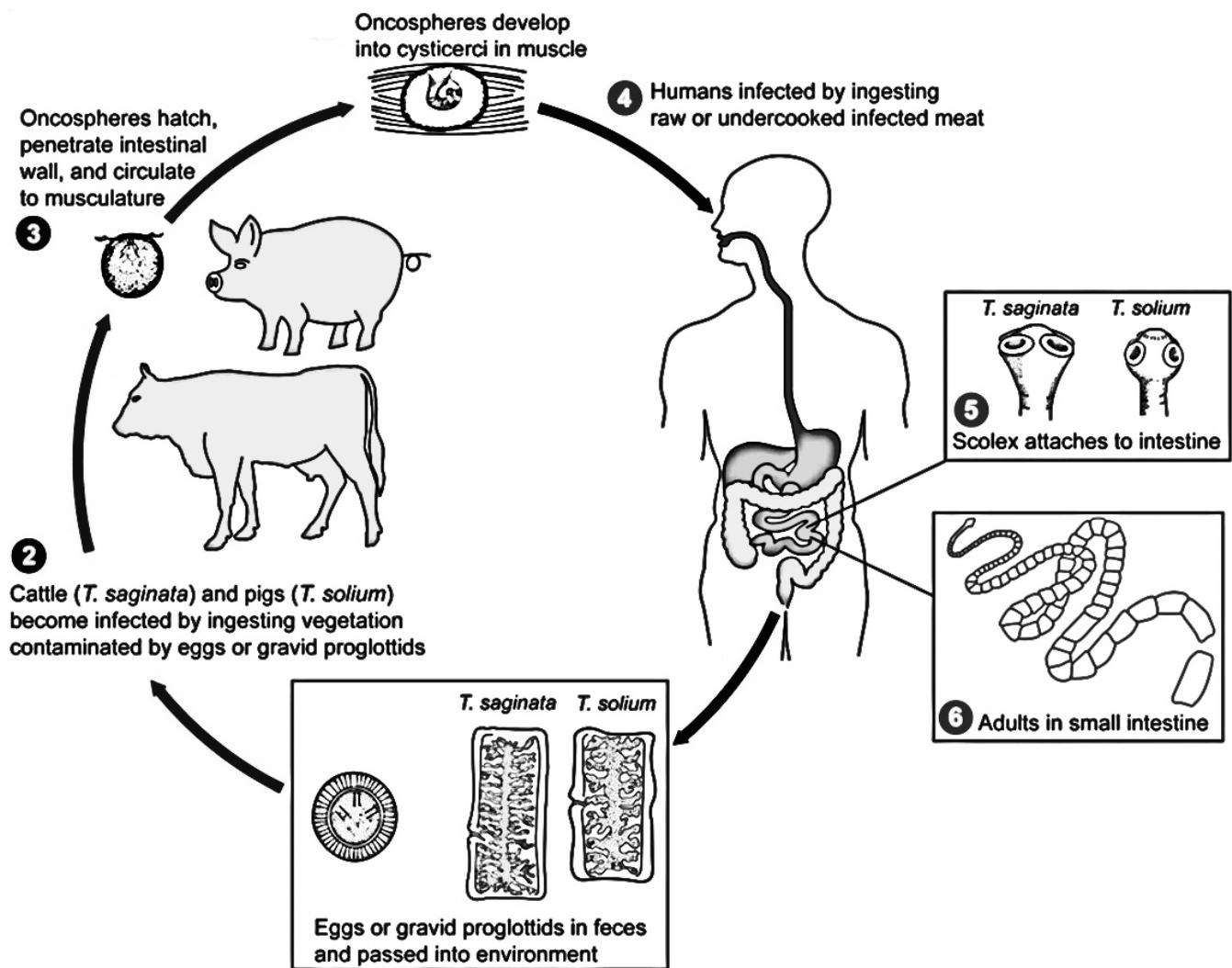


Fig 1.44 Life cycle of tape worm



Activity 1.19

Understanding the life cycle, structure and control of worms

You will need

a pen, notebook, reference textbooks and internet

Key question: what is the life cycle of internal worms, their structure and controlled on the farm?

In groups: using the internet or any other agriculture reference books;

1. Discuss the the life cycle and control of internal parasites that affect cattle.
2. Using a hand lens, carefully observe the worms presented to you by your teacher. Describe the structre of each. States how their structure help them to servive in the host.
3. Make a labeled drawing of the liver fluke, roundworm and tapeworm

Present your finding to the rest of the class and record conclusions.

Methods of controlling internal parasites in livestock

The methods that can be used to control internal parasites in cattle include the following:

- i. Regular drenching, deworming or dosing using appropriate drugs.
- ii. Proper sanitation in livestock houses.
- iii. Providing livestock with clean feeds and water.
- iv. Proper disposal of human wastes to avoid infection of animals.
- v. Practicing rotational grazing to break the lifecycles of parasites.
- vi. Draining or fencing off swampy areas.
- vii. Burning pasture to kill eggs of parasites on the ground.
- viii. Ploughing pasture to bury eggs deep in the ground.
- ix. Peripheral fencing to keep away other animals, which could be infected.
- x. Spraying swampy areas with copper (II) sulphate to kill snails.
- xi. Inspection of meat for worms and deep cooking of meat to kill bladderworms in meat.



In the life cycle of a beef tapeworm, man is the primary host and cattle the secondary host. The eggs ingested by cattle release oncospheres in the intestines which usually penetrate the intestines into the circulatory system into the muscles. However, oncospheres may remain in the intestines of cattle and grow to maturity



Activity 1.20 Practicing control measures for internal parasites in cattle

You will need Protective gears, drenching gun, recommended anthelmintic drugs

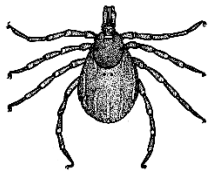
Key question: how are internal parasites controlled on the farm?

In groups:

1. Write a letter to the farm manager seeking for the permission to visit the farm.
2. Put on protective gears, ensuring the entire body is well protected.
3. Move to the farm and observe the farm manager or veterinary officer administering worm control measures.
4. With the supervision of the farm manager or verinary officer, prepare the anthelmintic drug and load it in the drenching gun. What precaution are being taken at this stage?
5. Restrain a calf with worm infection and go a head to drench it. Which other tools can you use to perform the same task if incase you do not have a drenching gun?
6. Clean the tools and equipment you have used. What environmental protection precautions and guidelines are you following?
7. Remove your protection gears . wash your hands thoroughly with water and soap.

Write guidelines about worm control that you would use to help cattle farmers in your community

Activity of Integration 	In your community, it is repeorted that there is an outbreak of foot and mouth disease (FMD) that is killing animals. foot and mouth diseases is a severe, highly contagious viral disease of cattle that has significant economic impact. You want to help your that very much depend on their animals for livelihood to find solution to this problem.	
Task	Prepare a poster you will use to sensitise the animal farmers in your community on how to overcome the problem.	

Exercise 1.7		Self-Check Questions	
	As an animal health expert, you are provided with two parasites of economic importance on the farm. 1. (a) Give the name of the parasite (b) Name 2 effects of the parasite on the host (c) Draw and label the parts of the parasite, and give the likely sites of attachment of the parasite on the host (d) Give suitable control measure for the parasite on the farm		
	2. (a) Give the name of the parasite (b) Name the primary and secondary hosts of the parasite (c) Draw and label the parasite (d) What are the symptoms of the parasite infection on the host (e) Give 2 methods of controlling the parasite on the farm (f) Classify the two parasites according to the places where each is found on the host animal		
	3. (a) Explain how internal parasites spread on the farm. (b) Other than using drugs, how can a farmer effectively control Tapeworms on the farm? (c) Describe the life cycle of a two-host tick		



Safe and Clean Milk Production on the Farm

Milk is a yellowish-white opaque fluid secreted from the mammary glands of an animal, which contain fat as an emulsion, protein and some mineral matters in colloidal suspension, and lactose with some minerals and soluble proteins in solution. Clean milk is the milk drawn from the udder of healthy animals that is free from diseases and extraneous materials such as dust, dirt, flies, manure, extraneous water, added solids and drug residues. Observing good health of the milking cows and proper hygiene during milking will allow you to produce milk that is clean and safe for human consumption.

Importance of clean milk production on the farm

- i. Clean milk protects the health of the consumer. Raw milk is known to be associated with pathogenic bacteria, which cause milk-borne diseases such as tuberculosis, brucellosis or typhoid fever, etc.
- ii. It fetches high price in the market which boosts the financial condition of the farmer.
- iii. High-quality milk has longer keeping quality, allowing transportation of milk and milk products over long distances.
- iv. Milk is a raw material that is processed to make many dairy products. Hence, good quality of milk should be preserved to get good-quality milk products (cream, cheese, ice cream, yoghurt etc.)
- v. Poor-quality milk can be rejected by consumers, causing heavy losses to the farmer.

Factors that make milk unfit for human consumption or marketing

- i. Zoonotic diseases such as tuberculosis, brucellosis, leptospirosis and salmonellosis.
- ii. Presence of drugs used to treat animal diseases in the milk.
- iii. Milk adulteration with water and solids.
- iv. Mastitis characterised by salty taste of milk, clots, blood stain, and abnormal colour.
- v. Contamination of milk with dust, dirt, manure, hair, and flies.

Milk Contamination

The milk can be contaminated before, during or after milking. High-quality milk should have longer keeping quality, proper nutritive value, normal taste, colour, odour and be free from extraneous matter. Milk contamination may occur in one of the following ways;

- i. Animal faeces and urine coming into contact with the milk during milking.
- ii. Infection of the udder (mastitis)
- iii. Cow diseases e.g. tuberculosis, brucellosis
- iv. Bacteria that live on the skin of the animals
- v. Dirty contaminated milking utensils
- vi. Unsanitary conditions in the milking parlour and milk stores
- vii. Dirty unhygienic milker getting in contact with the milk
- viii. Extraneous materials that find their way into the milk such as dust, dirt insects manure
- ix. Drugs given to lactating cows that find their way into the milk



It is advisable to keep the milking environment clean. How does this contribute to production of clean milk on the farm?

Clean Milk Production on the Farm

It is important to remember that quality control must begin at the farm. That way, milk will have fewer bacteria that cause spoilage and diseases. In order to ensure good quality and protect the health of consumers, you must always carry out milking in accordance with good hygienic practice. You should observe the following points:

- i. Maintain clean and healthy cows. Sick animals can transmit diseases to humans through contaminated milk.
- ii. Keep a clean milking environment, free of dust and mud. Garbage and dung in the milking area provide a good breeding ground for rats, flies and cockroaches that may transmit dirt and bacteria to the milk.

- iii. Ensure that the milker is free from communicable diseases like diarrhoea or typhoid.
- iv. Do not mix colostrum with normal milk.
- v. Wash the hands with soap and clean water and then, dry the hands with clean cloth.
- vi. Wash the udder with warm water and, dry the udder with a clean dry cloth.
- vii. Use clean containers for milking.
- viii. Make the first draw into a strip cup to check for mastitis and throw the milk away from the milking area even if the milk appears clean.
- ix. Cows with mastitis should be milked last and their milk discarded.
- x. Milk from cows under antibiotic treatment should not be sold or consumed until 3 days after last treatment or as advised by the veterinary doctor.
- xi. After every milking, dip the teats into an “antiseptic dip”.
- xii. During milking, the milker should not: have long nails, sneeze, spit or cough, smoke.
- xiii. After milking, sieve the milk through a strainer or muslin cloth to remove solid particles that may have fallen in during milking.
- xiv. Cover the milk to avoid contamination.
- xv. Move the milk to a clean and cool area.



Activity 1.21

Skills in the production of clean milk on the farm

You will need

a pen, notebook, reference textbooks and internet

Key question: how can clean milk be produced on the farm?

In groups: using the internet or any reference agriculture text books,

1. Research and brainstorm and list the common contaminants of milk on the farm, and give the importance of producing clean milk on the farm.
2. How can you produce clean milk that is fit for human consumption
3. How do you handle milk in accordance with good hygienic practices

Present your finding to the rest of the class and record conclusions.

Milking of Cows

Milking is the removal of milk from the animal's udder. Milking of farm animals can either be done by hand milking or by using a milking machine. Milking should be performed by the most capable person on the farm. This person must be familiar with the process of milk secretion and milk let-down in a cow.

The milk let-down process; As the animal is prepared for milking by washing and drying the udder and teats, the nerves in the teats send message to the pituitary gland located at the base of the brain. The hormone **oxytocin** is released into the blood stream, and transport to the udder. in the udder, this hormone causes the muscle fibre surrounding the alveoli (milk secreting cells) to contract and force the milk out into the ducts and the udder cistern, making milking possible

Milk holdup process; When a cow is mistreated or unduly disturbed or excited just before or during milking, the hormone **adrenaline** is released into the blood stream by the adrenal glands. This hormone reverses the effects of oxytocin (causes alveoli muscles to relax and sphincter muscles to contract) and stops milk let-down. This reaction can cause incomplete milking and reduced milk yield.

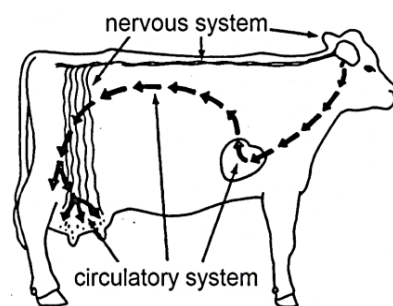


Fig 1.45 Milk let-down response in a cow

General rules in milking farm animals

1. Milk cows at regular intervals; cows can get used to time of the day when they are usually milked. This is vital in reflex conditioning of cows to let down their milk at milking time.
2. Provide quiet and peaceful environment; do not upset or disturb or beat the animal at the time of milking. Milking is a systemic reflex and any annoyance or excitement to the animal can lead to milk holdup.

3. Ensure that the milking cow is well stimulated to let down her milk before starting to milk her. This can be proven by filled up teats and dripping of milk from the teats.
4. The milk shed should be well-lighted and clean with ample temperature and humidity to keep the milkman and the animal comfortable.
5. The animal should be properly trained; properly trained animals do not require much restraints during milking, and it reduces the chances of injuries to the milkman.
6. Use individual towel for each animal to the chance for cow to cow transmission of infections.
7. Use a strip cup to test for condition of milk and mastitis. Affected animals should be milked and the milk discarded.
8. Milk quickly; milk let-down is a short lived condition lasting for about 7 minutes only. Any delay in the milking may reduce milk production and yield.
9. Follow the manufacturer's instructions properly when using milking machines. For hand-milking, the milkman should be clean and should follow the proper hand milking technique.

Preparation for milking

- i. Clean the milking parlour; it should be dust free and free from odours that can taint milk.
- ii. Clean all milking equipment; these should be washed, disinfected and dried.
- iii. Prepare dairy meal; it should be placed in a clean feeding trough in the milking parlour.
- iv. Assemble all milking cows in the assembly yard 30 minutes before milking starts. This gives the cows time to rest and to stimulate the milk let-down process.
- v. Assemble all the necessary milking equipment at the parlour.
- vi. Allow cows to enter the milking cubicles one at a time. Apply restraints if necessary.
- vii. Groom all cows with loose dirt and dust particles and falling hair. This prevents hair and dust from contaminating milk. Also groom stimulates milk let-down in cows.
- viii. Wash the udder and teats with clean water and then dry them with individual towel.
- ix. Test for mastitis. All suspected cows should be isolated, milked and then the milk discarded.



Activity 1.22 Understanding the milking practices

You will need a pen, notebook, reference textbooks and internet

Key question: what are the milking practices?

In groups: using the internet or any reference agriculture text books,

1. Describe the process of milk let-down in a dairy cow and state the conditions that stimulate milk let-down in a cow.
2. Outline the general rules in milking farm animals
3. What preparations should a farmer make before milking farm animals.

Present your finding to the rest of the class and record conclusions.

Procedure of hand-milking

With hand-milking, you use the hands to squeeze out milk from the teats and a milk bucket under the udder to collect milk.

1. Assemble all the required equipment. The equipment should be clean and dry.
2. Ensure a stress free environment for the animal. If the animal is abused in any way, it will not cooperate, and milk production will be reduced.
3. Move the milking animal in the parlour and provide some feeds to reduce anxiety.
4. For safety, to stop the cow from kicking the milk pail, tie the hind legs lightly with the milkman's rope.
5. Wash the udder and teats with warm water containing sanitising solution. And then dry the udder and teats with a towel.
6. Apply a milking salve to reduce friction during stripping and to keep the teats pliable.
7. Strip the foremilk from each teat into a strip cup to verify whether the milk in each teat is normal. If the milk appears abnormal (contains clots, flakes, blood, or serum), it should be milked into a separate container and discarded.

8. Milk out the remaining milk. Use the appropriate hand-milking technique to remove the milk from the teats.
9. Strip the teats to remove the remaining milk completely from the teats.
10. Immediately dip each teat in an effective tea-dip solution. This reduces infections of mastitis and helps to keep the teats pliable and guard against sunburn.
11. Untie the animal and release her back into the herd.
12. Strain the milk and store it in a cool dry place.

Hand Milking Techniques

Whole-hand/pressure technique:

In this technique, hand pressure is applied at the upper portion of the teat with the thumb and the first finger. This pressure trap milk in the teat and then the entire teat is squeezed with all the fingers and the palm. This causes milk to squirt from the canal. After squeezing the milk out of the teat, pressure applied at the upper portion of the teat by the fingers is released, letting milk free to flow again from the cistern of the udder into the teat. The process is then repeated again and again until all the milk has been removed.

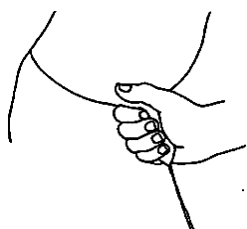


Fig 1.46

Stripping:

In this technique, pressure is applied at the top of the teat by the thumb and the first finger, and then the teat is pulled down, letting it slide between the thumb and the first finger. When the end of the teat is reached, the pressure is released and the procedure started over in the same manner. This technique is used especially in animals with very short teats, where there is no room for the rest of the fingers.

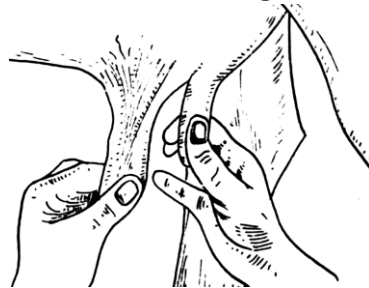


Fig 1.47



Activity 1.23

Participating in performing clean milking

You will need

a pen, notebook, protective gears

Key question: what are the different milking methods and how are they carried out?

In groups: research on the internet or agriculture reference materials;

1. Write a letter to the farm manager requesting to visit the farm. Specify the purpose of the visit.
2. Move to the farm. Put on protective gears.
3. Watch the milking process; which hand milking techniques is being used?
4. Participate in the milking process. With the guidance of the farm manager, apply the different milking techniques. Follow the correct steps of clean milk production.
5. Clean the equipment used, wash your hands and remove the protective gears.
 - a) List the equipment used in the milking process and how each is used.
 - b) Write the steps that should be followed to obtain clean and safe milk from cattle.

Present your findings to the rest of the class and record conclusions.

Procedure of machine-milking

- i. Assemble all the required equipment. The equipment should be clean and dry.
- ii. Ensure a stress free environment for the animal. If the animal is abused in any way, it will not cooperate, and milk production will be reduced.
- iii. Move the milking animal in the parlour and provide some feeds to reduce anxiety.
- iv. Wash the udder and teats with warm water containing sanitising solution, then dry the udder and teats with a towel.
- v. Pre-dip all teats with germicidal dip to reduce mastitis.
- vi. Strip the foremilk from each teat into a strip cup to verify whether the milk in each teat is normal. If the milk appears abnormal (contains clots, flakes, blood, or serum), it should be milked into a separate container and discarded.

- vii. Apply teat cups on each teat and ensure the correct vacuum and flow of milk in the hose
- viii. When the milk flow stops, turn off the vacuum and gently remove the teat cups.
- ix. Strip the teats to remove the remaining milk completely from the teats.
- x. Immediately dip each teat in an effective teat-dip solution. This reduces infections of mastitis and helps to keep the teats pliable and guard against sunburn
- xi. Release the animal

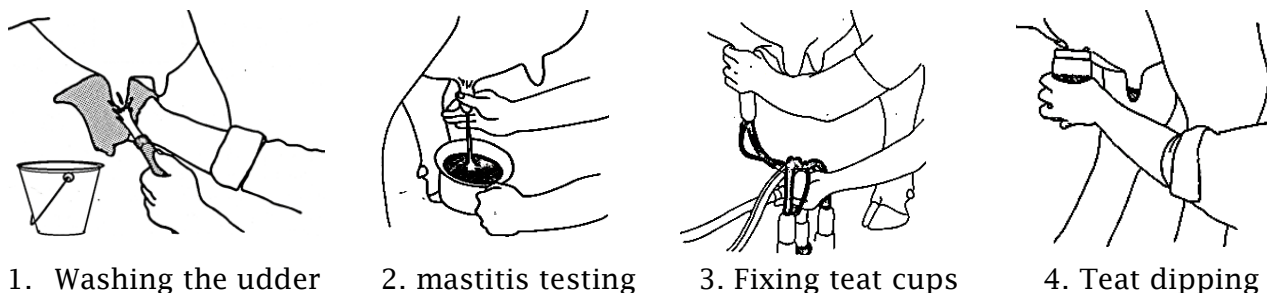


Fig 1.48 Machine milking procedure



Use the internet to identify the different types of milking machines, and state the roles of different parts of the milking machine.



Activity 1.24 Finding out how machine milking is carried out

You will need a pen, notebook, protective gears

Key question: what are the different milking methods and how are they carried out?

In groups: research on the internet or agriculture reference materials;

1. Write a letter to the farm manager requesting to visit the farm. Specify the purpose of the visit
2. Move to the farm. Put on protective gears.
3. Participate in the milking process using a milking machine. follow the correct steps of clean milk production.
4. Clean the equipment used, wash your hands and remove the protective gears.
 - a) List the components of a milking machine and for each, give its function.
 - b) Write the steps followed when using milking machine to obtain clean milk from cattle.

Present your finding to the rest of the class and record conclusions.

Handling and Storage of Milk

Poor hygiene during handling of milk and undesirable practices like addition of water and other substances can introduce the bacteria that cause milk to go bad. The following guidelines will help to keep the numbers of bacteria in milk low and avoid milk spoilage:

- i. Always handle milk in clean metal (aluminium or stainless steel) containers.
- ii. Use only clean and disinfected containers
- iii. Ensure that the milker's hands and the udder are well cleaned and dried with clean cloth
- iv. When transferring milk between containers, pour the milk instead of scooping. Scooping may introduce spoilage bacteria.
- v. Do not milk cows or handle milk if you are sick. Seek medical treatment and resume your work only when the doctor says you are fit to do so.
- vi. Do not store milk at high temperatures. Why?
- vii. Avoid keeping milk for a long time before it is delivered to the collection point.

Procedure for cleaning of milk handling containers

Metal containers such as milk cans, milk pail, milk strainer and milk scoop cups are preferred because these are easy to clean and sterilize.

Before re-using the milk container:

- i. Pre-rinse the container soon after use.
- ii. Thoroughly scrub the container with warm water and a foodgrade detergent or soap (using a stiff bristled hand brush or scouring pad).

- iii. Rinse the container in clean running water.
- iv. Dip-rinse the container in boiling water for at least one minute to kill germs. You may also rinse the container by pouring hot water into it.
- v. Air-dry the container in inverted position on a clean rack in the open.

Factors that influence the milk quality and quantity produced on the farm

Feeding; a well-fed and watered animal will produce more milk of good composition. If cows are fed a diet that is low in forages and high in starch, the butterfat content may fall. Thus, a good balance of forage and concentrates is important. Do not feed the cow with silage during milking or shortly before milking, as this will give rise to off-flavours in the milk.

Health of the cow; unhealthy cows will feed less and produce less milk of poor quality. Diseases like mastitis can permanently destroy milk secreting cells thus reducing milk yield. Diseases like tuberculosis and brucellosis can affect man when consumed in milk. Sick cows should be treated immediately. When the cow is being treated with antibiotics, the milk should not be sold or consumed until the withdrawal period is over.

Breed (genetic variation); local cattle produce less milk but of higher butter fat content whereas, exotic breeds produce more milk of less butter fat content. Also, within the exotic breeds, Jerseys produces less milk of higher quality than Friesians.

Age of the animal; The amount of milk increases with age due to an increase in body weight, which results in a larger digestive system and a larger mammary gland for the secretion of milk. However, milk quality declines as the animal ages.

Pregnancy; pregnant cows produce less milk compared to non-pregnant cows. This is due to the inhibitory effect on milk yield due to the increase in oestrogen and progesterone level.

Milking intervals and frequency; Cows are usually milked at equal intervals i.e. two milkings at 12 hours interval. The more the cow is milked, the higher the milk yield. Incomplete milking for several consecutive days can permanently reduce milk yield for the entire lactation.

Stage of lactation; at the beginning of lactation, milk yield is higher and the butter-fat content of milk is usually high. Milk yield and butter-fat content decrease as lactation time progresses.

Season; during rainy season, there is plenty of forage and water available to cows and as a result, milk yield is usually higher but of less butter fat content as compared to dry season when there is high environmental temperatures and less forage and water available to cows.



Activity 1.25

Finding out how milk is stored and the factors affecting the quantity and quality of milk produced on the farm

You will need

a pen, notebook, reference textbooks and internet

Key question: how is milk stored and what factors affect the quantity and quality of milk produced on the farm?

In groups: research on the internet or agriculture reference materials;

1. Explain the handling of milk that preserves the quality of milk on the farm.
2. What factors affect the quality and quantity of milk produced on the farm.
3. Participate in the cleaning of milking equipment and state the correct procedure that should be followed when cleaning milking equipment on the farm.

Present your finding to the rest of the class and record conclusions.



Activity 1.26

Detecting mastitis in milking cows

You will need

a pen, notebook, strip cup, restraint

Key question: how are internal parasites controlled on the farm?

In groups:

1. Write a letter to the farm manager seeking for the permission to visit the farm.
2. Put on protective gears, ensuring the entire body is well protected.
3. Move to the farm and with the guidance of the farm manager or veterinary officer, identify a lactating cow suspected to have mastitis.

4. Restrain and examine the cow and record your observation in the following table.
5. Feel the temperature of the udder and record your observation.
6. Touch and massage the udder and observe the cow for sign of pain in the udder. Usually the cow will kick when it experiences pain in the udder. Pain is a sign of mastitis.
7. Strip a few drops of milk from each quarter into the strip cup. Look at the surface of black pad and record your observation

Check	Observation
Presence of clots or pus	
Presence of blood stain	
Colour of the milk	
Viscosity of milk (thin and watery)	

8. Dip the infected teats in the teat-dip solution for at least 30 seconds
9. Wipe the treated teats with a single service towels.
10. Release the cow and keep monitoring.
11. Remove your protection gears. Wash your hands thoroughly with water and soap.
12. Write guidelines about mastitis test, the signs and control of mastitis on the farm.

Note; causes, symptoms, treatment and control measures for mastitis have been fully covered under cattle diseases.

Activity of Integration 	The Dairy Development Authority is worried that milk adulteration is on increase. Farmers reportedly add water and flour in bid to increase the quantity of milk and fetch higher prices. Yet, milk adulteration lowers its quality and at times the milk goes bad before reaching the final consumer. You have been selected to represent your class on a sensitisation campaign to help eradicate the vice among the farmers.	
Task	Prepare a poster you will use to sensitise the farmers in your community on the consequences of milk adulteration.	

Exercise 1.8	Self-Check Questions
	As an expert in clean milk production on the farm, (a) Explain the factors which influence the quality of milk produced by the cow (b) Describe how you can ensure production of high quality milk for selling (c) Describe how the mastitis test is carried out using a strip cup. (d) Why is it vital for the farmer to test milk before selling it to consumers. (e) Why is it advisable for milk to be cooled soon after milking?

Clean Meat Production

Meat is composed of muscles (flesh), bones, fats and connective tissues and viscera. It is the major product obtained from beef cattle. Meat is highly perishable product and should be handled carefully and sold or preserved immediately after slaughter. Production of clean meat requires good care of cattle during rearing such as controlling cattle diseases and parasites, and care for the meat during and after slaughtering the animal. Clean meat should be free from diseases, parasites, dirt, off flavours and fresh.

Observing good slaughtering procedure in production of clean meat

Procedure of slaughtering cattle

1. *Pre-slaughter handling:* this involves the following activities;
 - Putting animals in a holding yard and allowing them to rest to conserve glycogen in the body which at slaughter is converted into lactic acid that has a preserving effect on meat.
 - Starving the animals for 16-24 hours and given plenty of water for drinking. Starving reduces the amount of undigested food and faeces in the digestive tract which would contaminate meat.

- **Antemortem inspection** which involves examining the animal to find out whether it has any deformities or serious injuries.
2. **Stunning:** this involves rendering the animal senseless just before slaughter to reduce pain. It is done by use of a hammer, gun or electric shock.
 3. **Actual slaughter:** the neck is cut and the animal immediately hoisted using ropes with the head down and hind legs up to ensure complete bleeding which is essential to producing meat of good keeping quality. The blood can be collected and processed into blood meal.
 4. **Skinning:** Skinning (flaying) is the removal of the hide from the carcass. It is done carefully to avoid damage to the hide and to avoid leaving a lot of meat that can lead to deterioration.
 5. **Evisceration:** this is the process of opening the carcass to remove internal organs. The internal organs are carefully removed such that their contents do not spill to contaminate the meat.

The remaining weight of the carcass after removing the weight of the internal organs, head, hide, feet and blood is known as dressed weight. Dressed weight, divided by the live weight of the animal, multiplied by 100 equals the dressing or killing out percentage.

Dressing percentage

$$= \frac{\text{Dressed weight}}{\text{Live weight of the animal}} \times 100$$

For examples; a bull weighing 250 kg was slaughtered and after complete bleeding and removing the skin, internal organs, the head and feet the carcass weight was 175 kg. Calculate the dressing percentage.

Dressing percentage

$$= \frac{\text{Dressed weight}}{\text{Live weight of the animal}} \times 100 = \frac{175 \text{ kg}}{250 \text{ kg}} \times 100 = 70 \%$$

Post-mortem inspection: meat is examined after slaughter to check for dangerous diseases such as tuberculosis and cysts of internal worms. If the meat is found contaminated, it is then declared unfit for human consumption (condemned) and hence disposed of by burying or incineration. Condemned meat is not supposed to be fed to animals as they are likely to become infected and act as a source of infection to man.

Grading: meat is graded according to amount of fats and degree of marbling (distribution of fats within the muscles), texture and colour of the meat.

Cooling: after grading, meat is quickly stored in a cold room to reduce multiplication of microbes. Keeping of meat at 2°C for about six weeks makes it palatable and tender (tenderization). The enzymes in the meat break down some of the tissues making the meat soft. Tenderization can also be achieved by pounding meat in machines or by using protein splitting enzymes e.g. papain and trypsin to soften the meat by applying them on the surface of meat or injecting them into the animal before slaughter.

Sanitation of the slaughter place: care should be taken to ensure that the slaughter place is clean and dry before and after slaughter to reduce chances of meat contamination. Control of rodents and insects is vital.

Meat By-Products

Bone: these are burnt and crushed to produce bone meal rich in minerals (mainly Ca and P) for feeding poultry and cattle.

Blood: this is collected from the animal when fresh, boiled and dried under the sun, then crushed or broken into small bits. It can also be mixed with feeds as a source of iron and protein.

Condemned meat: this is sterilized and converted into meat for dogs and pigs as long as it is free from diseases like anthrax. It can also be boiled, dried and milled into meat powder for use in animal feeds.

Horns and hooves: these are collected and used in the industries for manufacture of buttons, combs, handles, glue etc.

Hides: may be processed and eaten or tanned into leather for use in manufacturing industry
Ruminal contents: the undigested food stored temporarily in the rumen, reticulum and omasum. They are dried and fed to other animals after mixing with other feed ingredients.



Activity 1.27 Understanding the process of slaughtering cattle for meat

You will need a pen, notebook, reference textbooks and internet

Key question: how is slaughtering of cattle done in order to produce clean meat?

In groups:

1. Research on the internet and agriculture reference books on the procedure of slaughtering farm animals for meat production.
2. Discuss and list the by-products of meat obtained from the abattoir.

Present your finding to the rest of the class and record conclusions.

Meat Quality

The following criteria are used to assess the quality of meat;

Appearance; (colour and texture); fresh meat of good quality is transparent in appearance, reddish and with smooth muscle fibres.

Odour and flavour; meat should have a pleasant fresh smell. Flavour related to rottenness as a result of attack by putrefying bacteria is an indication of poor quality.

Juiciness; meat should be moist but not dripping. Very stiff dehydrated muscles are indication of poor quality.

Tenderness (softness); good meat is that one which is tender-easy to cook and palatable. Tough meat especially from older animals has yellow fat deposits.

Percentage of fat/marbling; meat should be well marbled. High fat proportion lowers the quality of meat. Good quality meat has moderate fat and the fat is well mixed with muscles.

Fleshiness; meat should not be too bony. Good quality meat should have high proportion of flesh to bones.

Level of contamination; good quality meat should be free from contaminants such as soil, vegetation and dung.

Factors that lower the quality of meat

- i. Age of the animal; too old or very young animals produce meat of poor quality.
- ii. Chemical poisoning of the animals.
- iii. Incomplete bleeding of the animal: this increases the rate of putrefaction.
- iv. Infections by internal parasites such as larval stages of tapeworms in the meat and liver flukes in the liver.
- v. Poor feeding of the animal.
- vi. Poor storage of meat: this increases the rate of spoilage of meat.
- vii. Treatment of animals with drugs before slaughter: some drugs may be harmful to humans when consumed in the meat.
- viii. Unhygienic handling of meat i.e. dirty houses: this leads to contamination of meat and increases microbial infection of meat which eventually results into putrefaction of meat.

Before slaughter meat quality is affected by the following factors

- i. Feeding: animal fed on concentrate are more fatty than animals fed on only herbage
- ii. Sex of animal: carcass from bulls gives meat with fewer bones than heifers.
- iii. Age of animal: meat from young stock is tender and juicy.
- iv. Breed of animal: beef type breed produce good quality meat than dairy type.
- v. Exercise subjected to the animal: heavy exercise during growth increase muscle size and tenderness.
- vi. Climate: animals in cold climates have more of fat deposits in their muscles. Also, good climate promote growth of pasture thus weight gain in grazing animals.

Factors that can make meat unfit for human consumption

- Presence of parasites such as tapeworms' larvae in meat and liver flukes in the liver.
- Infection with diseases such as anthrax, tuberculosis, rabies, foot and mouth disease and brucellosis which attack man.
- Contamination from flies, soil, vegetation and stomach contents.
- Presence of prophylactic drugs in the meat such as arsenic compounds and dieldrin.
- Greening of meat due to putrefaction.
- Blood clots or haemorrhages in the meat.
- Rotting and bad odours due to rapid multiplication of putrefying bacteria and fungi.

Factors influencing the rate of meat spoilage

Diseases: meat of diseased animals usually gets spoilt faster than that of healthy animals.

Temperature: high temperature encourages rapid multiplication of putrefying bacteria that causes rotting of meat.

Slaughter conditions; unclean slaughter house leading to meat contamination.

Handling conditions; transportation of meat in unhygienic containers allows bacteria attack hence increasing rate of spoilage.

Moisture content of meat: high moisture content encourages growth of putrefying microbes hence increasing rate of meat spoilage.

Level of bleeding; incomplete bleeding of slaughtered animals leaving much blood in meat increasing rate of meat spoilage.

Preservatives used; application of preservatives like salt reduces the rate of meat spoilage significantly.

Contamination with Rumen contents and soil at slaughter: rumen contents and soil are sources of putrefying organisms.

Indicators of Meat Spoilage

Surface slime: this is caused by bacteria and fungi infection. Is an early sign of spoilage.

Changes in color of meat pigment: the red colour of meat, called "bloom", may be changed to shades of green, brown or grey as result of high bacterial activities in meat

Green fats: this occurs due to decomposition of fats.

Off odours and off taste: undesirable odours and sour tastes that appear in a meat due to decomposition of meat protein with the production of foul smelling compounds

Growth of moulds on the surface of the meat.



Activity 1.28

You will need

Understanding the meat quality and spoilage

a pen, notebook, reference textbooks and internet

Key question: what are the factors that affect the quality of meat?

In groups:

- Discuss the ideal qualities of meat for marketing and human consumption. Mention the factors that make meat unfit for human consumption.
- Explain the factors that affect meat quality before and after slaughter of an animal.
- From the discussion, outline the indicators of meat spoilage and conditions that accelerate meat spoilage.

Present your finding to the rest of the class and record conclusions.

Meat Inspection

At all the slaughter houses, meat is inspected to ensure that the meat is of good quality and fit for human consumption. The objectives of meat inspection are twofold:

- To ensure that only healthy, physiologically normal animals are slaughtered for human consumption and that abnormal animals are separated and dealt with accordingly.
- To ensure that meat from animals is free from parasites and diseases mainly zoonotic diseases, wholesome and of no risk to human health.

Antemortem (before slaughter) Inspection

Antemortem examination should be done within 24 hours of slaughter and repeated if slaughter has been delayed over a day.

Objectives of antemortem inspection

- i. To screen all animals destined to slaughter.
- ii. To ensure that animals are properly rested and that proper clinical information, which will assist in the disease diagnosis and judgment, is obtained.
- iii. To reduce contamination on the killing floor by separating the dirty animals and condemning the diseased animals if required by regulation.
- iv. To ensure that injured animals or those with pain and suffering receive emergency slaughter and that animals are treated humanely.
- v. To identify reportable animal diseases to prevent killing floor contamination.
- vi. To identify sick animals and those treated with antibiotics, chemotherapeutic agents, insecticides and pesticides.
- vii. To necessitate and ensure the cleaning and disinfection of trucks used to transport livestock.

The abnormalities which are checked on antemortem inspection include:

- i. Abnormalities in respiration
- ii. Abnormalities in behaviour
- iii. Abnormalities in gait and posture
- iv. Abnormal discharges or protrusions from body openings
- v. Condition of the coat
- vi. Body condition i.e., whether emaciated
- vii. Abnormal colour of mucus membranes
- viii. Condition of the of dung i.e., whether hard or watery or blood stained or excessive slime.
- ix. Parasites or parasite eggs in dung.

Postmortem (after slaughter) Inspection

Postmortem examination of carcass should be carried out as soon as possible after the completion of dressing in order to detect any abnormalities so that products only fit for human consumption are passed as food for consumption.

During dressing, the meat is exposed to contamination from:

- i. Abattoir environment including implements used, and the hands of the operators. A variety of bacteria and fungi are in the abattoir environment.
- ii. Hides of the animals; hides are heavily contaminated parts and can reach up to 3×10^6 bacteria per cm^2 or more.
- iii. Stomach and gastrointestinal contents; gastrointestinal contents have the heaviest load of microorganisms. Dung contains a lot of bacteria, and various yeasts and moulds.

Thus, during meat inspection it is an important duty of the inspecting officer to ensure that:

- i. The tools used during slaughtering, dressing and meat inspection are well sanitized.
- ii. During cutting into the hide and exposure of the carcass, the external surface of the hide does not contact the carcass meat.
- iii. The viscera are not accidentally opened during the dressing or evisceration. The opened viscera should be separated from the rest of the carcass as quickly as possible.
If any part of the carcass is contaminated with dung or visceral contents, it should be washed immediately or the contaminated part be trimmed off.

Bruises and haemorrhage

Bruises are frequently found on antemortem and postmortem checks in slaughter animals. Bruises caused by transportation or handling are commonly found in the hip, chest and shoulder areas.

Judgement: bruised carcasses affected with localised bruising are *approved* after the affected part being trimmed off.



Activity 1.29

Carrying out the slaughter process and meat inspections for quality

You will need a pen, notebook, protective gears, reference textbooks and internet

Key question: what are the parameters for meat inspection?

In groups,

1. Write a letter to the abattoir manager seeking for the permission to visit the facility.
2. Put on protective gears. What will they include?.
3. Move to the abattoir and participate in the slaughter process of cattle.
 - a) The cattle should have been food starved for at least 24 hour but supplied with water. How is this prctice important in ensuring clean meat production?
 - b) Restrain the cattle. how is this important?
 - c) Stun the cattle. what are the methods used? And why is stunning of the animal done?
 - d) Lay the animal on a clean surface to prevent contmination by soil and dirt.
 - e) Wash your hands and use clean slaughter equipment.
 - f) Use a slaughter machine or a sharp machete or panga cut the neck and major blood vessels at once to allow quick bleeding. Collect the blood in a clean container. Why do you think complete bleeding of the carcass is important.
 - g) Remove the skin and appendages such as the feet and tail. The feet should be cut from the knee or hock.
 - h) Open the carcass longitudinally from the bottom end to remove the internal organs.
 - i) Separate the digestive system and open the stomach chambers to remove waste, wash with clean water and rinse thoroughly to get rid of undigested materials.
 - j) Clean the intestines by squeezing out undigested food, and then rinse in clean water. Check for any signs of worms. Which worms are you likely to find in the intestines?
 - k) Inspect the liver for flukes, poison and infections and place it in a clean container.
 - l) Inspect the meat for bladder worms, bruises and haemorrhage and infections.
 - m) Cut the carcass into desirables pieces and place in clean containers or hoist on hooks.
4. Store meat in a clean cool place away from dust and flies awaiting consumption in 24 hours.
5. Outline the;
 - i) reasons for inspecting meat before taking it to the market for selling.
 - ii) abnormalities which are checked on the animal before and after slaughter.
 - iii) contaminants of meat in the abattoir

Present your finding to the rest of the class and record conclusions.



Diseased carcasses are usually condemned. For parasites, only the affected part or organ is condemned. For haemorrhagic, only affected part is cut and condemned. Carcasses contaminated with soil or dung are usually washed and passed for human

Activity of Integration



The people in your community have a complaint about the selling of unhygienic meat from most abattoirs and butcheries in the area. You have been selected by your class teacher to represent your class on a sensitisation campaign to help those people overcome the problem.

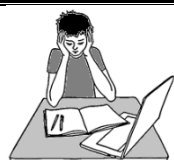


Task

Prepare a poster you will use to sensitize the butchers and butcheries in your community on potential risks of contaminated meat and how to produce clean meat for human consumption.

Exercise 1.9

Self-Check Questions



As an expertin clean meat production and marketing on the farm,

- (a) Describe the slaughtering process I the production of clean meat.
- (a) Outline the factors that make meat unfit for human consumption.
- (b) How can the quality of meat be wenhanced on the farm before and after slaughtering.



Keeping Records in Cattle Management

In senior one, you learnt about the different types of records kept on the farm, and the importance of keeping accurate regular records on the farm. Record keeping involves keeping of all important daily activities and events that occur on the farm. Record keeping is necessary as the records reveal the herd's current production level, the income and expenditure of the farm, identifies inefficiencies and provides information to make improvements. Records will also help to plan for selection and breeding of the herd and to provide vital information for government planning.

Records are kept for each unit for instance, cattle register, health records, breeding records, milk production records, financial records and feeding records. Records can be kept in record books, or using ICT devices.

Table 1.5 Sample for health record

ANIMAL ID NO _____ DATE OF BIRTH _____

Date	Symptoms/Injury /Test	Treatment Given	Cost	Remarks

Table 1.6 Sample for calf management record

ID No.	NAME	SEX	SIRE ID	DAM ID	BIRTH		WEANING		12 MONTHS	
					Date	Weight	Date	Weight	Date	Weight

Table 1.7 Sample for milk production record record

Cow No. _____ Date of Birth _____ Lactation _____ Last Calving Date _____							
Date	Milk Yield litres		Total litres	Date	Milk Yield litres		Total litres
	morning	evening			morning	evening	

Table 1.8 Sample for cattle register

ID No.	Sex	Birth date	Birth weight	Purchase date	Defects	Sire	Dam



Activity 1.30 Finding out the different cattle records kept on the farm

You will need a pen, notebook, protective gears, sample of cattle record and a camera

Key question: what are the different cattle records kept on the farm?

In groups:

1. Write a letter to the farm manager of a nearby cattle farm requesting to visit the farm. Specify the purpose of the visit.
2. Move to the farm.
3. Put on protective gears.
4. Ask the farmer to provide you with records from the different enterprises on the farm. Observe the records and take photographs.
5. How do the records help in cattle management?
6. Discuss how you would improve on the contents of the records observed on the farm.
7. On a paper, copy the records that you have observed and include the information that you would suggest for a farmer to improve on his/her records.

Present your finding to the rest of the class and record conclusions.

Activity of Integration	While attending a briefing by production officer at your sub county about the state of agricultural production, you had him say that many farmers don't know the value of keeping accurate records of their farming activities and as such, don't keep any records on the farm. This statement has left you wondering how irresponsible these farmers are.
	
Task	Prepare a presentation to be delivered at a sub county meeting sensitising farmers to keep records on the farm.

Exercise 1.10	Self-Check Questions
	You might have seen some farmers write some of their farming activities. (a) State how record keeping is important in cattle rearing. (b) Suggest the types of records which should be kept on a cattle farm. (c) Give reasons why most farmers in Uganda don't keep farm records.

Proper Handling of Cattle Products and By-products

Cattle provide a diverse range of products. The primary ones being meat and milk used as food. Also, there are by-products obtained from cattle during the process of rearing or when cattle are slaughtered. By-products from cattle include hides, blood, horns, bones, dung and urine. Fill table 1.8 below, indicating the use of each by-product from cattle.

Table 1.8

By-product	Use
Bones	
Hides	
Horns	
Blood	
Urine and dung	

After you have obtained your cattle products and by-products, you need to handle them carefully as most of them are perishable and can easily deteriorate and lose their economic value. Most cattle products and by-products should be utilised almost immediately or else, they should be preserved if they are to stay for a long period of time. Also, processing cattle products and by-products into other products can help to increase their longevity and economic value.

	Activity 1.31	Finding out how Preservation and processing of cattle products and by-products is done
You will need	a pen, notebook, reference textbooks and internet	

Key question: what methods are used to preserve and process cattle products and by-products?

In groups, research on the internet and reference text books

1. Identify the methods of preservation and processing of cattle products and by-products.

Table 1.9

Product/By-product	Preservation required	Processing required	Product
Beef			
milk			
Bones			
Hides			
Horns			
Blood			
Urine and dung			

2. Which health and safety protection precautions and guidelines are you going to follow?
 3. Outline the guidelines on how cattle products and by-products should be handled.
- Present your findings to the rest of the class and record conclusions.

Marketing of Cattle Products and By-products

Marketing is part of the entire process of producing, promoting and pricing of cattle products and by-products. Cattle products and by-products can be sold raw, or may be processed to decrease their perishability and increase their value in the market. In your community, there are many people who sell cattle products and by-products. Can you mention some of the products they sell in your community? How do they prepare the products to ensure that the products get the highest price on the market? Successful marketing of cattle products require good planning. First of all, you should do a market research; where is the market, when is the product required, how is the product required, level of demand of the product, the product desired and whether raw or processed and market price for the products. Secondly, prepare your product in a form that is most desired by the buyers; clean, sort, process and pack the product. Thirdly estimate the costs of marketing and processing the product, moving the product to the market and other cost such as market dues. Then deliver the product to the market and sell to the customers. Pay attention to the packaging and display of your products to attract the customers. In Activity 1.32, you will learn how to market cattle products and by-products.



Activity 1.32 Understanding the marketing of cattle products and by-products

You will need a pen, notebook, processed cattle products and by-products

Key question: how do you market cattle products and by-products?

In groups,

1. Make a market survey to gather information including;
 - a) what cattle products or by-products are needed in the market, that you will sell.
 - b) where the cattle products and by-products will be sold (market place).
 - c) the form in which the products or by-products will be sold (processed or un processed).
 - d) who to sell the products and by-products to (targeted buyers)
 - e) when the products and by-products should be supplied to the market
2. Prepare the most demanded product you have identified suitable for sale.
3. Set the selling price of the product making sure that there is a profit.
4. Deliver the product to the market.
5. Sell the product following the market survey guidelines.
6. Take note of the customers' advice so as to supply a better product next time.
7. Discuss the challenges you faced during the marketing of the products.
8. Set strategies to overcome the challenges in the next marketing process.

Write a report and share with the rest of the class with the guidance of the teacher.

Activity of Integration



During your class visit to the abattoir, you noticed that the slaughter house was poorly managed. Cattle were being slaughtered on dirty floor, the entire floor was soiled by blood and stomach contents, the carcass and meat was being placed on dirty surfaces and there was no observance of safety and environmental protection guidelines.

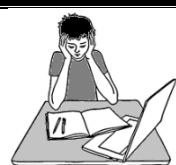


Task

As an agriculture student, design a poster you can use to sensitise the staff at the abattoir on proper handling of meat and by-products to ensure their safety, the safety of consumers and the environment.

Exercise 1.10

Self-Check Questions



As a skilled person in cattle products and by products handling,

- (a) Give the methods which can be used to process cattle products or by-products
- (c) For either meat or milk, describe the handling that can improve the quality.
- (c) What are the challenges of marketing cattle product. How do you overcome these challenges.

Livestock Feed Making

Competency: you should be able to formulate animal feeds based on nutritional requirements, using available feed ingredients for maximum nutrition and optimal economic production



KEY WORDS

- Pelleting
- Rationing
- Steaming
- Crude protein
- Basal feed
- Roughages
- Concentrates
- Feed supplements
- Feed additives
- Feed blending
- Compound feed
- Mineral lick
- Feeding standards

LEARNING OUTCOMES

By the end of this chapter, you should be able to:

1. know the range and types of livestock feeds
2. know the ingredients and understand the technique for making livestock feeds according to specific animal requirements
3. know how to:
 - a. prepare and process a single material feed for livestock
 - b. formulate air-dried livestock compound feeds
 - c. make dry season livestock feed
 - d. blend formulated livestock feeds (k)
4. know how to make mineral supplements



Introduction

A **Feed** is an edible material which when ingested, provides nutrients to the body of the animal. A **feedstuff** is a material containing one or more nutrients and used in formulation of feeds.

In your community there are many farmers keeping different types of farm animals. Farm animals are fed on a variety of feeds to meet their nutritional needs. Different classes of farm animals require different nutrients. Also, nutrient requirement changes at different stages of growth and production levels of the animal.

Range and Types of Livestock Feeds

Livestock feeds are classified into three main types: roughages, concentrates, and mixed feeds or compound feeds.

1. Roughages include pasture forages, hays, silages, root crops, straw and byproduct feeds that contain a high percentage of fiber.
2. Concentrates are feeds which have high nutrient concentrations. They are also classified as:
 - a. *Energy concentrates (Basal feeds)*: are feeds high in energy but often low in protein value. They include fat, cereal grains and their by-products (barley, corn, oats, rye, wheat),
 - b. *Protein concentrates*: are feeds high in protein but low in energy value. They include animal products e.g., blood meal, fish meal, dried skim milk, dried whey, dried buttermilk and oil meals or cakes (soybean, cottonseed, groundnut, oil palm and sunflower seed).
 - c. *Mineral and vitamin concentrates*: rich in vitamins and minerals such calcium, iron, phosphorus. Mineral sources include bone meal, blood meal, oyster and phosphate rock. Vitamin sources include yeast, fish solubles, distillers' solubles, liver meal, alfalfa meal, and milk by products, and some green vegetables.

Concentrates are used in mixing concentrate feeds to provide balanced deity to farm animals.

3. Compound feeds are produced by carefully selecting and blending different feedstuff (concentrates) to provide highly nutritional diets that both maintain the health of the animals and increase the quantity and quality of products such as milk, meat, wool and eggs.

Major nutrients in protein concentrates

- Blood meal: 80% protein, 1.6% fat
- Fish meal: 60-72% protein, 8-10% fat
- Sunflower seed meal: 43% protein, 2.8% fat
- Groundnut meal: 40-48% protein, 1.9% fat
- Cotton seed meal: 41% protein, 1.9% fat
- Hydrolyzed feathers: 84% protein, 2.5% fat
- Poultry-by-product meal: 58% protein, 13% fat
- Soy bean meal: 40-45% protein, 0.9% fat
- Simsim (Sesame) meal: 48% protein, 5% fat

Major nutrients in mineral supplements

- Bone meal: 29% calcium, 14% phosphorus, 51% protein.
- Limestone: 33-38% calcium,
- Oyster shell: 37-39% calcium
- Iodized table salt: iodine, sodium, and chlorine



Soybean has trypsin inhibitor. The substance prevents the action of protein digestive enzymes. Feeding value of soybean can be improved by roasting or cooking of seeds to destroy trypsin inhibitor contained in the seeds.

Groundnut meal has a major problem of contamination with aflatoxins which makes it dangerous to poultry especially ducks and turkeys.

Cottonseed meal has a major problem of having gossypol which is poisonous to most livestock. But, the process of oil extraction and heating reduce gossypol to safe levels.

Sorghum meal has a problem of prussic acid poisoning to mainly ruminants. However, the risk can be reduced by using mature grains and fermentation of grains before use.

Feed Additives

Feed additives are special substances added to livestock feeds to promote growth or prevent diseases. They are normally added to commercial animal feeds. Feed additives include hormones, antibiotics, antioxidants, tranquilizers and coccidiostats. They are important for the following;

- i. They stimulate growth.

- ii. Maintain feed quality during storage.
- iii. They improve food conversion efficiency.
- iv. Improve palatability.
- v. Improve the quality of products such as yellow colour of the egg yolk.
- vi. They protect livestock against disease and parasite attacks.



Concentrates have low crude fibre, have high energy and protein content, are highly digestible and palatable to livestock.



Activity 2.1

Understanding the types of livestock feeds

You will need

a pen, notebook, reference books, internet, different samples of feeds

Key question: What are the different types of animal feeds?

In groups:

1. Develop an observation list of the feeds provided and note the following the identity, physical nature and major nutrient supplied by each feed. Classify the feeds into types of livestock feed (roughages and concentrates).
2. Group the feeds according to their feeding values
3. Research on the internet and list the concentrates used in feeding farm animals, and their percentage composition of the major nutrients.

Present your findings to the whole class and compare your work with that of other groups.



Knowing the types of feeds is useful in livestock feeding. It allows you to feed animals with the right type of feeds and substitution of feeds in a ration should there be a need.



Activity 2.2

Identifying the ingredients for making livestock feeds

You will need

Protective gears, a pen, notebook, maize/wheat bran, soybean meal, fish meal, crushed oyster shells, vitamin premix and mineral salt

Key question: which ingredients are used in making of livestock feeds?

In groups:

1. Put on protective gears.
2. Develop an observation list of the feeds provided and note the following; the identity, physical nature and major nutrient supplied by each feed. Classify the feeds into types of livestock feed (roughages and concentrates).
3. Group the feeds according to their feeding values
4. Which other ingredients are used in your community to make livestock feed other than the one provided to you? Which nutrients do they provide to livestock?
5. Giving reasons for your choices, which ingredients would be used in greater amounts when mixing feeds for; (i) lactating cows, (ii) laying hens, (iii) young calves, and (iv) fattening piglets.

Present your findings to the whole class and compare your work with that of other groups.



Preparing and Processing Livestock Feedstuffs

Feed preparation involves practices like collection, shelling, sorting and cleaning of the feedstuffs. Feed processing means altering the physical (and sometimes chemical) nature of feedstuff to optimise utilisation by animals and to enhance mixing and stability of the feedstuff in the mix.

Objectives of feed processing

- i. To make the feed more palatable.
- ii. To improve digestibility and increase nutrient content and availability.
- iii. To destroy inhibitors and toxins in the feed.
- iv. To facilitate the use of a wide range of raw materials in diet formulations.
- v. To ensure destruction of pathogens and production of hygienic feed.

- vi. To reduce feed wastage during feeding
- vii. To make the storage of feeds easy and safe.
- viii. To change the particle size or density of feed.

Forms in which Livestock feeds are prepared

Livestock feeds can be processed into different forms to give them desired texture and shape.

Meal or mash: the feed is ground into fine particles. Its advantage is that it mixes well with other ingredients during feed mixing, but it also creates a dusty condition in the animal house and there is too much loss from wind blow. In poultry, it encourages selective pecking of the ingredients and the feed tends to stick on troughs and on the animal making them uncomfortable during feeding.

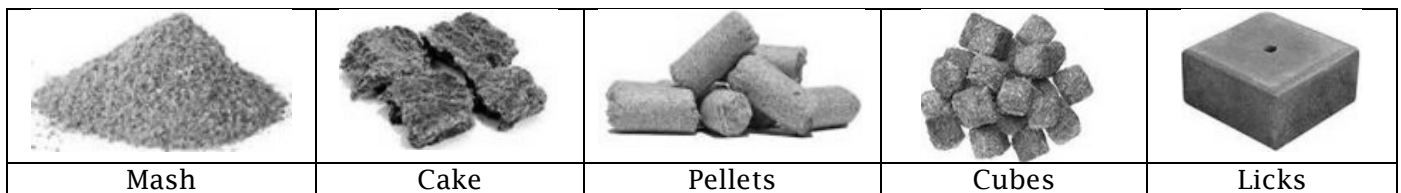
Cake: cakes are usually made from the remains of cereals after oil extraction.

Pellets: formulated ground feed is mixed with a binder e.g. sodium bentonite or cooked starch and then pressed into small cylindrical pieces. Pellets are most preferred form of feedstuffs by most farm animals. They make serving easy and reduce feed wastage during serving and feeding and can easily be picked by the animal from the troughs.

Cubes: a feed is pressed into cubes and then served to animals. It improves intake of feeds. At times, poor forage is mixed with concentrates and then pressed into cubes for feeding ruminants.

Chops: a feed mainly forage, peelings, vegetables, fruits and tubers, are cut into small pieces which can easily be picked by animals.

Licks: minerals are mixed with bulking material usually clay, then put in moulds and pressed into a solid block. The block containing minerals is provided to animals to lick out minerals.



Activity 2.3

You will need

Finding out the form in which livestock feeds are prepared

Protective gears, a pen, notebook, mash, pellets, cubes, broken maize grains, cotton seed cake, chopped forage and mineral block

Key question: in which form are livestock feeds prepared?

In groups:

1. Put on protective gears.
2. Observe the feedstuff provided. Describe the nature of each.
3. How does the nature of each help to increase its palatability and feeding efficiency?
4. Research from the internet and agriculture textbooks on the importance of feed preparation and processing.

Present your findings to the whole class and compare your work with that of other groups.

Methods of Processing Animal Feeds

The method of feed processing depends on the type of feed

Methods of Processing Forage

Dry processing method; in this method, water content is reduced to a desired level. The forage is cut and dried under shade. Dried forage is then baled. This method makes storage and handling of forage easy and convenient. Hay is an example.

Chopping: also known as chaffing: it is either done by hand or machine known as chaff cutter. The forage is chopped into small pieces as fine or coarse particles. Chopping avoids selective feeding thus reducing wastage, facilitates easy handling due to increased density and improves digestion due to exposure of relatively large surface area of roughages to microbial digesting

Drying: It is a process of reduction of moisture content in a desiccator using heat. Dehydrated forage retains a lot of dry matter and crude protein. There is slight loss of carotene content.

Cubing: forage is broken down into rough particles and pressed into cubes and feed to animals. It improves intake of poor quality forage. At times concentrates are added to forage before cubing.

Soaking: is a process of mixing or spraying water on forage such as hay to soften the tissues which improves the feed intake and digestibility of the forage.

Fermentation: anaerobic fermentation of forage helps to soften the plant tissues, break down toxic substance, kill harmful organisms, increase nutrient content, and improve feed intake..

Processing of Grains

Grinding: involves reduction of feeds into fine particles using pressure and shearing. Grinding of ingredients generally improves feed digestibility, acceptability, mixing properties, increases the bulk density of some ingredients, and facilitates further processes such as pelleting, avoids selective feeding of grains and increases compactness and reduces space requirement for storage

Soaking: Grains are soaked in water for 6 to 24 hours. Soaking softens the grains making highly palatable and digestible removes toxic substances and increases the solubility of the grain protein.

Steaming: grains are cooked under very high temperature and pressure. It softens the grains. The process also helps in gelatinizing starches, adding hydration, and partially breaking down proteins for easier digestion.

The process ruptures starch granules improving physical texture and nutrient utilization.

Extrusion: A process of cooking in which feeds are also expanded by the application of adequate pressure. It aids in gelatinization of starch in grains. It is also used for the incorporation of urea in starchy feeds and control of pathogens in feeds of animal source.

Pelleting: The ground feed material is forced to pass through the holes of specific size by a mechanical process of application of optimum amount of heat, moisture and pressure, giving rise to cylindrical-shaped pellets. The purpose of Pelleting is to change dusty and unpalatable feed material into more palatable easy to handle large particles.

Gelatinization: Complete disintegration of starch granules: by application of moisture, heat and pressure is known as gelatinization. It improves the digestion of feed by increasing water absorption ability and rate of action of amylase on soluble carbohydrates (starches).

Roasting: It causes expansion in volume due to heating and generally increases digestibility. Roasting of whole soybeans inactivates enzymes inhibitory factors thus improving nutritive value.

Decorticating; removing the shell covering the grain to enable further processing

Dehulling; removing the seed coat too improve digestibility of the grain.

Steaming Feedstuffs

Steaming some feedstuffs may be required mainly to give the mixture a sticky consistency by gelatinizing the starches present and thus producing a feed which is more stable. It can also help to improve the digestibility of some feeds, making them more useful to the animal, and can help to kill off any unwanted organisms in the feed material.

Steaming of feedstuffs can be done by using a home-made cooker. Mix well together various feedstuffs such as brewery waste and rice bran. Steam this wet mixture for 30 minutes to obtain a moist cake, which may then be blended with other dry feedstuffs such as oilcakes and blood meal with a further addition of water.

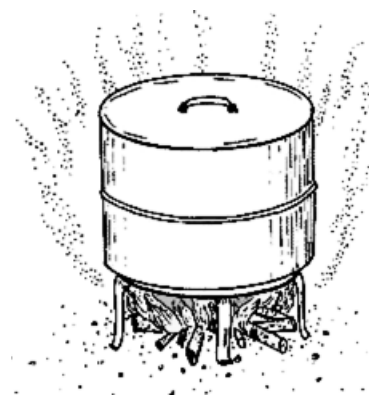


Fig 2.1 Home-made steam cooker



Activity 2.4

Understanding the processing of different feedstuffs

You will need

A pen, note book, forage, panga, protective gears, textbooks, internet

Key question: How to process feedstuffs for livestock feeding?

In groups:

1. Put on your protective gears
2. Harvest and process forage for feeding ruminants on the farm.
3. Research and brainstorm on the objectives of processing animal feeds on the farm.

4. Discuss the different method used to process grains and forage for feeding livestock. Present your findings to the whole class for discussion and record in learning logs

Drying Liquid and Wet Feedstuffs

Liquid and wet feedstuffs such as animal blood, cattle rumen contents, fish and brewery waste must be dried if they are to be stored or used in mixing of a dry feed. Remember that you should:

- Dry feedstuffs as fast as possible to avoid spoilage by fermentation and mould.
- Achieve a final moisture content of 10 percent or less.
- Use clean drying surfaces.
- Protect the drying feedstuffs from excessive humidity and rain by using removable trays or netting if necessary.
- Keep the feedstuff out of the reach of birds and other animals to prevent contamination.

Processing fresh blood

To process blood, proceed as follows;

1. Put finely powdered lime into the bottom of a blood-collecting bucket. One percent by weight of quick lime is preferred. You may also use 3 percent by weight of slaked lime. Assume 1 litre of blood equals 1 kg.
2. Fill the bucket with blood.
3. Stir well until the blood hardens into a black, rubber-like mass.
4. Spread the hardened blood in a thin layer on a clean surface in a sunny place. You may use sacks, mats or a platform. You may also use a solar dryer.
5. Meal the dry blood to fine texture for easy mixing with other ingredients.

Note: lime-treated blood does not stick or decompose. It does not attract flies. Its food value is enhanced by the additional calcium present.

Processing fresh rumen contents

The undigested mixture found in the first three stomachs of herbivores such as cattle, sheep and goats, called rumen contents, is usually available from local slaughterhouses. It can be used in preparation of feeds for fish and pigs. Collect the material and process it as follows.

1. Put on your protective gears and ensure good environmental protection.
2. Select an area exposed to direct sun.
3. Spread the rumen contents well over mats, corrugated iron sheets, trays or a concrete platform. The thickness of the layer should not exceed 5 to 7 cm.
4. Rake the drying mass over from time to time.
5. Once dried, you may easily pound the material with a pestle and mortar.

You may also use a dryer to make this process faster, even in the absence of sun.



Activity 2.5

Processing of livestock feedstuffs of animal origin

You will need

A pen, notebook, reference books and internet

Key question: How to process livestock feedstuffs of animal origin?

In groups:

1. Research and brainstorm on the correct procedure of processing fresh blood, rumen content and silver fish for use in mixing of livestock feeds.
2. Discuss the precautions to be taken when processing animal source feedstuffs.

Present your findings to the whole class for discussion and record in learning logs

Making Livestock Feeds

In our communities, there are many feed dealers who make animal feeds for selling to farmers, and farmers who make their own feeds for feeding their animals, and sometimes, selling excess to other farmers. Farmers and feed dealers, all use different ingredients to make feeds for different types and classes of animals. Most of these ingredients were discussed at the beginning of this

chapter. After identifying the ingredients, they are prepared and processed and then used in formulation of livestock rations. Different techniques are used to make livestock feeds according to specific animal requirements. However, similar steps are followed in all techniques. Be careful to weigh accurately the different ingredients.



Activity 2.6

Understanding the technique for making livestock feeds

You will need

Feed ingredients, weighing scale, tarpaulin, shovel, notebook and pen

Key question: what technique is used to make livestock feed?

In groups:

1. Select the desirable ingredients to use in making of a feed depending on the specific animal requirements. What is the use of each ingredient in the diet of livestock?
2. Describe how you would prepare the feed for the following animals; chick mash for feeding week old chicks and calf pellets for feeding weaning calves.
3. What safety precaution are you taking when mixing the animal feeds?
4. Make a poster showing the steps of mixing feeds on the farm and display it in class.

Present your findings to the whole class for discussion and record in learning logs

Preparing and Processing a Single Material Feed For Livestock

A feed ration can be made up of a single nutrient source (feedstuff). A single nutrient source should be highly nutritious to supply all the required nutrients to the animal.

Preparation of Maize Grain for Feeding Poultry

1. Measure the required quantity of clean dry maize grain.
2. Sort the grains to remove any foreign material.
3. Crush the maize grains to the appropriate size using a hammer mill or a crusher. Don't crush too finely, as birds prefer somehow large grain particles.
4. Serve the birds or bag the excess and store.

Preparation of Fresh Fodder for Feeding Ruminants

1. Harvest the fodder at the proper stage of growth i.e. at flowering stage. At this stage food has high content of proteins, carbohydrate and highly digestible by animals.
2. Remove unpalatable plant species that may have accidentally been harvested with the fodder.
3. Transport the fodder from the field to the farm.
4. Chop the fodder into small pieces to allow easy picking of fodder by the animals, and to increase palatability.
5. Wilt the forage slightly to reduce succulence so as to prevent incidence of bloating in animals
6. To increase palatability and digestibility, add some molasses or urea to the fodder.
7. In all steps, cleanness must be observed not to contaminate fodder since most animals will reject contaminated feed.



Activity 2.7

Skills in preparation of single material feed for livestock

You will need

Maize grain, forage, milling machine (crusher), chaff cutter, tarpaulin, panga, reference books, notebook and a pen.

Key question: How to prepare and processing a single material feed for livestock?

In groups:

1. Identify nutritious fodder at flowering stage. Why do you think it is recommended to harvest fodder at flowering stage?
2. Remove any foreign materials from the fodder.
3. Move the fodder to the farm.
4. Using a panga or chaff cutter, cut the fodder to small pieces of about 3 to 5cm.
5. Put the forage in the feeding trough and serve the farm animals.

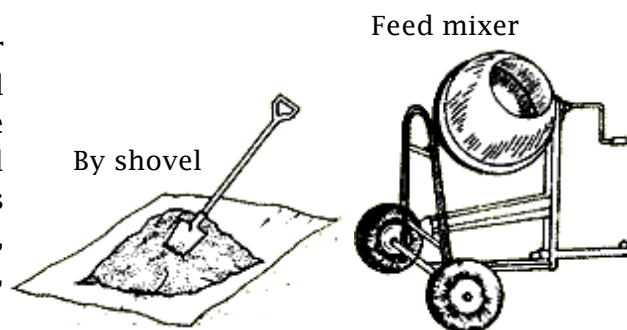
6. Research on how to prepare and process maize grains for feeding livestock. Present your findings to the whole class for discussion and record in learning logs

Preparation of Air-Dried Compound Feeds

Compound feeds are mixtures of two or more feed concentrates in suitable proportion. Compound feeds are feeds of high energy and protein value. The different ingredients in the mix supply different nutrients in different quantities. Compound feeds are normally formulated on the basis of the animal's nutritional requirements. Commonly used ingredients include, grains, brans, protein meals/cakes, minerals and vitamins. Controlled air-drying of ingredients and feeds minimises nutrient loss and preserves the quality of the feed.

Processing compound or mixed feeds

Compound feeds are made by mixing together different feedstuffs in various proportions. A feed mixer is usually used, but also, mixing is done manually using a shovel. Processing of compound feeds involves a wide range of operations such as pretreatment of ingredients (drying, shelling, dehulling and roasting), grinding, proportioning, mixing, pelleting, packaging and storage.



Procedure of hand-mixing compound feeds

1. Clean the floor where mixing is to take place. Mixing should be done on cemented floor or wooden boards or tarpaulin.
2. Proportion the ingredients; Measure out the required energy concentrates and divide it into two batches. Spread one batch on the mixing floor
3. Prepare the premix (get a given amount of vitamins, antioxidants, and minerals and mix with 1 kg of energy concentrate) Spread the premix on top of the energy concentrates.
4. Measure out protein concentrates and spread on top of energy and premix.
5. Spread the second batch of energy concentrate on top.
6. Spread out mineral pre-mix on top.
7. Mix/stir up the concentrates evenly using a shovel by turning the concentrates over and over again until a homogenous feed of uniform texture is attained.

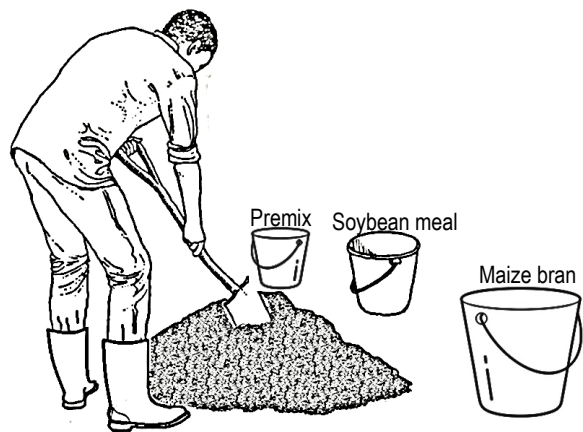


Fig 2.3 Processing a compound feed



Activity 2.8

You will need

Preparing an air-dried compound feed

Maize bran, air dried fish meal, crushed oyster shells, vitamin premix, mineral salt, cemented floor, shovel, weighing scale, notebook, pen

Key question: How air-dried compound feed is prepared on the farm?

In groups:

1. Weigh the desirable quantity of maize bran (energy source) and heap it on a clean surface
2. Weigh the desirable quantity of fish meal (protein source) and pour it on top of maize bran.
3. Add crushed oyster shells and mineral salt to the heap. What is the purpose of adding oyster shells and mineral salt to the feed?
4. Add vitamin premix to provide vitamins in the feed.
5. Mix the contents thoroughly using a shovel until a uniform colour and texture are obtained.
6. Serve the animal in a feed trough according to the specifications. Observe and judge the palatability of the feed. Make necessary adjustment in the choice of ingredients take care of the nutritional demands.

7. Store excess feed in a waterproof container on a raised platform.

Present your findings to the whole class for discussion and record in learning logs



Other than fish meal, which other sources of protein are available for mixing compound feeds for livestock?

Making Dry Season Livestock Feeds

During the dry season, there is usually too little pasture and pasture is of poor quality. This affects the performance of livestock. During the wet season, there is usually excess forage either from the natural pasture or from planted pasture. Thus, it is necessary for farmers to prepare feeds that animals will feed on during dry season. Farmers conserve the excess herbage during wet season in form of hay and silage. Silage is prepared mainly from Maize, Sorghum and good quality planted pastures such as Star grass, Guatemala and Napier grass. Farmers also feed their animals on crop residues, concentrate supplements and herbage from marginal lands such as swamps.



Activity 2.9

Making dry-season livestock feed

You will need

Maize or sorghum plants, grass or mixed grass-legume pasture, panga, molasses, chaff cutter, hay box, strings, weighing scale, notebook, pen

Key question: how to prepare **Silage** for feeding livestock during dry-season?

In groups:

1. Put on protective gears.
2. Harvest a suitable grass or mixed pasture. Instead, you can use maize or sorghum plants. At what stage of plant growth do you think it is best to harvest plant for conserving? And why?
3. Move the plants to the farm, and allow the forage to wilt in the sun for about 3 to 6 hours if too succulent.
4. Chop the plants into small pieces of about 1 to 5cm. why is this important?
5. Spread the chopped forage on a tarpaulin.
6. Add some molasses to the forage and mix thoroughly. Why is adding of molasses done?
7. Pack the material into a big polythene bag while compressing as much as possible to remove as much air as possible. Why is this important? Tie the mouth of the bag tightly with a string.
8. Keep the bag in the shed.
9. Clear the area, clean the equipment and move them to the store. Remove the protective gears.

Record the steps and activities and present them to the whole class for discussion and record the conclusion in your notebook.

Key question: how to prepare **Hay** for feeding livestock during dry-season?

In groups:

1. Put on protective gears.
2. Harvest a suitable grass or mixed pasture. Instead, you can use maize or sorghum plants. At what stage of plant growth do you think it is best to harvest plant for conserving? And why?
3. Leave the cut pasture in the field to wilt in the sun for 1-2 days depending on the weather conditions.
4. Move the plants to the farm and place them in a shed away from direct sunshine and rain for them to dry further.
5. Turn the plants one or twice a day. Why is this important?
6. Check for moisture content of the plants by squeezing the leaves between the fingers and the thumb. Continue to dry in a shade if still wet.
7. Once dry, collect the hay and bale it; place the ropes in the hay box, pack the hay in the box and press it as much as possible and then tie the bales tightly with the ropes.
8. Store the hay bales in a shed away from direct sunshine and rain

Record the steps and activities and present them to the whole class for discussion and record the conclusion in your notebook.



Blending Livestock Feeds

There is no single feed that contains all the nutrients in amounts and proportions required by the animal. Therefore, it is necessary to mix different feedstuffs to get a ration that provides all the necessary nutrients to the animal. A blended feed (also known as compound feed) is intended to improve the nutrients composition and palatability of a feed. Calculation of the proportions of the available ingredients that should be mixed or blended together is termed as *feed formulation*.

Advantages of blended (compound) feeds

- i. The feed is better balanced in terms of nutrients.
- ii. Locally available feedstuffs are better used.
- iii. Liquid feedstuffs such as molasses, liquid urea (bio-activator) can be used more efficiently by absorbing them on to dry materials.
- iv. Increases palatability and digestibility of feedstuffs.

Advantages of mixing animal feeds on the farm

- i. It prevents introduction of diseases in the feeds brought from other farms.
- ii. It is cheap than buying already made feeds from suppliers.
- iii. It ensures proper utilisation of material on the farm; old maize in stores can be utilised.
- iv. Feeds can be mixed according to animal nutritional requirements.
- v. Many feeds from suppliers are usually adulterated with non-food materials, such problems are solved by mixing feeds on the farm.
- vi. Excess feeds mixed on the farm can be sold to generate income on the farm.

Limitation of mixing animal feeds on the farm

- i. Not all required ingredients required in the feed can easily be obtained on the farm
- ii. Limited skills in feed formulation among many farmers.
- iii. Some feedstuffs need first to be processed before they can be used in feed formulation.
- iv. Lack of proper storage facilities for the feeds on the farm.
- v. Lack of proper equipment used in feed formulation.

Basic Factors to Consider in Feed Formulation

When formulating livestock feed, the fundamental aspects to consider are;

Nutrient Requirement of the Animal: In formulating feed, it is essential to consider the nutrient requirement of your livestock. The nutrient requirement of various classes of livestock differs. For example, the nutrient requirement of broilers differs from that of layers. Also, as the animals grow, their nutrient requirement differs. For example, in formulating feed for broiler from day-old to 3 weeks, their nutrient requirement differs from those of 3 to 6 weeks and above. The feed ingredients at your disposal have to meet the nutritional requirement of the animals.



Why do livestock need the nutrients? What is the role of proteins, carbohydrates, vitamins and minerals in the diet of farm animals?

Also, it is vital to avoid overloading animals with nutrients. Of course, animals need nutrients, but what quantity of these do they need? You must supply them the exact nutrient they will need.

Availability of the feedstuff: Apart from the nutrient requirement of animals and the digestibility of the feed by the animals, the feed ingredients at your disposal also determine your feed formula. Use locally available feed sources to minimise costs.

Cost of feed production: When formulating a feed, there is a need to take note of the cost. To ensure profitability, the cost of the feed formulation should not be too high. So, during feed formulation, the formula chosen and the required ingredients to use should be supplied cheaply.

Wholesomeness of the feedstuff in the mixture: the final mixture of the feedstuffs should not be toxic to animals or should not cause digestive disorders when eaten by the animals.

Type of the animal: non-ruminants for example cannot easily digest roughages thus their ration should contain very little roughages than that for ruminant animals.

Feeding Standards

These are statements or quantitative descriptions of the amounts of one or more nutrients needed by animals to remain healthy and highly productive. Or, they are tables, which indicate the quantities of nutrients to be fed to the various classes of livestock for different physiological functions like growth, maintenance, lactation, egg production and wool growth.

Safety Factor or Safety Margins

This is an allowance of feeds/nutrients added when formulating animal rations. Animal feeds are formulated with some level of safety when it comes to nutrient requirements. The use of safety factor is recommended to cater for losses due to spillage during feeding, weighting errors, losses due to insufficient mixing and processing of the feed, and deterioration of nutrients during storage, to cater for animals with requirements above the herd average, and to cater for variability in nutrient composition of feedstuffs used in feed formulation.

Methods of Feed Formulation

After knowing the class of the animal you are formulating feed for, their nutrient requirement, and ingredient available to use. Then you can formulate the feed.

There are 3 main methods for formulating the diet;

The Computer Method: Also known as Linear Programming, it is a simple method of feed formulation. All you need to do is input your ingredient in the software to get the desired nutrient requirement. There is no standard formula you have to follow, so you can play with it depending on the needs of the animals (whether you want to increase or decrease their weight), the ingredient that is available to you at that time or whether the nutrients are digestible.

The Trial and Error Method: The method takes into account all the needed nutrients, calculated and put into proportions that can be adjusted to the required nutrient level.

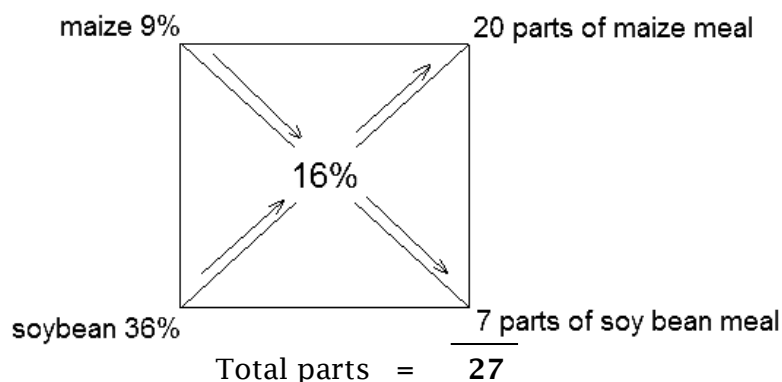
The Pearson Square Method: This method is cumbersome because it involves manual calculation.

Formulating animal feeds using Pearson Square

Procedure

- Draw a square
- Place the protein percentage of the desired ration in the center of the square
- Place the protein percentage of energy concentrate in the upper left corner
- Place the protein percentage of protein concentrate in the lower left corner
- Subtract diagonally across the square, the small number from the larger number irrespective of the sign
- The digits got in the right hand corners gives the parts of ingredient to mixed in the ration

For example, formulate a diet for laying birds containing 16% protein using two ingredients, maize containing 9.0% protein and soybean meal containing 36.0% protein. Additives form 2%



Thus a ratio of 20 parts of maize meal is required to mix with 7 parts of soybean meal to obtain a diet containing 16% protein. Expressed in terms of percentage;

But additives form 2% of the total ration, Thus a percentage of 100 - 2 (98%) will be used in the calculation;

$$\frac{20}{(20 + 7)} \times 98 = 74\% \text{ maizemeal}$$

$$\text{and } \frac{7}{(20 + 7)} \times 98 = 25\% \text{ soybean meal}$$

Vitamins and minerals, antibiotics, coccidiostats and antioxidant are applied as pre-mix.

To ensure that an even mix is obtained, the part should be put into the mixer in the following order; Cereals-Vitamin premix-Protein feed-Cereals - Mineral pre-mix.

Calculating the Mixing Ratio of industrial concentrates

If you notice, some concentrate feed manufacturers write percentages on the packaging sack of their concentrates. For example, 40% Broiler Finisher Concentrate. This shows the proportion/ratio of the concentrate that must go to the quantity of feed to be prepared. In other words, it is the mixing ratio. For example, if you want to make 100 kg of broiler finisher using 40% broiler finisher concentrate, it means you need to use 40 kg. The calculation is as follows:

$$(40 \div 100) \times 100 \text{ kg} = 40 \text{ kg of the finisher concentrate}$$



Activity 2.10

You will need

Calculating proportions of feedstuffs using Pearson Square Method

a pen, notebook, calculator, agriculture reference books, internet

Key question: how is the Pearson Square Method used in the formulation of animal rations?

In groups:

1. Calculate the required amount of fish meal with 65% crude protein, and maize bran 10% crude protein, that you would need to blend 100kg of a feed with 16% crude protein for broilers.
2. Mention other ingredients and the nutrients they provide, that should be included in the above ration to ensure that it supplies the broiler chicken with the desired nutrients.
3. During formulation and blending of animal feeds, it is recommended that you use a *safety factor*. Why is this important?
4. Suppose you did not have the fish meal and maize bran, what other ingredients would you substitute to blend a ration for broiler chicken?

Make a report of your work and share it with the rest of the class.

Making Mineral Supplements

Minerals not provided by feed can be easily and cheaply be supplied with a mineral supplement. Mineral supplements are prepared in form of powder or lick block. The powdered mineral supplements are used as premix in animal feed formulation.

Ingredients used in lick blocks and their roles

- Powdered bone meal; provision of calcium and phosphorus in the mix.
- Termite mound soil; provides mainly iron and other trace minerals. It also used as a bulking material and the high clay content in termite mound soil binds the ingredients together.
- Clay; used as a bulking material and as a binding agent for the ingredients
- Rock salt to provide iodine and contains additional essential minerals and trace elements. Salt also acts as a preservative and an appetizer for the animals to lick the block.
- Cement; used in absence of clay. It acts as a binder to make the block more stable.
- Anthelmintic used for medicated mineral lick blocks to provide extra benefit of controlling internal parasites in farm animals.
- Urea; a bio-activator that stimulates micro-bioactivities in the Rumen thereby increasing feed fermentation and production of volatile fatty acids and protein content of the feed.
- Molasses; Molasses provide some sugars and trace minerals thereby increasing fermentation of food, reduces the dust of finely ground feeds. Makes the lick blocks palatable to livestock.

Formulation of Mineral Lick Block

Materials required; Powdered bone meal, Termite mound soil, Rock salt, nutritional tables (feeding standards) Mould, Water, 2 Buckets

Steps involved;

1. Assemble all the necessary ingredients and equipment.
2. Put on your protective gears.

3. Obtain termite mound soil.
4. Pound and sieve the soil from a termite mound. Then,
5. Mix two parts of rock salt, four parts of bone meal and one part of termite mound soil.
6. Add water gradually as you mix to create a paste of desirable consistency.
7. Mix the ingredients thoroughly until a homogeneous paste is attained.
8. Put the paste in the mould and press to remove air spaces.
9. Leave to dry for in a shed for some hours. Then after, remove the mineral block from the mould and leave it to dry indoors for 1-2 weeks depending on the weather conditions.
10. Use you mineral block for feeding your livestock or for marketing.

Formulation of Medicated Mineral-Vitamin Salt Lick

Materials required; Molasses, Urea (fertiliser grade), Mineral premix, Maize bran (or other bulking agent), Cement, table salt, Water, moulds, Anthelmintic (benzimidazole). The most used drugs are Zerofen and Panacur.

Steps involved;

The procedure of making 20 blocks each weighing 5 kg is as follows:

1. Assemble all the necessary ingredients and equipment.
2. Pour 5 litres of water and 10 kg of urea in the mixer. Run the mixer until urea is dissolved.
3. Add 1 litre of anthelmintic (a 10% formulation of the appropriate drug e.g. Zerofen, Panacur).
4. Weigh out 12 kg of cement and slowly mix in 3 litres of water until a homogeneous paste is attained. Pour this into the running mixer and
5. Add 3 kg of common salt and 5 kg of mineral premix to the mixer.
6. Measure out 25 kg of molasses and put in the running mixer. Use 2 litres of hot water to rinse the cement and molasses containers and add this to the mixer.
7. Measure out 35 kg milled maize bran or any other bulking agent and add it to the mixer. Let the mixer run until the thick paste is homogeneous. The resulting paste is ready for moulding into any preferred shapes and sizes.
8. Oil the inside of the moulding boxes for ease of removal of the blocks after setting. Any edible salad oil can be used for this purpose. Pour the paste into the moulds of 5 kg each.
9. Put the blocks in the sun for about two hours. After this, they should be removed from the moulds and left to dry indoors for 1-2 weeks depending on the weather conditions.



If animals (mainly sheep) consume too much of the blocks at once they can suffer from urea poisoning. Introduce the blocks gradually and the consistency of the blocks should not be too soft otherwise animals may eat too much. Suspending the blocks instead of putting them on a flat surface also discourages animals from biting on the block



Activity 2.11

Skills in making mineral supplements

You will need

Feed concentrates, premixes, mixing equipment and tools, reference books or internet, animal nutritional tables (feeding standards)

Key question: How to make mineral supplements for feeding livestock?

In groups:

1. Using the internet or agriculture reference books, research, and state the important of mineral in the diet of livestock.
2. Why is it advisable for a farmer to prepare his own mineral supplements on the farm?
3. Prepare mineral supplements for feeding your livestock. State the materials used and the quantities of each ingredients used and the procedure followed.
4. What safety measures did you take to protect yourself, the animals and the environment during the preparation of the mineral supplements?

Present your findings to the whole class for discussion and record in learning logs

Storing Animal Feeds

Feedstuffs should be stored with special care to prevent excessive deterioration in quality and feed losses. The most important factors to control are the following:

- Moisture content of both air and feedstuffs should be maintained as low as possible. You should try to keep the relative humidity of the air below 75 percent.
- Temperature of both air and feedstuffs should be kept as low as possible. At temperatures above 25°C, the rates of deterioration and loss may rapidly increase.
- Moulds (fungi) and insects (beetles and weevils) may cause great losses and may contaminate feeds with their wastes. High temperature and moisture favour their development.
- Rodents (mice, rats, etc.) and birds consume feeds, Their wastes also contaminate the feeds.
- Store feeds away from farm chemicals and petroleum products like fuel and lubricants.



Moulds (fungi) may produce toxic metabolites called mycotoxins. The best known being the aflatoxins produced by moulds of the genus *Aspergillus*. These toxins usually occur in by-products of sorghum, maize, groundnut, cottonseed and coconut. The levels of these toxins greatly increase as moulds develop in the presence of favourable temperature and moisture conditions. Some animals can be very sensitive to these toxins: animals stop feeding, and they soon die. Do not buy mouldy feedstuff! Throw out any mouldy material and try to keep storage conditions so that fungal toxin levels remain insignificant.

Improving the storage of feedstuffs

In order to reduce such deterioration and loss of feedstuffs to a minimum during their storage on the farm, pay special attention to the following points.

1. Keep the storage areas as dry and cool as possible:
 - provide a good roof to protect against rainstorms and sunlight;
 - protect the floor from flooding or rising moisture (a raised concrete floor or a wooden platform are best);
 - provide good ventilation through the building (it is best to have its length facing prevailing winds and cut openings in these walls, low down on the front wall and high up on the opposite wall).
2. Close all openings: small wire mesh to prevent the entry of rodents and birds. Make sure the netting is well embedded, to prevent animals passing under it. The presence of a few cats can be very effective in achieving total control of these animals. Lock the door properly. Use ledges to deter ants or termites from entering.
3. Buy and store only materials that are dry, not mouldy and not infested by insects. Sieve a sample to check for mites, insect larvae, pupae, animal droppings, etc. The materials should be loose, free-flowing and should not have been overheated inside storage piles.
4. If bags are used to store feedstuffs, do not stack them too high. If possible, raise these stacks off the ground on wooden platforms.
5. Improve air circulation around individual stacks by: leaving a space between stacks and walls; And separating individual stacks from each other.
6. If materials are stored loose, turn them over from time to time to make sure they are dry and do not overheat in the centre of the pile.
7. Manage your stocks well, limiting their storage time to a minimum by:
 - buying reasonable quantities and avoiding excessive stocks;
 - rotating your stocks carefully, using the oldest feedstuffs first,
 - storing the newest materials behind, if necessary;
 - labelling your stocks clearly, with name and date of entry; and
8. Keep the storage areas clean by sweeping floor and walls regularly and removing spilled materials and broken bags.
9. Keep good records.



Group Project

Aim: Making mineral supplements for livestock using locally available materials

Justification: there is an increase in the counterfeit mineral supplements on the market. The available mineral supplements are of low quality and expensive for an ordinary farmer to afford.

Materials required;

Powdered bones, Rock salt, Anthill soil, Mineral premix, Molasses, Cement, Urea, Rice bran, water, Anthelmintic e.g. Zerofen and Panacur (for medicated blocks), Block Mould, Containers and a Mixer

In groups,

1. Make a plan and design a budget for formulation of mineral and vitamin lick blocks.
2. Obtain the materials to use including the equipment and ingredients.
3. Mix the ingredients in various proportions, and following the correct procedure and Observing high level of hygiene.
4. Use the moulds you designed to shape your lick blocks.
5. Cure the lick blocks using the sun and use them to feed farm animals or for marketing.
6. Write a report indicating the materials you used, the steps you followed to make the mineral and vitamin lick blocks, the people involved, lessons learnt, challenges and recommendations on improvement in the production process.
7. Individually keep records at each stage in the process and produce a report.

Activity of Integration 	There is a general outcry from the farmers on the quality of feeds present in the market; the feeds are highly adulterated, contaminated, mouldy and of low feeding value. The situation affects their livestock by reducing growth and output, and increases incidence of diseases and at times death of their livestock.	
Task	Using your knowledge of nutrition in farm animals and feed making, write a message you will deliver to the livestock farmers on how to prepare their own quality feeds using locally available ingredient. Your message should also show them how they can process pellets and mineral supplements for feeding their animals and safe storage of animal feeds.	

Exercise 2.1	Self-Check Questions
	As an animal nutritionist and an expert in feed making, - List the major nutrients require by farm animals. - Discuss the benefits of processing animal feeds on the farm - Explain the meaning of the following terms as used in animal feeds - Feeding standards - Safety factor - Describe the steps in hand mixing of concentrates during the formulation of compound feeds for layers. (c) Discuss the advantages of mixing feeds on the farm and what advise can you give to farmers to ensure the production of quality feeds safe for feeding farm animals. (d) Describe how fresh blood is processed for use in feed formulation.

Pasture Management and Conservation

Competency: You should identify, establish, manage and conserve nutritive pasture



KEY WORDS

- Deferred grazing
- Ensilaging
- Fodder
- Hay
- Inoculation
- Legume pasture
- Ley crop
- Mixed pasture
- Oversowing
- Rotational grazing
- Silage
- Soil conservation
- Stall grazing
- Standing hay
- Stocking rate
- Undersowing

LEARNING OUTCOMES:

By the end of this chapter, you should be able to:

1. identify and describe different types of pasture
2. evaluate the quality of natural pasture land and improve it
3. select appropriate plant species to improve the botanical composition of a pasture land
4. select appropriate planting material for growing pastures
5. show skills in managing and conserving pastures
6. show creativity in using available waste resources to grow irrigated crops
7. understand the causes of soil erosion and how soil can be conserved, and apply learning in practice
8. understand and apply learning in relation to the importance of the conservation of soil moisture

Introduction

Pasture is land covered with edible plants and used for grazing of livestock. A pasture also means plants animals graze on. Such plants are also called forage. Pastures plants are nutritious edible plants for grazing of livestock.

Importance of pastures

- They are the cheapest source of animal nutrition.
- They protect the soil against soil erosion.
- Legumes pastures add nitrogen and organic matter to the soil.
- The roots of pastures bind soil particles together, hence improving soil structure.
- Preserved pastures such as silage and hay can be sold to earn income.

Types of Pastures

Pasture is classified according to the mode of establishment. Some pastures grow naturally while others are planted specifically for feeding livestock. Thus, we have two types of Pastures; the natural pastures and cultivated pastures or leys.



Activity 3.1

Identifying the different types of pastures

You will need

a pen, notebook, protective gears, camera, observation list

Key question: what are the different types of pasture?

In groups:

- Write a letter to the farm manager seeking for permission to visit the farm. Specify the purpose for the visit.
- Put on you protective gears. Which protective gears are you going to put on and why?
- Visit the farm. With the guidance of the farm manager, move to different grazing areas of natural and planted pastures.
- Observe and record the features of natural and planted pastures.
- Take note of the plant species composition of the natural and planted pastures.
- Discuss how the pastures;
 - are important in the environment
 - botanical composition and growth are influenced by the soil, climate and management.

Present your finding to the rest of the class and record conclusions.

Natural Pastures

These are open areas with a dense cover of native legumes, grasses and herbaceous plant species with scattered trees. Natural pastures include the following types:

- Rangeland:** this is an extensive area of short grasses with scattered shrubs. In the drier areas, these shrubs are usually thorny and fire resistant.
- Grass land:** this refers to natural or planted vegetation in which the dominant species are grasses with a few or without woody plants.
- Bush pasture:** this comprises of shrubs and small trees scattered among medium to tall grass.
- Woodland pasture:** this is comprised of large trees and other herbaceous plants mixed with tall grasses. The trees and shrubs provide browse plants for animals.

Advantages of natural pastures

- Natural pastures are cheap to maintain since they don't require a lot of care and inputs
- Natural pastures contain plant species of different botanical composition, thus, increasing their chances of survival in the environment, palatability and nutritive value.
- They can support a large numbers of livestock i.e. cattle, sheep, goats.
- They can grow even in unproductive soils.
- They require less inputs such as fertilizers if there is need to improve them.

- vi. Pasture species found in a natural pasture are well adapted and can persist under poor management conditions.

Limitations of natural pasture

- i. Natural pastures are less productive in terms of herbage yield and nutritive value
- ii. Grasses mature very fast become fibrous and woody leading to low palatability.
- iii. They have a shorter growing season than that of improved pasture. Thus may not provide a continuous supply of forage to livestock for a long time.
- iv. Output per unit area is generally low as compared to planted pasture.
- v. They are heavily infested by animal parasites.

Improvement of Natural Pastures

Sign that show a natural pasture is in need of improvement

- i. The presence of poor unproductive grass species. Some species of grass may be less nutritious or may have low digestibility.
- ii. High proportion of dead plant tissues on the ground which may prevent free movement of water and air into the soil.
- iii. Increased selective grazing due to increased less palatable or unpalatable pasture species.
- iv. Presence of parasites such as liver flukes and ticks in the herd. Liver flukes may be as a result of poor drainage in the pasture.
- v. Reduced milk and meat production in animals in terms quantity and quality
- vi. Reduced weight gain in especially growing stock. This is most applicable in meat animals.
- vii. Overgrown pasture with less foliage high stalk and less palatable leaves.
- viii. Presence of weeds and poisonous plants such as Sodom and thorn apples in pasture.
- ix. Presence of stagnant water in the pasture. The water acts as a breeding place for parasites such as liver flukes and mosquitoes.

Methods of improving natural pasture

- i. Oversowing with legumes to fix nitrogen into the soil.
- ii. Removal of all weeds and poisonous plants.
- iii. Applying fertilizers to prevent mineral deficiencies limiting pasture growth and nutrient value
- iv. Irrigation of pastures especially during dry season.
- v. Control of run off by ridging and terracing to increase infiltration of water into the soil.
- vi. Clearing of shrubs and trees to avoid shading of grasses.
- vii. Fencing of pastures to control diseases and pests and to facilitate controlled grazing.
- viii. Keeping the right number of animals on the pasture to avoid over grazing
- ix. Provision of drinking water in the pasture.
- x. Controlled burning to destroy a mat of undecayed plant residues and to stimulate regrowth of pasture and also to destroy animal parasites in the pasture.
- xi. Draining of all wet spots to control parasite such as mosquitoes and liver flukes.
- xii. Regular topping of overgrown pasture to stimulate fresh growth.
- xiii. Use of baits to pick and control parasites in the pasture. Baits are animals usually resistant to parasite infection which are released into the pasture to pick up parasites. Baits are later treated to kill all the parasites they picked from the pasture.
- xiv. Provision of animal shades in the pasture to protect animals from bad weather such as rains, sunshine and winds.



Activity 3.2

Understanding natural pastures

You will need

a pen, notebook, agriculture reference textbooks and internet

Key question: how is the quality of natural pastures evaluated and improved?

In groups: using the internet and agriculture textbooks;

1. Brainstorm and discuss the different type of natural pasture.
2. State the advantages and limitations of natural pastures

3. State the signs that indicate poor state of a natural pasture, and the different ways of improving a natural pasture.

Present your findings to the rest of the class and record conclusions.

Planted Pasture

These are pastures planted following the basic principles of crop production.

Selecting Appropriate Plant Species to Grow In a Pastureland

In your community and the farms you have visited, you might have seen different plant species that are fed on by farm animals. These plants are well adapted to the climate and soils in the area. You have to select plant species that are adapted to the local climate, soils and that will give the best production from livestock feeding on them. A good pasture should have grasses and legumes growing together in a mix.

Important Pasture Species

Pasture species are classified into two categories; grasses and legumes.

Common Grass Pasture Species

Signal grass *Brachiaria brizantha*: it is a loosely tufted perennial growing up to 1m high. It has weak rhizomes with struggling decumbent stems. It has poor seeding qualities and is usually propagated by root division. It is fairly palatable when young.

Rhodes grass *Chloris gayana*: it is a leafy perennial growing up to 1.3m or more in height. It spreads by stolons. It is propagated from seeds. Palatability is very good even when mature.

Bermuda grass / Star grass *Cynodon dactylon*: It requires dry to medium rainfall and prefers a high pH. It is rhizomatous and stoloniferous growing up to 90cm high. It withstands heavy grazing and palatability is good when young. It is propagated by root cuttings.

Bahia grass *Paspalum notatum*: It is a deep rooting, rhizomatous perennial growing up to 60cm or more in height. It is established by cuttings or seeds.

Guinea grass *Panicum maximum*: It is an erect tufted perennial grass growing up to 3m high, it forms turfs up to 1.2m in diameter. It is very palatable especially when young. It is usually propagated by root divisions but can also be propagated by seeds.

Jaragua grass *Hyparrhenia rufa*: It is tufted perennial growing up to 3m high, it is drought resistant. It is fairly palatable and fairly nutritious especially when young. It is propagated by seeds or tuft divisions.

Elephant grass / Napier grass *Pennisetum purpureum*: It is a tall tufted grass growing up to 7m high. It is deep rooted and palatability is very good when young and highly nutritious. Propagation is from rhizomes, and stem cuttings. The grass is good for fodder production and silage making.

Kikuyu grass *Pennisetum clandestinum*: It requires fertile well drained soil and a cool climate with high well distributed rainfall. The grass is low-growing deep rooting, rhizomatous and stoloniferous. Palatability and nutritive value are good.

Guatemala grass *Tripasum laxum*: It is a tall tufted grass growing up to 5m high. It is deep rooted and has broader leaves with good palatability and highly nutritious when young. Propagation is from rhizomes, and stem cuttings. The grass is good for fodder and silage production.

Common Legume Pasture Species

Centro *Centrosema pubescens*: It is a prostrate, trailing leafy perennial. The leaves are trifoliate with petiole 3-10cm long and ovate leaflets. The flowers are yellow, red or purple and the seeds are enclosed in a pod. It is propagated by seeds. It mixes well with Napier and Jaragua grass.

Green leaf desmodium *Desmodium intortum*: It is a coarse, spreading or trailing perennial rooting at nodes. It tolerates water logging and drought. It mixes well with many grasses if well managed palatability is good propagation is by seeds.

Siratro *Macroptilium atropurpureum*: It is deep rooting, creeping and climbing by twinning, stoloniferous perennial. It is persistent, compatible with many grasses e.g. Rhodes, Guinea and Napier, palatability is good and propagation is by seeds.

Glycine *Glycine wightii*: It is a trailing deep rooted perennial, requires fertile soil that is neutral to slightly acidic and well drained. 700-1700mm rainfall is ideal for proper growth. It mixes well with tall grasses such as elephant grass and Jaragua. Palatability is good, propagation is by seeds.

Stylo *Stylosanthes gracilis*: It is erect tufted shrubby perennial tolerant of acidic infertile and poorly drained soils. It is drought resistant and tends to dominate many grasses. Palatability is fair and propagation is by seeds.

Lucerne or Alfalfa *Medicago sativa*: It is deep rooted, grows well in high lands with cool climate. It possesses small leaves which are shed during drought. The stem and leaves are hairy.

Silverleaf desmodium *Desmodium uncinatum*: It is a coarse, spreading or trailing perennial rooting at nodes. Leaves are pale green with a silver patch at the centre. It tolerates water logging and drought. It mixes well with short grasses. Palatability is good and propagation is by seeds.

Ecological Success of Grasses

Grasses are the main species among pasture plants found everywhere in east Africa this is due to several reasons outlined below:

- They produce very many viable seeds thus multiplies very fast.
- Seeds of grasses are light and easily dispersed, and others have hooks to attach them to animals and humans. This helps them to quickly spread and colonize new areas.
- Grasses are highly adapted to defoliation by regenerating quickly after being grazed
- They demand less nutrients thus they can survive on poor/infertile soils.
- They have short life cycles. This insures them against unfavourable weather.
- Some have underground vegetative structures such as rhizomes that are able to sprout after the shoot has been destroyed.
- They are quite resistant to drought.
- Their extensive root system allows reach to nutrients and moisture from wider soil sphere.
- Grasses are less affected by pests and disease as compare to other species of pasture.
- The seeds can remain viable for a very long time in the environment.



Activity 3.3

Identifying pasture species of economic value in Uganda

You will need

a pen, notebook, reference textbooks and internet

Key question: What are the common pasture species in Uganda?

In groups:

- Write a letter to the farm manager seeking for permission to visit the farm. Specify the purpose for the visit.
- Put on your protective gears and go ahead to visit the farm
- With the guidance of the farm manager or a teacher, observe and identify pasture species on the farm, and give the description and growth of each. Which species is dominant and why?
- Group the pasture species identified into legumes and grasses. Why is this important?
- From the growth habits of legume and grass species identified, select one legume and a grass species that should be grown together to make a mixed pasture. How is this practice beneficial?
- Curry some plant parts from the pasture and use them to prepare a pasture album.

Present your finding to the rest of the class and record conclusions.



It is evident that grass pasture species are the most common pasture species identified in different areas. What factors do you think have made grasses the dominant species?

Factors that influence the choice of pasture species to grow

- | | |
|---|--|
| • Adaptability of the species to the climate. | • Topography of area/relief. |
| • Intended use of pasture. | • Resistance to pests and diseases. |
| • Resistance to drought. | • Palatability of the pasture species. |
| • Life span of the pasture. | • Ability to re grow after hard grazing/cutting/topping. |
| • Nutritive value of the pasture. | |
| • Soil type. | |

Qualities of a good pasture species for planting

- i. It should be easy to establish.
- ii. It should be tolerant to drought.
- iii. Resistant to grazing
- iv. Highly digestible or easy to digest and high in essential nutrients.
- v. It should be able to re grow quickly after grazing
- vi. It should be long lasting
- vii. It should be palatable i.e. animals should voluntarily and readily eat it.
- viii. It should not contain toxins
- ix. It should have a good compatibility with other pasture species.
- x. It should be resistant to pests and diseases
- xi. High yielding / high dry matter productivity.
- xii. Legumes should be able to fix enough Nitrogen in a wide range of conditions.



Activity 3.4

Participating in selecting suitable pasture species to grow in an area

You will need

a pen, notebook, reference textbooks and internet

Key question: what factors should you consider when selecting a pasture species to grow?

In groups:

1. Write a letter to the farm manager seeking for permission to visit the farm. Specify the purpose for the visit.
2. Put on your protective gears and go ahead to visit the farm
3. With the guidance of the farm manager, observe and identify the dominant pasture species.
4. Find out the condition of each pasture species growing in the area.
5. Find out if there is any grass and legume pastures growing together. How are their growing habits related?
6. Find out the aspects of the pasture species growing in the area in terms of palatability to livestock, ease of propagation, nutritional value, tolerance to drought and growth rate.
7. From your evaluation, which pasture species would you recommend for growing in the area and give reasons.
8. Brainstorm and discuss the factors that you should consider when selecting a pasture species to grow.

Write a report of your findings and present it to the rest of the class.

Selecting Appropriate Planting Materials for Growing Pasture

Different planting materials can be used to establish pastures. They can be established from seeds and vegetative parts like tuft splits and stems cuttings. Legume pastures are mostly established using seeds, and grass pastures mainly established using vegetative parts and some times, using seeds. The quality of planting materials used determine the initial establishment of the pasture. Planting materials should be free from pests and diseases, viable i.e., able to germinate and clean.



Activity 3.5

Participating in selecting suitable pasture species to grow in an area

You will need

a pen, notebook, pasture seed; guinea grass (A), signal grass (B), Jaragua grass (C), stem cuttings (D) and tuft splits (E), hand lens and internet

Key question: which planting materials are used in the establishment of pastures?

In groups:

1. Put on your protective gears and go ahead to visit the farm
2. Examine each of the following planting materials provided A, B, C, D and E. Name the materials and pasture species they are used to establish.
3. Comment on the suitability of each for use as a planting material
4. What benefits and disadvantages are associated with the use of planting materials A and D?
5. Brainstorm and discuss the factors considered in selecting appropriate planting materials

Write a report of your findings and present it to the rest of the class.

Agronomic Practices in Pasture Growing

Seedbed preparation: Land should be cleared of any vegetation including trees. Then the land should be deeply ploughed and harrowed to get a fairly fine seedbed. Fine seedbed is vital especially when small seeds are to be sown.

Seed selection: this involves selection of the species to be sown. Species to select should be:

- i. Ecologically well adapted to the local climate and soil conditions.
- ii. Suitable for the planned length of time for which the pasture is being established. Species are selected for short term or long term ley, and for permanent pasture.
- iii. Legume to be planted in the mixture to fix nitrogen for grass species.
- iv. Leafy plant species to provide good canopy and protect the soil, increase organic matter and improve soil structure.
- v. Highly palatable and nutritious

Seed treatment: seed treatment may be done to break dormancy, improve germinability, protect seed from microbial and pest attack and to carry out Rhizobial inoculation.

Planting: planting should be done at the beginning of the rains because at this time soil moisture, soil temperature and nutrient level are optimum. Small seeds are broadcasted using hands or machines or planted using seed drills

Weed control: Occasionally some annual weeds may invade the young pasture. These should be controlled immediately as they compete with young seedling for growth factors. Weed control can be achieved by hand pulling of weeds and use of selective herbicides.

Irrigation: this is mainly important during dry periods or when rainfall is inadequate to provide the moisture for vigorous pasture growth. Sprinkler irrigation is the most conducive method to use.

Fertilizer application: phosphatic fertilizers are applied during seedbed preparation or they are broadcasted together with seed during sowing. In mixed pasture, application of nitrogen fertilizers is not important as the legumes fix nitrogen in the soil. In pure stand pasture, nitrogen fertilizers should be applied as top dress.

Controlled grazing/ harvesting of fodder: once the pasture has fully established, they can be harvested and fed to animals, or animals are allowed to graze on the pasture lightly to stimulate fresh growth of especially lateral buds, allowing full growth and coverage of the ground.

Establishing a mixed pasture

1. Clear the existing vegetation by burning to ensure that the new pasture stand starts off without off types.
2. Remove all tree stumps and other obstacles such as stones.
3. Plough the land deeply to improve soil aeration and tilth.
4. Harrow the ploughed area for a fine seedbed. i.e, Carryout secondary tillage.
5. Apply basal fertilizer (phosphate-based fertilizer) for a good start for the pasture.
6. Select good seeds or vegetative materials from suitable grass specie.
7. Plant the grass. Broadcast the seeds or make shallow well-spaced drills if seeds are to be used as planting materials by drilling. If vegetative parts are to be used, make wider furrows.
8. When grass has germinated and fully developed, select a suitable legume that is compatible to the grass species. Inoculated the seeds. Make well-spaced drills in between the grass and plant the legume seeds. (oversowing).
9. Irrigate the pasture adequately especially if rainfall is insufficient.
10. Carryout gap filling where there was scanty germination.
11. Fence off the pasture and carry out the subsequent management practices for proper growth.

Factors limiting pasture establishment

The following factors limit successful establishment of pasture:

- i. Physical loss of the seed, perhaps due to predator or due to physical movement of seeds to inaccessible depths on cracking soils. The above situation leads to uneven growth of pasture.
- ii. Loss of seed viability which is often associated with seed weathering.
- iii. Failure to germinate due to environ-mental stresses or to soil impenetrable hard soil surface. These have been discussed in previous chapters.

- iv. Mortality of emerged seedlings, due to environmental stress, severe competition among pasture plants for growth factors such as moisture, numerical salts and light.
- v. Early invasion of seedlings by weeds and pests and diseases.
- vi. Grazing the pasture too early before full development and establishment of roots and shoots.
- vii. Drought immediately after germination that results into moisture stress with in the plant.
- viii. Unfavourable soil condition such as extreme pH, salinity and poor drainage.



Activity 3.6

Understanding activities involved in establishing a pasture

You will need

a pen, notebook, agriculture reference textbooks and internet

Key question: what activities are involved in establishing a pasture?

In groups: using the internet and agriculture reference textbooks;

1. Brainstorm and discuss the steps involved in establishing a pasture.
2. It is recommended that grass pasture and legume pastures species are planted together in the pasture land. How is this practice beneficial? Summarise how you can establish a grass-legume mixed pasture.
3. Sometimes despite all the efforts to grow a good pasture, the pasture may fail to establish. Outline some factors that may cause a pasture to fail to establish successfully.

Prepare your findings and present them to the rest of the class and record conclusions.



Activity 3.7

Participating in planting of pastures

You will need

a pen, notebook, protective gears, garden tools, planting materials, fertilisers

Key question: what practices are involved in establishing a pasture?

In groups:

1. Put on your protective gears such as the gumboots, overalls, hand gloves and face mask.
2. Observe the environmental protection measures.
3. Prepare the land. Which tools are going to use? Which activities are involved in the land preparation? How will the planting material influence the way you prepare the final seedbed?
4. Apply fertilisers. Which fertiliser are you going to apply and why?
5. Plant the pasture. Which method of planting are you going to use and why?
6. Clean the tools, remove the protective gears and wash up. Why is this done?

Prepare a report and present it to the rest of the class and record conclusions.

Managing Pastures

Once the pastures are fully established, they should be properly managed to allow them to provide quality forage to livestock. The following are management practices that should be carried out;

- i. Controlling weeds: weeds in this case may involve any intruder whether palatable or non-palatable dangerous weed species.
- ii. Maintain correct stocking rates to avoid over grazing.
- iii. Regular application of fertilizer to maintain/ improve the fertility of the soil.
- iv. Oversowing with legumes to maintain balance between grasses and legumes and to increase nitrogen fixation.
- v. Control over growth by topping. This also ensures fresh regrowth of pastures.
- vi. Irrigation during droughts to ensure good growth of pasture during dry spells.
- vii. Practice a correct grazing system e.g. rotational grazing or strip grazing and stall grazing. This is aimed at avoiding incidences of overgrazing and trampling of pastures.
- viii. Harvesting and conserving the pasture during time of plenty mainly during rainy season. The conserved forage can then be fed to animals during time of scarcity especially during dry spells. Pasture can be conserved as hay (dry form) or silage (fermented form).



Activity 3.8 Understanding management of pastures

You will need a pen, notebook, agriculture reference textbooks and internet

Key question: how is a pasture managed?

In groups: using the internet and agriculture textbooks;

1. Brainstorm and discuss the effects of poor pasture management. How you can manage an already established pasture.
2. Describe how you can effectively control;
 - (a) Poisonous weeds in the pasture
 - (b) Bare patches in the pasture
 - (c) Stagnant water
3. State why is necessary to
 - (a) Control poisonous weeds in the pasture
 - (b) Apply fertiliser in pasture
 - (c) Fill bare patches in the pasture
 - (d) Remove Stagnant water
 - (e) Create paddocks in the pasture
4. Apply the principles of pasture management to manage the pasture at school

Present your findings to the rest of the class and record conclusions.

Oversowing

This is the planting of legume pasture seeds in already established grass pasture. Oversowing is required where protein limit stock production or where soil nitrogen limits pasture growth.

Procedure of Oversowing

1. Select the legume seeds which can compete favourably in the mixture. Select climbing legumes for perennial tall grass species and creeping legumes for short-low growing grasses.
2. Prepare the grassland for planting: the success of oversowing usually depends on some (often minor) modification of the grassland. This can be achieved by:
 - i. Heavy grazing before seeding to suppress grass. Also heavy concentration of animals provides hoof cultivation that favours successful germination of broadcasted seeds.
 - ii. Burning of the pasture especially at the start of rains, and when shooting (regrowth) has commenced legume seeds are sown without cultivation.
 - iii. On sloping land where burning and cultivation can lead to erosion of soils, a sod seeder or jab planter can be used to sow seeds in a stand of pasture. Also narrow strips can be made between undistributed grassland
3. Planting/seeding: timing of seeding is the most critical factor in the establishment of legume crop in grass land. The seeding should be made at the beginning of rains or after rains have established. Seed placement can be done by chop and plant method or by broadcasting.
4. Irrigate and control the weeds up to the time when the mixture is fully established.
5. Light grazing should start after the mixture has fully established.



There should be systematic criteria for selecting grass-legume combinations for mixed pasture; Grass-Legume plant species should be compatible and their existence should be complementary rather than competitive. They should have different growth habit i.e. erect stem versus creeping to provide good ground cover, erect stem versus climbing stem, same growth height to reduce competition for sunlight.

Undersowing

Sometimes it is necessary to sow pasture seeds or to plant vegetative materials under a crop cover. The crop should be that one that allows at least some light to reach the ground.

Procedure of under sowing

First plant the nurse crop at proper spacing in a well prepared seedbed at the beginning of rain season. After the nurse crop has fully established immediately after the first weeding (usually by cultivation) pasture seeds or vegetative materials are planted in the nurse crop.

No further weeding should be done up to when the nurse crop is harvested. Harvesting should be done as soon as possible to allow enough sunshine to reach the pasture seedlings.

Benefits of undersowing	Disadvantages of undersowing
• Young seedlings are protected from heavy drops of rainfall, and scorching sunshine. • Cases of soil erosion and wind damage to the seedlings are minimized. • It allows multiple productions of food crops and pasture in the same field. • The cover crop residues after harvesting may provide forage to animals before grazing can begin. • Nutritional benefits may arise if one of the species involved is a legume	• Competition for growth factors such as nutrient and moisture may occur leading to poor yield especially in cover crops • There may be easy spread of pests and diseases especially where the species involved are of the same family e.g. smut disease in maize may also affect grasses. • The shade from cover crops may lead to etiolation of under sown pasture leading to poor growth. • During crop harvesting, trampling of pasture may occur resulting into death of pasture.



Activity 3.9 Finding out the meaning of terms used in pasture improvement

You will need a pen, notebook, reference textbooks and internet

Key question: how is seed inoculation, seed dressing, undersowing and oversowing done?

In groups:

1. Research and write down the meaning of the following terms; seed inoculation, seed dressing, undersowing and oversowing.
2. Describe the steps involved in oversowing an already established grass pasture

Present your finding to the rest of the class and record conclusions.

Mixed and Pure-Stand Pastures

A *mixed pasture* is a pasture that is composed of grasses and legume growing together in the field. A *pure stand pasture* is a pasture made up of only one plant species growing as a sole plant in the field. It may be a pure stand of a legume or a grass.

Advantages of mixture pasture

- i. Legumes in the mixture help to fix atmosphere nitrogen into the soil. The nitrogen fixed is used by growing grasses in the mixture, thus reducing expenditure on nitrogenous fertilizers.
- ii. Legumes are deep rooters; this helps them to absorb nutrients from deeper layers of the soil, thus helps in nutrient recycling especially when they shade their leaves on the surface soil.
- iii. Legumes in the mixture increases palatability and nutritive value of the pasture thus minimizing the need for supplementary feeding.
- iv. Because of the rooting habit of legumes they are more tolerant to drought thus extend the grazing period into dry seasons.
- v. Legumes provide good soil cover and supply more organic matter to the soil; this helps to minimize soil erosion and conserves soil moisture.
- vi. A mixture of legumes and grasses gives a higher yield of herbage than pure stand pasture
- vii. The mixture reduces management costs by reducing the frequency and intensity of weeding and fertilizer application.

Advantages of pure stand pasture

- i. Seed harvesting is made easy, especially when pure seeds of a given specie are needed (i.e. no contamination/mixing up of seeds)

- ii. Forage harvesting by especially machines is made easy since a pure stand pasture is uniform in growth and maturity.
- iii. Cases of selective grazing are reduced since the animal will have no option but to graze on the available specie in the field.
- iv. Pure stand pasture is easy to manage; practices such as fertilizer application, irrigation and weed control are made easy to apply.
- v. Pure stand pasture may be easy to establish requiring a uniform seedbed throughout, which demands less labour as compared to preparing different seed beds for different species.



Activity 3.10

Participating in planting of mixed pasture

You will need

Garden tools and equipment, planting materials, fertilisers, protective gears, a pen, notebook, reference textbooks and internet

Key question: What agronomic practices are required to manage an established pasture?

In groups:

1. Prepare all the tools and equipment to use.
2. Put on protective gears.
3. Move to the land where initial ploughing has been done
4. Prepare the seed bed making sure to provide a suitable soil tilth. Apply fertilisers
5. Select a suitable method of planting depending on the planting materials available.
6. Plant the grass pasture first and then after three weeks, plant legume pasture. Why is it not advisable to plant grass and legume at the same time in the field?
7. Keep proper records. Which records do you keep and why?

What is the benefit of mixed pasture?

What other management practices do you intend to adopt for your mixed pasture?

Keep a learning log at every step and share with the rest of the class.



Group Project

Aim

To Investigate the effect of growing grass and legume pasture species together on quality and yield of pasture

Justification: many cattle farmers rarely plant pastures and the few who plant them are reluctant to mix them. As a result, farmers use a lot of inputs to improve pastures and usually, the quality and yield of pasture is low.

Materials required;

Two plots of land, grass pasture seeds, legume pasture seeds, necessary tools and equipment, protective gears, record book, notebook, a pen

As a class,

1. Develop and document a budget for establishing pasture gardens, buying seeds, fertilisers, pesticides and equipment.
2. Divide the land into two plots; Plot 1 plant a grass pasture and Plot 2 plant grass and legume pastures together.
3. Grow a pasture garden carrying out the different agronomic practices (land preparation, botanical composition selection, planting, applying fertilizer, weeding, irrigation, utilising shade trees and water points)
4. Write a report indicating the materials you used, the steps you followed to establish and manage the pastures, results of the investigation, discussion of the results, the lessons learnt, challenges, recommendations and conclusions.
5. Individually keep records at each stage in the process and use a word processing program to produce a report.
6. Present your report to your teacher and the rest of the class

Pasture Conservation

The supply of forage to live stock is seasonal because of rainfall pattern. During the rainy season, there is abundant growth of pasture and during the dry season, pasture is very scarce with a resultant decline in animal performance. Thus forage conservation helps to bridge the nutrition gap during period of plenty and period of scarcity of pasture.

Objectives of forage conservation	Problems of pasture conservation
• To sell forage to other farmers during drought. This act as an alternative source of income on the farm. • To increase the total amount of herbage produced per unit area. • To get feeds for use in seasons when forage may be scarce. • To conserve forage in succulent form. • To avoid wastage of forage during periods of plenty. • It increases the number of livestock that can be kept per unit area. • It enables animals to eat plant materials that they would not eat in fresh form.	• High costs involved in acquisition of equipment and construction of storage structures. • High labour requirements in harvesting forage, curing and storage • Low nutritional values in forage. • High technical skills involved • High risks of losing the material especially when a mistake is made in the process of preservation or fire as in hay. • Unfavourable weather mainly rainfall which affects the quality of preserved forage.

Methods of Forage Conservation

1. Drying (hay)
2. Fermentation (silage)

Hay

This is pasture conserved in dry form. Forage is dried to very low moisture content to prevent deterioration of the forage.

Hay making procedure

1. The crop is cut when about 50% of the plants have flowered
2. The crop is spread out evenly on the ground to dry for 2-3 days. It should be dried under controlled conditions in order to retain its nutritive value and the original crop colour.
3. The hay is windrowed and then gathered or baled.
4. The bales of hay are then stored in a shed out of reach of rainwater and sunshine.



Note: Rapid drying is recommended to ensure high quality hay. Slow drying results in oxidation of soluble carbohydrates hence poor quality. Prolonged exposure to sun results in the breakdown of chlorophyll and carotene.

Factors affecting the quality of hay

- i. The type and nutrient value of the forage
- ii. The stage of growth of the forage
- iii. The weather conditions such as rain and sunshine
- iv. The rate of drying of the forage
- v. The method used in drying the forage sun drying vs. air drying
- vi. The method of storage of hay
- vii. The amount and type of additives added

Losses in quality and quantity of hay in the field is due to;

- i. Plant respiration: respiration converts sugars to water and carbondioxide, decreasing the energy content and increasing the cell-wall content.
- ii. Physical loss: this results at the time of mowing, conditioning, raking and baling. Physical

loss represents forage not picked up during the harvesting and loading and also breaking of forage to pieces especially when it is too dry.

- iii. Rain damage: rainfall especially during field drying of forage not only prolongs the drying time but also results in a direct loss of soluble nutrients through leaching.
- iv. Microbial breakdown of forage: this is true especially when the moisture of the swath is high. Moulds tend to grow on the forage reducing the nutrient value and palatability of the forage.

In the activity 3.6 below, you are going to learn practical skills involved in the making of hay from fodder available around the school.



Activity 3.11 Participating in the process of making Hay

You will need

Sickle, sisal ropes, tripod stands, bale box, fodder at flowering stage, good sunny weather, protective gears

Key question: What are the practical steps involved in the making of hay?

Safety precautions

- Wear protective gears; overall, eye shield, hand gloves, face mask, and gum boots.
- Be careful when using the knife and sickle not to cut yourself.
- Keep safe distance from colleagues when cutting fodder to avoid harming them.

What to do

In groups:

1. Select a nutritious fodder plant preferably at flowering stage. You can use any of the following plants Guinea grass, Star grass, Congo signal grass, Kikuyu grass, etc.
2. Harvest the fodder using a sickle.
3. Place cut forage on tripod stands or spread them evenly on the ground to dry 12 -24 hours depending on the weather condition.
4. Transfer the forage to a shade and spread it evenly to be further dried by air for 3 to 6 days depending on the weather condition.
5. Bale the hay.
6. Store the hay under shade or in a well-ventilated store.
 - (a) Discuss and write down the objectives of conserving pasture on the farm.
 - (b) Describe the steps involved in making of high-quality hay on the farm
 - (c) Brainstorm on the factors that affect the quality of hay made on the farm

Compile a report of your finding and present it to the rest of the class.

Silage

This is forage preserved in succulent form by fermentation.

Silage making

The process involves fermentation of forage to convert soluble carbohydrates to lactic acid which drops the pH to a level sufficient to inhibit any biological activity in the ensiled forage.

Procedure of making silage

1. Prepare the silo. The silo should be clean and enough to accommodate all the forage. The silo can be stack silo or pit silo.
1. Select good, palatable forage at flowering stage. At this stage the plant is fairly edible, tender and has high protein content.
2. Cut the forage and allow it to wilt for 4 - 48 hours depending on the weather conditions. This helps to reduce the moisture content in the plant tissues.
3. Transfer the forage to the silo site and chop it into small pieces (4-10cm). This allows easy piling and compaction of the forage in the silo. Chopping the forage also allows easy feeding out of forage from the silo and increases palatability and silage intake in feeding animals.

4. Pack the forage evenly in the silo. The silo should be filled about 1m in the first day and should be allowed to warm up to 40°C but not more, before the next layer is added. This encourages lactic acid producing bacteria to multiply. Add water soluble carbohydrates, e.g. molasses which provide energy to microbes involved in the fermentation process. Also, add urea so as to increase the nitrogen content of the silage. The filling of the material in the silo should be completed in 3-4 days.
5. Consolidate the forage by trampling or running a tractor on the material to remove excess air from the forage. Excess air in the forage may lead to high temperature in the silo resulting into rotting and severe losses of nutrients in the silage. On the other hand, if the material is filled in too quickly and compacted too much, oxygen will be low. This may encourage anaerobic butyric acid producing bacteria to multiply and produce rancid-smelling unpalatable silage. Adding molasses can help prevent this.
6. Seal off the silo to exclude air and water from the silo. The material should be covered with a layer of straw, or grass or a polythene sheet and the 15cm layer of soil should be covered on top to weight the material down.
7. If pit silo was used, it is necessary to make a trench around the pit to keep surface runoff away from the pit

Low temperature in the silo can be avoided by:

- i. Partial wilting of the material before ensiling to reduce moisture in the herbage.
- ii. Addition of fermentable water soluble carbohydrates e.g. molasses.
- iii. Filling the silo rapidly so as to seal and keep away air and water from the silo.
- iv. Compressing the material to exclude much of the air from the material.
- v. Sealing of the silo immediately after packing to keep away air and water from the silo.

Silage Quality

The relative proportions of organic acids in the silage are an indication of its quality. In good quality silage, the order of predominance should be; lactic, acetic, succinic and formic acids. Poor silage compaction leads to low temperature which results in excessive production of Butyric acid instead of Lactic acid. Good quality silage should be,

- i. From high quality forage cut at the proper stage of growth.
- ii. Have 5-9% lactic acid
- iii. Have a pH of 4.2 or below.
- iv. Be free from moulds and bad odour such as ammonia and butyric acid.
- v. Be greenish to yellow in colour not brown or black
- vi. Have a fine texture with no sliminess.

Factors that affect the quality of silage

1. *The stage of maturity of the plant:* forage should be harvested at flowering stage when the nutrient content is high and the dry matter is fairly good.
2. *The type of forage crop:* for instance, legume-grass mixture gives high quality silage than silage from pure stands.
3. *The moisture content of the forage ensiled:* high moisture content leads to very high temperature which leads to deterioration in quality of silage. Moisture in plant can be regulated by partial wilting of the material before packing it in the silo.
4. *Amount of air in the silo:* too much air allows long aerobic respiration and the multiplication of clostridia responsible for secondary fermentation which converts lactic acid to butyric acid and amino acids to amines and ammonia. Thus air should be excluded as much as possible to allow anaerobic fermentation of material by lactic acid producing bacteria.
5. *Rate at which the silo is filled:* delayed or prolonged filling of the silo may lead to decomposition of the material at the bottom due to aerobic activity of bacteria and fungi.
6. *The amount and rate of bacterial inoculants and additives added:* additives such as molasses increase fermentation process and palatability of silage.
7. *The pH of the material ensiled:* pH of 4 is the most ideal. High pH encourages the growth of undesirable organisms in the forage.

8. *Temperature inside the silo*: the degree to which the temperature rises during ensiling; Very high or very low temperature can result into rotting of the forage in the silo.

Advantages of making silage

- i. It is less weather dependent unlike hay.
- ii. Forage remains succulent, which improves palatability and intake.
- iii. Provides high level of lactic acid, important for lactating animals.
- iv. Handling of silage is easier and shattering losses are minimized.
- v. Minimizes fire risks unlike hay which can easily catch fire.
- vi. It minimizes nutrient loss during preservation as compared to hay making.
- vii. Silage making requires relatively less land as compared to hay making.
- viii. The process of fermentation breaks toxic compounds in plant tissues making most plants palatable to the animals without fear of intoxication.

Methods of maintaining feed supply to livestock on the farm

- i. Conservation and preservation of pastures in form of silage, cut or standing hay.
- ii. Use of supplementary feeds such as concentrates.
- iii. Use of pasture species with wide tolerance to varying climatic conditions.
- iv. Adjusting animal numbers to the available feed supply (use of correct stocking rate).
- v. Adjusting feed supply through irrigation and application of fertilizers.
- vi. Improving natural pastures by over sowing with legumes and fertilizer application.
- vii. Use of crop stubbles such as residues from cereals and other crops after harvesting.
- viii. Grazing in marginal land such as river bank and swamps during dry season.

In the activity 3.12 below, you are going to learn practical skills involved in the making of silage from fodder available around the school.



Activity 3.12

Participating in the process of making high-quality silage

You will need

sickle, panga, polythene sheets or tarpaulin, hoe, spade, molasses, fodder at flowering stage, protective gears

Key question: What are the practical steps involved in the making of silage?

Safety precautions

- Wear protective gears; overall, eye shield, hand gloves, face mask, and gum boots.
- Be careful when using the knife and sickle not to hurt yourself.
- Keep safe distance from colleagues when harvesting and chopping fodder.

What to do

In groups:

1. Select a clean leak-proof polythene bag of about 60 litre capacity.
2. Select a nutritious fodder plant preferably at flowering stage. You can use any of the following plants Guinea grass, Napier grass, Guatemala grass, Maize, etc.
3. Harvest the fodder using a sickle or a panga.
4. Wilt the cut fodder for 3-6 hours depending on the weather conditions.
5. Chop the forage with a panga to small pieces of about 5 to 10 cm.
6. Fill the chopped fodder in the bag.
7. Sprinkle molasses between the layers of the forage during filling.
8. Press and compact the fodder in the bag to exclude as much air as possible.
9. Tie and seal the mouth of the bag.
10. Store the bag under shade until the forage is required for feeding animals. usually the silage is ready after 10 days to 3 weeks.
 - (a) Brainstorm and explain the factors that affect the quality of silage made on the farm.
 - (b) Suggest Ways in which a farmer can maintain feed supply to animals during dry season.

Compile a report of your finding and present it to the rest of the class.

Grazing Management

The following are the objectives of grazing management;

- i. To increase animal productivity.
- ii. To increase the quantity and quality of forage.
- iii. To allow pasture to persist for a long time
- iv. To maintain favourable legume and grass balance.
- v. To protect young pasture and to allow enough time for recovery between grazing periods.
- vi. To avoid overgrazing of pasture.
- vii. To control building up of parasites in the pasture

Terms related to grazing

Stocking rate: this is the total number of animals per unit area of land.

Carrying capacity: this refers to the maximum number of animals a particular area of land is able to support or carry, without causing overgrazing. For high pasture productivity, stocking rate must be adjusted to the carrying capacity of the pasture. When there is plentiful herbage, stocking rate is increased and with reduced herbage, the stocking rate should be reduced.

Grazing pressure indicates the number of animals on a given amount of pasture.

Importance of maintaining optimum stocking rate on a farm

- i. Avoids overgrazing and the associated effects of over grazing
- ii. Maintains optimum productivity of pasture exhibited in form of increased productivity in grazing animals on the farm.
- iii. Ensures optimum utilization of resources on the farm thus increasing profits on the farm.
- iv. Reduces spread of diseases and parasites on the farm.
- v. It allows proper management of animals on the farm.

Factors influencing stocking rate

- i. *Rate of pasture growth:* More animals can be stocked as the rate of pasture growth increases.
- ii. *The nutritive value of pasture:* Highly nutritious pasture can support more animal's nutritional demands than less nutritious pasture.
- iii. *Botanical composition of the pasture:* when the botanical composition is poor the pasture can only support small number of animal.
- iv. *Seasonal variation in feed supply:* Optimum stocking rate varies with feed supply thus in some months when rainfall is high, feed supply is also high and stocking rate is also high.
- v. *The level of nutrients needed by individual animals;* For example a lactating animal needs comparatively much nutrients than wool producing animals.
- vi. *Size of the animals:* Small stocks such as goats and sheep require comparatively less space and herbage than large stocks such as cattle donkeys and horses.
- vii. *Level of supplementary feeding:* In a situation where pasture is supplemented by feeds from other places outside the farm, stocking rate is likely to be high.
- viii. *Pasture management practices:* Pasture management practices such as irrigation, fertilization and paddocking increases the rate of pasture growth and in turn supports high stocking rate on the farm



Activity 3.13 Understanding grazing management

You will need a pen, notebook, reference textbooks and internet

Key question: What are the objectives of grazing management and terms related to grazing?

In groups:

1. Brainstorm and research about the objectives of grazing management.
2. Discuss the meaning of the following term as used in grazing management; stocking rate and carrying capacity of a pasture, and factors that influence carrying capacity of a pasture.

Present your finding to the rest of the class and record conclusions.

Grazing Systems

1. Rotational Grazing

This refers to practice of allowing livestock to feed on a part of pasture for a period down to certain level before they are moved to the next. This gives time for pasture to properly regenerate.

Objectives of rotational grazing

- i. To provide animals with highly nutritious forage all the time.
- ii. To control and maintain botanical composition of pasture.
- iii. To provide for particular animal needs.
- iv. Improve utilization pasture by animals.
- v. Allow convenience of farm operations.
- vi. To break the lifecycle of parasites by keeping hosts away.
- vii. To allow ample regrowth of pasture.

Advantages of rotational grazing

- i. Livestock make maximum use of pasture
- ii. Reduces the buildup of parasites and diseases.
- iii. Animal waste is distributed evenly in all fields/paddocks.
- iv. Pasture area is given time to re-grow before it is grazed on again.
- v. Excess pasture can be harvested for conservation
- vi. It's possible to apply fertilizers in parts of the pastures are not in use
- vii. It facilitates reseeding and weeding.
- viii. Controls selective grazing and maintain the botanical composition of pasture.
- ix. Rotational grazing makes it possible to spare pasture for during dry period.
- x. It reduces the possibility of overgrazing.
- xi. Animals get fresh pasture throughout the year.

Methods of Rotational Grazing

- a) **Paddock grazing:** paddocks are areas of fenced land used for confining animals. It is a form of rotation grazing in which the land is divided into small areas where animal grazes in a systematic arrangement. Animals start grazing in one paddock then move onto another leaving the previous paddock to rest and grasses to regrow.
- b) **Strip grazing:** this is another form of rotational grazing where animals are allowed to graze pasture strips. The pasture is divided by an electric fence powered by a battery. The amount of grazing required by animals each day is calculated and a fence is moved each morning to allow the correct area for grazing.
- c) **Tethering:** This involves tying the animal to a post with a rope such that it feeds within a restricted area. The rope may also have a metal ring that slides along a strong wire supported by strong poles.

Principle of Rotational Grazing

- i. Animals of the same age group should be moving together.
- ii. Young animals should be moved first in new paddock and later old ones
- iii. Pasture should be given ample time to regrow.
- iv. Lactating and pregnant animals should not mix up with bulls.
- v. Sick animals should be isolated in special paddocks for supplementary feeding
- vi. Animals with parasites e.g. ticks should come last to avoid spread to others
- vii. Only the right number of animals should be allowed to graze a particular area or strip.
- viii. Animals should not be left in paddocks for a long time otherwise overgrazing will occur.
- ix. Animals should be dewormed and dipped before transfer is done to avoid introduction of worms and ecto-parasites.

2. Continuous Grazing (Herding)

In this system the animals are allowed to graze a piece of land for long periods of time. This system allows animals to have free access to any part of the pasture. For this system to be effective, proper control of stocking rate is essential because the land may be under grazed during rainy season and over grazed during dry season.

Advantages of continuous grazing

- i. Labour is required is low as one herder can take care of many animals.
- ii. It is cheap.
- iii. Animals select the pasture of their choice.
- iv. It requires less skill to manage compared to the other systems of grazing.

Disadvantages of continuous grazing

- i. It encourages random mating which leads to inbreeding.
- ii. Pasture is not allowed time to regenerate because grazing is continuous.
- iii. It is difficult to keep records on production, health and breeding of the livestock.
- iv. There is poor management of pastures, thus lowering their quantity and quality.
- v. Production is low because animals walk long distances and sometimes feed on poor pastures.
- vi. it encourages the build-up of parasites such as ticks and worms because there is no resting period to break their life cycles

3. Zero Grazing (Stall Feeding)

In this system, animals are not allowed out to graze at all. Instead fresh forage (legumes and grasses) is cut daily and brought to the animal in the kraal or yard. Fodder grasses e.g. Elephant grass, Guatemala grass, Giant Setaria and legumes e.g. Lablab, Calliandra, Gilicidia and Leucaena are specifically grown for cutting and carrying to the animals in stalls. This method is mainly practiced in highly populated areas where there is scarcity of land and the demand for livestock products, particularly milk is high.

Advantages of zero grazing

- i. Since animals are confined, selective grazing and trampling of forage are eliminated.
- ii. Animals are safe from adverse weather conditions such as strong sunshine and storm.
- iii. Since there is limited movement by the animal, there is good conservation of energy which is reflected by higher production in terms of milk yield and live-weight gain.
- iv. Manure is easily collected and returned to the garden or used as fuel to produce biogas.
- v. Animals are protected from pests and diseases that are spread by other animals or that can be picked from the grazing area.
- vi. Animals can best make use of land in different areas e.g. forage can obtained from very steep slopes and swamps.
- vii. It is possible to keep animals even where land is limited as is the case in urban areas.
- viii. It is possible to attain very high stocking rate on the farm.
- ix. Closer supervision of animals is possible.
- x. It allows proper rationing of feed by only giving what is enough for the animal. This eliminates over feeding in animals.
- xi. Animals are well protected against theft and attack from wild predators.

Disadvantages of zero grazing

- i. The system needs high initial capital investment in construction of stalls and buying forage harvesting equipment.
- ii. It needs much more labour in feeding the animal and cleaning of the kraal.
- iii. It may lead to soil depletion if manure from animals is not returned to the fodder gardens.
- iv. The method is only feasible where the market for animal and animal products such as milk is readily available.
- v. The system requires high level of managerial skills.
- vi. Because of limited space, animals lack exercise. Thus animals become physically weak and prone to many complications and diseases.
- vii. Diseases such as foot rot are common especially if the kraal is too wet at most times.

4. Deferred Grazing

This means setting aside of certain area of pasture (paddock) to be used at a later stage in future time. The pasture provides just a maintenance ration for livestock during dry season when pasture is scarce.



Activity 3.14 Understanding the grazing systems

You will need a pen, notebook, maker, manila, reference textbooks and internet

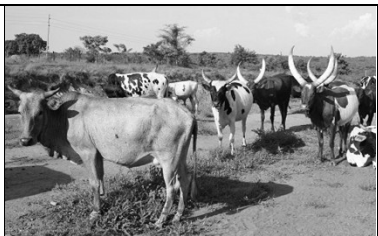
Key question: What are the different grazing systems, and the merits and demerits of each?

In groups:

1. Brainstorm and describe the grazing systems used on pasture.
2. Discuss the advantages and disadvantages of each grazing system discussed.

Present your finding on manilas and hung the in class compare your work with other groups.

Exercise 3.1	Self-Check Questions
	As an expert in pasture management and conservation, 1. (a) List the signs that indicate a natural pasture needs improvement. (b) Describe the practices used to improve a natural pasture. 2. (a) Describe how you can establish a pasture from seeds. (b) What are the possible causes of failure of a pasture to establish? 3. (a) Discuss the different systems used in the grazing of farm animals. (b) Outline the advantages and disadvantages of rotational grazing and stall feeding systems of grazing farm animals. 4. (a) Describe the steps of preparing a good quality silage on the farm. (b) Outline the factors that affect the quality of silage.

Activity of Integration	In Basongora community in Kasese district, cattle rearing is the main activity. Farmers are semi-nomadic and graze their cattle communally along the fringes of Queen Elizabeth national park. During rainy season there is excess pasture, and during dry spells, the grass land is usually left bare by grazing animals. 
Task	Prepare a speech you will use to sensitise the community on how to ensure continuous pasture production.

Using Available Waste Resources to Grow Irrigated Crops

On the farm, there are several waste resources which can be used to grow irrigated crops. The waste resources include; plastic bottles and containers, crop residues and wastewater. *Explain how the different waste resources are used to grow irrigated crops.* Water is a major factor in crop growth. The water is mainly provided by rainfall. At times the rainfall received is inadequate or unreliable leading to poor growth of crops such as pasture. This problem can be solved by recycling wastewater and using it to irrigate crops. Wastewater is often contaminated with disease-causing organisms and chemical agents that can harm the health of the farmers, the traders who handle crops and the crop consumers. It is therefore very important for you to be aware of the health-risks associated with using wastewater for irrigating crops and to know how to use wastewater safely at farm level to reduce those health risks. *How can waste water be made safe for use on the farm?*

Using available waste resources to grow irrigated crops is helpful in the following ways; it is a form of recycling of farm wastes thus, reducing pollution of the environment and minimises the cost of production on the farm.

Risks in using waste resources in irrigated crops

Salts, pathogens, heavy metals and pesticides are commonly found in wastewater, manure, bottles and containers and are harmful to people and the environment.

There are a number of approaches you could use to reduce health risks from using wastewater on the farm. Some of these can be combined for even greater reduction in contamination.

- i. Use irrigation methods that reduce contact between edible parts of crop and irrigation water.
- ii. Use only bottles for mineral water and other beverages. Bottles and containers to use for irrigation should be washed carefully with soap and plenty of water.
- iii. Stop irrigating some days before harvest; most pathogens are easily killed by harsh environmental conditions such as heat, sunlight and lack of water.
- iv. Use sedimentation ponds; most organisms that cause diseases are attached to silt and other particles and will settle to the bottom of ponds. Some others such as worm eggs can settle on their own because they are heavy.
- v. Use filtration techniques; filtration systems remove disease-causing microorganisms from polluted water by trapping them in the filtration media.
- vi. Use manure with care; do not use fresh manure on your crops. Composting removes harmful organisms. Do not apply manure on the edible parts of crops.



Activity 3.15 Participating in identifying waste resources for irrigated crops

You will need a pen, notebook, reference textbooks and internet, protective gears

Key question: What are the available waste resources for irrigated crops?

In groups:

1. Put on your protective gears.
2. Move around the farm and collect the waste materials and wastewater. State how each waste material collected will be utilized in the growing of irrigated crops.
3. Treat the wastewater. Which methods are you using to treat the wastewater?
4. Clean the bottles and containers you are going to use.
5. Decompose the organic waste materials to make manure for use.
6. Remove the protective gears and wash up.

Present your finding to the rest of the class.

Methods Used to Irrigate Pasture

Two types of irrigation are suitable for irrigating pastures: sprinkler irrigation and drip (or trickle) irrigation. Furrow irrigation method is be used to irrigate fodder plants that are planted in rows.

Several factors must be put into consideration when planning an irrigation system:

- Cost of the irrigation system
- Irrigation requirements of targeted pastures;
- Slope and terrain of the area;
- Soil type (intake capacity/soil water holding capacity);
- Water source;
- Operational requirements in terms of labour and cost.

Sprinkler or Overhead

In this system water is sprayed over the soil surface just like rainfall through nozzles from a pressure system or under gravity force. The system is made up of a pump which pumps water from the source and pipeline which delivers and distributes water to the field. Finally at specific intervals, the pipes are fitted with sprinklers bearing nozzles which break water into fine spray or droplets at a constant pressure into the air similar to rainfall.

Advantages of sprinkler irrigation

- i. It can be used in any topography.
- ii. No wastage of land as no water channels are constructed.
- iii. It does not interfere with machinery and animal movement.
- iv. The system ensures uniform distribution of water in the pasture.
- v. Distribution points can be adjusted as the lateral can be moved from one point to another.
- vi. Can be used in any type of soil even in the porous sandy soils.

Disadvantages of sprinkler irrigation

- i. It is expensive to install initially.
- ii. Nozzles can easily get blocked making the system ineffective.
- iii. Requires skills to install.
- iv. Labour intensive when transferring the pips/laterals.
- v. In windy conditions, it may be ineffective as water will not be evenly distributed.
- vi. Water infiltration is shallow hence poor root development.

Drip or Trickle Irrigation

The water from the water source is supplied under low pressure through pipes that bears small holes that releases water to plants. Usually these holes are adjacent to the plants. Water keeps trickling slowly drop by drop around the root zone of the plant. It is mainly used to irrigate forage and horticultural crops.

Drip irrigation using plastic bottles: is this method, small hole is put on either the cover or the bottom of the bottle and then, the bottle is filled with water, then the bottle is tied with the hole facing downwards on a stick near the stem of the plant to be irrigated. Water from the bottle will gradually drip to the soil near the plant, hence nourishing it.

Advantages of drip irrigation

- i. It permits use of small amounts of water.
- ii. The amount and flow of water going to a crop can be regulated as required.
- iii. Fertigation can easily be done.
- iv. Can be used in any weather conditions.
- v. Only a small portion of the soil receives water and the rest of the land remains dry. This control the growth of weeds.
- vi. Evaporation of water is minimized since only a small portion of moistened soil is moistened reducing the evaporative surface.
- vii. Soil erosion is checked since just enough water is released slowly to the soil for plant use.
- viii. High yields are obtained and the crop quality is improved than in sprinkler irrigation system.
- ix. Leaching of nutrients is minimized since there is less deep percolation of water into the soil.
- x. Reduces incidence of crop diseases since the part of plant above ground is usually dry.

Disadvantages of drip irrigation

- i. Only a small portion of the root zone is wetted this restricts root growth to a small portion of the soil thus reducing absorption of nutrients from the soil.
- ii. Can lead to build-up of salt around the root zone leading to poor growth or death of plants.
- iii. It can lead to invasion of plants by weeds and some bacteria as the system create a conducive micro-climate for their survival around the plants
- iv. The system needs high initial capital to install the tank, pipes, applicators etc.
- v. The system can easily be clogged by particulates and algae.



Activity 3.16

You will need

Skills in construction of a simple drip irrigation system

pole stands, plastic bucket, pipes, used water bottles; strings; pipe fittings, a pen, notebook, maker, manila, reference textbooks

Key question: How is drip irrigation system designed and laid out?

In groups,

Procedure 1; using a bucket and pipes

1. Design a layout plan for a simple drip irrigation system
2. Put on your protective gears
3. Place a bucket stand on a stable ground and place a bucket on the stand. The bucket should be at least 1 metre above the ground.
4. Fit a pipe of a bout 5m.
5. Perforate the pipe and fit it with drippers or emitters, fitted at 10 cm interval. Seal the open end of the pipe.

6. Fill the bucket on the stand with clean water. Water from the bottle will gradually drip to the soil to nourish the plant.

Procedure 2; using water bottles

1. Design a layout plan for a simple drip irrigation system.
2. Put on your protective gears.
3. Collect used water bottle.
4. Put a small hole using a three inch nail on either the cover or the bottom of the bottle.
5. Erect a shot stick near the stem of the plant to be irrigated.
6. Tie the bottle with a string on the stick making sure the hole is facing downwards.
7. Fill the bottle with treated wastewater. Water from the bottle will gradually drip to the soil to nourish the plant.

Remove the protective gears and wash up.

Groups present an evaluative report of the system's effectiveness.

- (a) What could be the advantages of using the irrigation system you have created?
 - (b) What are the challenges of the irrigation system you have created?
 - (c) What recommendations do you give to improve the effectiveness of the irrigation system?
 - (d) Make an annotated illustration of your irrigation system and briefly describe how it works.
- Put your work on a manila and display it in the corner of the class for the rest to see.

Present your findings to the rest of the class and records class conclusions in your notebook



Soil and Water Conservation in Pasture Area

Soil conservation is the application of a set of farming techniques and practices that help to reduce soil erosion, conserve soil moisture, keep soil fertile, avoid degradation, and minimize environment pollution.

Objectives of soil and water conservation

- i. To reduce soil losses by taking remedies against erosion.
- ii. To reduce degradation of soil and vegetation.
- iii. To retain enough water in the soil essential for healthy pasture and crop growth.
- iv. To maintain a sustainable soil productivity through good soil management.
- v. Reduce chemical pollution which would be dangerous to soil flora and fauna and aquatic life.

Soil Erosion

In your school compound or farm, do you see any indicators of soil erosion? These could include exposed subsoil and tree roots, bare ground or plant buried under the soil. Soil erosion creates gullies that limit movement of livestock and machinery, leads to loss of fertility by removing the top soil, which is the natural feeding zone for plants, leads to siltation of water reservoirs such as dams, lakes, and fish ponds, leads to the spread of weeds on the farm and from one farm to another, leads to the spread of disease on the farm, pollution of the water bodies may take place; soils from farmland containing agricultural chemicals such as pesticides and fertilizers are carried and deposited in the water bodies, harming aquatic life and eutrophication of the water that also lead to blooming of water weeds and algae.

Causes of soil erosion

- i. Improper farming practices, e.g. overgrazing, forest clearing, bush burning.
- ii. Climatic factors such as high wind velocity or high rainfall intensity.
- iii. Land tenure and land fragmentation.
- iv. Very steep slopes that increase the speed of runoff to erode soils.
- v. Soil factors: soil erodibility and relief of the area.
- vi. Intensified movement of livestock as the case with nomadic herding. The animal hooves dig trample vegetation and loosen up the soil, making easy for water and wind to erode.



Using the internet and reference books research about different types of soil erosion. Present the work to your teacher and record the conclusions in your note book.



Activity 3.17

Finding out the causes of soil erosion

You will need

protective gear, 3 1.5L soda bottles, mulch, soil samples, 3 beakers, a 5 inch x 3 inch rectangular furrow slice with root and soil intact, a stool

Key question: what are the causes of soil erosion?

In groups:

1. Put on protective gears.
2. Using a sharp hand trowel, Scoop a 5 inch x 3 inch soil slice with to vegetation, roots and soil intact.
3. Collect loose forest soil.
4. Using a scapel and scissors, cut each soda bottle longitudinally into two halves.
5. Place the cut bottles on a stool.
6. Place a beaker under the neck of each bottle.
7. Fill one bottle with loose arable soil, one with intact soil with a top layer of grass with roots and one with compacted forest soil.
8. Pour water slowly into each soda bottle and watch the water flow into the beaker.
9. Observe how much soil erodes from each bottle and flows into the beaker.
Which soil condition had the least amount of runoff? Which had the most? Why?
10. Share you results with other groups.

Arable soil Forest soil Grassland soil

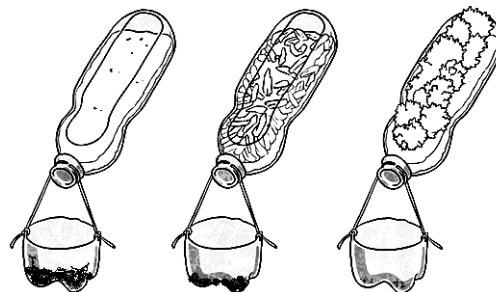


Fig. 4.1 Demonstrating cause of soil erosion

Methods of soil and water conservation

Cultural practices used in soil and water conservation

Mulching: this is only possible in planted fodder. Mulching increases water infiltration and reduces the impact of both wind and raindrop on the soil.

Optimum stocking rates and rotation of animals in the pasture: this helps to minimize overgrazing and destruction of vegetation cover.

Planting along contours: The grass between the crop strips help to trap eroded soil and reduce the speed of runoff.

Contour ploughing: The furrows across the slope trap water moving down the slope which increases infiltration instead of run-off.

Ridging: ridges reduce the speed of runoff thus reducing the ability of flowing water to erode the soil. Also ridges trap water and encourage infiltration in to the soil.

Crop rotation: rotation of crops with poor ground cover with those of good ground cover provides an efficient control of erosion especially the rain drop impact.

Intercropping: In this method, crops which provide poor soil cover are grown with cover crops. This ensures that the soil is well covered against erosion agents.

Strip cropping: this involves growing of a variety of crops and pastures in alternate strips along the contours of the slope.

Use of wind breaks: these comprises of thick trees planted perpendicular to the direction of the prevailing winds. They help to trap and reduce the speed of wind.

Agroforestry: trees can be planted on steep slopes to reduce both wind and water erosion. Also, tree roots help to hold soil particles together to resist soil erosion.

Minimum tillage: minimum tillage helps to reduce destruction of soil structure. it also allows accumulation of crop residues which help to protect the soil against rain drop erosion.

Deep ploughing: deep ploughing helps to break hard soil surface underneath, increasing water infiltration, thus, reducing surface run-off and soil erosion.

Fallowing of the land: fallowing ensures good vegetative cover of the ground to protect the soil from erosion agents.

Proper spacing of crops: close spacing of crops is important to provide adequate soil cover to protect the soil against the impact of rain drops and wind.

Application of manure in the soil: manure provides humus to the soil which binds soil particles together making the soil firm and able to resist erosion. Also manures promote bioactivities within the soil, which in turn improves drainage and water infiltration into the soil.

Early planting of crops: early planting ensures that crops grow faster to provide adequate soil cover before rains become heavy.

Use of cover crops (also called living mulch): cover crops protect the soil from the impact of rain drops and reduce the speed of water flow thereby increasing water infiltration into the soil.

Qualities of a good cover crop;

- It should be highly vegetative and leafy.
- It should be drought resistant.
- It should be shade-tolerant and therefore compatible as an intercrop.
- It should be resistant to pests and diseases.
- It should be easy to establish.
- It should be highly adapted to the local environmental conditions.
- It should be able to fix nitrogen and grows even in very poor soils.
- It should have faster growth.
- Grow vigorously to suppress weeds.
- Should not compete with main crop.
- Tolerant to shading.
- Should not produce toxic substances that affect crop growth.

Other benefits of cover crops

- i. Deep-rooted cover crops are able to recycle nutrients acquired from deeper in the soil profile thus acting as nutrient “pumps”.
- ii. The decomposing residues from cover crops form humus which also cement soil particles together thus improving soil structure.
- iii. By keeping the soils cool cover crops prevent excessive evaporation of water from the soil.

Physical or mechanical methods of soil and water conservation

Use of Gabion/porous dams: gabions are wire net baskets filled with stones which are positioned along the sides of a gully or across the gully. Gabions trap and retain the soil particles in the water and reduce the speed of the water

Use of trash and stone lines: plant residues such as stems from cotton, maize stalks or stones are heaped along the contours to reduce the speed of runoff or trap the soil carried down the slope.

Use of bunds: these are heaps of soil built along the contours of the slope to reduce the speed of run-off. Grass such as *Paspalum notatum* can be planted on top so as to stabilise the soil.

Cut off drains/diversion channels: these help to divert or cut runoff into a valley or any natural water way such as a river, or into non-erodible stony-rock.

Use of terraces: these help to reduce the length and gradient of the slope. It reduces the volume and the speed of run-off. Different types of terraces can be constructed depending on the gradient of the slope; bench terrace used for slope gradient of about 35-55%, narrow-based terrace for slope of about 10-20% and broad-based terrace used for slope gradient of about 2-12%.

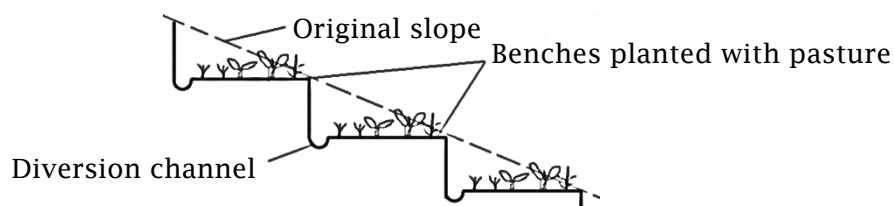


Fig. 2.15 Bench terrace

Use of graded banks: runoff is reduced by dividing the land into smaller pieces of land through the construction of banks which leads the water away from the field to prevailing rivers or valleys.

Use of absorption banks: these are channels constructed across the slope to trap and retain water, encouraging sedimentation and reducing the volume of surface runoff.

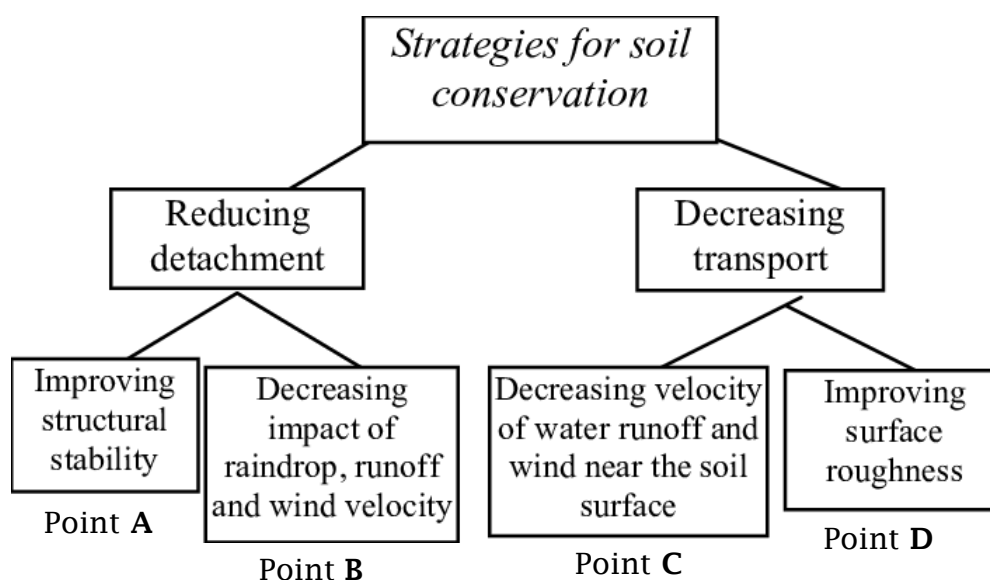
Construction of barrages (dykes) across the gullies: these are made up of stones or wood constructed across the gullies to reduce the speed of water flow and allow sedimentation of the soil. Also, when barrages are filled with soils the slope gradient is reduced.

Dams and reservoirs: These store excess water and reduce the volume of run off. They are also advantageous that they provide water for other purpose.

Complete the Soil and water conservation word puzzle below

I	W	J	R	E	C	W	R	N	X	Z	B	U	P	D	I	A	E	R	F	C	A	K	J
W	O	X	C	A	I	O	B	I	P	B	X	T	H	A	Q	G	M	E	O	C	F	L	W
Q	Z	N	M	J	A	R	N	U	W	I	N	V	L	Q	T	X	O	N	Z	K	T	A	Q
F	E	E	A	Z	A	Q	V	T	B	B	U	F	F	E	R	S	S	N	E	W	T	U	M
F	N	J	N	E	C	W	V	Q	O	C	G	Y	W	F	I	E	E	U	U	E	U	B	C
J	J	T	U	W	O	D	U	H	L	U	K	J	P	E	R	V	T	V	R	P	C	D	A
E	T	O	R	T	M	F	F	O	N	U	R	L	M	V	T	X	X	Q	R	Z	Y	E	N
D	H	I	E	L	A	Q	J	R	G	A	S	S	A	T	G	L	U	U	Q	A	S	L	A
O	W	A	T	N	E	K	V	J	M	N	P	T	T	N	J	A	A	R	D	P	H	A	T
R	N	B	N	Z	I	V	O	N	Q	W	I	A	V	R	L	P	N	N	Y	D	Y	Q	U
E	I	Z	W	I	O	O	R	Y	U	O	M	E	X	I	I	C	K	A	D	S	M	G	R
J	Q	P	U	C	I	V	G	V	N	G	L	P	T	D	J	P	Y	X	K	S	Z	D	A
I	A	N	A	F	O	D	A	Z	Q	U	U	Y	G	F	F	Q	S	H	X	F	E	C	L
D	G	S	K	R	L	V	N	K	N	Y	F	E	R	T	I	L	I	Z	E	R	D	U	R
S	R	G	P	O	I	P	I	X	J	U	F	M	D	F	Z	X	V	N	S	X	C	L	E
R	I	O	O	X	L	A	C	I	X	S	T	E	R	O	S	I	O	N	X	Q	N	T	S
J	C	A	R	V	L	W	N	E	G	V	B	R	J	C	D	O	L	I	Q	E	E	G	O
X	U	G	C	G	N	C	V	B	S	C	K	R	I	H	R	T	T	Q	E	Y	T	S	U
H	L	Z	R	G	E	O	G	I	U	G	R	J	O	E	Q	V	T	S	G	K	O	Q	R
X	T	Q	E	H	A	T	Z	L	O	F	B	Z	P	T	N	R	V	J	B	I	S	J	C
K	U	B	V	T	E	P	I	F	V	X	F	Q	Z	N	C	T	Z	K	L	A	G	O	E
G	R	B	O	Q	S	Y	L	W	T	X	B	E	I	Z	J	A	S	V	K	R	D	X	T
F	E	S	C	I	Z	U	Z	G	T	R	K	L	R	N	X	T	R	M	C	V	M	D	Q
R	R	E	T	A	W	W	I	R	W	Y	C	M	E	D	J	B	P	T	Z	H	I	N	J

Agriculture	Cover crop	Fence	Riparian buffer	Runoff
Manure	Nutrient	Soil	Natural resource	Harvest
Wetlands	Water	Erosion	Water quality	Organic
Conservation	Fertilizer	tractor	Contour strips	Buffers



What soil management practices should be used at points A, B, C and D to conserve the soil?

Activities at point A

Activities at point B

Activities at point C

Activities at point D



Activity 3.18 Showing skills in soil and moisture conservation in pasture areas

You will need a pen, notebook, maker, manila, reference textbooks and internet

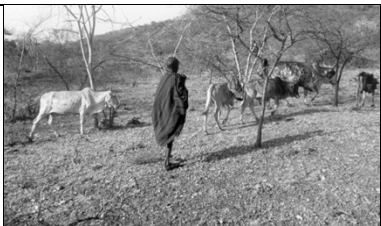
Key question: What are the different methods used to conserve soil and moisture in pasture areas?

In groups:

1. Research and give the objective of soil and moisture conservation in pasture areas.
2. Discuss the different methods used in soil and water conservation.
3. Brainstorm and explain the factors that affect the rate of soil erosion and outline the methods that can be used to control soil erosion on steep slope.

Present your finding on manilas and hung the in class. Compare your work with other groups.

Exercise 3.2	Self-Check Questions
	As a student who has studied about soil and moisture conservation in the growing of crops, 1. (a) Discuss how waste resources on the farm can be used to grow irrigated pasture. (b) Outline the health hazards likely to be faced by the farmer when using wastewater on the farm. (c) Describe how waste water can be treated to make it safe for use in irrigation of crops on the farm. 2. (a) Discuss how high stocking rate can cause soil erosion on a cattle farm. (b) Outline the effects of soil erosion on the farm. (c) Suggest the remedies that can be engaged to prevent soil erosion on the farm?

Activity of Integration	The recent livestock census indicated that Karamoja is the leading region with the highest number of livestock. This region receives the least amount of rainfall. During the dry season, the grassland is left bare by the grazing herds and most animals starve. 	
Task	Prepare a poster you will use to sensitise the people in the area on how overcome the above problem using the available waste resources on the farm.	

Hydroponics Farming

Competency: be able to establish, manage and produce crops using hydroponic systems.



KEY WORDS

- Hydroponic system
- Growth media
- Nutrient solution
- Aquaponics
- Passive system
- Closed system
- Open system
- Static solution
- Active systems
- Agar

LEARNING OUTCOMES

By the end of this chapter, you should be able to:

1. know the meaning and understand the principles of hydroponics
2. be able to distinguish between and explain open and closed hydroponic systems
3. understand how to select an appropriate hydroponic system and media to suit the requirements of different plants
4. understand and apply the practical steps involved in growing crops in the hydroponic farming system
5. be able to budget and estimate the construction costs of a selected hydroponic farming system
6. know how to apply agrochemicals in the hydroponic garden for increased production
7. understand how to market crops produced in hydroponic farming system

Introduction

You are likely to have seen some wild plants growing on water where there is no soil. These plants look very healthy, and growing throughout the year. Plants obtain the nutrients they need to grow from the water and the air around them. *Where do nutrients in the water come from?*

Hydroponics is a method of growing plants without soil, by instead using nitrified solution (water solution). Or it is the growing of plants in a liquid nutrient solution with or without the use of artificial media. Crops are grown with only their roots exposed to the nutritious solution, or the roots may be supported by an inert medium.

Principles of hydroponics farming

Plants require light, oxygen and carbon dioxide while the plant root system requires water, nutrient and oxygen. When plants are grown normally in soil, water dissolves nutrients from the soil and carries them to the roots. Water and nutrients are absorbed by the roots and used for plant growth.

With hydroponics, nutrients are dissolved in water. The soil is replaced with inert growing medium to provide the roots with anchorage. The nutrient solution can be fed drop wise into each plant, or by regularly flooding the root chamber, and subsequently draining it.

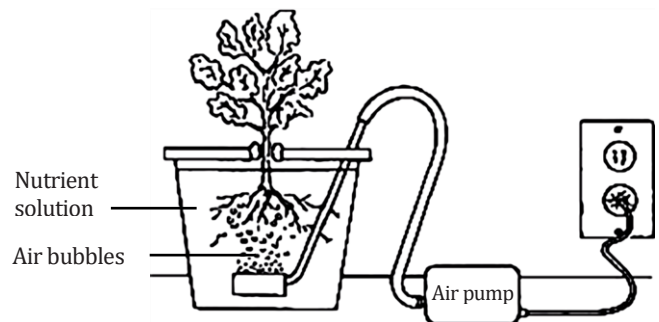


Fig 4.1 a hydroponic growing system

Advantages of Hydroponics

- Stops the spread of soil borne pests and diseases
- It can be done in a small space.
- Nutrient and other growth requirements can easily be manipulated to allow faster growth of crops.
- It allows production of crops in areas where soil conditions don't favour crop production.
- It allows production of crops throughout the year since it does not rely on the seasonal rain water.
- It reduces production costs.
- Allows hygienic handling of the crop.
- It makes it easy to monitor the progress in the development of tubers or roots of crops.
- Crop yields are relatively higher due to adequacy of plant nutrients in the growing solution/media.
- Labor for tilling, cultivating, fumigating, watering, and other traditional practices is largely eliminated.

Disadvantages of hydroponics

- Skill and knowledge are needed to operate properly.
- Initial and operational costs are higher than soil culture.
- Requires more labour to setup and monitor the system.
- The system is favourable for growing of all types of crops
- Not feasible on low value crops
- Introduced diseases like can spread quickly through the system. However, many varieties resistant to the above diseases have been bred.
- Cannot be used in the growing of all types of crops.



Discuss and select the crops which can be grown in the hydroponic system and give reason for the choice of crops selected



Activity 4.1

Understanding the meaning of hydroponics farming

You will need

a pen, notebook, maker, manila, reference textbooks and internet

Key question: What is hydroponics farming?

In groups:

- Brainstorm and discuss the meaning of hydroponics.
- How different is hydroponics farming different from the usual way of growing crops?
- How could hydroponics farming solve the problem faced in the growing of crops?
- What challenges do you think farmers practicing hydroponics are likely to face.

Present your finding to the whole class and compare your work with that of other groups. Use Microsoft word program of the computer to type and print out your presentation.

Open and Closed Hydroponic Systems

Hydroponics may take different forms but they all work on the same principles. Hydroponic systems can either be liquid or aggregate. **Liquid systems** have no supporting medium for the plant roots; whereas, **Aggregate systems** have a solid medium of support. Hydroponic systems are further categorized as **Open** where the nutrient solution is used only once on crops and never reused. A fresh nutrient solution is introduced for each irrigation cycle. For example, the drip system. **Closed** where the used nutrient solution is replenished and reused on several other crops.

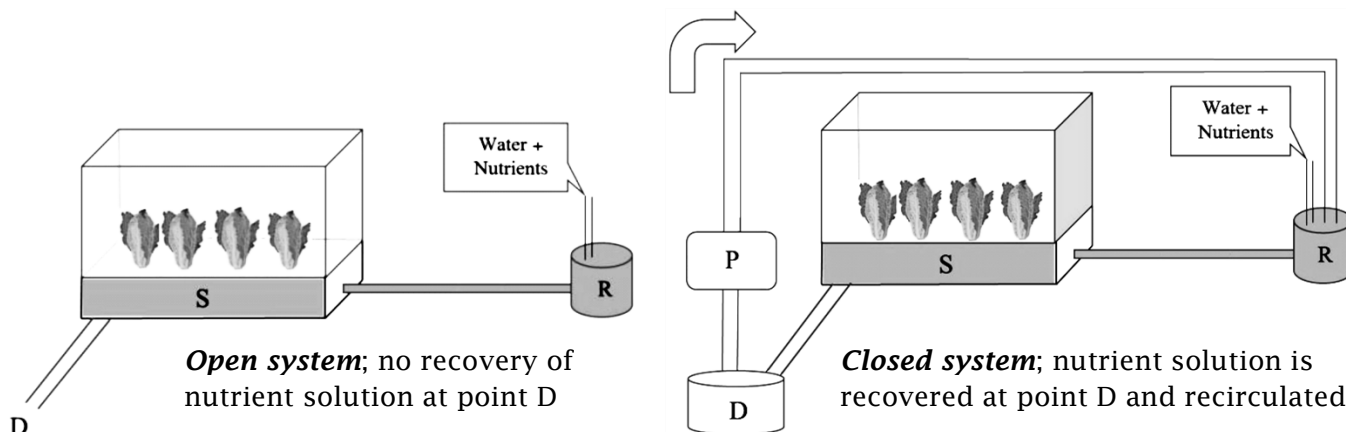


Fig. 4.2 Open and closed hydroponic systems

Selecting the Appropriate Hydroponic System to Use

The type of hydroponic farming system to use may be influenced by the initial capital investment. It may also depend on the availability of the system, maintenance cost, skills required to operate the system, and the type of crop to be grown.

The following are the common hydroponic systems;

1. **The Ebb and Flow System;** the system works by temporarily flooding the grow tray with nutrient solution and then draining the solution back into the reservoir. This action is normally done with a submerged pump that is connected to a timer. The timer is set to come on several times a day, depending on the size and type of plants, temperature, humidity and the type of growing medium used.
2. **NFT (Nutrient Film Technique);** this system of hydroponics has a constant flow of nutrient solution so no timer is required for the submersible pump. The nutrient solution is pumped continuously into the growing tray (usually a tube) and flows over the roots of the plants, and then drains back into the reservoir. There is usually no growing medium used. Usually, the plant is supported in a small plastic basket with the roots dangling into the nutrient solution.
3. **Deep Water Culture System;** This method is normally operated as an “open” system with the nutrient solution not recovered or reused. The platform that holds the plants is usually made of Styrofoam and floats directly on the nutrient solution. An air pump supplies air to the air stone that bubbles the nutrient solution and supplies oxygen to the roots of the plants.
4. **Drip System;** this is probably the most widely used type of hydroponic system in the world. A timer controls a submersed pump. The timer turns the pump on and nutrient solution is dripped onto the base of each plant by a small drip line.
5. **Aeroponics;** in this, plant roots hang in the air and are misted with nutrient solution. The misting is usually done every few minutes. Because the roots are exposed to the air, the roots will dry out rapidly if the misting cycles are interrupted. The method does not require substrate. Excellent aeration is the main advantage of aeroponics.
6. **Wick system;** it is an open system. The set-up makes the use of a cotton string referred as a wick. The wick absorbs nutrient solution and conveys it to the plant roots or growing media.

7. **Aquaponics** which is the integration of aquaculture, which is fish farming with hydroponics or the growing of crops without soil. How the system works

- Waste water from the fish tank is pumped up and collected in the top tank.
- A valve similar to that one in the toilet flush tank periodically flushes the waste water into gravel-filled grow bed where the plants use the nutrients in the water.
- Clear aerated water flows from the grow bed down into the fish tank.
- Water and the fish wastes are again pumped up back into the top tank for recirculation.

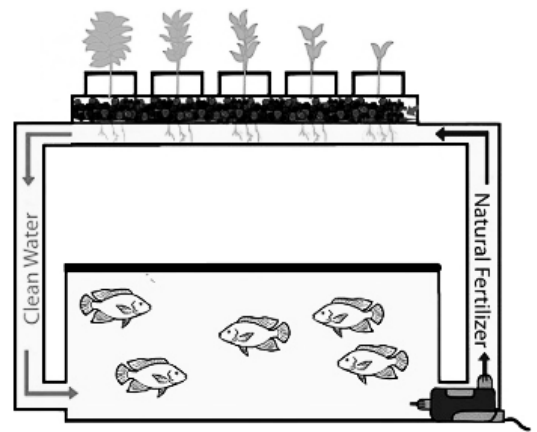
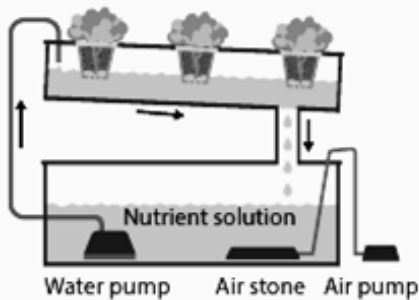
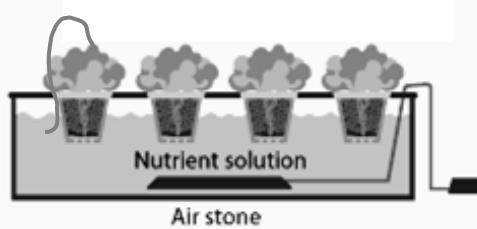


Fig 4.3 Aquaponic system

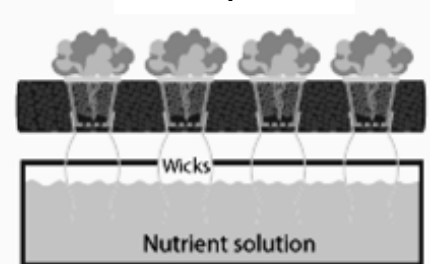
Nutrient film technique



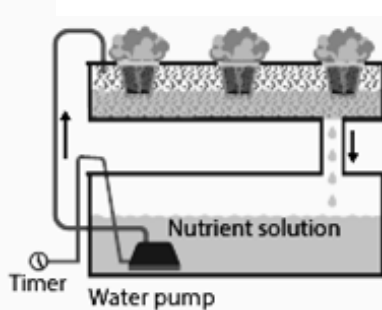
Deep water culture



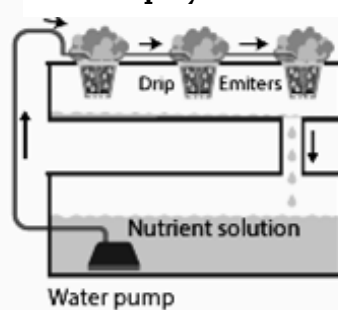
Wick system



Ebb and flow



Drip system



Aeroponics

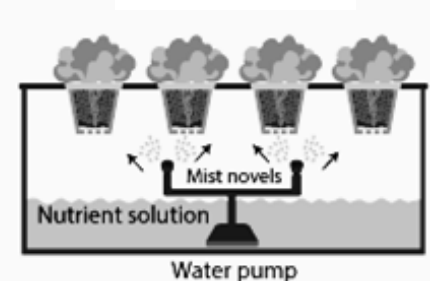


Fig 4.4 Hydroponic systems.



Activity 4.2

Understanding the hydroponics production systems

You will need

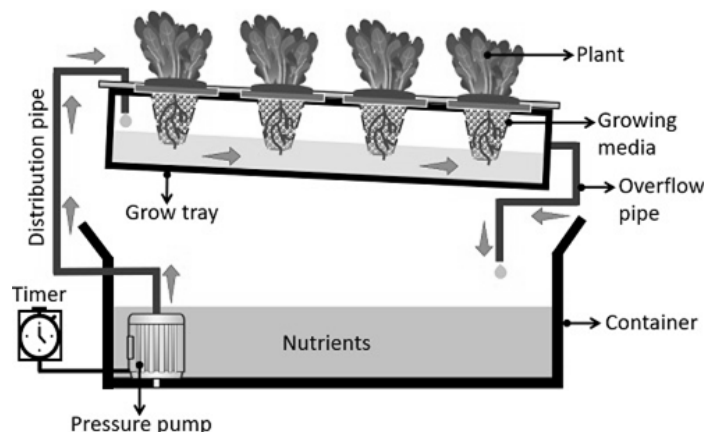
A pen, notebook, maker, manila, reference textbooks and internet

Key question: What are the different hydroponics production systems?

In groups:

1. Study figure 4.5 showing an example of a hydroponic system

Figure 4.5



- (a) Name the hydroponic system shown in fig 4.3, state whether it is open or close system.
- (b) Name the components that make up the system.
- (c) How does each component contribute to the working of the whole system?
2. Other than the above hydroponic system, describe other hydroponic systems.
3. Choose the system(s) you would prefer to use at home, and give reasons for your choice.
4. What are the limitations of hydroponics.

Present your finding to the whole class and compare your work with that of other groups.
Use Microsoft word program of the computer to type and print out your presentation.

Nutrient Solution Formulation for Hydroponics

The nutrients needed by crops to grow are the same. Plants require 16 elements for growth and these nutrients can be supplied from air, water, and fertilizers. *Can you mention nutrients?* You should ensure adequate concentrations of all nutrients throughout the life cycle of the crop. Inadequate or excessive amounts of any nutrient result in poor crop performance. Excessive amounts can damage the crop, waste money and fertilizer resources, and pollute the environment when fertilizer is released during flushing of the nutrient delivery system.

There are mainly two sources of nutrients for formulating nutrient solutions; premixed formulas and grower-formulated solutions. Premix formulas can be readily obtained from agro-shops. Please follow the manufacturer's instructions when using them. The common premixes are Epsom salt or Magnesium sulphate, calcium nitrate and nitrogen, phosphorus, potassium and micro nutrient. Numerous 'recipes' for hydroponic solutions are available. Many use different combinations of chemicals to reach similar total final compositions

Recipe for the solution to be mixed in 20 litres of water.

- 25 ml of calcium nitrate
- 1.7 ml of potassium sulfate
- 8.3 ml of potassium nitrate
- 6.25 ml of monopotassium phosphate
- 17.5 ml of magnesium sulphate
- 2 ml of trace elements



When formulating the nutrient solution for hydroponic plant consider the following; the age of the plant, stage of growth i.e., seedling, vegetative, flowering or fruiting, the type of the plant and feeding technique or media used.

Management of Nutrient solution

Selective absorption of nutrients by plants often imbalances the amount of ions in solution. This imbalance can rapidly affect solution pH and the ability of plants to absorb nutrients of similar ionic charge. Depending on the pH, nutrients such as iron can precipitate from the solution and become unavailable to plants. A routine adjustment of pH and buffering the solution is necessary. As in soil, nutrients should be adjusted for specific plant variety. However, most nutrient solutions have concentrations of 1,000 to 2,500 ppm. Values below these ranges often lead to nutrient deficiencies while exceeding these ranges can lead to toxicity. Unlike soil, hydroponic nutrient solutions do not have cation exchange capacity (CEC) from clay particles or organic matter. The absence of CEC means the pH and nutrient concentrations can change rapidly in hydroponic setups than is possible in soil. Thus it is important to;

- i. Ensure proper pH of the solution by regular checking and adjusting the pH 5.5 - 6.5.
- ii. Ensure the correct level of water in the reservoir.
- iii. Ensure proper balance of cations in by regular checking and adjusting the cations
- iv. Control any leakage in the solution holding and delivery system.
- v. Ensure that the solution is well aerated.
- vi. Ensure that the solution delivery system is functioning properly.

- vii. Disinfect and the solution regularly.
- viii. Prevent any contamination by the dust by keeping the solution covered all the time.
- ix. Clean the reservoir and replace the nutrient solution regularly.

Organic Hydroponics Solutions

Organic fertilizers can be used to supplement or entirely replace the inorganic compounds used in hydroponic solutions. However, using organic fertilizers has a number of challenges such as:

- i. Organic fertilizers differ much in their nutritional compositions. Even similar materials can differ significantly based on their source (e.g. the quality of manure varies based on an animal's diet).
- ii. Organic fertilizers are usually obtained from animal wastes, making disease spread a serious concern for plants grown for human consumption or forage.
- iii. Organic fertilizers are often particulate and can clog substrates and growing equipment.
- iv. Some organic materials particularly manure can further degrade to emit foul odors.

Equipment used to Manage Hydroponics Solutions

Common tools used to manage hydroponic solutions include:

1. Electrical conductivity meters, a tool which estimates nutrient ppm by measuring how well a solution transmits an electric current
2. pH meter, uses an electric current to determine the concentration of H^+ ions in solution
3. Litmus paper, disposable pH indicator strips that determine hydrogen ion concentrations by color changing chemical reaction.
4. Balances for accurately measuring materials.
5. Graduated cylinders or measuring spoons to measure out premixed nutrient solutions.
6. Laboratory glassware, such as burettes and pipettes, for performing titrations.
7. Colorimeters for solution tests which apply the Beer-Lambert law.



Activity 4.3

You will need

Participating in formulation of nutrient solutions for hydroponics

a pen, notebook, water, fertilisers, pH down, pH up, trace-elements, pH meter, premix, beaker, pipette, stirring rod, protective gear, hydrogen p

Key question: how do you formulate nutrient solution for hydroponics?

In groups:

1. Assemble all the required equipment, water and chemicals.
2. Put on protective gears.
3. Place twenty litres of rainwater or distilled water in a bucket. Why do you use distilled water?
4. For nutrient premix formula, open the pack and empty the contents in the bucket containing water. For home formulation, measure and add 25 ml of calcium nitrate, 1.7 ml of potassium sulfate, 8.3 ml of potassium nitrate, 6.25 ml of monopotassium phosphate, 17.5 ml of magnesium sulphate and 2 ml of trace elements.
5. Stir the contents thoroughly until all the contents dissolve completely.
6. Adjust the pH of the solution using drops of pH down and pH meter.
7. Add some drops of hydrogen peroxide to the solution to inhibit bacteria and algae growth.
8. Store the solution in a plastic container and away from heat and sunshine.
9. Clean all the equipment and store
10. Remove protective gears and wash up.
 - (a) Why would you advise a farmer to mix his own nutrient solution on the farm?
 - (b) Why would you not recommend using organic nutrient source in hydroponic systems?
 - (c) Suggest the ideal qualities of nutrient solution for hydroponics.
 - (d) How can you manage the nutrient solution to ensure its efficiency in supporting crop growth?

Present your findings to the whole class and compare your work with that of other groups.

Selecting Appropriate Substrates (grow media or support materials)

Different materials are used to provide support/anchorage to growing plants in different hydroponic systems. Ideal substrate should possess the following qualities;

- i. Should be inert not to produce elements or alter pH of the nutrient solution.
- ii. Should not decompose.
- iii. Able to provide support.
- iv. Should have mild water retention.
- v. Should not release particulates that can clog the water ways.
- vi. Clean and free from animal and plant diseases.
- vii. Cheap and readily available.

Different support material/media are used for different growing techniques.

Expanded clay: clay is formed into round pellets and fired in a kiln at 1200 °C. This causes clay to expand, like popcorn, and become porous. It is light in weight, and does not compact over time.

Perlite: this is a volcanic rock that has been superheated into very lightweight expanded pebbles. It is used loose or in plastic sleeves immersed in the water.

Vermiculite: this is a mineral that has been superheated until it has expanded into light pebbles. The pebbles have natural property that draws water and nutrients in a passive hydroponic system.

Sand: this can be used after washing with plenty of clean water. Sand is inexpensive, easy to keep clean, drains well and will not become waterlogged.

Organic fibres: e.g. Wood shavings, coconut coir, rice husks, peat and saw dust. These should be well processed to remove salt, tannins and phenolic compounds before being used.



Activity 4.4

You will need

Participating in selecting and preparing media used in hydroponics

a pen, notebook, lake sand, wood shavings, water, containers, reference book, internet, protective gears.

Key question: What types of substrates are used in hydroponics?

In groups:

1. Put on protective gears.
2. Follow the right procedure to prepare the substrates provided. Why should the substrate be treated before use in aquaponics.
3. Store the treated substrate safely in a polythene bag to prevent contamination.
4. Brainstorm, research and give the characteristics of suitable substrate to use in hydroponic farming and list the common substrates used in hydroponics systems.

Present your findings to the whole class and compare your work with that of other groups.

Setting Up and Managing Hydroponic Systems

Hydroponic growing can be done indoors or outdoors. In either setting, plants will need good access to light, air, water, nutrients, ambient temperature and space.

Choose an area that is flat and free from flooding, the area should be easily accessible, has access to electric supply and full sunshine (only important for outdoor production).

Use standard materials to set up the hydroponic system. The equipment you need for hydroponics will depend on the hydroponic system you plan to use. For wick system, you will need a reservoir, growing medium, nutrient solution, grow trays and tables, temperature control devices, lighting, an air pump and the wicks that will dip into the nutrient solution. Drip system will require on top of the above, plastic tubing, drippers, a water pump, and a timer to properly schedule watering.



Activity 4.5

You will need

Setting up a simple hydroponics farming system for study purpose

a pen, notebook, tools, equipment, green bean seeds, lake sand, nutrient solution obtained in activity 4.3, protective gears, plastic bottle, bucket

Key question: What is the procedure for setting up a hydroponic farming system?

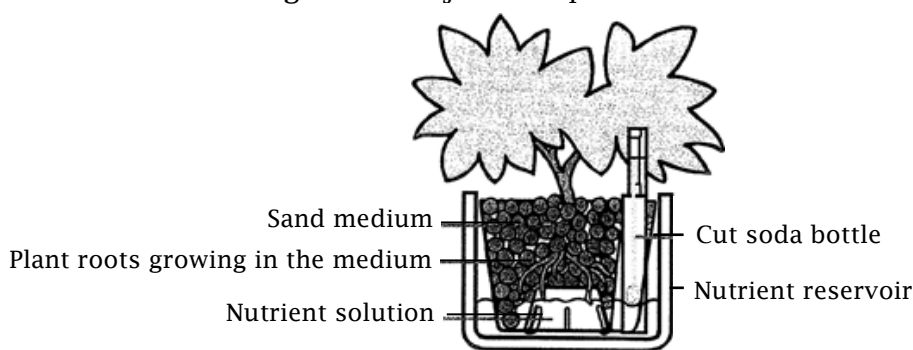
In groups:

1. Assemble the necessary tools and equipment and materials to use.
2. Put on protective gears. What will they include?
3. Remove the top of a 1.5-liter plastic soft drink bottle by cutting around the bottle at the upper label level
4. Drill a 1/4 in. diameter hole in the center of the bottom of the bottle.

5. On the inside of the bottle, cover the hole in the bottom of the bottle with plastic mesh. The plastic mesh will prevent the loss of sand from the opening in the bottle.
 6. Fill the bottle with lake sand all the way to the top.
 7. Using the prepared nutrient solution, leach the sand until the nutrient solution freely flows from the hole in the bottom of the bottle.
 8. Place two seeds about an inch deep into the moist sand. It may be necessary to add a small amount of water (half a cup) daily to the top of the bottle to keep the sand moist until the seeds germinate and the cotyledons appear.
 9. Place the bottle into a small plastic bucket and fill the container to a depth of about 2 in. with nutrient solution. With a black marker pen, put a scribe mark at the nutrient solution level on the side of the bucket. When adding nutrient solution, always fill to that mark. Cut an opening in the bucket lid sufficient to accommodate the bottle. The nutrient solution in the bucket will also fill the bottle with nutrient solution at that same level.
 10. Place the bottle in its bucket in full sunlight. Add nutrient solution when needed (usually every day) to maintain the level in the bucket at the scribe mark using a 50 mL graduated cylinder so that water use can be monitored.
- When the seeds germinate, remove one of the seedlings to leave just one plant in the bottle.



Fig. 4.4 the setup



Monitor and manage the system and present a report of your findings to the rest of the class

Agronomic Practices in Hydroponics

Preparation of the grow bed, formulation of the nutrient solution, plant species selection, planting, provision of enough nutrient solution, provision of supports/staking, supplemental nutrification of solution, provision of light, provide suitable temperature, ensure proper aeration regular checking and control of the condition of solution and pH, and control of pests and diseases, harvesting of ready crops.



Activity 4.6

Caring for the hydroponics garden

You will need

A pen, notebook, thermometer, hygrometer, pH meter, garden tools and equipment, water, protective gears

Key question: how do you manage the hydroponic garden to ensure good performance?

In groups:

1. Put on your protective gears.
2. Go to the hydroponic garden and do the following activities;
3. Check and ensure that there is the recommended amount of nutrient solution in the reservoir. Top up with clean water if the level is below the recommended minimum level.
4. Check for the pH of the solution is within the recommended range of 5.5-6.5. Nitric sulfuric and hydrochloric acids can be used for lowering the pH. Solutions of either sodium or potassium hydroxide (NaOH and KOH, respectively) are suitable alkalis for raising the pH.
5. Check and ensure that the temperature of the solution is within the acceptable range of 20°C - 21°C. for green houses it may be necessary to warm the solution if the temperature of the solution is too low. What would be the effect of too low or too high temperature?
6. Check and ensure that the air pump and water pump are functioning properly if they are part of the system. Of what importance is the air pump and water pump?

7. If your garden is indoors, ensure proper ventilation to allow good air circulation in the room so that plant can get enough carbon dioxide for proper growth.
8. Also check and ensure that humidity level is optimum. i.e. 40% to 60%. How does humidity level affect the system?
9. If you are using artificial lighting, ensure that the system is functioning properly and enough light is being provided to plants
10. Check plant for signs of infections and nutrient deficiencies and take appropriate measures. How can infection spread in indoor hydroponics?
11. Check and repair any leakage within the system.
12. Keep proper record of all the activities, and share with the rest of the class.

Budgeting for the Hydroponic System

Whatever hydroponic farming plan you make, be sure to make clear budget and estimate of the costs. Estimate the cost of construction materials, labour, and chemicals. Use the locally available materials. The materials should be of good quality and relatively cheap. The costs involved will depend on the type, size, whether production is indoors or outdoors and control system especially if light, temperature and air have to be controlled. The budget should be designed to minimise the overall cost of production.



Activity 4.7

Participating in budgeting and estimating construction cost of the hydroponic system

You will need

A pen, notebook, calculator, price quotations for different materials

Key question: how to budget and estimate the construction cost of a hydroponic system?

In groups:

1. Decide on the type of hydroponic system you are going to construct
2. Make a list of materials that you will need to construct the selected hydroponic system.
3. Check and sort out the material that you already have.
4. Attach the price to the material you intend to buy.
5. Identify other costs involved such as transport cost and loading and offloading costs.
6. Identify the labour requirements and attach the cost.
7. Calculate the overall cost of the material.
 - (a) How does making a budget help your hydroponic project to be successful?
 - (b) What are some limitations in making a budget for your hydroponic project?

Present a typed final work to the rest of the class and record the conclusions

Developing a Marketing Plan for Crops Produced in Hydroponic System

Crops produced under hydroponic are mainly vegetable. These crops are highly perishable and need to be sold immediately after harvesting. Therefore, A good marketing plan should be put in place to ensure that there is market for the products. The plan should not be costly so as to obtain good profits. The market plan should indicate product, price, place and promotion. use the following promotional methods; advertising in local papers and magazines, reaching out to local website, signs and billboards, flyers, social media, local radio advertising.

You can sell your products in the local markets or, you can cooperate with other producers in the area to increase the volume of your products and share the costs involved in marketing. This will also help you to have big volumes of products for export market.



Activity 4.8

Developing a marketing plan for hydroponic products

You will need

A pen, notebook, marketing journals, internet, communication devices such as flyers and phones

Key question: how do you manage the hydroponic garden to ensure good performance?

In groups:

1. Develop a plan of how you are going to market your produce. What will you include in the plan and why?

2. Which distribution channel do you intend to use and why?
3. How will you market plan increase sales of your hydroponic products?
4. What challenges are you likely to face as you market your hydroponic products?

Share and discuss your plan with the rest of the class and record conclusions



Group Project

Aim: Sustainable production of high quality forage using hydroponics

Justification; It is increasingly becoming hard to have pasture supply all year round. Soils are unproductive and rains are inadequate and unreliable. Thus use of hydroponics systems can help to produce the required pasture all year round.

Materials required;

Inorganic rooting medium, plastic baskets, bucket with lid, solution premix, PVC pipes, air pump, water pump, platform, construction tools

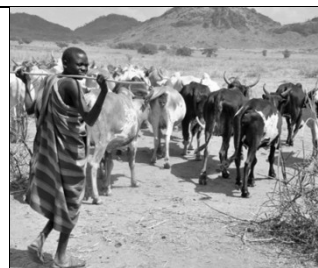
In groups or as a class,

1. Make a plan and design a budget for establishing a hydroponics unit.
2. Procure the materials required to establish the hydroponics garden.
3. Set up the hydroponics garden unit, select appropriate crop, plant and manage the unit, keeping proper records and learning logs, find the effectiveness of the system by comparing the records with those of conventional crop production systems.
4. Use a word processing program to produce a report indicating the materials you used, the steps you followed to make the mineral and vitamin lick blocks, the lessons learnt, challenges and recommendations on improvement in the production process.
5. Individually keep records at each stage in the process and produce a report.

Activity of Integration



The residents in Nadunget, Moroto district are complaining of poor climate in their area. The area receives short rains followed by prolonged periods of droughts. They have also pointed out another problem of unproductive soils. Quite often, the area experiences famine and the people are in dire need for solutions to enable them produce food for themselves throughout the year.



Task

Task; using your knowledge of hydroponic farming, write a message you will deliver to the local on how the effect they are experiencing can be overcome by hydroponics farming and what should be done to improve the climate and the condition of the soil in their area.

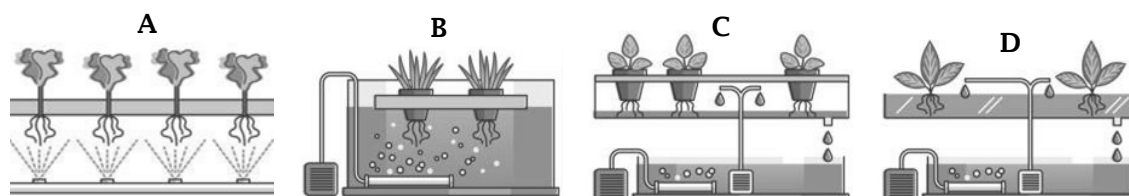
Exercise 4.1

Self-Check Questions



As a student with good knowledge of hydroponics,

1. (a) Define hydroponics farming and give its benefits and limitations as a method of growing crops
(b) Make an annotated sketch of a nutrient film system of hydroponics
(c) Explain how hydroponic can improve crop production in your area.
2. (a) Discuss the agronomics practices involved in hydroponics.
(b) Outline the ideal qualities of grow media in hydroponics system.
(c) Below are different hydroponic production systems. Name each system and indicate whether it is a closed or open system.



Financial Services and Money in Agriculture

Competency: You should understand the importance of prudent financial management and be able to use knowledge and skills to manage financial resources effectively.



KEY WORDS

- Needs
- Wants
- Agriculture finance
- Budgeting
- Financial Literacy
- Good debt
- Bad debt
- Investing
- financial literacy
- Electronic Money
- Electronic Payments
- Mobile Banking

LEARNING OUTCOMES

By the end of this chapter, you should be able to:

5. understand the difference between needs and wants
6. be able to explain the five core pillars of financial literacy
7. know the difference between investment and saving
8. understand the importance of the saving culture
9. know the different types of banks and bank accounts
10. demonstrate the ability to open a bank account
11. show skills in budgeting using spreadsheets/Excel, or books and calculators

Introduction

In senior one, you learnt that, agriculture is the main source of income for many rural people. Relative to other sectors, agricultural growth can reduce rural poverty rates faster and more effectively. One relevant vehicle to achieve growth in the sector may be finance. Farmers' decisions to invest and to produce are closely influenced by access to financial services. Improving access to finance can increase farmers' investment choices and provide them with more effective tools to manage risks.

Importance of agriculture finance

Agricultural finance is needed to help farmers to meet various demands like buying of quality seeds for planting, supply of power, paying for workers' wages and salaries, procurement of fertilisers and pesticides, agricultural marketing, post harvesting storage, processing and transport of produce, meeting the risks like damage due to pests, diseases and drought and procurement of land for expansion and meeting other obligations like paying taxes.

Understand the difference between needs and wants

Needs are the basic requirements or necessities that are essential for your survival. In other words, they are the things you have to have, or cannot do without. So, what are the basic needs for a human being's survival? As we all know, a human being cannot survive without Oxygen, water and food. In addition to these three elements, there are some other needs that are essential to live a healthy and happy life. Some of these needs include basic health care products, clothing, tools to work, and protective housing and security for you and your property.

Wants are unnecessary but desired items or items which increase the quality of living. But, they are not essential to your survival. You can live without them, but you enjoy your life more when you have them. Examples include a car stereo, CD's, car, and designer clothes, a big house, fancy food, branded clothes; a new car, trip to a foreign country etc. Wants can change over time. For example, when you were a child, you might have wanted to buy a big toy castle. But as you grow older, your tastes and preferences change, so, your desires change with them.



Activity 5.1

Understanding the difference between needs and wants

You will need

a pen, notebook, agricultural reference books, internet

Key question: What is the difference between needs and wants?

In groups Research and brainstorm;

1. Giving examples in each, explain the difference between needs and wants.
2. Why do you think agriculture financing is a necessity to agricultural development?
3. As a poultry farmer, what do you consider as wants and needs on your farm?

Present your findings to the whole class and compare your work with that of other groups.

Financial Literacy

Financial literacy is the process of building knowledge, skills and attitudes to become financially literate. It introduces you to good money management practices with respect to earning, spending, saving, borrowing, and investing. It is the ability to understand and effectively use various financial skills, including personal financial management, budgeting, and investing.

Pillars of Financial Literacy

1. **Earning;** make sure you have some income from your farming enterprises. Change from subsistence farming to market or commercial farming. Ensure that you make the most out of what you earn by having good financial discipline to use your earning carefully.
2. **Saving;** saving is an essential part of financial wellness, a secure present, and a happy future of the farmer. Wealth is built through spending less of your income so that as a farmer you can

achieve the following: Buying more land to expand the farm, buying machines to process the farm produce, paying off the loans, and buying other basics that make your life comfortable. Putting your savings into an interest-yielding bank account not only keeps your money safe, and out of temptation's reach, but also allows you to grow it over time.

3. **Budgeting;** Budgeting is the life skill of planning and managing your money. By understanding exactly where your money goes every month, you are empowered to create an actionable plan by which you can spend less, by curtailing those unnecessary expenses and saving more for the things you need and want. The rule here is that money coming in (your total income) should always be greater than money going out (your total expenses). The difference between the two values is what you should be keeping as savings.



Budgeting helps you plan for short, medium, and long-term expenses, enabling you to save accordingly to afford all three. It is, therefore, entirely necessary for financial

4. **Spending;** Be sure you are getting good value for your purchases by shopping around and comparing your options. When considering large purchases, determine whether it is a need or a want and make sure you are living within your means.
5. **Investing;** Investing is all about creating and growing the wealth you have now so as to enjoy a financially secure and happy future. It's about putting your money into something that will make you a profit over time, such as property, retirement funds, and buying shares in business companies. The growth of your investment's value can make an added income for you.
6. **Borrowing;** Borrowing means to take money from a bank or other financial institutions and pay it back over a period of time. Usually an interest is levied on the borrowed money. Debt is money you spend that isn't yours. Borrowing is necessary because sometime you may have emergencies that require funds which are above your savings. It can also help you grow your wealth by providing liquidity needs, while your investments grow.

As a farmer, you can borrow from the following Sources;

- Commercial banks
- Microfinance institutions
- Development banks like Uganda development bank
- Credit from friends and relatives
- Private money lenders
- Savings and credit co-operative societies if you are a member.
- Government sources e.g. the small production credit such Emyoga, youth livelihood projects, parish development model funds and "Entandikwa" scheme.
- Credit from your suppliers, where you get inputs on credit and pay in later.

The funds you have borrowed can be utilised in the following ways;

- i. To purchase equipment like tractors and other equipment used in agricultural production.
- ii. To pay workers' salaries and wages.
- iii. To buy inputs like fertilizers, improved seeds and pesticides.
- iv. To expand the farm by acquiring more land and acquiring better farming technologies.
- v. To add value to output by buying processing machines.
- vi. Enable efficient marketing of farm produce by buying packaging materials construction of stores and acquisition of transport trucks.
- vii. Money may be used to avert risks and the effects of uncertainties on the farm.
- viii. To pay back other loans which are due.
- ix. To plan and budget promptly for the farm business.
- x. To install irrigation system and soil conservation structures.



Good debt is money borrowed for things that are absolutely necessary for making a life e.g. a house and for advancing your money-making potential.

Bad debt is borrowing money to pay for things you don't need, such as expensive clothes, hi-tech electronics, eating out at restaurants, going on holidays, etc.



Activity 5.2

Understanding the core pillars of financial literacy

You will need

a pen, notebook, reference books, access to computer and internet

Key question: What are the five core pillars of financial literacy?

In groups:

1. Research on the internet, define financial literacy, and explain and explain how they can led to good financial management.
2. As a student, why would you need to have good financial literacy?
3. Type your findings using word program (observe the following when typing your work; use Arial font, font size 12 and Line spacing of 1.5).

Present your findings to the whole class and compare your work with that of other groups.



Understanding Investment and Saving

Investing means putting money into business with the hope of earning profit. An **investment** is the purchase of goods that are used to generate other goods or wealth. Or an investment can refer to any mechanism used for generating future income.

Saving is setting aside money you don't spend now for a future use. Saving is often deposited into a bank savings account or a fixed deposit account. Saving does not automatically include interest.

An investment is generally different from savings as an investment is a more active way of using money to earn profits or dividends, while saving is generally storing of wealth with or without earning interest.

Methods of Saving

Methods of saving include putting money aside in, for example, a deposit account, a pension account, an investment fund, or as cash. Saving also involves reducing regular expenditures.

Strategies to save money (reduce expenses)

1. Grow your own feeds on the farm to avoid buying expensive feeds from outside
2. Keep detailed records of your farm business to identify and remove enterprises that make the least profit and are costing you the most to maintain.
3. Use good husbandry practices to control parasites and diseases on the farm this reduces expenses on veterinary and drugs, increase yield and avoid loss of animals.
4. Make your own fertilizer. Consider using of farm made organic manures to reduce the money spent on buying fertiliser and other soil conditioners.
5. Use of cultural pest and weed control measures to reduce costs on pesticides and herbicides.
6. Invest in alternative energy sources. For future savings on energy costs, consider investing in wind turbines and solar panels for smart energy.
7. Use cheap family labour if possible to reduce salary and wage expenses.
8. Buy Seeds during off season to avoid the high seed prices during planting season.
9. If possible buy your own machines instead of hiring to cut on the future expenses.

Reasons for saving

There are several reason why saving is important.

Financial independence: if you have enough savings to fall back on, you can take major life decisions without worrying or stress.

Living debt-free: you don't need to rely on credit for all your expenses. By relying on your savings to make expenses, you save a lot of money by avoiding unnecessary interests and penalties.

Unforeseen expenses: If you have some savings in your bank account, you will be able to manage unforeseen or emergency expenses much more prudently. It is always better to be prepared for these expenses and have some savings to fall back on.

Buying or paying for necessities: Saved money can be used to buy other necessities such as a house for your own, a car, land, medical bills and paying for school fees.

Planning your retirement: saving money can ensure that you retire when you want. Set a target, work hard to achieve that target and then retire to do other things.

It offers peace of mind: Knowing that you have a certain amount accumulated for times of your need, gives you peace of mind. You can lead a stress-free life with the knowledge that you will not have to struggle if things take an unexpected route.

It gives you a better future: Your savings can be the answer to a number of your goals. You can buy a house, accumulate funds for your retirement, or purchase a vehicle. You can secure your future, indulge in the best of things that life has to offer and live a very fulfilling life.

It provides for your children's education: With a considerable amount of savings, you can fuel your children's dreams and pay for the best schools and colleges across the world.

Advantages of savings money in the bank	Disadvantages of saving money in the bank
• bank give interest on savings • It's easy to open and access your savings. • Your money is safe and secure • By keeping money away from you, it controls your spending.	• They require minimum balance so you cannot get all the money in times of need. • They give lower interest rates • They limit saving withdrawal.

Reasons for Investing

Investing your money can allow you to grow it. Putting your money into profit making business can help to grow your money. The profit allows your money to build, creating wealth over time.

Long-term returns: If you invest in the right places, there is a lot of potential for long-term returns. One of the main benefits of investing is that the money you invest has the potential to grow substantially over time. Rather than just putting your money into a savings account to save for the future, investing is can be a much smarter way to make your money work for you.

Building wealth: In addition to earning enough money to cover your basic needs, investing your money in a variety of assets can be a great way to potentially build your wealth. If you start investing now, just imagine how much money you could save up by the time you reach retirement.

Plan for retirement: Investing can help grow your savings, so that your money can help at the time when you're not actively working. you may be able to actually achieve those retirement dreams of spending your golden years with few financial problem.

Meeting personal and financial goals: Another important reason to invest is that it can help you achieve your personal and financial goals such as:

- A education fund for your children
- Buying a home
- Going for Vacations and for Muslim comrades, for Hijja and Hijra
- Overall financial stability
- Big purchases

Staying ahead of inflation: the rate of inflation can vary each year, so it's important to find investments that can generate enough return to outperform inflation.

Multiple streams of income: with investing, you could generate multiple streams of income right from home.

Have savings to fall back on: things can happen unexpectedly like the lock down that happened during COVID 19 outbreak or you can lose your job. Having sufficient savings can help you go through these unexpected life situations smoothly.

Making an Investment Plan

An investment plan refers to a process of matching your financial goals and objectives with your financial resources. It is considered to be a central component of financial planning and it is impossible to have one without the other. There are various types of different investments and the most commonly used are cash, equities, bonds as well as property.

Steps to creating investment plan

1. Set specific and realistic goals.
2. Calculate how much you need to save each month.
3. Choose your investment strategy.
4. Develop an investment policy statement with your adviser.
5. Review your plan regularly.



There are multiple investment plans available today. Each has its levels of risk and reward. Before making any commitment, make sure that the type of investment you choose aligns with your current and future financial goals.

There are many investment options available to you as a student as an agriculturalist; can you mention them?

Other investment options

1. **Stocks:** Stocks represent ownership or shares in a company. When you invest in a stock, you're buying a share or portion of the company's earnings and assets. If the company does well, the value of stocks increases and vice versa.
2. **Bonds:** A bond is an investment where you lend money to a company, government, and other types of organization. In return, the bond issuer pays you interest for the borrowed money, and, at the same time, repay you the original amount you paid for the bond (principal). In general, bonds are a fixed-income investment.
3. **Property:** This includes housing, real estate, land, and other rental properties. This type of investment is appealing to many primarily because it's tangible.
4. **Money Market Funds:** Not to be confused with the normal savings account, money market funds allow the investor to leave a certain amount in a bank for a predetermined period of time. By the time it's over, you get your principal back but for a slightly higher rate of interest.

While investing has many advantages, there are some disadvantages to investing as well.

- i. Losses due to risks and uncertainties: There is no such thing as a total risk-free investment and there is always the risk of a loss of your investment.
- ii. Requires Investing Knowledge: Investing requires specialized knowledge about finance and different types of asset classes. Thus there is need to hire a financial advisor. Choosing the right financial advisor is also hard for farmers.



Activity 5.3

Understanding investment and saving

You will need

a pen, notebook, reference books, internet, different feed samples

Key question: What is the difference between investment and saving?

In groups:

1. Brainstorm and differentiate between saving and investment
2. State the reasons why you may need to save and invest.
3. Outline the different methods you would use to save.
4. Develop a saving and an investment plan
 - (a) What guides you in deciding the method of saving and what you are going to invest in?
 - (b) Identify your sources of income and
 - (c) Decide on how much you want to save daily or monthly.
 - (d) Determine how much you need to invest.
 - (e) Decide your investment strategy or rules
 - (f) How would you evaluate the progress of your investment from time to time?

Prepare your report and present it to class using a power point presentation program.



Roles of Different Types of Banks and Bank Accounts

There are several banks and other financial institutions all providing financial services. The services provided may include saving deposits, issuing loans, payments and settlement of debts.

Types of Financial Institutions

There are many types of financial institutions in Uganda, including the following:

1. The Central Bank
2. Commercial banks
3. Microfinance institutions
4. Saving and Credit Cooperative Societies (SACCOs)

The Central Bank

We have one central bank in Uganda, which has the responsibility of supervising all the other financial institutions.

The Central Bank is defined as a government bank established to supervise, guide and assist commercial banks and other financial institutions in the country. It provides banking services and financial advice to the government and commercial banks. Bank of Uganda, based in Kampala, is the central bank of Uganda.

The central bank performs a number of functions, including:

- i. Issues currency in the country in form of bank notes and coins. All the money you see being used in the country is issued by the central bank. It also replaces all worn out bank notes.
- ii. It is a bank for all commercial banks; it is where they keep part of their total deposits. Commercial banks are expected to keep part of their deposits with the central bank.
- iii. It licenses and supervises all banking activities of commercial banks.
- iv. All government bodies receive their funds through the central bank.
- v. It lends to commercial banks as lender of last resort.
- vi. It controls money supply in the country by using interest rates, open market operations, selective controls, variable reserve requirements and other tools of monetary control.
- vii. It advises government on the monetary policies meant to guide the economy.

Commercial Banks

This is a type of financial institution which deals directly with the public and offers a number of services. In Uganda, we have a number of commercial banks which provide these services.

Example of commercial banks currently operating in Uganda include; ABC Bank

Absa Bank, Bank of Africa, Bank of Baroda, Bank of India, Cairo Bank Uganda, Centenary Bank, Citibank Uganda, DFCU Bank, Diamond Trust Bank, Ecobank Uganda, Equity Bank, Exim Bank, Finance Trust Bank, Housing Finance Bank, KCB Bank, NCBA Bank, Opportunity Bank, Post Bank, Stanbic Bank, Standard Chartered bank, Tropical Bank and United Bank for Africa.

Commercial banks deal directly with the public and provide a number of services, which include the following:

- i. Accepting deposits from customers and safeguarding the money deposited;
- ii. Assisting in making payments and settlements by using cheques and payment orders;
- iii. Assisting in transferring money by using bank drafts, standing orders, travelers' cheques and credit transfers;
- iv. Providing finance to customers in form of loans and overdrafts;
- v. Buying and selling foreign currency on behalf of their customers;
- vi. Assisting customers in international trade by selling travelers' cheques, bank drafts, letters of credit, etc.
- vii. Providing night safe custody services to their customers;
- viii. Assisting customers to access their money any time they want it, through
- ix. Automatic Teller Machines (ATM).

Micro-Finance Institutions (MFIS)

These are financial institutions which provide credit to customers. The amount of money normally given is usually small. Many of these financial institutions are located in rural areas. Examples in Uganda include Pride Micro-Finance and FINCA.

Savings and Credit Cooperative Societies (SACCOs)

These are small financial institutions started by private people on their own. They mobilize savings from members and lend some of these savings to those who need loans. SACCOs are not categorized as banks.



Identify any financial institution in your community and outline the services it offers to the public. Advise how the finance institution can be helpful to farmers.



Activity 5.4

Finding out about the different types of financial institutions

You will need

a pen, notebook, reference books, internet, manila paper and markers

Key question: What are the functions of different types of financial institutions?

In groups:

1. Research on the internet and agriculture reference books, and identify the different types of financial institutions in Uganda.
2. Outline the services offered by each type of financial institution.
3. IF you wanted a loan to finance your project, which type of bank would you go to and why?
4. Where do you keep the money you collect from the sale of products produced from your project and why.

Present your findings using a slide presentation program or a chart, to the rest of the class.

Types of Bank Accounts

A bank account is a record of transactions between you and the bank. This relationship can be shown by opening any of the following accounts: savings account, current account, fixed deposit account, collection account and loan account.

Savings Account

A savings account is an account that can be opened by an individual who would like to accumulate money for a particular purpose e.g. buying a feed processing mill, starting a poultry farm, construction of a house, buying of tractor etc.

Features of a Savings Account

- A minimum initial deposit is required at the time of opening it.
- You can deposit money on this account any time during working hours.
- You are paid interest on the balance on account at the end of a given period, which could be a month, quarter of a year, half year or a year.
- You can withdraw and deposit money at any branch within the country.
- You are not allowed to withdraw more money than what you have on the account.

Current Account

This is an account available in financial institutions and is mostly operated by agribusiness people. This account is ideal for farmers and agribusinesses with large sums of money, wishing to make large and frequent deposits and withdrawals.

Features of a current account

- A minimum initial deposit is required at the time of opening it.
- No minimum balance is required. You can withdraw all your money from the account and the account remains open, unless you tell the bank to close it.
- You can deposit any amount of money on it.
- No interest is paid on deposits kept on your current account.

- Overdrafts are allowed on a current account on special arrangement with the bank.

Fixed Deposit Account

This account is opened by an individual with a given amount of money for a stated period and promises not to withdraw it until the expiry of that period. It has the following features:

- A minimum amount is required for a specific period.
- No further deposits or withdrawals are to be made on the fixed account until the expiry of the agreed time.
- Higher interest is paid to the account holder in comparison to other types of accounts.
- You are issued a receipt at the time of placing the deposit.
- When you withdraw your money before expiry of the fixed deposit period, you forfeit the interest expected on the account.

This type is ideal in case you have large sums of money you may not need over a specified period.



You can have more than one account in the same bank or in different banks. Periodically you may ask for a document that shows your transactions. This is a *bank statement*



Activity 5.5

Finding out about the different types of bank accounts

You will need

a pen, notebook, reference books, internet, manila paper and markers

Key question: What are the different types of bank accounts?

In groups:

1. Research on the internet or your teacher can invite the bank manager of a nearby bank. Identify the different types of bank accounts and how they operate.
2. Outline the advantages and disadvantage of each type of bank account.
3. From your research finding, which type of bank account is suitable for your project?

Present your findings on a manila and display it at the back of your class for other to see.

Opening up a Bank Account

To open an account at the bank, you'll need to choose a bank and provide the bank with some necessary information, including your identification and proof of address. You'll usually need to be 18 to meet the bank's eligibility requirements. Finally, you might need to fund the account or have a minimum balance to avoid fees.

1	Identification Documents (Acceptable IDs) Ugandans: National IDs, Passports, Employment ID, Driver's License and NSSF Card. Foreign nationals: • Passports are the ONLY acceptable form of identification proof for foreign nationals and • Work Permit/Certificate of residence. FATCA form for US Nationals Minors: birth certificate/Health cards and photos must be obtained. While the minor's Guardian who will operate the account MUST provide Individual requirements. Students: National ID/Student ID
2	Proof of residence/ address verification include any of the following. • Letter from a public authority or embassy or consular office • Utility Bills in the customer's name and not older than 3 months • Where utility bill is not in the name of the customer obtain a copy of the tenancy agreement • Letter of recommendation from employer, • LC 1 Letter confirming residence • Letter from School/University administration in case of students • Bank statement from existing or previous bank bearing the current customer's address. • Certificate of residence for foreigners
3	Proof of Income/Source of Income • Letter of recommendation/appointment/confirmation from employer • Pay slip not older than 3 months • Declaration of Income form • Trading License • Work Permit (only applicable to Foreign Nationals)

Please complete in black ink only.

Controlling branch _____

 Date
(YYYY-MM-DD) _____

Section 1 - Entity type

- ☐ Sole Proprietor ☐ Body Corporate ☐ Partnership ☐ Informal Body e.g. Club, society ☐ Close Corporation ☐ Trust/Attorney Trust
☐ Company (including incorporated companies) ☐ Deceased Estate ☐ Other (Specify) _____

Section 2 - Accounts required

- ☐ Current account/Business Transaction account _____ ☐ Call account _____
☐ Fixed deposit _____ ☐ Other (specify) _____
☐ Notice deposit _____ ☐ Currency _____

Section 3 - Bank location

- ☐ Stanbic Bank Botswana Limited 1991/1343 ☐ CFC Stanbic Bank Limited C9520 (Kenya) A separate account opening form is required for each of the following banks:
☐ Stanbic Bank Ghana Limited 54,199 ☐ Stanbic IBTC Bank PLC RC 125097 (Nigeria)
☐ Standard Bank Limited Malawi 1246 ☐ Stanbic Bank Uganda Limited P525 Standard Bank of South Africa Limited
☐ Standard Bank Namibia Limited 78/01799/06 ☐ Stanbic Bank Zambia Limited 6559 Standard Bank SARL Mozambique
☐ Stanbic Bank Tanzania Limited 22443 ☐ Standard Lesotho Bank Limited 94/165 Standard Bank (Mauritius) Limited
☐ Stanbic Bank Zimbabwe Limited 3387/89 ☐ Standard Bank Swaziland Limited 40/1974 Stanbic Bank Congo s.a.r.l.

Section 4A - Applicant information
Residency ☐ Resident ☐ Non-resident If non-resident - Country of incorporation _____

Registered name _____

Name of account/Trade name _____

Physical business address (if applicable) _____

Postal code _____

Postal address _____

Postal code _____

Head office address/Registered address (if applicable) _____

Postal code _____

Identity/Trust/Registration no. _____

Registration date (YYYY-MM-DD) _____

Income Tax number _____

VAT number _____

Establishment date (YYYY-MM-DD) _____

Financial year end (MM-DD) _____

Anticipated Annual turnover (in local currency) _____

Reason for opening account with this Institution _____

Type of activity expected on the account (e.g. cash deposits, debit orders) _____

Source of funds (e.g. Donation, Third Party loan) _____

Source of income (e.g. profit from sales) _____

Type of business conducted _____

No. of employees/members _____

Business telephone number _____

Business fax number _____

B - Contact details

Contact person	Capacity	Contact telephone numbers	E-mail

C - References

Record details of two business/trading associates the bank may contact to obtain trade references

Name of firm/company	Name of contact person	Telephone number

D - Accounts held at other financial institutions

Name of financial institution and branch	Type of account	Account number

Fig. 5.1 Sample of bank account opening for

A Deposit Slip

A deposit slip is a paper form that a bank customer includes when depositing money into a bank account. A deposit slip, by definition, contains the date, the name of the depositor, the depositor's account number, and the amounts being deposited.

CASH / CHEQUE DEPOSIT SLIP

____/____/____ **CUSTOMER COPY** **CAIRO BANK**

Date

Account No.

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

(Please mention this Account Number on the reverse of the cheques also)

Name of Account Holder _____

Cash/Cheque Detail	Cheque No.	Amount sh.
	Total sh.	

AMOUNT IN WORDS: Sh. _____

Deposit by _____ Teller _____

Fig 5.2 A sample of a deposit slip.

How to use Cash Deposit Slip?

One can deposit money into the bank either by cash or cheque. Below is a step-by-step process to use the cash deposit slip.

Step 1	Ask for a deposit slip and fill up all the details, like date, account, type, account number, account holder name, amount and finally sign on the deposit slip.
Step 2	Remember to fill all the details in the receipt attached to the deposit slip.
Step 3	If the deposit is by cash then enter the denominations in the denomination column.
Step 4	If the deposit is by cheque, then enter the bank details like the name of the bank, cheque number, date etc.,
Step 5	Turn over the deposit slip and the cash or the cheque to the teller in the bank
Step 6	Remember to collect the receipt (also called as counterfoil).

Cash Deposit Slip Process

The completed cash deposit slip is bundled with the checks, bills and coins itemized on the form and presented to the cashier at the bank. The cashier or the concerned bank person processes the deposit and matches the total processed to the total stated on the deposit slip to ensure that they match; thus, the cash deposit slip is a cash processing control for the bank.

Once the cashier processes the deposit, he gives the customer a receipt, which states the total amount of the deposit, along with the date and time. The receipt given by cashier to the depositor also called a counterfoil act as proof that the deposit was made.

Benefits of Cash Deposit Slips

Cash deposit slips offer protection to both the bank and the customer. Banks use them to maintain a written ledger of funds deposited throughout the day and to ensure that no deposits are unaccounted at the end of the business day.

For the person depositing cash, a cash deposit slip serves as a de facto receipt that the bank well accounted for the funds and deposited the correct amount and into the correct account. If the depositor later checks the account balance and discovers the deposit was not entered properly, the deposit slip serves as proof that the bank acknowledged receiving the funds from the customer.



Today, banking system has evolved to include electronic mobile and ATM banking. You need to have ATM or VISA card to use the services. Consult any bank personnel for info



Activity 5.6

Opening a bank account and filling a deposit slip

You will need

a pen, notebook, deposit slip and account opening form from a bank, school ID, passport photos

Key question: How to open a bank account and filling a deposit slip?

Individually, Use templates and dummies from a financial institution to imitate opening a bank account and depositing money on the saving bank account with the school bursar's office. Share your experiences with the rest of the class members.



Bank account can be opened up online. This save time and money as you don't have to go to the bank in person.



Budgeting for Farming Initiative

Farm Budgeting

Any profitable farming business starts with proper financial planning. You need to make a detailed quantitative statement of a plan of action for a farm business highlighting projected incomes and expenses this is called a **budget**. **Farm budgeting** is the process of estimating the future income and expenses of a farm bussiness.

Farm budgets address several managerial questions some of which include;

- What to produce?
- Which available resources can best be employed?
- What enterprise or combination of enterprises will bring the most returns to the farm business?
- What are the resource requirements of each enterprise?
- How much of each resource (land, labour, capital can profitably be allocated to each enterprise?
- What are the best production practices for each enterprise?
- What are the expected costs and returns?
- What are the expected losses and profits?

Importance of farm budgeting

- i. It acts as a test kit to predict the outcomes before resources are committed in production thus improving allocation of scarce resources.
- ii. Helps to estimate the need and availability of resources.
- iii. It helps to estimate the cash in or out flows and liquidity.
- iv. It encourages efficiency in order to meet projected targets.
- v. It helps in periodic assessment and analysis of the farm business.
- vi. It uncovers over looked costs and revenue items.
- vii. It communicates to lenders and stakeholders the anticipated revenues and expenses.
- viii. It assists in estimating future taxes on farm income.
- ix. Provide a basis for performance appraisal.
- x. Budgets are pressure devices to achieve set targets for each department.
- xi. It motivates staff from different enterprises to work together in harmony with the goals of the enterprise.
- xii. Enables remedial action to be taken as variances emerge.

Characteristics of good farm budget

- i. Should be comprehensive to cover all functional activities of the enterprise.
- ii. It should be flexible enough to be easy to adjust to fit the existing situations and resources.
- iii. It should be a collective participation of all staff on the farm as this motivates the staff and make them more responsible in achieving the set goals.
- iv. It should act as a standard against which to compare actual and anticipated performance. This will provide a means of indicating where and when control action may be necessary.
- v. It should facilitate feedback so that actual results are compared with budget projection to induce corrective action if necessary.
- vi. It should clearly spell the set goals/objectives of the farm business.
- vii. The objectives should be realistic and achievable.
- viii. Should show clearly expected sources of income and expenditure.



Activity 5.7

Understanding the importance of budgets on the farm

You will need

a pen, notebook, reference books, internet, manila paper and markers

Key question: what are the roles do budget play on the farm?

In groups:

1. Research and brain storm on the importance of budgeting in farm business and list the managerial questions that farm budgets should address
2. What do you consider as the characteristics of a good farm budget?

Present your findings with other groups and record conclusions..

Making a Farm Budget

As you prepare to make a farm budget, you should first know that farm resources are limited. Which are these resources? Some resources must be share by many farm enterprises, so as to have more returns from their use. It is also important to know that situation at the farm may change with time. Thus several budgets should be drawn to determine the most profitable alternative.

Budgeting Principles

Budgeting principles enable the comparison of costs and returns of alternative plans of action for a farm business. There are five basic budgeting principles.

1. Add units of an input as long as the value of the resulting output or added return is greater than the added cost i.e. invest more as long as returns increase.
2. Substitute one input for another as long as the cost of added input is less than the cost of replaced input and output is maintained i.e. invest as little as possible.
3. Substitute one product for another for as long as the value of the added output is greater than the value of the output that is replaced and the cost is constant i.e. invest in different product if returns are greater.
4. Use each unit of resources where it gives the greatest return i.e. invest money only where it earns the highest return.
5. Discount for time and risks.

Data Requirement for Budgeting

Before preparing a farm budget, some form of information is required to make a holistic budget. In agricultural, data required is input-output data; the estimates of the physical quantities of different resources that will be required to produce different products. Data required include; input prices, product prices, cost of capital, yield estimates, current crop conditions, methods and timing of input use etc.

Sources of Data (Information)

The value and reliability of a budget is only as good as the quality of the input data and hence should be based on the best information available. Sources of information in farm budgeting are;

1. **Own farm production records;** a farmer may use records obtained from his own farm. This is only possible if the farmer has a culture of keeping accurate records on his farm.
2. **Records from farmers' organizations;** co-operative societies normally keep accurate records and these could be used as a reference when making a budget.
3. **Internet data;** published data on the internet can be used as reference.
4. **Sub county production data;** in Uganda and possibly Kenya and Tanzania, sub county agriculture office collects data from individual farms, villages or parishes and compiles a production record report. This information can be used as a reference when making budgets.
5. **National bureau of statistics;** all national agricultural production records are provided by the government that could serve as an important source of information in budget formulation.
6. Data from lenders: through credit card
7. Data from other farms within the area.
8. Data from farmers meetings or workshops.
9. Data from agricultural or experimental stations research.
10. Data from typical sample budgets.
11. Farm literature; farmers' magazines or other agricultural publications.

A combination of sources should be used to collect and verify data used in preparing budgets.

Types of Farm Budgets

These are four types of farm budgets used in the farm business management process. All budgets use the same principles but each in specific in its application.

Complete or Whole-Farm Budget

This is a detailed listing of resources of the entire farm business which need a plan of how to employ resources to achieve the farm long-term goals. A budget for each enterprise on the farm is prepared to come up with the entire farm budget. Whole-farm budget is recommended when;

- The farmer wants to start a new farm business.
- The farmer wants to evaluate the whole farming operation.
- Comparing the profitability of alternative plans.
- Estimating the requirements for labour, capital, compared to suppliers.
- Changes are likely to affect all enterprises on the farm.

Procedure of making a whole- farm budget

1. Formation and listing the goals and objectives of the farm business e.g., maximize profit, long term land productivity, optimizing input and output.
2. Take farm inventory (including physical, human and financial) available for use in production, including accessibility to markets.
3. Show how the inventory resources (including physical, human and financial) are to be utilized in production under different enterprises.
4. Identify enterprises e.g. maize growing, fish farming, dairy production, poultry production, bee keeping, mushroom growing etc.
5. Identify reliable estimates of input, output prices, and calculate the expected variable and fixed costs and all returns.
6. Conduct a gross margin analysis i.e. estimates the gross margins for each enterprise.
7. Find out and correct the structural and operational weaknesses.
8. Make a number of alternative farm plans and choose one for adoption and implementation.
9. Assess the budget by determining whether performance levels are realistic, the level of risk involved is manageable, whether better alternatives exist, adequacy of management capabilities and effect on cash flow.

Partial Budget

This is a type of budget used in planning a small or minor change in the farm organisation. It assists management to analyse the economic effect of a minor change in a portion of the farm

business. Partial budgets evaluate changes in resources which are non – fixed. A partial budget is relevant whenever a change does not require complete farm reorganization. Such changes include: expanding an enterprise, changing to an alternative enterprise, changing products or practices, purchasing rather than hiring equipment, making capital improvements like replacing hired labour with machinery.

Enterprise budget is a statement or listing of all estimated incomes and expenses associated with a specific enterprise to provide an estimate of its profitability. The enterprise budget is a physical and financial plan for specific enterprise. The enterprise budget can be developed for a single enterprise to represent different alternative combinations of inputs and output enabling comparison of profits across alternative and competing enterprises.

Reasons for developing enterprise budgets

- i. They provide estimates of enterprise profitability.
- ii. They enable estimation of the profitability of a new production technique relative to existing technique.
- iii. Analyse the revenue and expenses associated with an enterprise. This enables the farmer to make a final decision on which enterprise to expand or eliminate.

Steps in developing enterprise budget

1. Form the goals / objectives
2. Estimate total production (output /yields) and expected output price.
3. Estimate total expenses.
4. Assess fixed assets
5. Calculate the net receipt. This is income earned less all costs and expenses.

Constraints in budgeting

- i. Lack of information; budgets are constructed to reflect future actions and it is difficult to accurately forecast future markets, inputs prices, weather conditions
- ii. Input and product price fluctuations limit budget reliability..
- iii. Production and marketing risks limit budget reliability.
- iv. Budget making is time consuming, actions such as data search and analysis.
- v. Limited skills among many farmers on how to prepare a farm budget.
- vi. There is a chance to over-estimate returns and underestimate costs, reducing reliability.
- vii. Numerous risks and uncertainties that makes budgeting unrealistic in agricultural situations.
- viii. Failure to stick to the set objectives and goals of the farm.
- ix. Dishonest farm managers may use the budget to cover their mistakes by over estimating costs and underestimating returns.



State when it may be necessary to prepare a partial budget on the farm



Activity 5.8

Skills in preparing a budget for farming initiative

You will need

a pen, notebook, reference books, computer, chart, projector

Key question: How do you prepare a budget for farming initiative?

In pairs:

1. What are the goals of the farming enterprise you plan to make a budget for?
2. Which products do you expect to get?
3. How much do you expect to earn from the products?
4. Which resources do you need in the production process? What is the cost of resources? required in the production process?
5. How would you increase the earning from these products?
6. Outline the sources of information used in preparing a budget.

7. What limitations are there in making farm budgets?

8. List the sources of information you have used in the making of the budget.

Draw a budget for your project using a spreadsheet or a book and calculator.

Present your findings using a slide presentation program or a chart, to the rest of the class.



Group Project

Aim: To identify students' attitudes in my school towards saving and investment

Materials required;

- Questionnaire
- Pen
- Notebook
- Folder
- Copies of your resume
- Business cards
- Portfolio/work samples
- References
- Introduction letter

In groups,

1. Decide the information required.
2. Define the target respondents.
3. Choose the method(s) of reaching your target respondents.
4. Decide on question content.
5. Develop the question wording.
6. Put questions into a meaningful order and format.
7. Evaluate your question
8. Collect responses
9. Analyze the responses.

Make a typed report of your findings and present it to the teacher and to the rest of the class

- (a) What are the advantages of using a questionnaire to collect information?
- (b) What challenges did you face while collecting information in the school
- (c) How best can the saving culture be cultivated in the students?
- (d) How do you think the saving culture can help to improve farming and the welfare of farmers

Activity of Integration



In your community, it is reported that most farmers have poor saving culture and just a few have saving accounts with financial institutions in the area. Farmers keep money from the sale of their products in their houses under the beds and keep spending it slowly until it is finished. When the season starts, many of them run to private money lenders to borrow money for preparing land and buying planting seeds.



Task

Design a speech you will use to sensitise farmers in your community on how to manage their farm income well.

Exercise 5.1

Self-Check Questions



As a scholar of financial services and money in agriculture,

1. (a) Give the difference between needs and wants.
(b) Explain the five core pillar of financial literacy.
2. (a) Differentiate between saving and investing.
(b) Describe the different methods of saving and investing money.
(c) Describe the types of financial institutions in Uganda.
(d) Discuss the requirements for opening up a bank account.
3. Budgeting is an important practice in farm management. ;
(a) Outline the importance of making farm budget.
(b) Describe the types of budgets.
(c) Outline the step involved in making a complete farm budget.

Farm Buildings and Farm Structures

Competency: you should be able to establish cost effective and functional farm buildings and farm structures.



KEY WORDS

- Concrete
- Mortar
- Curing
- Batch mixer
- Seasoning wood
- Wood conversion
- Flushing-up
- Grouting
- Lorrying
- Aggregate

LEARNING OUTCOMES

By the end of this chapter, you should be able to:

1. show skills in handling building tools correctly and safely
2. understand and perform the processes of concrete mixing, mortar mixing, brick making, plastering and painting of farm structures and farm building
3. know appropriate types of housing for different animals
4. be able to budget and procure materials for establishing farm buildings and farm structures
5. show skills in constructing farm building and structures
6. know how to use locally available materials for roofing

Introduction

On the farm, you have seen different physical constructions. These are called **farm structures**. Can you mention some examples of farm structures? These constructions help in the farm production process. Examples of farm structures are:

- Animal handling structures such as fences, milking parlour, dips, spray races, and crushes.
- Farm buildings such as stores, machinery/crop/animal sheds, workshops, calf pens, piggery, poultry houses, green houses, animal stalls and farm houses.
- Other structures such as check dams, fish ponds, bee hives, soil conservation structures, farm roads, water storage structures and manure heaps.

Uses of farm structures

Farm structures form an important part of farming enterprise and are used for the following;

- To establish farm boundaries to stop wrangles on ownership, for example, fences
- To protect livestock from extremes of heat and cold, for example, bull and calf pens
- To facilitate separation of animals according to sex and age groups e.g., paddocks.
- Building rooms are needed for a workshop or office from which the farm is managed.
- To provide a residence for the farmer and his employees, for example, building.
- To control movement of animals and consequently to control diseases and pests.
- To restrain animals when spraying, treating and identifying animals e.g., crushes.
- To restrict unwanted visitors coming into the farm, for example, perimeter fences.
- Providing security and protection to farm tools and equipment e.g., machinery sheds.
- To facilitate control of diseases and parasites, for example, double fences and crushes.
- To facilitate proper storage of produce, for example, crop store and milk store.



Different farm structures have different importance. As a learner, make sure that you make your own findings about the importance of different farm structures.



Activity 6.1

Finding out the different types of farm structures and their uses

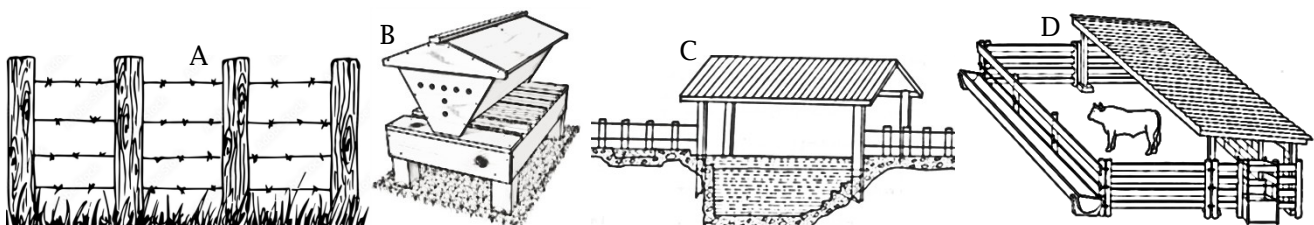
You will need

a pen, notebook, observation list, protective gears

Key question: What are the roles of different types of farm structures?

In groups:

- Write a letter to the nearby farm manager requesting for permission to visit his/her farm.
- Develop an observation list of the farm structure in your community.
- Put on your protective gears
- Visit the farm. Guided by your teacher/farm manager, conduct a tour around the farm.
- Examine all the different farm structures and note the following:
 - The different types of farm structures present and outline the importance of each.
 - What type of materials have been used to construct the different structures and why?
- Brainstorm and suggest how the different farm structures identified help to improve production and management of the farm.
- Below are some of structures typical of those you saw on the farm. Identify each structure



Present your finding to the whole class and compare your work with that of other groups.

Tools used in Building Construction

In senior one, you learnt about different tools found on the farm and how they should be handled for safety. This time you are going to specifically look at tools used in construction work on the farm. Can you name them? It is important for the farmer to have these tools on the farm.

Fig 6.49 Building tools and equipment

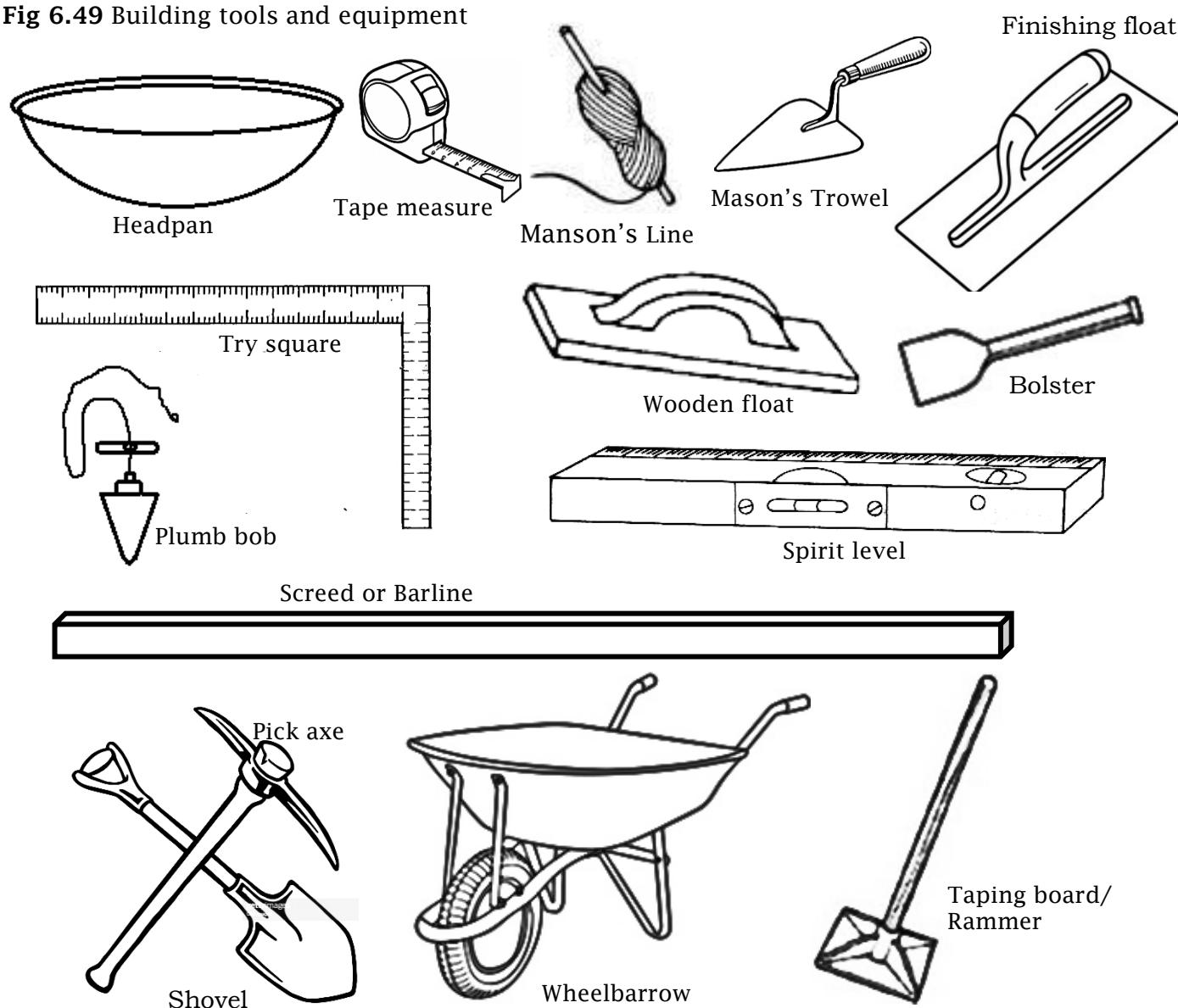


Table 6.1 Building construction tools

Tool	Use(s)	Maintenance
Spirit level	Used to check the vertical and horizontal straightness or level of the wall surface	Store in dry cool place during Coat with a oil film to avoid rusting Store in a tool rack to avoid damage
Screed/ Barline	Used to cut the plaster mortar flat or used to check the flatness of a newly laid piece of wall	Scrape off mortar, wash and dry in air every after work
Plumb bob	Used to check the vertical straightness (perpendicular) of the wall	Clean every after use, Oil the body to prevent rusting. Keep the string away from cutting objects
Try square	Used to check whether or not the angle formed by two walls meeting one another is at right angle (90°)	Oil the entire tool during storage to prevent rusting, Clean every after use

Trowel	Used for picking up mortar from the mortar board and placing it on the wall during plastering or on the block during block laying	Clean every after use to remove dried mortar, Smear with oil during storage to prevent rusting Replace damaged handle
Tape measure	Used to measure distances during construction work	Keep in the tool rack to avoid damage
Float	Used to place mortar in line on bricks Used for holding mortar during brick laying or plastering Used as a finishing tool to smoothen the surface of the walls	Scrape off the mortar on the surface every after work, clean with water and then dry, apply a film of oil on metallic parts to prevent rusting
Head pan	Used to measure the volume of sand and cement and gravel during mixing of mortar or concrete	Clean with water to remove any mortar or cement on the surface Oil to prevent rusting
Shovel	Used to mix mortar or concrete. Used to scoop and load concrete or mortar on the wheelbarrow or head pan	Clean the blade and the handle Ensure that the handle is firm
Wheelbarrow	Used to transport of building materials and soil. Also used to measure the volume of sand and gravel to use in the mixing of mortar or concrete	Scrape off the mortar on the surface every after work, clean with water. Paint to prevent rusting. Repair any damaged part. Grease the wheel pivot
Spot board	Used to hold mortar at a comfortable working height during brick laying or plastering	Scrape off the mortar on the surface every after work Clean with water every after work
Taping board	Used to ram, compact and level concrete or mortar laid on the floor	Clean with water every after use
Pick axe and mattock	Digging out rocks and breaking up hard ground during construction of the building foundation	Clean every after use. Replace broken handle
Batch mixer	Used to mix concrete or mortar	Clean every after work. Replace faulty parts. Ensure appropriate load



Use the internet to identify more tools and equipment used in building construction and share with the teacher and the rest of the class.



Activity 6.2

Understanding the roles of building tools and equipment

You will need

A pen, notebook, display of tools, computer, textbooks and internet

Key question: What are the roles of different tools and equipment used in building construction?

In groups:

1. Put on the appropriate protective gears.
2. Move to the agriculture lab or a place where the different building tools have been displayed.
3. Examine the tools provided, identify them and give the role of each tool in construction work
4. Brainstorm and discuss with the group members about other building tools and equipment that have not been provided, giving their names and roles in building construction work.
5. Find out the structural and functional difference between a spirit level and a plumb bob.
6. Select any three tools of your choice, make a labeled drawing of each tool and state how each should be maintained on the farm.
7. Typeset your findings and share with the rest of the class.

Handling and Using Building Tools and Equipment Correctly and Safely

In senior one, you learnt how to use and take care of tools and equipment used on the farm. You are now going to concentrate on those used in construction work on the farm. Building tools and equipment can pose a risk when they are misused or mishandled during work or storage. You are going to practice safe procedures for handling and working with these tools and equipment.

Why is it important to handle building tools and equipment correctly and safely?

Using the building tool and equipment in the wrong way can be dangerous, decreases efficiency, damages the tool/equipment and the workpiece and increases the likelihood for accidents. Using the correct tool and equipment for work not only ensures your safety but also increases your productivity and efficiency. The right tool and equipment can make the construction process faster and saves you from injuries.

The following are safety procedures for handling and using building tools and equipment;

- i. Select the right tool and equipment for the job. Using the wrong tool and equipment can lead to decreased efficiency, damage to the tool or workpiece, and an increased likelihood of accidents.
- ii. Use personal protective equipment such as helmet, face mask, reflectors, hand gloves, overall and boots.
- iii. Keep tools and equipment in good working condition. Inspect tools for faults and repair regularly. A loose handle or damaged tool is a recipe for disaster.
- iv. Training and education on how to handle and use tools and equipment correctly
- v. Read and follow the user manual.
- vi. Keep the work area clear of dangerous materials to allow easy movement and ample working space.
- vii. Ensure a firm hand grip of the tool/equipment and good feet balance.
- viii. Always select tools and equipment that are easy and safe for you to handle and use.
- ix. Ensure good lighting and visibility especially when working indoors or at night.

Examples of Safety procedures when using construction tools and equipment

Wheelbarrow; ensure appropriate load, keep the back, arms and feet straight, ensure good hand and feet grip.

Ladders; ensure proper equilibrium of the ladders; it should be well placed on a firm support. Ensure good hand and feet grip.

Bolsters and chisels; ensure good hand grip of the tool, put on protective gears like hand gloves, eye goggles, ensure that the cutting edge is sharp, use appropriate hammer to hit the tool.

Hand saws; secure the work piece securely on the work bench using a clamp, hold the work piece securely with one hand, ensure good hand grip, ensure well set and sharp teeth, cut across the wood grains to avoid the blade from sticking in the wood.

Hammers; ensure the handle is tightly fixed in the head, the work piece to hit should be well secured with one hand, ensure good hand grip, use protective gears such as hand glove and eye shield, and a helmet.



Use the internet and reference books research about safety procedures when using other construction tools and equipment.

Proper Use of Construction Tools and Equipment

Practice makes perfect. As a learner, make sure you practice working with different building tools and equipment to acquire the necessary skills. In whatever you are doing, Work under the supervision of your teacher and observe the safety precautions

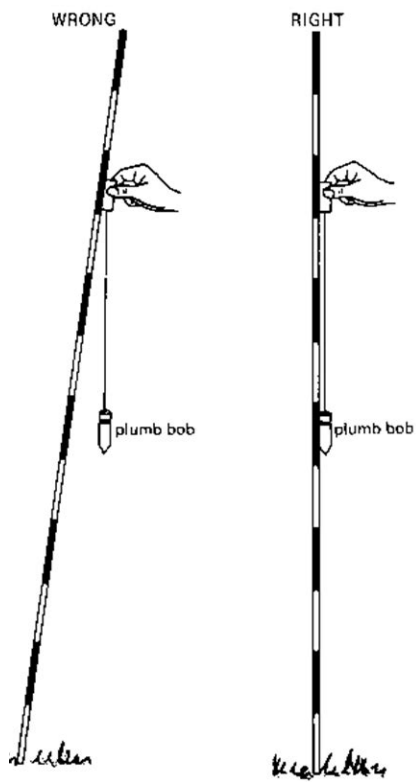


Fig 6.2 Using a plumb bob to set the wall vertically at angle 90° with the floor.

Practice using the plumb bob using a piece of wood and on different building walls present. Hold the wood vertically and slide the plumb bob up and down while adjusting the wood until the wood is perpendicular to the floor.

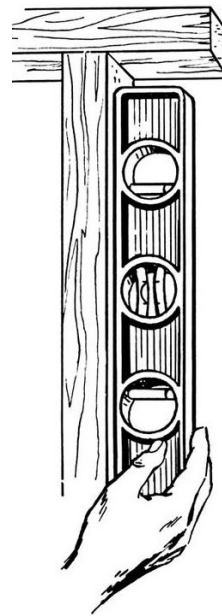


Fig 6.3 Using a spirit level to set the vertical and horizontal angle of the wall and floor.

Practice using the spirit level using a piece of wood and on surfaces of building walls and floors. Place the spirit level vertically or horizontally on the work piece, adjust the work piece until the bubble inside the mercury tube is in the centre.

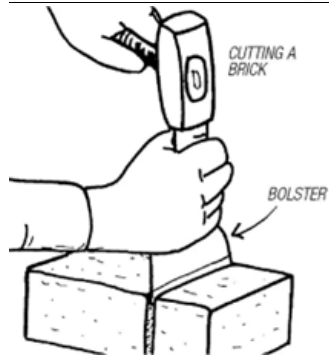


Fig 6.4 Using a Bolster to cut a brick. Practice using a bolster to cut bricks. Mark the brick and place it on a firm surface. Hold the bolster handle firmly with one hand and place the cutting edge at the point where you want to cut the brick. Hold a hammer with another hand and hit firmly the bolster to cut and split the brick

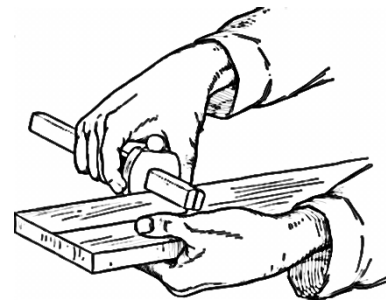


Fig 6.5 Using a spirit marking gauge to scribe a mark on work piece

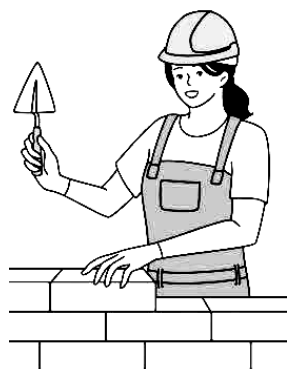
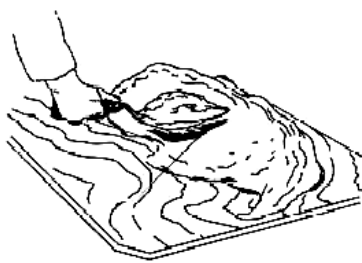
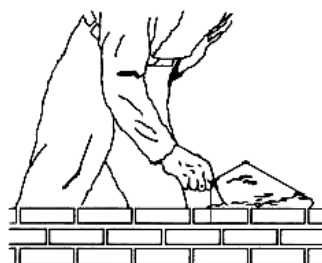


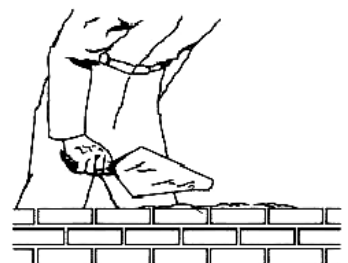
Fig 6.6 Using a trowel to pick and place mortar



Step 1 Pick the mortar



Step 2 Place the mortar over the bricks



Step 3 Spread the mortar over 3 to five brick at a time



Use the internet and reference books research about safety procedures when using other construction tools and equipment.



Activity 6.3

Skills in proper use of building tools and equipment

You will need

A pen, notebook, building tools and equipment, textbooks and internet

Key question: What skills do you need to handle building tools correctly and safely?

In groups:

1. Put on your protective gears; what will they include?
2. Choose the tools and equipment to use in construction of a building foundation. Identify and state the function of each.
3. Demonstrate to the members of the group how to use a plumb bob, try square, marking gauge and spirit level correctly during construction work.
4. In your own opinion, why is it advised to handle building tools and equipment correctly and safely?
5. What safety precautions do you need to observe when using building tools and equipment?
6. How do you handle tools and equipment after using them?

Share your findings the rest of the class and record conclusions.

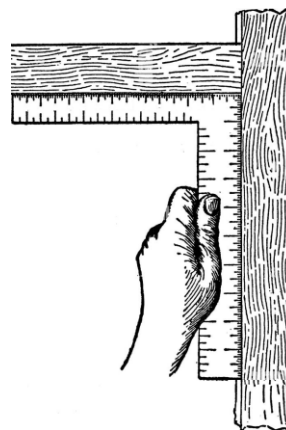


Fig 6.7 Using a try square to set right angles

Activity of Integration 	In your community, many builders don't use the necessary building tools and equipment when constructing farm buildings. It is difficult for them to appreciate the diversity and availability of numerous building tools and equipment and the importance of using them in construction work.	
Task	Prepare a poster sensitizing builders in your community on how they can improve the quality of their work through use of appropriate building tools and equipment.	

Exercise 6.1	Self-Check Questions
	As a learner who has completed a course in building tools and equipment, 1. (a) Give the example of tools and equipment which are needed in the construction of building supper structure, and state the role of each. (c) Explain how you can ensure your safety when using the building tools and equipment. 2. (a) Describe how you can use a plumb bob to set the building wall. (b) Discuss the possible causes of builders' failure to use building tools and equipment during construction work.

Making Bricks for Construction Works on the Farm

A brick is a hard rectangular shaped walling unit made of baked clay or earth. Making bricks on the farm will help to cut down the cost of putting up farm buildings and structures. A brick mould is used to shape and determine the size of the brick to make it easy to use during construction work.

Types of bricks

1. Solid: Have no holes, cavities or depressions
2. Perforate; have very small and few holes.
3. Hollow: Have very large and many holes
4. Cellular: Have holes closed at one end.

Procedure in hand making of mud bricks

- i. Put on protective gears
- ii. Clear the surface vegetation and organic matter from the site.
- iii. Dig to loosen up the soil. Remove foreign materials such as stones, plastics, metals and roots.
- iv. Spread out the soil and some cement if available in the ratio of 1:25 or add chopped cereal (millet, rice, wheat) straws. This aims at providing extra strength to the brick.

- v. Add water and mix the contents thoroughly, breaking hard soil lumps until a uniform mass of soil of desirable consistence is obtained.
- vi. Wet the working table and place the brick mould on top.
- vii. Fill the wet mixture in the mould while ramming the paste to remove any air pockets.
- viii. Wet and smoothen the top of the wet brick and then lift the mould with the wet brick from the working table. Slide the mould as you lift to get a flat and smooth bottom surface.
- ix. Shake out the wet brick from the mould and place it on a smooth surface
- x. Place the wet bricks under shade or cover with dry grass to protect them from rain damage and direct sunlight that could cause rapid drying and cracking of bricks.
- xi. Leave the brick to dry slowly under a shade. The time it takes for the bricks dry depend on the weather conditions and can be shorter in dry season or longer in wet season.
- xii. During drying period, turn the bricks regularly to ensure uniform drying.
- xiii. After the bricks have dried well, they can be used in construction work, or the dry bricks can be cured by firing in the kiln.
- xiv. Allow the fire cured bricks to cool and then use them for construction work

Summary of stages of preparation of earth bricks

Clearing vegetation and organic residues

Digging the soil

Mixing/blending the soil

Moulding

Drying

Burning

Brick



Fig.6.8 Using brick press

Defects in unbaked bricks

1. Cracked: rapid drying under direct sunshine, poor mixing of the soil, high clay content
2. Misshapen: trampling by animals and humans, drying on unlevelled surface.
3. Worn out edges rough surface: rainfall water erosion as a result of drying out in open

Defects in baked bricks

1. Black core or hearting (black center): caused due to rapid burning of the bricks causing the surface to vitrify and the interior to remain black.
2. Bloating or swelling: caused by excess carbonaceous matter in soil and bad burning or excessive heating.
3. Burnt bricks stick together: caused by excessive heating. Occurs especially in bricks adjacent to the flues.
4. Chuffs: these are badly cracked and misshaped bricks produced due to excessive heating. Occurs especially in bricks adjacent to the flues.
5. Crazeing: a brick having small/fine cracks. Results as a result of differential swelling and shrinking in soil components.
6. Lamination: thin laminae (bursts) or swellings on the faces of the brick caused due to entrapped air in clay.
7. Nodules in the faces of a brick caused by uncrushed stones and clods in soil.

Increasing the strength and performance of mud bricks (unbaked brick)

- i. Application of cement to increase the strength and to make earth water proof.
- ii. Application of lime to improve the strength and reducing cracking tendency.

- iii. By mixing with clay soil to increase cohesion and strength.
- iv. Mixing natural fibres such as straws, sisal and saw dust to make earth more stable.
- v. Mixing earth with resins, gypsum and cow dung to improve strength.
- vi. Mixing earth with sand to reduce shrinkage and to prevent cracking during drying.



Activity 6.4

Skills in making of bricks for use in construction work

You will need

Protective gears, Non-rocky soil, brick mould, working table, water, hoe, drying shade, covering materials, wheelbarrow

Key question: How are mud bricks made?

In groups, practice making bricks using locally available materials: sun dried earth bricks, soil cement bricks, fire-baked bricks

- How are bricks utilised on the farm?
- How can the strength of earth bricks be increased?
- Discuss the defects that can occur in brick during the making process.

Individuals keep the learning log and share their findings with the rest of the class.



Mixing Concrete and Mortar for Construction Works

Concrete is made by mixing cement, sand, aggregates (small stones) and water in the right proportions. Sand fills up the spaces between stones and cement fills up spaces between sand and binds the ingredients together. The amount of each ingredient affects the strength and quality of concrete.

Cement: this is the substance which binds together the aggregates (usually sand and gravel) to form concrete. The strength of the cement paste depends on the relative proportion of water to cement. Too much water produces weaker cement.

Cement is sold in bags of 50kgs (approximately 37 litres). Cement must be stored in a dry place, protected from moisture and for periods not exceeding two months. Cement should be in powder form when used. Caked cement with lumps is of low quality but can still be used if the lumps can be crushed between the fingers.

Aggregates: coarse aggregates or ballast or pebbles are either gravel or crushed stones and fine aggregates are sand. Coarse aggregates and fine aggregate are mixed with cement and water to make concrete. Ideal aggregates for concrete mixing should be;

- i. Hard; made up of silica or quartz or granite
- ii. Should be clean and free from any contamination such as silt, vegetable matter and oils.
- iii. Should be of desirable sizes and textures. They should not be of exaggerated sizes.
- iv. They should be dry at the time of mixing (batching)

Water: water is needed to set up the chemical reaction in cement and to give the mix workability and will evaporate from the mix during curing. Water should be clean and free from heavy metals. Too much water in the mix will make concrete porous on curing and reduce its durability and strength. The water to use should not be salty and should not contain a lot of heavy metals.

Batching

Batching is the process of proportioning the quantity of cement, sand, coarse aggregates and water to use in preparing the concrete mix. The components are measured on the basis of either weight or volume. Proper batching improves the quality and workability of concrete.

The following are some of the concrete mixes and their application in building work.

1. 1:4:6, 1:3:6 and 1:4:5 are weak concrete mixes normally used for plain concrete work like foundation, stanchion bases, floors with very light traffic.

- 1:3:5, 1:3:4, 1:2:4 and 1:3:3 are fairly strong concrete used for normal reinforced concrete work such as reinforced floor slabs (floors for dairy/beef cattle, pigs, poultry, machinery stores, pavement, roads and water tanks), lintels and foundations.
- 1:2:3, 1:1.5:3 and 1:1:2 are very strong concrete work in milking parlor, silage silos, feed and water troughs, floors for heavy machinery, bridges, dams, suspended floors, beams and lintels, fencing posts, precast concrete components.

Hand Mixing of Concrete

Components are measured, placed on a clean surface and a hand hoe and a shovel are used to turn and mix the components until a homogeneous mix is obtained.

Procedure of hand mixing of concrete

- Clean the surface where the mixing is to take place. Remove all the organic material and loose soils to reduce contamination of concrete.
- Measure the suitable quantity (weight or volume) of clean sand and spread it on the surface.
- Measure the suitable quantity of cement and spread it on top of sand.
- Mix sand and cement very well by turning over and over with a shovel.
- Spread a known volume of aggregates on top of the sand-cement mixture.
- Add some water and then mix up the contents with a shovel by turning over and over until the mixture becomes homogenous and workable.
- Use the concrete within 30 minutes of mixing.



Fig. 6.8 Hand mixing of concrete.

Machine Mixing of Concrete

Here mixing of concrete is carried out using a mixer (engine driven or hand operated). In this method, all the ingredients of predetermined volumes/masses are added in the drum at once and then mixing is carried out by the rotation of the drum for at least 2 minutes.

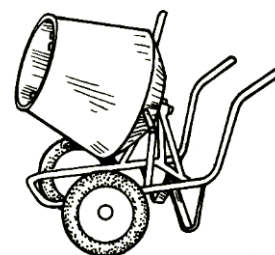


Fig. 6.9 Concrete mixer

Placing Concrete

This is the laying of concrete to its final point of deposition such as foundation, floor, columns, lintels and beams. Concrete should be placed on its final point of deposition within 30 minutes. After placement in its final position concrete should be leveled and compacted to remove entrapped air and spaces in the concrete.

Curing Concrete

This is the chemical reaction that leads to setting and hardening of concrete. The chemical reaction is between water and cement. If water dries up quickly, the chemical reaction will stop and concrete will lose strength and cracking will occur. It is therefore important to keep concrete damp for at least 7 days. Watering concrete should begin 10 to 12 hours after placement. Cover concrete with sacks or grass or sand or polythene prevents rapid drying.

The strength of concrete is affected by a number of factors;

- The proportion and type of cement used in the concrete.
- The quality and proportion of sand used in the concrete
- The proportion of water used in concrete mixture; too wet concrete produces low quality concrete with low strength.
- The method of mixing used and the amount of compaction

- Care during curing such as watering and covering of concrete to prevent rapid drying.
- The type and amount of admixture used to add protection against water penetration, and fungal attack and to prevent concrete from chemical attack.
- The amount of organic matter contamination of the concrete.

Types of concrete

Plain concrete: this is a mixture of cement, fine aggregates, coarse aggregate and water without any structural reinforcement in it. It is often used in places where there is no tensional forces acting. For example ground floor and some foundations.

Reinforced concrete: this is concrete with steel bars and iron mesh providing it with tensile strength. Concrete is strong in compression but weak in tension. Thus, where tension occurs it is necessary to introduce steel bars to provide the tensile strength which the concrete lacks. Reinforced concrete is used in columns beams and lintels over doors and windows.

Advantages of using concrete as a building material

- It is a poor conductor of heat thus it has high insulating value to keep the inside of the building independent of the outside temperature.
- Concrete is usually made up of nonflammable materials, making it resistant to fire.
- It is made from durable materials which are resistant to weathering and most chemicals. This makes concrete more durable material to use in building of permanent structures.
- Concrete has good resistance to fungal and pest attack which are a big problem in wood.
- Concrete, if well reinforced is stronger than wood and plastics.
- Concrete is easy to mould and shape making it easy to use during construction work.
- Concrete is relatively cheaper in terms of initial installation than metals, and in terms of maintenance it requires low maintenance work and costs than wood and plastics.
- Concrete has a good sound insulating property

Disadvantages of using concrete as a building material

- Concrete especially if not well reinforced is low in tensile strength.
- Concrete is bulky and a heavy material requiring more labour force to work with.
- Concrete is costly to make on the farm or buy as pre-cast concrete from the suppliers
- Concrete requires time to cure and to gain its strength thus cannot be used in the quick construction of structure which is required immediately for emergencies on the farm.
- Use of concrete requires skilled labour that may not be readily available on the farm.
- Concrete can easily absorb moisture either from the ground or from the rain. The moisture can find its way into the building making the building unnecessarily very cold and also, the moisture can affect produce in stores.
- Concrete because of its rigid nature can easily be affected by earth tremors such as earthquakes which lead to cracking of the concrete.



Activity 6.5

Understanding the process of mixing concrete

You will need

A pen, notebook, cement, sand, gravel, water, shovel and a hoe

Key question: What is the correct procedure of mixing concrete?

In groups:

1. Put on appropriate protective gear
2. Clean the surface where the mixing is to take place. Remove all the organic material and loose soils to reduce contamination of concrete.
3. Measure 2 buckets of sand and spread it lightly on the ground.
4. Measure 1 bucket of cement and spread it on top of sand.
5. Measure 3 buckets of aggregate and spread it on top of the sand-cement mixture.

6. Add water gradually and mix up the ingredients by turning over and over with a shovel until a paste of acceptable consistence is obtained.
 - State the possible places on the farm where this concrete can be used.
 - Suggest how this concrete strength can be improved.
 - Describe how this concrete will be cured.
7. Place the concrete in a wooden box 25cm x 13cm x 10cm to shape it into a block. Make sure the concrete is properly rammed and compacted to remove any spaces.
8. Leave the block to dry for seven days, pouring water on it every morning to cure it
9. After one week, remove the block from the mould. State;
 - How the block can be utilised on the farm
 - The advantages and disadvantages of using concrete as a construction material.

Individuals keep the learning log and share their findings with the rest of the class.

Mortar

This is a mixture of cement, sand with or without lime and water. Cement and lime act as binder or glue to hold sand particles together or to bind sand with the surface (where it is applied). Mortar is used in brick or stone laying to hold the bricks or stones together in place. Mortar is also used for plastering of the surfaces (walls, floors, concrete roofs and ceilings) and for roughcast of walls.

Qualities of good mortar

- i. It should have adequate strength
- ii. It should have good workability
- iii. It should have a reasonable amount of plasticity, improved by adding lime to the mix.
- iv. It must provide a good and strong bond with brick or blocks or stones
- v. It should be able to be mixed at an economic cost
- vi. It should be durable over a long period

Mixing mortar

- i. Clear the surface to remove any contaminants.
- ii. Measure a known quantity of clean sand and spread it slightly on the clean surface.
Proportioning of cement or sand can be done by volume or by weight; 1:3 (cement: sand) suitable for brick work in exposed conditions and for the brick work in foundations.
- iii. Measure a known volume of cement and add on top of sand.
- iv. Mix the content thoroughly by turning over with a spade for several times until the mixture becomes homogenous.
- v. Add water a little at a time as mixing proceeds until mortar of suitable consistence and workability is obtained.
- vi. Use the mortar within two hours of mixing or else should be discarded.



Activity 6.6

Understanding the process of mixing mortar

You will need

a pen, notebook, cement, clean sand, water, shovel and a hoe

Key question: What is the correct procedure of mixing mortar?

In groups:

1. Put on appropriate protective gear
2. Clean the surface where the mixing is to take place. Remove all the organic material and loose soils to reduce contamination of concrete.
3. Measure 3 buckets of sand and spread it lightly on the ground.
4. Measure 1 buckets of cement and spread it on top of sand.

5. Add water gradually and mix up the ingredients by turning over and over with a shovel until a paste of acceptable consistence is obtained.
 - How can the paste be utilised on the farm?
 - Why is water an important element in the mixing of mortar?
 - Describe how a building surface should be prepared for plastering.
6. Practice using your mortar in brick laying and plastering of the building wall.

Individuals keep the learning log and share their findings with the rest of the class.

Exercise 6.2	Self-Check Questions
	Given the skills so far acquired in making bricks and mixing mortar and concrete for use in building, 1. (a) Explain why it is advisable to make your own bricks on the farm. (b) Discuss the different mechanisms that a farmer can employ to increase the strength and durability of earth bricks. (c) Procedure in hand making of bricks from mud. (d) Describe four defects that make brick unfit for use in the construction work on the farm. 2. (a) Describe the procedure of hand mixing of concrete (b) Outline four factors that affect the strength of concrete.



Locally Available Construction Materials

When you look at the farm structures and buildings, you will notice that they were built with materials of different types. Can you identify some of those materials?

The common materials used in construction work on the farm include the following:

- Wood/Timber
- Bricks
- Blocks
- Stones
- Metals
- Glass
- Plastics
- Cement
- Gravel
- Sand
- Concrete
- Mortar

Proper selection of construction materials to use is important because:

- i. It influences the original cost of construction
- ii. It influences the ease of maintenance and the cost of maintenance.
- iii. It influences the ease of cleaning
- iv. It influences the durability of the structure where the material is used
- v. It influences the final appearance of the structure
- vi. It influences the type of labour to use in construction
- vii. It influences the length of the time the construction work is to take.

Factors to consider in choosing the materials for construction work

1. *Type and function of the building* and the specific characteristics required of the materials i.e. strength, water resistance, wear resistance, attractive appearance etc.
2. *Cost of the material*: The material chosen should be cost effective i.e. the material should be relatively cheaper to buy
3. *Availability of the material*: The material to use should be one that can easily be obtained within the surrounding areas. This reduces the cost of transportation
4. *Ease of maintenance*: The material chosen should be easy to maintain, requiring less labour and less cost of maintenance.
5. *Quality and durability*: the material should be of high quality and durable so as last for a long time without deteriorating and to reduce the cost of replacements and repair.
6. *Cultural acceptability or personal preference*: the materials selected should be socially accepted in the area and should also conform to the farmers own taste and preference.

7. *Fire resistance*: a good material to choose for construction work should be less flammable to reduce the risks of fire destruction of the building. Plastics and untreated wood are highly flammable while metals and glasses have good resistance to fire.
8. *Skills required in using the material*: the materials should not require highly specialized skills to use. Labour to use such materials should be readily available in the area.
9. *Thermal properties*: the material to select should be able to withstand high or low temperature, and also the material should be able to keep the inside of the building cool despite high atmospheric temperature and vice versa.
10. *Sound or noise insulation property*: the material to select should be able to prevent the free transmission of sound from inside to the outside of the building. This property is important especially where the farm is surrounded by residential buildings.



Activity 6.2

Identifying suitable material for construction of farm structures

You will need

a pen, notebook, maker, manila, reference textbooks and internet

Key question: What factors should be considered when selecting suitable materials to use for constructing farm structures and buildings?

In groups:

1. Brainstorm and state the common types of materials used in construction work on the farm
2. Discuss the reasons for selection of construction materials.
3. Outline the factors to consider when selecting construction materials for use on the farm.

Present your finding to the whole class and compare your work with that of other groups.

Wood as a Construction Material

From the farm tours we've so far had, we have seen that most structures are constructed from wood. Wood is used in form of poles, timber or timber offcuts.

Wood has several advantages over other building materials in the following ways;

- i. Wood is relatively light in weight requiring minimum labour to carry and install. Also the light weight of wood reduces the final weight of the structure where it is used.
- ii. Wood is relatively cheap to buy and to install thus reducing the construction cost
- iii. Wood can be easily obtained on the farm this saves time to look for and deliver wood to the site of construction and also saves costs of transporting wood from very far places.
- iv. Wood can easily be worked using simple tools which can easily be obtained on the farm.
- v. Wood requires less skill to work with: shaping of wood and installation of wood during construction work can be done by semi-skilled or ordinary labourers on the farm.
- vi. Wood has a good thermal insulation property and is capable of excluding heat from the building to maintain ambient temperatures inside the building.
- vii. Wood if well sawn and free from defects, it is relatively strong to withstand both compression and tensional forces than plastic and glass.
- viii. Wood if well treated with suitable preservative and well protected from bad weather it is durable and can be used in construction of permanent structures.
- ix. Wood has a pleasant appearance especially if well finished.

Disadvantages of wood as a construction material

- i. Wood can easily catch fire. This makes wood a fire risk in construction of buildings
- ii. Wood especially if not properly sawn is very low in strength and cannot be used to support heavy loads on the building.
- iii. Wood if not treated with preservatives can easily be damaged by termites and beetles
- iv. It can easily be damaged by moisture and is highly prone to decay due to fungal attack.
- v. Because of low durability of wood, frequent replacement and repairs are required which in the long run increase the cost of maintenance.

Factors to consider when selecting wood for construction work on the farm

Strength: The resistance to breaking under heavy loads. This is important in wood to use in beams and column. Wood strength is related to the species, defects and the size of wood.

Hardness: the hardness of the wood is its ability to resist wear and denting. Hardwood is commonly used in flooring and in making of tool handles.

Resistance to decay: This is important in humid climate and in a situation where wood is to be used in the open or where wood is to touch the ground.

Paint holding ability: the wood should allow painting and should provide a strong bond with different types of paints and varnishes.

Nail holding ability: good wood should allow penetration by the nail without splitting and should be able to firmly hold the nail in position.

The **final look** of the wood after finishing: different wood have different colors and textures when finished. The look should be impressive and should conform to the farmers taste.

Workability: a good wood should be easy to saw, shape and nail and also should have high polishing ability.

Ability to bend/deflect without breaking: woods that are too brittle resist deflection or bending when loaded but rather such wood break suddenly when overloaded.

Resistance to warping: wood should be resistant to twisting, bending and bowing distortion.

Resistance to pest and fungal attack: the resistance to pest and fungal attack increases durability of wood and make wood more suitable to use on permanent structures. Wood can be treated with preservative to enhance this property.

Freedom from defects such as shakes, knots, sloping grains and upsets such defects reduces the strength of the wood and make the finishing less attractive.

Common defects in wood

Shakes such as star shakes, heart shakes, cup shakes and radial shakes. These reduce the strength of the wood and make the work piece unattractive.

Knots; these are the end sections of branches where they grow out of the trunk. Dead knot usually fall off leaving holes in the wood that should be refilled.

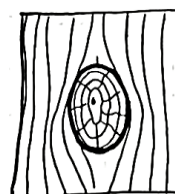
Decay: results from high moisture level in the wood. Decay reduces the strength of the wood and spoils its appearance.

Insect damage caused by borers or termites. These affect the strength of the wood.

Upsets: this is the fracture of the timber fibres across the grains. Upsets in wood can be as a result of lightning striking of the tree or as a result of poor felling of trees.

Twisting: this often occurs in timber which is not sawn parallel to the pith of the log.

Splitting: wood grains or fibres separate mainly due to rapid drying or shrinkage of wood cells especially in immature wood.



Knot



Splitting



Shake

Fig. 6.16 Wood defects



Activity 6.2

Understanding the popularity of wood as a construction material

You will need

a pen, notebook, maker, manila, reference textbooks and internet

Key question: what are reasons for the popularity of wood as a construction material?

In your groups, research and discuss,

1. The reason for the popularity of wood as a construction material on most farms.
2. The common defects that render wood unsuitable for use in the construction work.
3. The factors that should be considered when selecting wood for use in construction work.

Present your findings to the whole class and compare your work with that of other groups.

Seasoning of Wood

The term **seasoning** refers to the controlled drying of wood by natural or artificial means. Wood is usually seasoned to reduce the moisture content to about 15-20%.

Reasons for seasoning timber

- To reduce fungal attack and rotting of wood.
- To ensure that shrinkage takes place before the timber is used
- Dry timber is stronger and stiffer than wet timber
- Dry timber will readily accept glue, paint or polish
- It reduces splitting and distortion and makes timber more stable and reliable to use.
- Minimizes pest attack by beetles and termites and increases durability.
- Wet timber is unnecessarily heavy and demands more labour to use.

Methods of Seasoning Timber

Timber may be seasoned in one of the three ways and these include: the natural means (air seasoning), the artificial means (kiln seasoning) and the smoke seasoning. The rate at which wood dries depends on the size of wood, type of wood, the air temperature, humidity, speed of wind, air circulation around the wood and the age of the tree from which timber was got.

Air Seasoning (Natural Means)

In this method, timber is stacked in open sided covered shades which protect timber from rain while still allowing a free circulation of air around the timber.

Procedure of air seasoning timber

1. Clearing and leveling the ground.
2. Use brick or timber joists to keep the bottom of the stack well above the ground to avoid moisture from the ground from rising into the timber and to ensure good air circulation underneath.
3. Lay timber on the joists horizontally, largest at the bottom, smallest at the top one piece above the other
4. Board column are spaced at about 22mm apart
5. Place rods in between each board spaced evenly to effectively support the boards and to allow free air circulation around the boards.
6. Paint the ends of the boards or cover with strips of timber to prevent them from rapid drying and splitting.
7. Provide a shade to keep away rainwater and sunlight. The shade should be well raised and should allow inflow of air from all directions.

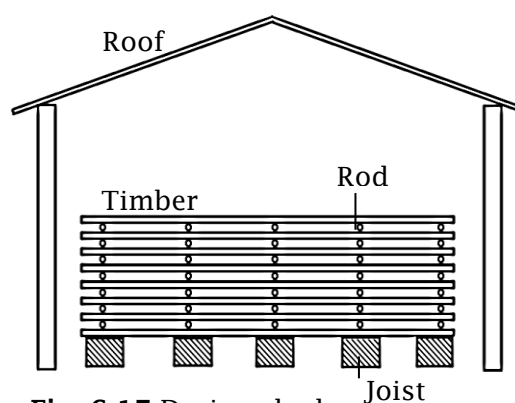


Fig. 6.17 Drying shed

Precautions when air seasoning timber

- i. Provide a shade to keep off rain water and direct sunshine
- ii. Timber should be separated from each other by placing rods in between the timber.
- iii. The piling sticks should be of the same type of timber as that being seasoned to avoid staining of the timber.
- iv. Timber must be supported off the ground to avoid absorption of moisture from the ground and to allow free air circulation underneath the pile.
- v. The ground should be leveled to avoid bending and distortion of the timber.
- vi. The rods or stickers should be spaced closely to each other to avoid warping of timber
- vii. The ends of the boards should be covered with strip of timber to prevent splitting
- viii. Heavy boards should be placed at the bottom and lighter boards at the top of the pile
- ix. The Board columns should be spaced at about 25mm apart to allow air to reach at the board in the middle of the pile.

Kiln Seasoning

This is the most effective way to reduce moisture content of timber to the required level, without any damage to timber. It is used to season timber for internal use. In this method, hot air is blown into the chamber containing timber, and the moisture from the timber is then forced out by the fan.

Smoke Seasoning

In this method, timber is placed over the fire and moisture is gradually evaporated from the timber. The method is not reliable since it can lead to rapid drying and splitting of the timber. Also the smoke changes the natural colour of timber making it less attractive. The smoke that enters the wood greatly discourages insect and fungal attack.

Storage of seasoned timber (care for seasoned timber)

Since drying is a reversible process, great care must be taken in the storage of dried timber. The following precautions should be taken when storing timber.

- i. Keep timber off the ground to avoid re-absorption of moisture from the ground.
- ii. Timber in the pile should be stacked to separate them from each other and to permit free air circulation so as to allow further drying of timber.
- iii. Store the timber under a well roofed shade or cover with tarpaulins to keep away rain water and direct sunshine as these can damage and cause distortion.
- iv. Apply preservatives to prevent pest and fungal attack.



Activity 6.1

Understanding the reasons for seasoning timber and the methods used to season timber.

You will need

a pen, notebook, reference textbooks and internet

Key question: What are the methods and reasons for seasoning timber?

In groups, Brainstorm and discuss:

1. The reasons for seasoning timber.
2. Common methods used to season timber.
3. Precautions when air seasoning and storing timbers.
4. Use Microsoft word program of the computer to type and print out your presentation.

Present your findings to the whole class and compare your work with that of other groups.

Wood Preservation

You have probably seen wooden structures being attack by termites or boring pests, or wood decaying due to high moisture or attack from fungi. To avoid this, wood to be used in construction of permanent structures should be treated to resist wood pests, fungi and fire. There are various preservatives used for this purpose. Ideal preservative should possess the following qualities:

- i. It should be highly poisonous to fungi and insects.
- ii. It should be able to stay in the wood permanently i.e. it should be resistant to leaching.
- iii. It should be environmentally friendly and bio-degradable.
- iv. It should penetrate the wood readily and sufficiently.
- v. It should be relatively cheap to use.
- vi. It should be readily available.
- vii. It should not render wood more flammable by fire.
- viii. It should allow the wood surface to hold/bond with paint or glue.
- ix. It should be easy to apply on wood using a variety of techniques.
- x. It should not produce an offensive odour preferably it should be odourless.
- xi. It should be non-corrosive to metals such as nails used on wood.

Examples of wood preservatives

Creosote: is a general purpose preservative used mainly for exterior work. Creosote increases fire risk, evaporates easily, doesn't allow painting and has unpleasant smell.

Coal tar: works the same way as creosol but coal tar is less effective and less poisonous and does not penetrate deep into the wood because of its high viscosity.

Unleachable metallic salts: these are mostly based on copper salts. The copper and arsenical salts are the toxic and non-leachable. The preservative is that it last longer, it is very toxic to fungi and insects, it is odourless and allows wood to be painted.

Water soluble preservatives: these are used only on wood for internal use only, or else, the preservatives are liable to be washed away by rain e.g. Dieldrin. The advantage is that it is odorless, colorless, highly toxic, penetrate wood easily and allows painting.

Old engine oil: old engine oil contains many residual products from combustion and some of them act as preservatives, but it is not highly toxic thus less effective than other preservatives.

Ammonium phosphate and **Ammonium sulphate** are used to protect wood and other building materials against fire.

Treatment of Wood

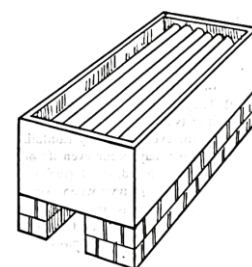
Wood treatment is best done before wood is used on structures to allow proper application of the preservatives to all parts of the wood.

Methods of Wood Treatment

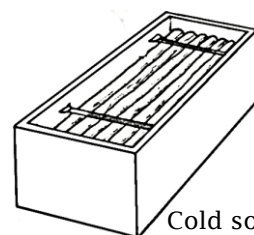
1. **Open tank treatment:** this method is commonly known as steeping or soaking and is used for relatively small quantities of timber.

a. **Hot and cold soaking:** seasoned peeled posts or sawn timber of desired sizes and lengths is placed in a drum containing a suitable preservative. The drum is then fired until the preservative is about to boil for one to two hours and then allowed to cool. During heating, air in the wood cells and conducting tubes expand and some of the air is expelled from the cells and conducting tubes. As the preservative cools down, the air in the wood cells and conducting tubes contract creating a vacuum inside the cells and the tubes. This vacuum draws the preservative into the wood.

b. **Cold soaking:** in this method seasoned peeled posts or timber of desired sizes and lengths is soaked in cold preservative for several days. The wood cells and conducting vessels slowly absorb the preservative.



Hot and cold treatment



Cold soaking

2. **Pressure impregnation:** in this method seasoned peeled posts or sawn timber of desired sizes and lengths is placed in a horizontal steel cylinder where the chemical is forced into the wood cells under very high pressure. This is one of the best ways to apply preservatives into the wood and can only be done on large commercial farms. This method is used to apply effectively creosote and unleachable metallic salts.

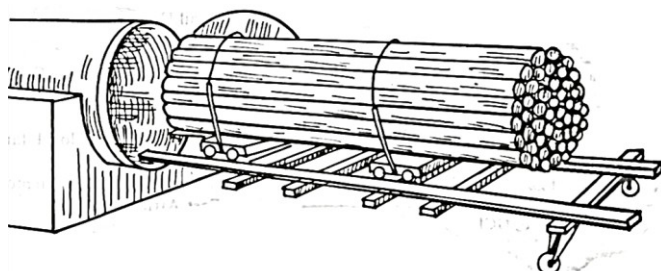


Fig. 6.19 Pressure impregnation of wood



Fig. 6.20 end diffusion treatment.

3. **End diffusion or sap displacement method:** in this method, freshly cut posts are immediately packed in a vessel containing a suitable preservative with their bottom ends down in the preservative for several days. As the sap in the vessels dries up, the chemical is drawn up by capillarity to replace the lost sap. This method makes use of the natural transpiration stream to replace the water in the conducting tubes with preservatives. It is particularly effective on freshly cut posts from living trees.
4. **Superficial treatment:** in this method the preservative is applied to the wood by spraying or brushing the wood with the preservative. Only the surface of the wood is treated and the preservative only penetrates the wood slightly. The wood should first be seasoned and the surface should be dry and clean before application.



Which one of the wood treatment methods do you recommend for wood to use on permanent farm structures and why?



Activity 6.1

Understanding the reasons for preserving wood and the different methods of treating wood with preservatives.

You will need

a pen, notebook, reference textbooks, observation list

Key question: What are the reasons for preserving wood and what are the methods of applying preservative on wood?

In groups:

1. Visit a wood treatment plant near your school and observe the methods and processes in the treatment of wood.
2. Brainstorm and research about the ideal qualities of wood preservative.

Present your findings to the rest of the class and keep learning logs.



Construction and Maintenance of Farm Buildings

Farm buildings are constructed from building plans. A **building plan** is a drawing showing the location, arrangement and fitting out of building and estimates of quantities of materials.

A **Structural design** is the way in which the different structural elements and the finished structure will appear. Both the building plan and design are essential help to prescribe to the builder what is to be done from the starting of the building to the finishing of the building.

Structural Elements

Structural elements are units which make up the building when assembled. They include;

The Foundation: this is the element of the building in direct contact with the ground which supports all other elements and loads on the building. It transmits the weight of the building to the ground.

Steps in construction of the foundation

1. Select the site: the site should have a firm ground and well drained
2. Lay out the foundation plan on the ground using pegs and strings. The foundation plan is obtained from the building plan and setting out drawings.
3. Grade the site by removing the top loose soil, tree stumps and rock outcrops
4. Dig the trench where the foundation is to be constructed. It should be deep and wide enough depending on the soil type, drainage and the weight of the building.
5. fill the trench with hard core layer of about 30cm. put big stones first and then put in small stones to fill the gap between the big stones.

6. Pour a concrete layer of about 10 inches in the trench, the concrete mixing ratios of 1:4:6 or 1:3:6 or 1:4:5 are ideal. Leave the concrete to cure.
7. Construct a plinth wall. It should be raised 15-20cm above the ground. The plinth wall can be made of concrete blocks or baked bricks bound together with mortar, and it is usually wider than the building wall that is constructed above it.
8. Lay a damp proof course on the plinth wall to stop upward rise of water into the walls.
9. Fill the gap left between the plinth and the walls of the trench with compacted sub soil mixed with termite poison. This helps to stabilize the foundation and to keep termites away from the building foundation.

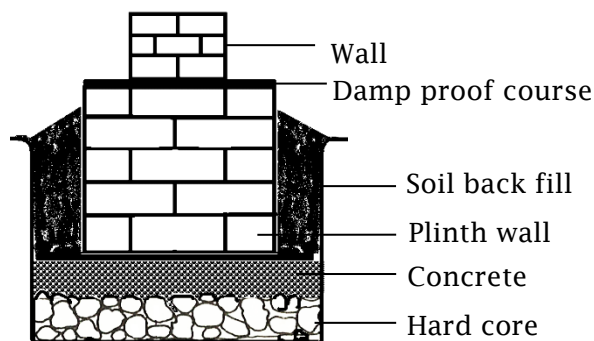


Fig. 6.16 parts of the building foundation

The strength of the foundation will depend on;

- The materials used in building foundation.
- The mixtures used in concrete formulations.
- The nature of the soil on which the foundation is constructed.
- Site drainage: well drained site produces strong and stable foundation.
- The depth and width: deep and wider foundation is more stable and strong.

Walls: also called the super structure; a wall is a horizontal structure which support load from the roof and resist side pressure from wind and stored materials in the building.

Lintel: this is a horizontal bridge over a window or door opening usually formed of reinforced concrete, block with concrete or brick and steel and sometimes timber of suitable dimensions. A lintel is usually short in length as compared to the beam.

Column: this is a vertical component and transfers weight to the foundation. Columns are made of reinforced concrete or masonry or bricks. Other members in this category include: stanchions which are columns made of steel. Piers column made of concrete and masonry or bricks. Pillars columns made of timber or wood.

The beam: this is a horizontal unit of reinforced concrete running across the whole length of the wall of building. It is used to support and stabilize the walls and also used along the windows and door to support the weight of the materials above.

The Floor: building floor may be made up of compacted soil present on the site, or hardwood, or concrete. Floors may be built at ground level or supported above the ground.

Qualities of a good building floor

- i. It should be strong and able to support the loads without cracking or sinking
- ii. It should be termite and rodent proof and should not allow growth of vegetation
- iii. It should be easy to clean and should allow free drainage
- iv. It should be fire resistant
- v. It should allow easy movement of both machines and animals
- vi. It should be durable and less prone to erosion by either abrasive forces or water.
- vii. It should allow a noise free movement of objects on the surface of the floor.
- viii. Should provide acceptable surface finish which meets the user's need.
- ix. It should prevent capillary water from rising into the building.

Roofs: a roof is the top most part of the building. The roof frame (truss) is used to support the thatching material. The roof structure must be designed to endure the load imposed by the thatching material. Examples of thatching material used on farm buildings include; Corrugated iron/aluminium/plastic/asbestos sheets, thatch and tiles.

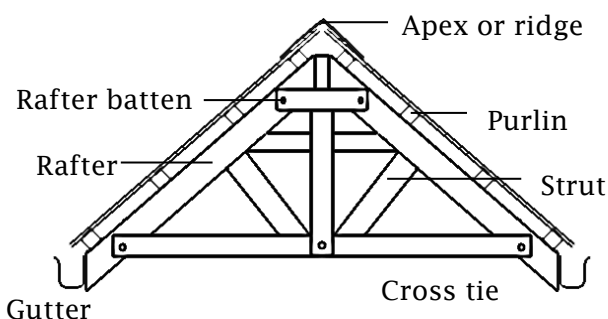


Fig. 5.9 Parts of a building truss

Function of parts of a building truss

Wall plates: They support the roof.

Rafters: provide support for roofing material.

Struts: They support the rafters.

Purlin: It holds the trusses in position and receives the roofing materials.

Gutter: It collects rainwater for storage. It also prevents raindrops from falling on the wall.

Cross tie or tie beam: It supports all other parts of the roof. It is attached to the wall plate.

Quality of a good roof frame (truss)

- Have a well raised pitch to enable free draining of water from the roof.
- It should be made up of durable material such as metal or treated wood.
- It should be made up of strong material such as metal and hardwood to be able to support the roofing material to and to withstand strong winds and snow fall.
- It should be well designed in such a way that the dead load and the load of the roofing material is properly distributed on the whole building structure.
- It should have an extended roof overhang to prevent strong sunshine and rain water splashes from reaching the walls or inside the building.

Qualities of good roofing materials

- It must be leak proof
- Should have good thermal insulator or high in thermal capacity
- Must be durable and able to withstand different weather conditions.
- It must be fire resistant
- Easily obtained and cost effective
- It should not require specialized skills to apply on the building
- It should be light in weight to prevent damage of the walls due to over loading
- It should be hygienic particularly where roof rainwater is to be collected
- It should be vermin and rodent proof. This is important property for roofing materials to be used on crop stores.



Activity 6.19 Identifying the parts of farm buildings

You will need A pen, note book, reference books, access to internet

Key question: what are the basic parts of a farm building?

In groups,

1. Brainstorm and discuss the major parts of a farm building and give the desirable qualities of each part.
2. Give examples of materials commonly used in roofing of farm buildings and Outline the advantages and disadvantage of each
3. Make an annotated drawing of a roofing truss.
4. Why is a DPC important in building construction?

Type and print your findings and share with the rest of the class.



Activity 6.20 Demonstrating building skills

You will need Bricks or concrete blocks, mortar, concrete, DPC, hoe, spade, spirit level, plumb bob, trowel, wheelbarrow, brick layer's line.

Key question: How to set and construct a farm building?

In groups,

Show skill in the construction of a building foundation;

1. Make a building plan

2. Put on your protective gears.
3. Set the building on the ground using pegs and strings
4. Dig the trenches for construction of the foundation.
5. fill the trench with hard core layer of about 30cm. put big stones first and then put in small stones to fill the gap between the big stones.
6. Mix concrete 1:3:6 or 1:4:5 and lay it on top of the hard core. Leave the concrete to cure.
7. Use blocks or brick to construct the plinth until it is 15-20cm above the ground..
8. Refill the gap between the foundation and the trench wall with marrum or sub soil mixed with termite poison.
9. Fill the floor with hard core and apply rich concrete mixture on top.

Discuss the factors that affect the stability and strength of a building foundation.

Show skill in the construction of a building wall (super structure);

1. Lay a damp proof course on the plinth wall to stop upward rise of water into the walls.
2. Spread a layer of mortar on the DPC and lay the bricks horizontally over the mortar.
3. Lay the bricks on the mortar and press each brick by tapping it with the handle of the trowel or wooden mallet.
4. On completion of a course, the vertical joints are fully filled with mortar from the top. This operation of filling open joints is termed *flushing-up*.
5. All courses are laid truly horizontal. A plumb bob or spirit level is used to ensure the brick are laid vertically.
6. All vertical joints must be alternated in the subsequent courses of bricks. Vertical joints should never come directly, one over the other.
7. Construct the openings (windows and doors
8. Construct the beam using reinforced concrete
9. Construct the wall plate
10. Construct the roof frame (truss) and install the roofing materials.
11. Plaster the walls and then do the floor finishing
12. Fix the shutters (doors and windows



Individuals keep the learning log and share their findings with the rest of the class.



Plastering Farm Buildings

Plaster is a thin layer of mortar applied over the wall surface. it provides a protective coat to the wall from rain, heat and humidity, it covers the defective workmanship in brick work, it conceals any gaps in the bricks/blocks, increases the strength of the wall, prevents dust from the construction materials, increases beauty and provides a suitable surface for painting.

Steps in plastering a building wall

1. Preparation of surface for plastering

- Keep all the mortar joints of wall rough, so as to give a good bonding to hold plaster.
- Roughen the entire wall to be plastered.
- Clean all the joints and surfaces of the wall with a wire brush, there should be no oil or grease, loose mortar, dust etc. left on wall surface.
- If there is any cavities or holes on the surface, then fill it in advance with appropriate material.
- If the surface is smooth or the wall to be plastered is old one, then rake out the mortar joint to a depth of at least 12 mm to give a better bonding to the plaster.
- Wash the mortar joints and entire wall to be plastered, and keep it wet for at least 6 hours before applying cement plaster.

- If the projection on the wall surface is more than 12 mm, then knock it off, so as to obtain a uniform surface of wall. This will reduce the consumption of plaster.

2. Applying under coat or base coat

- In case of brick masonry the thickness of first coat plaster is in general 12 mm.
- Apply the first coat of plaster on the wall surface using a trowel.
- Level the surface by means of flat wooden floats and screed.
- After leveling, leave the first coat to set but not to dry and then roughen it with a scratching tool to form a key to the second coat of plaster.



Fig. 6.10 plastering the wall

3. Applying finishing coat

- The thickness of second coat or finishing coat may vary between 2 to 3 mm.
- Before applying the second coat, damp the first coat evenly.
- Apply the finishing coat with wooden floats to a true even surface and using a steel trowel, give it a finishing touch.
- As far as possible, the finishing coat should be applied starting from top towards bottom and completed in one operation to eliminate joining marks.

4. Curing of plastering works: After completion of the plastering work, it is kept wet by sprinkling water for at least 7 days in order to develop strength and hardness.

Note, improper curing may lead to development of cracks or efflorescence in plaster work.



Activity 6.7

Demonstrating skills in plastering farm buildings

You will need

Mortar, hoe, spade, spirit level, trowel, wooden and steel floats, mortar pan, wheelbarrow, screed and spot board.

Key question: What skills are required in plastering farm buildings?

In groups,

1. Put on appropriate protective gears.
2. Collect the materials, tools and equipment
3. Mix the mortar to use in plastering. The mixing ratio is usually 1:3 or 1:4
4. Prepare the wall surface to be plastered
5. Apply the undercoat plaster
6. Apply the finishing coat
7. Cure the plaster.
8. Clean all the tools and equipment used
9. Remove the protective gear and wash up.

Individuals keep the learning log and share their findings with the rest of the class.



Why would you recommend plastering of farm buildings. Why is it important to; prepare proper mortar, and to prepare the wall surface before applying plaster

Walls that are properly plastered will not hurt the animals that may rub themselves on it. They will not hide parasites and pests, they will make cleaning inside the building easy and they will reduce leakage and wastage of chemicals as in animal dips and spray races.

Painting Farm Buildings

Painting help to protect the building from weather elements and beautify the building. There are three main types of paints; water-based paint mainly used indoors and oil-based paint mainly used outdoors. An emulsion paint is a water-based paint with vinyl or acrylic added for durability. An example is weather guard paint.

The procedures of painting include;

1. Choosing the type of paint, primer and colour you want to use
2. Preparing the wall surface by scrapping and rubbing the wall to remove ant loose mortar, dust, oil and old paint. Use a scraper, brush and sponge to do the job
3. Applying the primer.
4. Applying the paint starting with the ceiling and then the walls. you can a brush or roller
5. Finishing up the corners using a hand brush.
6. Cleaning the paint from the floor, windows and doors
7. Cleaning the tools and protective gears used.



Fig. 6.11 Using a roller to paint

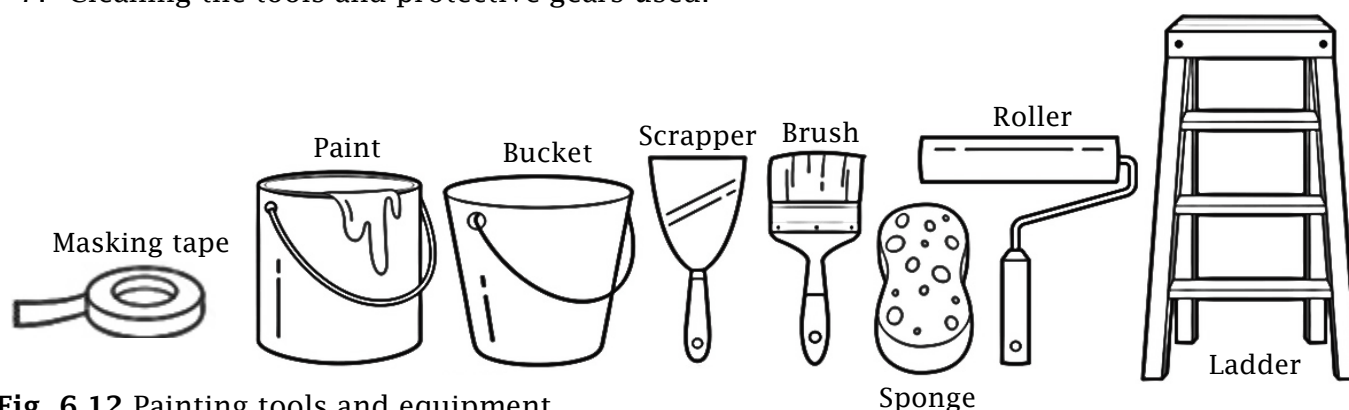


Fig. 6.12 Painting tools and equipment.



Activity 6.8

Participating in painting of a farm building

You will need

Paint, roller brush, hand brush, sand paper, filler, basin, stirring rod, hand scraper, ladder, puffy Knife, masking tape

Key question: What is the correct procedure for painting a farm building?

In groups,

1. Wear appropriate protective gear.
2. Assemble all the materials, tools and equipment to use.
3. Move to the school farm and identify a building that is in need of painting.
4. Clean the surface to be painted by scraping off any old paint that may be there. Which tool will you use? Explain why it is important to prepare the surface before painting?
5. Apply a primer to remove any dents
6. Mix the paint following the manufacturer's instructions. What are the main instructions?
7. Apply the first layer of paint using either a brush or roller and leave to dry and then after, apply the final layer of paint.
8. Clean the paint from the floor, windows and doors
9. Clean the tools and protective gears used and wash up.

(i) State why painting farm building should be a regular maintenance activity.

Individuals keep the learning log and share their findings with the rest of the class.

Fences

A fence is a top-open enclosure which forms a barrier to human and livestock movement.

Importance of fences on the farm

- Perimeter fencing helps to establish farm boundaries so as to claim ownership, and reduce land disputes between neighbors.
- Double fencing of the farm may help to control the spread of ticks and diseases particularly from neighboring farms.
- Perimeter fencing of the farm or the farmer's house helps to provide privacy and enhance security by keeping out trespassers and thieves
- Fences make it easy to practice mixed farming by helping to protect crops from animals on the farm or to stop animals from damaging neighbours crops.
- Internal fences help to establish paddocks which are important in rotational grazing, separation of the herd according to sex and age and in control of diseases and parasites.
- Fences may help to protect structures such as water ponds, drainage channels, wells, fish pond, silos and bins dams and farmers dwellings from being damaged by animals
- The hedges and tall woody plants used as live fences provides good wind brakes, supplies fresh air and provides shade to the animals.
- Since animals are kept in a confinement in a fence or paddock management practices such as rounding and culling of animals becomes easy to carry out.

Types of Fences

Dead Fences

These are made up of materials which have no life (dead). Examples include;

Woven wire fences: these are made up of either thin or thick gauged plain steel wires woven to make an open mesh. There are two types of woven wires depending on the gauge of the wire and the size of the diamond open.

- Heavy gauged woven wire or chain link used to fence animals such as pigs, goats, sheep and mature birds such as turkeys, goose, ducks and chicken.
- Thin gauged woven wire or chicken wire mesh used to fence mostly poultry, rabbits and guinea pigs. The wire is usually galvanized or plastic coated to avoid rusting.

Log or wooden fences: in this type logs or off cuts from timber or tree branches are well laid closely to each other to leave no gap where the animals can pass through.

Walls: these may be made of stones, blocks or concrete bound together with mortar. This type of fence is more durable but expensive to construct.

Trench fences: deep V-shaped trenches are made around the farm. It is usually used on crop farms near game reserves to stop wild animals from crossing into the farm.

Barbed wire fence: this is made up of barbed wires suspended on posts. It is the most commonly used fence in East Africa and effective in controlling the movement of animals.

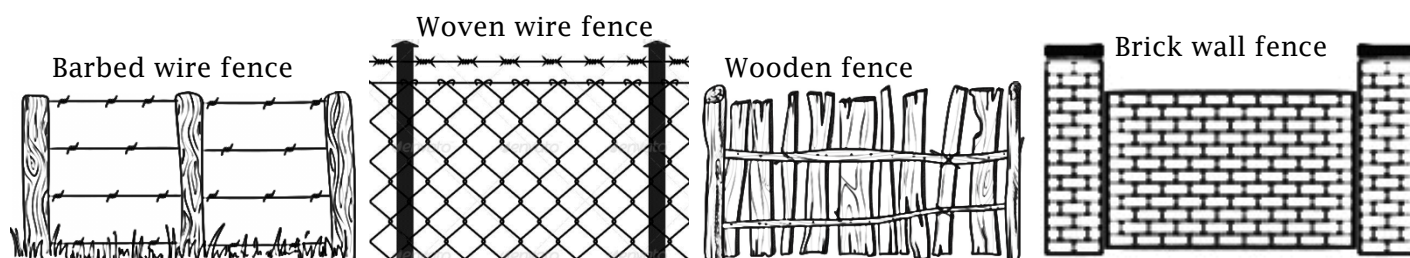


Fig. 6.23 Different types of fences



Which one of the above fences did you use for your crop or animal project and why? Explain how the fencing of your project has helped in managing of your project.

Advantages of barbed wire fence

- Barbs effectively discourage animals from passing through the fence.
- Barbed wire fence is relatively cheap as compared to chain link fence.
- It is easy to construct as it does not require much skills and materials.
- If the poles are well treated, the fence can last for long time.
- It does not occupy much space as is the case with hedge fence.

Disadvantages of barbed wire fence

- The barbs can prick the animal and cause damage to the skin/hide.
- The barbs can injure workers during construction of barbed wire fence.
- Cannot provide an effective fence to handle small stocks such as rabbits, kids, lambs and poultry.
- The wires break frequently requiring frequent maintenance.
- The poles can easily be attack by fungi and wood pests.

Components of a barbed wire fence

Strainer/King/Corner post: it is the biggest of all posts. It is 3.0m long and 25 - 30cm wide. It is used in corners or at the gate to take the strain of the wire pulling from one direction.

Standard/Ordinary post: It is smaller than strainer. It is 2.5m long and 15-25cm in diameter. It is placed between the strainers to hold up the wire pulling in both directions

Dropper: It is thin and short pole 2-5cm in diameter and 1.5m long. It is placed along the fence line to reduce the space between the posts and to reduce wire sag.

Strut/support: It is 3.0m long and 15 - 20cm in diameter. The strut is used to brace or support the strainer against the pull of the wire.

Dropper/batten: it is light and thin about 5cm in diameter. It is normally tied on the wires vertically to reduce wire sag. It prevents the animal lifting and passing in between the wires.

Barbed wire: this is made up of two plain steel wires twisted together with steel barbs installed on the wire at certain intervals. Barbed wire can be of low or high tensile strength.

Fencing nails or staples or U-nails: They help to attach the barbed wire to the posts.

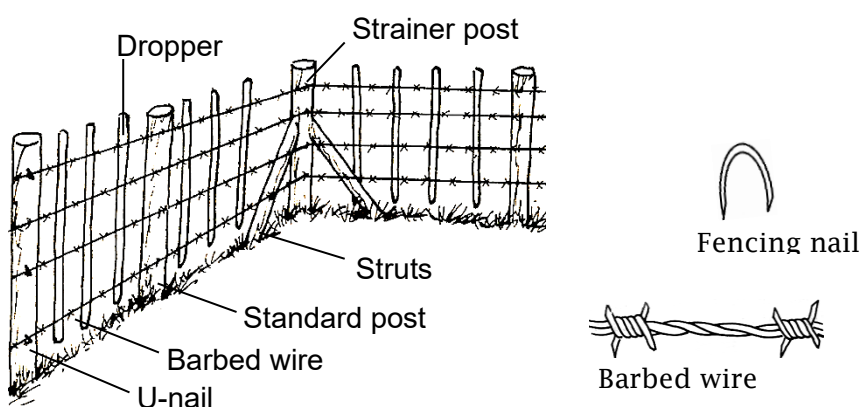


Fig. 5.2 Barbed wire fence

Fencing Tools

Pliers: for cutting fencing wires. It is also used to remove staples from the post.

Claw hammer: this is used for driving staples into the posts.

Posthole digger: used to make post holes.

Saw: for cutting the posts to the required length

Ramming rod: for firming and stabilizing the fill back soil around the post.

Splicing tool: for joining two ends of the fencing wire by weaving.

Tape measure: for measuring sizes of posts and distances between posts and wire strands.

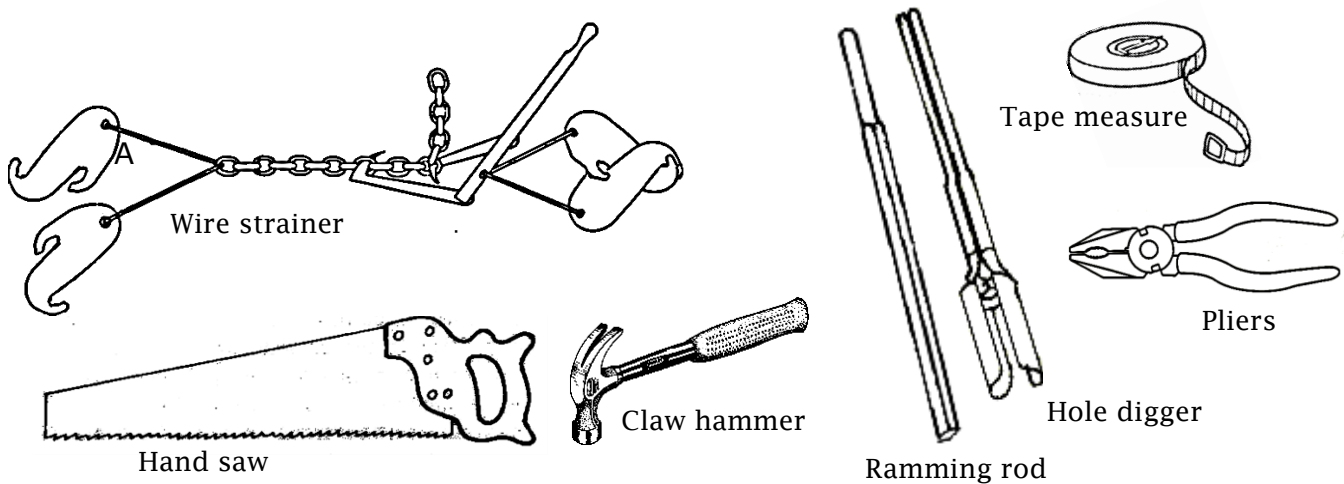


Fig 5.3 Fencing tools.

Construction of a barbed wire fence

Once the land to be fenced has been identified, you need to have materials; roll of barbed wire, posts, pliers, hammer, staples, hole digger wood chisel mallet, pegs, string and wire strainer.

Procedure constructing a barbed wire fence

1. Determine the area on the ground where the fence line is to pass.
2. Clear the area over where the fence line is to pass about 2m wide to give working space.
3. Measure and mark the positions of the holes for the posts with pegs the holes should be placed 4 - 6 m apart.
4. Determine the corners and gate post position.
5. Dig the post holes about 40-60 cm deep and just wide enough to take the size of the post and the earth fill or concrete fill.
6. Dig the gate and corner holes together with holes for their support posts. The holes for the corner and gate post should be 60-80cm deep, while the holes for the struts posts about 1.5 m away from the corner and gate post holes.
7. Place all the posts and support posts in their holes. All post should be 2 m high above the ground level.
8. Pour the concrete into the hole and ram very well to fill the hole.
9. Fix the struts to the corner and gate posts.
10. Fix the first strand of wire at the bottom about 30cm above the ground all-round the posts.
11. Hold the wire on the first post with a staple and stretch it with a wire strainer until the required tension is achieved. Staple the wire onto all the posts using a fencing hammer.
12. Fix the other strand of the wire 30cm apart. The number of wire strands to fix on the posts and the spacing between the wire strands depends on the type of the animals being fenced.
13. Fix the droppers on the wires.
14. Construct the gate.

How to use the chain-type wire strainer

Usually the chain type wire strainer has two grippers each at the end of the chain, the lever which operates the implement, the chain links and the two hooks attached to the lever.

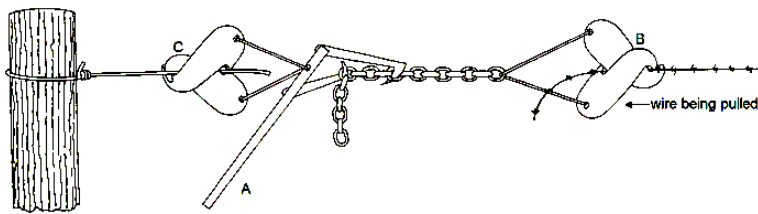


Fig. 5.4 using a wire strainer to stretch a barbed wire

The procedure of using the chain-type wire strainer

1. Fixed the grippers marked **C** on the wire stapled to the post and then fix the grippers marked **B** on the wire to be stretched.
2. Fix the short hook into the nearest chain link.
3. Pull lever marked **A** backward towards the grippers **C** to stretch the wire.
4. Fix the long hook into the nearest chain link to hold the tight chain in position.
5. Remove the short hook from loose chain and then hook it to the next chain link on the tight chain and then pull the level backwards again to pull and tighten the wire more.
6. Repeat step 3 to 5 until the wire is stretched to the required tension.
7. Fix the wire on the post or join broken wires by weaving them together with a splicing tool.
8. Remove the grippers from the wire and remove the strainer.

Budgeting for Materials in construction of Farm Structures



Project Work

Projects are assignments given to the learners to be done over a period of time. They are done either individually or in groups depending on the nature of the project. Learners are expected to come up with a tangible product.

Why use projects in learning?

We live in a problem-based world, so the use of projects in learning is important. The use of projects empowers learners to engage with 21st century challenges and to acquire 21st Century Skills such as innovation, time management, research skills, values, risk taking/willingness to fail, critical thinking, problem solving, creativity, collaboration, communication and technology.

Developing a Project

- **Identification of the project:** Title (aligned to the theme); objectives
- **Organisation:** Planning, Methodology, resources, Drafting, Implementation, Creating a portfolio and documenting

- **Report writing**

Features of Projects

- i. It should be unique depicting creativity; no two projects are exactly similar even if they are exactly identical or are merely duplicated.
- ii. A project has a life cycle reflected thus has a life span. In this context, it should not go beyond a term; one month is appropriate
- iii. Project should be efficient; efficiency is assessed by outputs obtained per the inputs used.
- iv. A project must meet or exceed the planned targets hence effective.
- v. A project calls for team-work; in this context, a team is constituted of members belonging to the same class, or may be selected by the teacher from different streams where they exist for purposes of avoiding unhealthy competition.
- vi. A project should be designed in a way that observes health, safety and environmental protection practices.
- vii. A project is expected to be disseminated for addressing the intended societal challenge.
- viii. It should exhibit a degree of innovation and creativity in addressing a societal problem.
- ix. Preferably, a project should be in line with the theme and topics covered in the syllabus.

Materials for the Project

- You are advised to identify projects which can be done using materials which are locally available and affordable. Low cost materials e.g. waste materials like papers, plastics, used packaging materials, manure from farm animals etc.
- Use materials which are in line with “Buy Uganda Build Uganda” (BUBU). By so doing, the project work will be promoting industrialization for employment, inclusive growth and wealth creation.
- Materials selected should be environmentally friendly

How many projects should you take in a year?

You have a maximum of two projects every term provided that by the time you sit for final UNEB examinations, the project registered should have been completed and submitted for assessment.

What is the role of the teacher in your project?

In project based learning and assessment, your teacher is expected to:

- Make observations
- Hold conversations
- Provide guidance and support the learner
- Keep records
- Receive a product and report

This is continuous throughout the project lifetime. As learners, you are expected to cooperate and work closely with your teacher(s)/project supervisor(s).

Assessing Your Project

- Scores for each parameter will be determined by the teacher. The total score for the project will be scaled to 10%. This will be added to the 10% score from the Activities of Integration to account for the 20% score of the end of cycle summative assessment.
- A learner who has not been assessed at school level does not qualify to be graded by UNEB.
- Even when the project has been conducted as a group, scores shall be awarded to members individually. You are expected to have individual records and project report.

Project Report

Project report is a detailed account of the project. The report should be a simple write-up depicting the following;

Title

The title should accurately reflect the scope and content of the project. In addition, it should be concise, simple and catchy in not more than 20 words. The title should be informative/descriptive yet discrete and contain the key words of the project.

The title page; this page should not be numbered. The title page should have the Logo and the name of school, the title of the project in CAPITALS, full names of the learner and the name of the supervisor(s). All information on the title page must be centred.

Purpose of the Project

Introduce your project with its purpose. For example, To make a low cost flavoured yoghurt or To discover the role of bacteria in the formation of yoghurt.

Leading Question

What do you really want to find out? What is at the back of your mind as you undertake this project? Examples, What is the role of bacteria in the milk industry? What is the use of bacteria in making yoghurt?

Prediction of the Project

Make a brief statement of your expectation during the project. For example, Bacteria in the mixture will ferment the milk to form yoghurt.

Material Selection

Write a list of items you will use in the project. Ensure that all materials are readily available.

Budget

This is the financial plan for implementation of the project. It should be clear, realistic and reasonable (affordable). It should be itemised according to the following:

- Equipment/tools
- Stationery
- Materials e.g. nails, wood, ingredients etc.

Methodology/Procedure

How are you going to use the material you selected? Write a short story of how these items helped you to perform the tasks. Include the time it took you to achieve results.

Results/Observations

What did you see, hear, smell, taste or feel during the performance of your procedure? Write about it. Showcase your output/product. Describe your product/output.

Conclusion

How do the results relate to the key question and predictions you made earlier.

Challenges and recommendations:

Challenges faced during the execution of the project, recommendations, and the way forward, should come as the last item.

ICT Tools

Please ensure that you use a computer to prepare your report or a presentation. (You can use micro soft word to process your report and power point to present the project report.

Questionnaire to Aid the Learners

Move around your community which may be your school, areas around your school, at home, areas near your home, or your entire village to identify one problem that the community is

facing. Make sure that the problem you identify can be solved by the knowledge you have gained in class using any of the chapters so far covered.

1. What problem have you identified in your community? (Write a problem statement).
2. Write the title of your project and State the chapter on which your project will focus.
3. Write the main aim, objectives and benefits that your project will have.
4. State the TANGIBLE PRODUCT that you as a person are going to make physically (not buying) to help solve the problem.
5. Think and design a plan of how your product will look like with a pencil.
6. Explain in details how your product will work to solve the problem.
7. Using Google/internet, book or skilled people, research on how that product is made.
8. Think critically and list all the materials you will need to make your product. After, make a budget for your project as follows.

No	Item	Description (Quality)	Quantity	Unit Price	Amount
1					
2					

9. Explain in details the step by step procedures of how you will make the product.
10. Explain how you will experiment to see if your product will work to solve the problem stated. Also predict challenges you may face in your project and provide their solutions.
11. Make a simple schedule or work plan/action plan for your project as follows.

No	Activity	Starting Date	Ending Date	Number of Days
1	Problem identification			
2	Planning for the project			
3	Project implementation			
4	Testing the product's efficiency and any adjustments/modification			
5	Report writing			

Project Report Template (*Hypothetical*)



GOMBE SECONDARY SCHOOL - BUTAMBALA

PROJECT REPORT

SUBJECT:

AGRICULTURE

THEME:

ANIMAL PRODUCTION

TOPIC:

ANIMAL NUTRITION

TITLE OF PROJECT:

IMPROVING NUTRITIVE VALUE OF ANIMAL FEED USING LOCALLY MADE MINERAL SUPPLEMENT

SUBMITTED BY;

BALUKU CHARLES
SENIOR THREE, REG. NO/LIN. 023/2023

PROJECT SUPERVISOR;

MR. ONGOM UMAR

BALUKU CHARLES;
Learner

Signature

Date

MR. ONGOM UMAR;
Project Supervisor

Signature

Date

NOVEMBER 2023

Names of Project Partners

S/N	Name	Position	Signature
1			
2			
3			
4			
5			
6			
7			

HYPOTHESIS/PROBLEM TO ADDRESS

.....

.....

.....

.....

JUSTIFICATION or why you carried out this project?

.....

.....

.....

.....

.....

MATERIALS USED state all the resources or materials you used in the project and if possible, have their pictures displayed and captioned

	Material /Items	Quantity	source
1			
2			
3			
4			
5			

BUDGET FOR THE RESOURCES/MATERIALS this is the financial plan for implementation of the project. It should be clear, realistic and reasonable (affordable)

	Material/Items	Quantity	Unit cost	Total cost	remarks
1					
2					
3					
4					
5					

PROCEDURE/METHODOLOGY methods, instruments and procedures used in the project. Ensure sequence of events. Captioned photos of events can be included

.....

.....

.....

.....

.....

.....

.....

Table1 *Photos of some of the stages / processes*

Photo here	Photo here
Figure ... caption	Figure ... caption
Photo here	Photo here
Figure ... caption	Figure ... caption

PRODUCTS / RESULTS / FINDINGS it's where you report the findings/results/products of your project as a result of the methodology or procedures applied. Simply state the findings in past tense. If possible provide captioned photographs of your products

1.
2.
3.

Captioned photo of the product

CONCLUSION this is the last part of the project report. It summarises the entire work, key ideas and the things which have been learnt as a result of conducting the project. It should states. In conclusion.....

CHALLENGES

The following challenges were faced during the course of implementation of the project;

1. Inadequate skills of group members.
2. Competition among the group members over positions of authority.
3. Lack of cooperation and commitment among the members.
4. Limited support from the school administration.
5. Limited time for project work due to the busy school routine (conflict over schedules).
6. High prices of materials and equipment.
7. Market price fluctuations for equipment and materials.
8. Vandalism from non-agriculture students.
9. Limited space/land available for the project.
10. Inadequate funds for running the project

FUTURE RECOMMENDATIONS

The following are recommendations to improve the project;

1.
2.
3.

Use the project report template below to assess your competence in preparing a project report for assessment

AGRICULTURE CLASS PROJECT REPORT

Name Class..... Stream..... Term..... Year

1.1 Title of the Project: Improving Nutritive Value of animal Feed Using Locally Made Mineral Supplement

1.2 Theme: Animal Production

1.3 Justification / Objective of the Project / Problem to Address

Most famers feed their animals using fresh forage, silage and hay which has low-mineral content. Mineral supplement is manufactured to supply minerals to livestock. However, mineral supplement is not commonly used by most farmers. Therefore, there is need for farmers to use locally made mineral supplement to improve the nutritive value of feed for livestock in our community.

1.4 (a) Materials used

The following materials were used to make 4 pieces of a 5kg mineral salt lick

1. Bone meal for Calcium	6. Feed pigment for colouring the ingredients
2. Ashes for Ca and K mineral	7. Empty water bottles of 2 liters to use as moulds
3. Rock salt for sodium, Ca, K & Cl	8. Weighing scale for measuring ingredients)
4. Cement for binding & setting the mixture in the moulds	9. Bowl for mixing ingredients
5. Molasses to binds and sweeten the mixture for improved intake	10. Polythene for lining the moulds
	11. Protective gear (gloves, overall & gumboot)

(b) Budget for materials

S/N	Material / item	Quantity	Rates	Total cost	Remarks
1	Bone meal	3kg	5,000	15,000	
2	Rock salt	12.5kg	3,000	37,500	
3	Ashes	1kg	1,000	1,000	
4	Cement	3kg	1,500	4,500	
5	Molasses	2liters	6,000	12,000	
6	Feed pigment	1 tin	5,000	5,000	
7	Water bottles	5	500	2,500	
8	Bowl	1	3,000	3,000	
9	Weighing scale	1	100,000	100,000	
10	Gloves	1	5,000	5,000	
11	Overall	1	20,000	20,000	
12	Gumboot	1	15,000	15,000	
	Transport		10,000	10,000	
	Total			230,500	

1.5 Procedure / Methodology

The materials and equipment required were mobilised and gathered through buying some according to the budget.

The process of making the mineral supplement involved

- Crushing rock salt and bone meal in to fine particles separately
- Sieving rock salt, bone meal and ash in to fine powder separately
- Weighing the ingredients separately (rock salt 12.5kg, bone meal 3kg, cement 3kg, ash 1kg and 1kg of feed pigment)
- Clean the cemented surface (tarpaulin) to avoid contamination of ingredients
- Pour the ingredients starting with rock salt, bone meal, cement, ash and feed pigment.
- Mix the ingredients thoroughly to ensure uniformity
- Transfer the mixture in to a big bowl (basin, saucepan)
- Add 1liter molasses and water in the right proportion and stir to make a thick paste
- Shape the 4 mineral water bottles to use as moulds
- Line the moulds with polythene to avoid mineral paste sticking on moulds
- Pour and compact the mineral paste in to the shaped moulds
- Drill a hole in the center of the mineral paste to be used for hanging the salt lick
- Keep the set up (moulds with mineral paste) under shade for proper curing
- Allow the moulds to cure for at least two weeks
- Remove the moulds to expose a hardened mineral salt lick

1.6 Products / Results / findings

- a) Solid mineral supplement / salt lick produced
- b) Weight of the of the mineral salt lick
- c) Colour of the mineral salt lick
- d) Shape of the mineral supplement
- e) Texture of the mineral salt lick block
- f) Composition of the mineral salt lick (ingredients)
- g) Palatability to the animal

h) Information leaflet attached

Photo of product (s)

1.7 (a) Conclusion:

The mineral supplement produced is palatable to the animals and can be used as a source of minerals to improve the nutritive value of livestock feed

(b) Challenges:

- It takes time for the mineral supplement to dry
- Difficult to get feed pigment

(c) Recommendation:

-

Name of teacher/Supervisor

.....

Signature

Date.....



Activity

Preparing a project report prototype

You will need

Notebook, pen, reference text books, internet, project report samples

Key question: What are the key contents and features of a project report?

In groups:

1. Study the project report template above.
2. Comment on the presentation and arrangement of contents.
3. Observe and study the project reports provided.
4. Prepare a project report prototype. Type set your report. Only one side of the page should be printed, the other side should be left blank. Project title should be in TIMES NEW ROMAN font, font size 16, capital and bold. Sub heading should be in TIME NEW ROMAN, font size 14. The font used in the body of the report should be TIME NEW ROMAN, 12 font size, plain and with 1.5" line spacing.
5. Present the prototype to the rest of the class.

Assessment of the Project

The following Aspects shall be examined in the Project Work

Note; marks allocations may change depending on the teacher's or school's discretion

	No	Aspect	Allocated Marks	Marks Scored
Planning	1	Title	2	
	2	Alignment to theme	2	
	3	Justification of the project	2	
	4	Methodology	14	
	5	Identification of materials and budget making	2	
Implementation	1	Organization (was the project work organized?)	1	
	2	Use of resources (how were the resources utilized?)	1	
	3	Expression of critical thinking and problem-solving skills	5	
	4	Expression of Creativity and innovation skills	5	
	5	Expression of Communication skills	5	

	6	Expression of Co-operation and learning skills	5	
	7	Expression of Calculation and ICT skills	5	
	8	Demonstration of Values in the project work.	6	
Product	1	Originality of the product.	5	
	2	Creativity of making the product	5	
	3	Innovation of the project	5	
	4	Accuracy / precision of the product	5	
	5	Testing of the product	5	
	6	Effectiveness and efficiency of the learner's product in solving the problem identified by the learner	1	
Report	1	Relevance of the report content	3	
	2	Accuracy of the report's content	3	
	3	Coherence of the reports content	3	
	4	Relevance of the report's format	3	
	5	Accuracy of the report's format	3	
	6	Coherence of the report's format	3	
	7	Excellence of the whole report	1	
		Overall total marks	100	

	Phase	Indicators	Max Score
1	Identification, planning, design	Title, alignment to theme, justification of the project, methodology, identification of materials	$\frac{X}{22}$
2	Project Implementation	Organization, Use of resources, focus on generic skills and values	$\frac{X}{33}$
3	Product	Originality, creativity and innovation, accuracy	$\frac{X}{26}$
4	Project report	Relevancy, Accuracy, coherence	$\frac{X}{19}$
		Total	$\frac{X}{100}$

Glossary of Terms

This glossary is designed to help you to understand the meaning of the terms commonly used in agriculture. While this list is not exhaustive of all the terms used in agriculture, it will be useful in providing basic foundation to better understand agriculture.



Acidic soil: A soil with pH of less than 7.0.

Acre: Unit of measurement for agricultural land. It is equivalent to 43,560 square feet.

Adsorption: The process by which atoms, molecules, or ions are taken up from the soil solution and retained on the surfaces of soil particles by chemical or physical binding.

Aeration: Involves the process of creating small holes in the soil, allowing air, water and nutrients to travel to the plant roots.

Agglomeration: a method of increasing particle size by gluing tiny particles together.

Agriculture: The science and art of cultivation of crops and rearing of livestock.

Agroforestry: Combining the operations of agriculture (cropping, grazing) with growing trees for sale or farm use.

Agronomy: The science of crop production.

Alkaline soil: A soil with pH of more than 7.0. The higher the number, the more alkaline (base) the soil will be. The pH scale ranges from 0 to 14, with 7 as neutral.

Allelopathic compounds: Chemical substances synthesised by plants to inhibit growth of other plant species growing in the vicinity.

Allelopathy: Chemical inhibition of one organism by another.

Alluvial soil: Soils formed from sediments deposited on land by streams.

Animal Husbandry: The breeding, feeding and care of farm animals.

Annual Plant: A plant that completes its life cycle and dies in one year or less.

Anoestrus: The state in a female animal when no oestrus occurs. The animal is not receptive to mating.

Anthelmintic: A medicine used to treat internal parasites or worms in livestock.

Anther: The upper part of the stamen of a flower in which pollen is produced.

Anthesis: The stage in reproduction when flowers open and pollen is shed.

Apiary: A place where honeybees are kept; colonies of bees in hives.

Aquaculture: The commercial production of aquatic plants or animals in a controlled environment.

Aquaponics: A system of aquaculture in which the waste produced by farmed fish or other aquatic animals supply nutrients for plants grown hydroponically, which, in turn, purifies the water.

Arable soil: Soil suitable for ploughing and cultivation of crops.

Arid: A term applied to regions or climates where lack of sufficient moisture severely limits growth and production of vegetation.

Artificial Insemination: The introduction of semen into the uterus of the female with a special apparatuses. The process begins with the collection of semen from the male.

Asexual reproduction: The establishment of a plant from any cell, tissue or organ of that plant. This includes by cuttings, grafting, budding, division or tissue culture.

Auction: Marketing farm produce through which an auctioneer sells to the highest bidder.

Auxin: A plant hormone promoting or regulating growth.

Available Water Holding Capacity: The quantity of water held in soil that is available for plants to use. As a rule, the deeper the soil, the greater its AWHC.

Avian: Pertaining to poultry and/or fowl.

Azonal soil: A soil lacking a well-defined profile.



Backcrossing: The mating of a crossbred animal with animals from one of its parent.

Bag silo: Long plastic bag that holds silage.

Balance Ration: A ration which contains all the necessary nutrients in the proportions and amounts needed by the animal for normal functioning and growth.

Bank: A financial institution licensed to receive deposits and give loans.

Barrow: Male pig castrated before sexual maturity.

Biennial: A plant that lives for 2 years, producing vegetative growth the first year, usually blooming and fruiting in the second year, and then dying. (Example: blackberry plants have biennial canes and perennial root systems).

Biocide: A chemical toxic or lethal to living organisms.

Biodegradable: Capable of being decomposed by natural processes.

Bio-control: A management practice in which predatory or parasitic organisms are used to control pests and pathogens.

Biogas: A mixture of gas (methane and CO₂) produced when animal waste is broken down by microorganisms in the absence of oxygen.

Biomass: refers to all organic matter derived from living or recently living organisms, plant and animal-based.

Biosecurity: A set of management and physical measures designed to reduce the risk of introduction, establishment and spread of animal diseases, infections or infestations to, from and within an animal population.

Bloat: A distension of the rumen due to an accumulation of gases. Can occur in animals grazing lush, wet, legume pastures.

Boar: Intact (not castrated) male pig.

Body condition score: An assessment of an animal's body condition based upon an estimate or measurement of the amount of flesh and fat tissue under the skin on certain body parts.

Bolting: When a plant produces flowers or seeds prematurely instead of a crop. This is usually the result of excessive heat and sun exposure.

Boot stage: A grass plant growth stage that occurs just before the seed head emerges from the leaf sheath.

Bovine: Family of animals that includes beef and dairy cattle.

Breaker: Highest conditioned cull cow, carrying heavy flesh and having a high dressing percent.

Breed: A group of animals descended from common ancestry and possessing certain inherited characteristics which distinguish it from any other group.

Breeding herd: The animals retained for breeding purposes to provide for the perpetuation of the herd or flock.

Breeding value: The 'true' genetic merit of an animal, which is passed on to its progeny through its genes. It measures the average advantage (or disadvantage) expected in a large number of progeny.

Breeding: controlled mating and of livestock or plants.

Briquettes: Small compact block of organic waste use for burning in the stove or fire.

Broadcasting: Random scattering of seeds, fertilizing or pesticide application over the surface of the ground.

Broiler: A chicken of either sex about 7 weeks of age.

Broken mouth: A sheep or goat whose incisor teeth have fallen out or badly worn or irregular, usually as a result of old age or hard grazing.

Brooding: Provision of warmth to newly hatched chicks.

Browsing: Grazing of shrubs and trees (rather than pasture)

Buck: A male goat. Also referred to as "billy."

Buffer capacity: The ability of a soil to withstand changes in pH. Soils with high levels of clay or organic matter have high buffering capacity.

Bull: A sexually mature, uncastrated bovine.

Bullock: Castrated male bovine (cattle species) over 30 months old.



Calf: A young male or female bovine animal under 1 year of age.

Calving interval: Period of time between the birth of calves from the one cow.

Calving percentage: An indicator of herd fertility. The percentage is usually calculated using number mated compared with number marked or weaned. It also applies to lambing.

Capillarity: Ability of water to move up in soil pores or a force that enables soil to retain water.

Capillary water: Water held in the small pores of the soil. Water can move in any direction.

Capon: A male chicken castrated before reaching sexual maturity.

Caponizing: Castration of male chicken.

Caprine: Of, relating to, or being a goat.

Career: Often composed of the jobs held, tittles earned and work accomplished over a long period of time, rather than just referring to one position.

Carrier: (1) Material used to dilute the active ingredient in a chemical formulation. (2) Material used to carry a pesticide to its target. (3) Plant or animal carrying an infectious disease agent internally, but showing no marked symptoms.

Carrying capacity: Refers to the number of animals a pasture can sustain without degradation during a given grazing period.

Cash crop: Any crop that is sold off the farm to yield ready cash.

Casings: Intestines and bladders that, after cleaning, are used in making sausages.

Castration: removal of testicles from a male animal.

Cellulose: A carbohydrate formed from glucose and a major constituent of plant cell walls. It is a colorless solid that is insoluble in water.

Certified Seed: Seeds which meet the standards of certifying agency (usually a state agency). Certification is based on germination, freedom from weeds and disease, and trueness to variety

Cervix: thigh strong circular muscles that separate the vagina from the uterus.

Chlorosis: Deficiency of chlorophyll in a green part of a plant making the leaves to appear yellowish or pale green, as a result of mineral deficiency, poor light or infection.

Clay: Soil separate consisting of particles less than 0.002 millimeters in equivalent diameter.

Claypan: A dense compact layer in the subsoil having a much higher clay content than the overlying material.

Clone: A group of plants, derived by asexual reproduction from a single parent plant. Such plants are therefore of the same genetic constitution.

Cockerel: A young rooster not more than a year old.

Colloid: An organic or inorganic particle in the soil that is extremely small, but with very large surface area. Soil colloids usually have a high cation exchange capacity.

colostrum: Also called the 'first milk'. During the "first 48 hours after birth the udder secretes antibodies in the milk that help to protect the calf from some diseases.

Communicable diseases: Diseases which spread from one animal to another by direct or indirect contact. e.g. foot and mouth disease, bird flu.

Companion crop: Also called nurse crop. A crop that is sown with another, for example, small grains are companion crops often sown with forage crops.

Complete Fertilizer: A fertilizer containing the three macro nutrients (Nitrogen, Phosphorous, and Potassium) in sufficient amounts to sustain plant growth.

Compost: Organic residues, or a mixture of organic residues and soil, that have been piled,

moistened and allowed to undergo biological decomposition for use as fertilizer.

Composting: Controlled decomposition of organic material into a fertilizer.

Concentrate feed: food materials that contain a high proportion of nutrients and are low in fiber and water.

Concrete: A mixture of cement, sand, aggregates and water in particular proportion used for construction work.

Conformation: Characteristic traits of an animal that combine to describe its shape or frame.

Conservation Tillage: crop cultivation method that maintains good ground cover throughout the years and minimal disturbance of the soil.

Contact herbicide: A herbicide that kills primarily by contact with plant tissue rather than as a result of translocation.

Continuous Grazing: Grazing an area without rest periods or rotation.

Controlled breeding: A planned program whereby livestock males and females are brought together for breeding purposes so that offsprings are born during a desired period.

Conventional Tillage: Refers to a form of field preparation and planting where the soil is ploughed prior to planting.

Cooperative: An organization formed for the purpose of production and marketing of goods or products owned collectively by members who share in the benefits.

Corrugated iron sheets: Iron sheets shaped into series of regular folds that look like waves.

Cotyledon: (1) First seedling leaves that serve as food-storing organs or may develop the ability to photosynthesise as the seed germinates. (2) Part of the placenta that supplies nutrients to the embryo.

Cover crop: A close-growing crop that provides soil protection.

Cow: A mature bovine usually having had at least one calf.

Cream: The thick white or pale yellow fatty liquid with rises to the top when milk is left to stand undisturbed.

Creep feeding: (1) Supplemental feeding of suckling livestock in such a manner that the feed is not available to the mothers or other mature livestock. (2) Special feeding of young

animals during the weaning period that helps them in transition to forage-based diets.

Crop Residue: The plant material left in the field following the harvest of a crop.

Crop Rotation: The practice of growing different crops in succession on the same land.

Cropland: Land used primarily for the production of cultivated crops.

Crossbreeding: The mating of animals (or plants) of different breeds, both of which are purebred, in order to combine the best characteristics of the two breeds. The progeny, known as crossbreds, possess hybrid vigour.

Cross-pollination: The transfer of pollen from the anther of a flower to the stigma of another flower in a different plant of the same or different species.

Crown slips: Sprouts arising from the buds in the leaf axils on the crown.

Crown: Terminal bud on top of the fruits.

Crude fiber: The coarse, fibrous portions of plant that is partially digestible and relatively low in nutritional value - for example, cellulose.

Crutching: The act of shearing wool from the breech area and hind legs.

Cryptorchid: A male animal with one or both testicles retained in the abdomen.

Culling: The removal for sale or slaughter of animals from the herd that are no longer economically productive or valuable as breeding stock. See also classing.

Culm: An aerial stem of a grass or sedge, terminating in a flower.

Cultivar: A plant bred from a cultivated variety that has originated and persisted under cultivation and is significant enough to be given a name.

Cultivate: digging the soil, growing of crops and care of crops.

Cuttings: Sections of plants taken from a parent plant used as planting materials.



Dagging: Shearing or clipping the dags from sheep, especially to prevent fly strike.

Dags: Locks of wool on the rear of wool sheep or goats heavily coated with dung.

Dam: The female parent of a calf.

Damping off: The rapid rotting of seeds or seedlings before they emerge from the soil or the rapid rotting of the stem bases and toppling of seedlings after emergence.

Decomposer: Organisms, chiefly microorganisms that break down dead, decaying matter.

Deferred Grazing: Refers to a grazing practice where grazing of a given pasture is suspended for a given period.

Dehorning: Removal of horns in farm animals.

Denitrification: The conversion of nitrate or nitrite to nitrogen.

Detritus: Fragmented particulate organic matter derived from the decomposition of debris.

Digestible dry matter (DDM): A unit that measures the amount of feed livestock consume minus the feces they produce. It is expressed as a percentage of the dry feed matter consumed.

Digestible energy (DE): A unit that measures an animal's feed intake gross energy minus fecal energy. It is expressed as calories.

Digestible protein (DP): A unit that measures an animal's feed protein minus feces portion (nitrogen x 6.25). It is expressed as a percentage of amount in feed.

Direct Sowing: Germinating seeds at the site you want them to grow, as opposed to growing them in nursery before transplanting.

Disbudding: In crops, the removal of vegetative or flower buds. Carried out mainly in flower growing to give a central single flower. Or the removal of horn buds in cattle and goats.

Disinfection: Application of sanitizers, after thorough cleansing of contaminated surface to destroy the infectious or parasitic agents.

Dispersion: The separation of soil aggregates into individual particles in the presence of water.

Docking: Cutting of animal's tail (mainly lambs)

Doe: A female of the goat or rabbit family. Female goats with kids are sometimes referred to as nanny goats.

Dormancy: A condition of a viable seed that prevents it from germinating even in the presence of otherwise favorable germination conditions. Can be physical or physiological.

Double cropping: Raising two different crops on the same area within one growing season.

Drafting: Separating animals into different groups. Drafting is often done using a race.

Drainage: The removal of excess water from the soil by means of surface or sub-surface drains.

Drenching: The application of liquid medications and vaccines to the animal's stomach via the mouth and throat.

Dressed weight: The weight of a carcass after removal of hide, head, feet and viscera.

Dressing percentage: The portion (percentage) of the live weight that remains in the carcass.

Dropper: A light post in a fence used to space wires, but not set in the ground.

Drought: A prolonged shortage of rainfall during which the soil water content is reduced to an extent that plants wilt.

Dry cow: A cow that is not producing milk, the period before the next calving and lactation.

Dry land farming: The practice of crop production without irrigation.

Dry matter: Refers to the part of feed that is not water.

Dry period: The length of time between the completion of one lactation and the start of the next lactation.

Drying off: stopping milking of a cow to stop milk secretion in the udder. Cows are usually forced to dry off after 300 days of lactation.

Dystocia: Abnormal labour or difficult birth.

E

Earthing up: The mounting of soil around the base of a plant.

Embryo transfer: Animals bred by taking a fertilised embryo and placing it in another cow's uterus.

Empty: An animal that is not pregnant.

Entire: A male animal that has not been castrated. Capable of breeding.

Entomology: The study of insects.

Enzyme: Protein catalyst which speeds up chemical reactions.

Erosion: Detachment and movement of soil by water, wind, ice, or gravity.

Evapotranspiration: Removal of moisture from soil by evaporation and by transpiration by plants growing in the soil.

Evisceration: The process of cutting the carcass open to remove internal organs.

Ewe: A mature female sheep.

Exotic species: A species that is not native to a region in which it is found.

Experiment: A procedure designed to test a hypothesis as part of the scientific method.

F

Fallow: A farming system in which land is left without a crop or weed growth for extended periods to accumulate soil moisture.

Farm: Any tract of land used for agricultural production (crop or animal production).

Farrow: To give birth to a litter of pigs.

First Aid: The initial treatment provided before professional medical personnel arrive to treat major injuries or illness.

Fat score: Indicates the depth of fat cover on the animal's rump. Ranges from 1(lean) to 5(fat).

Fecundity: Ability for regular reproduction in an animal (a measure of fertility).

Feed additives: Materials other than the feeds themselves added to diets; e.g., vitamins, mineral supplements, or antibiotics.

Feed conversion: Units of feed consumed per unit of body weight gained; the production (meat, milk) per unit of feed consumed.

Feedstuffs: Any material fit as animal food.

Fence: A structure that acts as a barrier to livestock, wildlife, or people.

Feral: Wild, untamed or undomesticated animal.

Fertigation: Application of fertilizers through an irrigation system.

Fertile land: Land capable of producing high crop yields.

Fertilizer: Any organic or inorganic material of natural or synthetic origin that is added to soil to supply certain elements essential to the growth of plants.

Field capacity: The amount of water remaining in the soil holds after gravitation has drained the free water.

Flash Grazing: Refers to the process of heavily stocking an area of pasture for a very brief period. This is done to prevent excessive growth of the forage.

Flocculation: The process where very fine clay particles, suspended in water, come together in larger groups and then settle out of suspension.

Floriculture: The cultivation of flowers.

Floriculture: The study of growing, marketing, and arranging flowers and foliage plants.

Flushing: Improving the nutrition of female breeding animals prior to and during the breeding season to stimulate ovulation.

Fodder: cut forage used for feeding farm animal. Includes Napier grass, Maize plant, Sorghum, Lablab, Guatemala and other plants that are easily harvested and fed to animals.

Foetus: The developing baby animal before birth.

Foliage: vegetative or green parts of a plant.

Food safety: The process of protecting the food supply against intentional or unintentional contamination by setting standards in the production chain and quality control.

Forage: Vegetative matter, fresh or preserved, that is gathered and fed to animals as roughage; includes hay, silage, pasture and fodder.

Forcing: spraying the crop hormone such as auxin to encourage uniform flowering.

Form pruning: selective removal of vertical branches from a plant to ensure that the plant grows in a desired shape.

Foster mother: A mother suckling another mother's young.

Fostering: attaching of a young animal a mother not its own.

Free martin: Female calf born as twins of which the other twin is a male. A free martin is sterile.

Full mouth: A sheep or goat that has all its permanent incisor teeth fully developed.

Fungicide: Any chemical agent that kills or inhibits fungi that cause plant or animal diseases.

Furrow: (1) A trench made in the soil by tillage equipment. (2) Small channel for conveying irrigation water downslope across the field.

Gapping: Refilling or reseeding of empty spaces/holes where seeds failed to germinate.

Genes: Basic units of hereditary material that govern the characteristics of individuals.

Genetic engineering: Manipulating genetic material in such a way to give certain desirable outcomes, or prevent undesirable outcomes.

Genetically Modified Organism (GMO): plants or animals whose genes have been artificially manipulated to give it desirable characteristics.

Germinate: The beginning of growth in a seed; to sprout.

Gestation period: Duration of pregnancy. Average gestation periods are: Cattle - 9 months, Sheep - 5 months and Pigs - 4 months

Gilt: A young female pig that hasn't given birth.

Grading up: Where a poor breed is improved by continued crossing with a better breed.

Grafting: The joining of two separate structures, such as a root and stem or two stems, so that by tissue growth they form a union and grow as one plant.

Grass tetany: Also called hypomagnesemia. A livestock disorder that can occur in cattle on lush pastures that have soils that are low in phosphorus but high in potassium and nitrogen. This combination inhibits magnesium uptake.

Grassland: Any land in which the dominant vegetation is grasses.

Gravitational Water: Water that either runs off or percolates through a soil under the influence of gravity. Not available for use by plants.

Graze: Feeding directly on pasture in the field.

Green Manure: Any crop or plant grown and plowed under to improve the soil, by addition of organic matter and the subsequent release of plant nutrients, especially nitrogen.

Gully erosion: the removal of soil by surface runoff resulting into formation of deep channels that cannot be smoothened over by normal tillage operations.



Hapas: shoots produced at the base of the peduncle (fruit stalk).

Hardening off: The process of increasing crop's stress tolerance by gradually exposing it to the



same or different stresses (for example, preconditioning for drought or heat tolerance).

Hay: Forage harvested and preserved by controlled drying using air.

Hazard analysis: Identifying the range of risks. It should reveal what can occur, how often, and how serious the effects might be.

Hazard: Anything that has the potential to cause injury or harm; a source of danger.

Headland: Land which is left uncultivated during the initial pass of the cultivating implement while it is turning. This area is usually 'cut out' or cultivated last.

Heat period: A period of time during which a female animal is sexually receptive.

Heat: A condition when a female animal is ovulating and ready to accept the bull.

Heifer: A young female cow that has not calved.

Hen: An adult female chicken or turkey.

Herbicide: Any agent used to destroy or inhibit growth of plants (mainly weeds).

Heritability: The degree to which a characteristic can be passed on genetically.

Heterozygous: having two different alleles of a particular gene or genes

Hogget: A young sheep of either sex from about nine to ten months until it is two-tooth. The hogget stage follows the weaner stage.

Homogenisation: the process of breaking milk fat into very small droplets such that the fat are uniformly distributed in the milk and the cream does not form or separate.

Homozygous: Having two identical alleles of a particular gene or genes.

Hormone: A chemical compound which stimulates changes to the growth and reproduction of an organism. Plant hormones include auxins and gibberellin.

Horticulture: the cultivation of fruits, flowers and vegetables.

Horticulture: The intensive cultivation of plants.

Host: Organism which is subjected to parasitism, predation or disease by parasite, predator, pest or pathogen.

Humus: Dark brown semi-solid substance that originates from decomposed organic matter.

Hybrid: The offspring of parents of different species, varieties or breeds of plants or animals.

Hydroponics: Growing crops without soil, using solutions enriched with nutrients and oxygen.



Identification: placing a mark on farm animals to identify them from the others.

Illuviation: The process of depositing soil material removed from one horizon to another, usually from upper to lower horizon.

Implanting: The placement of a growth-stimulating compound, in the animal's body.

Improved pasture: Pasture that has been fertilised, and usually with introduced species.

In calf: A pregnant cow.

Inbreeding: This occurs when a female is mated with a male to which she is related. The more closely related the two are, the more intense the degree of inbreeding.

Incubation: A process of holding eggs under controlled conditions of heat and moisture permitting the fertile eggs to hatch.

Indigenous: A plant or animal native to the location where it is growing or living.

Infection: The entry, development and multiplication of a pathogen in the body.

Infertility: A temporally inability of the animal to reproduce.

Inoculant: A seed or soil additive (especially for legumes) that contains nitrogen fixing bacteria that facilitate nitrogen fixation in the subsequent crop. See: rhizobia.

Insecticide: An agent that controls or destroys insects.

Integrated pest management: Controlling pest using a combination of methods while reducing harm to other organisms in the ecosystem.

Intensive Grazing: A grazing system by which a pasture is heavily stocked for a number of days then allowed to rest before being grazed again.

Irrigation: The artificial application of water to the soil for the benefit of growing crops.



Kid: A young goat.

Kindling: The act of giving birth in rabbits

L

Lactation Period: The length of time a female gives milk following birth of offspring-usually with reference to dairy cows and milk goats.

Lactation period: The time an animal is in milk, usually 10 months for cows.

Lactometer: A device for testing the density (or thickness) of milk.

Lamb: A young sheep.

Land degradation: The decline in land quality caused by improper land use.

Land fragmentation: The practice of splitting land into small pieces of land.

Land Reclamation: Restoration of land to a desired condition for agricultural production.

Landslide: Rapid movement of a mass of soil, rock, or debris down a slope

Land tenure: The rules and arrangements concerning the ways in which land is owned, used, transferred, and succeeded in a given society.

Land title: A certificate of ownership of land.

Layer: A hen producing eggs regularly.

Layering: A vegetative propagation of plants by making a new plant to produce roots before it is cut from the parent plant.

LD50: Measure of toxicity of a chemical. The lethal dose that will kill 50% of test animals. The lower the LD50, the more toxic is the chemical.

Leaching: The process of removal of soluble materials by the passage of water through soil.

Legumes: A family of plants with the ability to fix nitrogen in the soil and high in protein.

Ley pasture: the planting of pasture and crops in a mixture or in rotation with crops.

Lime: All limestone-derived materials applied to neutralize acid soils.

Liming: Refers to the addition of limestone or other basic (alkaline) material to raise the soil pH.

Line breeding: Breeding from the same bloodline or within the same family, but not from close relations.

Litter: Live birth of multiple offspring of animals at one time from the same mother.

Livestock: Any domestic animal produced or kept primarily for farm, ranch, or market purposes, including beef and dairy cattle, hogs, sheep, goats, and horses.

Liveweight: Total weight of an animal including fleece, gut "ll and contents of the uterus.

Loam: The name of a soil texture class with moderate amounts of sand, silt and clay.

Loan: The mount of money that an individual or company borrows from a lender.

Longevity: Length of life. In dairy animals, it is the level of milk production over a long lifetime.

Lysimeter: Device and monitoring system used to measure evapotranspiration.

M

Macronutrient: A nutrient that plants or animals need in relatively high amounts.

Manure: The excreta of animals or plant materials in varying stages of decomposition.

Mare: Mature female horse.

Marginal land: Land almost too unproductive to be farmed profitably.

Mass selection: Selection based on performance of appearance of individual animal.

Mastitis: A disease of the cow's udder resulting from infection by microorganisms. The infection may be caused by improper milking procedures.

Mature soil: A soil having well-developed soil horizons.

Metabolizable energy (ME): The gross energy of feed minus energy in feces, urine, and gaseous products of digestion.

Microorganisms: organisms that cannot be seen using naked eye or can only be seen using a microscope.

Milk curd: the part of milk that coagulate when milk sours or when it is treated with an acid.

Milk let down: The release of milk from the secreting cell to the gland cistern and finally to the teat cistern.

Milk stage: In grains, the growth stage that follows pollination in which the endosperm appears as a whitish, milk-like liquid.

Milking parlour: A special area where cows are milked from.

Minced meat: Meat that has been cut or ground into small pieces.

Mineralization: Decomposition of organic matter, thereby releasing mineral elements as inorganic ions.

Minimum tillage: A soil management system in which weeds are controlled mainly by chemicals instead of cultivation. Conservation farming and reduced tillage involve similar practices.

Mixed farming: Where several types of farming enterprises (such as crop cultivation and animal rearing) are conducted on the one farm.

Mortar: a mixture of cement, sand and water in particular proportions used to bind blocks, bricks or stones during construction, used to plaster walls and floors.

Mulch: (1) A layer of dead plant material on the soil surface. (2) An artificial layer of material, such as paper or plastic, on the soil surface.

Mycorrhiza: The association, usually symbiotic, of fungi with the roots of higher plants.

Mycotoxin: A toxin or toxic substance produced by a fungus.



Nematodes: Soil worms of microscopic size that may attack the root or other structures of plants and cause extensive damage.

Nitrification: The process by which ammonia is converted to nitrates.

Nitrogen Cycle: Sequence of transformations undergone by nitrogen in its movement from the free atmosphere into and through soils, into plants, animals and eventually back.

Nodule: Tissue enlargement or swelling on a plant root caused by nitrogen-fixing bacteria.

Non-selective herbicide: A herbicide that kills broadly any plant that it comes in contact with.

Notifiable disease: A disease listed by the Authority, and that, as soon as detected or suspected, should be reported to the Authority, in accordance with national regulations.

Notifiable diseases: diseases that should be reported to the authorities once noticed on the farm or in the area.

Noxious weed: An unwanted plant specified by laws as being specifically undesirable, troublesome, and difficult to control. It grows

and spreads in places where it interferes with the growth and production of the desired crop.

NPK: Letters used to designate the elements of nitrogen, phosphorous, and potassium in that order; usually expressed as a percentage by weight of fertilizer.

Nurse crop A temporary crop planted at or near the time primary plant species are planted to provide protection and otherwise ensure establishment of the latter. Syn. companion crop.

Nutrient solution: A carefully proportioned liquid fertiliser used in hydroponics.

Nutrient: Materials essential for normal body metabolism, growth and production.



Oestrus cycle: The interval from the beginning of one heat period to the beginning of the next.

Offspring: The progeny of parents.

Olericulture: The cultivation of vegetables.

Omnivore: Animals that eat both animal and plant origin feeds.

Organic farming: A production system totally excluding the use of synthetically compounded fertilizers, pesticides or growth regulators.

Organic matter: undecomposed or decomposed materials of plant or animal origin in the soil.

Overgrazing: Refers to the process of allowing animals to graze a pasture for long periods of time without giving the plants an opportunity to recover.

Overstocking: Placing a number of animals in a given area that will result in overuse if continued to the end of the planned grazing period.



Paddock: A subdivided section of pasture used to more efficiently graze a large field.

Palatability: The relish with which a particular species or plant part is consumed by an animal.

Pandemic disease: a disease that effects a large area, multiple countries or multiple continents.

Parent material: Unconsolidated material from which the soil is formed.

Parturition: Giving birth.

Pasteurization: Heating and cooling raw milk quickly to kill harmful bacteria without effecting the taste or nutritional value.

Pasture: (1) Grazing land comprised of planted or native forage species that are used primarily for the grazing of livestock. (2) Forage plants used as food for grazing animals. (3) Any area devoted to the production of forage, native or introduced, and harvested by grazing.

Peat: Soft dark substance formed by partial decomposition of organic matter in the wet or marshy conditions.

Perennial plant: One with a life cycle of three or more years.

Permaculture: A term for 'permanent agriculture'. A system of practicing farming in harmony with nature. Land, plants and animals are used in a diverse and balanced way in which they would occur in natural systems.

Permanent Wilting Point: That point at which a plant is dried so badly that even though put into a humid atmosphere and watered, it will no longer recover.

Pest: Any organism injuring plants or plant products.

Pesticide: Any substance used to control or kill pests, such as insects, animals, or microbes.

pH scale: The scale that measure the acidity or alkalinity of the soil.

Pheromone: A sex-attractant hormone or scent produced by animals or insects.

Photoperiod: Number of daylight hours in a day.

Pinching Off: The periodic removal of new, freshly grown leaves from a plant to encourage growth.

Pistil: The female sexual reproductive, seed-bearing organ of a flower, consisting of an ovary, style and stigma.

Plucking: The harvesting of the tips of the shoots as they appear above the plucking table.

Polled: Animal that is naturally without horns.

Pollination: The transfer of pollen from the stamen (male part of the flower) to the pistil (female part of the flower).

Pollution: The introduction of contaminants into the natural environment that causes adverse change.

Pomoculture: The cultivation of fruits.

Pomology: the science or study of growing fruit.

Pond: A dam or an embankment for rearing fish or to supply drinking water for livestock.

Porosity: Amount of pores or open spaces between soil particles.

Poult: A young turkey of either sex, usually of less than 2 months of age.

Poultry: Domestic birds raised for eggs or meat.

Predisposing factors: conditions inside or outside the body of an animal that make the animal vulnerable to diseases or injuries.

Pre-emergence herbicide: A herbicide applied prior to emergence of a specified weed or crop.

Pricking out: Transferring seedlings from nursery bed into another specially prepared bed to reduce congestion in the nursery.

Primary tillage: The first working of the soil by implements for cropping or pasture.

Processing: Any activity which enhances or maintains the quality or changes the form or characteristics of a farm product.

Progeny testing: Assessing the performance of an animal's progeny to assess the animal's value as a parent. Mainly used to assess sires.

Progeny: The offspring of the sire and dam.

Propagule: Any part of a plant that is capable of giving rise to a new individual.

Pruning: Cutting away dead or overgrown branches or stems to increase fruitfulness and growth or to maintain a certain size or shape.

Pullet: A female chicken less than a year old, or a young hen before she begins to lay eggs.



Quarantine: a restriction on the movement of animals and plants suspected to be carrying an infection, to control the spread of the infection.



Rainfall intensity: The amount of rainfall in a given time.

Raised Bed: An elevated garden bed offering better drainage, aeration and warmer soil than a conventional bed.

Ram: An uncastrated male sheep.

Rangeland: Land on which the climax plant community is predominantly grasses, grasslike plants, or shrubs.

Ration: The total amount of feed (diet) allotted to one animal for a 24-hour period.

Recycle: To process waste materials into useful products.

Relief: Elevations or nature of the land surface.

Rhizobium: Bacteria living in nodules on the roots of leguminous plants that are capable of removing nitrogen from the air and soil “fixing” it into forms that plants utilize for growth.

Rhizome: A horizontal underground stem that usually sends out roots and aboveground shoots from the nodes.

Rill erosion: A form of water erosion that produces narrow shallow channels in the soil.

Roaster: A young meat bird, 12 to 16 weeks used for roasting.

Rodenticides: Poisons used to control rats, mice, and other rodents.

Rooster: An adult male chicken, (syn. a cock).

Rootstock: The lower part of a plant graft. See grafting.

Rotation grazing: A type of grazing system and involves moving grazing animals from one pasture to another to achieve a desired management objective.

Roughages: Plant materials containing a low proportion of nutrients per unit of weight. Generally bulky and coarse, high in fiber, and low in total digestible nutrients.

Rounding up: The purposeful gathering of animals from a specific area.

Rumen: The large, first compartment of the stomach of a ruminant which stores food in which fermentation and digestion is aided by symbiotic action of microbes.

Ruminant: mammals that chew the cud and have a 4-chamber stomach.

Runoff: Rainfall that is not directly absorbed into the soil. Water flows across the soil surface and, depending on the soil condition, the runoff may pick up soil particles and cause erosion.



Sample: Part of a population taken to estimate a parameter of the whole population.

Scion: The upper part of the join in a graft.

Seed cake: Residues left after extracting oil from plant seeds like cotton, sunflower and ground nuts.

Seed dormancy: Live seed in a non-germinative condition because of internal inhibitions in the seed; i.e., hard seed, or unfavorable environmental conditions.

Seed dormancy: A state in which seeds will not germinate under normal germination conditions

Seed inoculation: Treatment of legume seed with rhizobium bacteria before planting to enhance subsequent nitrogen fixation.

Seed longevity: The ability of the seed to remain viable for a given period of time.

Seed purity: Percentage of the desired species in relation to the total quantity, including other species, weed seed, and foreign matter.

Seed scarification: Physical or acid treatment of seedcoats to improve moisture absorption and enhance germination the seed.

Seed viability: The ability of a seed to germinate and develop into a seedling once the conditions necessary for germination are provided.

Seedling: A plant that has just emerged from its seed with its first root, stem and leaves.

Selection: Choosing stock to keep for breeding or growing out. The opposite to culling.

Selective grazing: The grazing of certain plant species, individual plants, or plant parts on rangeland to the exclusion of others.

Selective herbicide: A herbicide that kills only certain groups of plants. For example, 2,4-D kills broadleaf plants, but not grasses.

Self-pollinating: Plants not requiring pollen from another plant to produce fruit.

Semen: The fluid containing sperms that is produced by the reproductive system of a male animal.

Senesce: The yellowing and withering of older, lower leaves of plants as they become shaded by higher, younger leaves. Nutrients in these older leaves are translocated to younger tissue.

Set stocking: The practice of allowing a fixed number of animals to graze on a fixed number of acres for a fixed amount of time, usually the grazing season.

Sexing chicks: Determination the sex of newly hatched chicks.

Sheet erosion: A more or less uniform removal of soil particles from an area without the formation of clear water channels.

Shoat: A weaned young pig of either sex.

Side-dressing: Placement of fertiliser alongside the root zone of a growing crop.

Silage: Forage preserved in a succulent form by organic acids (lactic acid primarily) produced by partial anaerobic fermentation of sugars in the forage.

Sire: The father or male parent of an animal.

Soil aeration: A diffusion process by which air in soil pore spaces is replaced by air from the atmosphere, replenishing soil oxygen and removing excess soil carbon dioxide.

Soil aggregate: Group of primary soil particles that adhere to each other.

Soil amendments: Any material, organic or inorganic, applied to the soil to make it more conducive to vigorous plant growth. Usually refers to added materials other than fertilizer.

Soil auger: A tool for boring into the soil and withdrawing a small sample for observation in the field or laboratory.

Soil degradation: Soil in which the structure has been damaged, compaction or erosion has occurred. It may also refer to soil acidity and salinity and the loss of nutrients from a soil.

Soil erodibility: The likelihood of a soil to erode. See erosion.

Soil erosion: The removal of top soil running water or wind.

Soil fertility: The ability of soil to sustain crop growth and maximum crop yield.

Soil Horizon: A layer of soil material nearly parallel to the land surface which differs from adjacent genetically related layers in color, structure, texture, or consistence. It also differs in biological and chemical characteristics.

Soil pH: The degree of acidity or alkalinity of the soil.

Soil plasticity: Capacity of soil to be moulded or shaped without cracking.

Soil profile: Vertical section of the soil through all its horizons.

Soil Reaction: The degree of acidity or alkalinity of a soil usually expressed in terms of pH value.

Soil structure: Combination or arrangement of primary soil particles into secondary units or aggregates.

Soil test: A chemical, physical, or biological analysis of a soil.

Soil Texture: Refers to the coarseness or fineness of a soil. It is determined by the relative proportion of various sized particles (sand, silt, and clay) in a soil.

Soil water (moisture): Water stored in the soil.

Soil water infiltration: Water movement through the soil surface into the soil medium.

Soil: The unconsolidated material on the immediate surface of the earth that serves as a natural medium for the growth of plants.

Solids not fat: A component of milk that consists of dissolved or suspended solids other than butterfat. It includes protein (casein), minerals and lactose.

Solum: (plural sola) The upper horizons of a soil profile. Usually consists of A and B horizons.

Sow: A female hog that has produced at least one litter of pigs.

Sphincter muscle: Ring like muscle surrounding and able to close or contract an opening.

Spray drift: The movement of airborne spray particles from the intended area of application.

Staking: The practice of providing support, using sticks or stakes, to crops that grow tall and are not capable of supporting themselves

Stamen: The male reproductive organ of a flower, consisting of a filament and a pollen-containing anther.

Standing heat is a condition in which an animal on heat stands still and accepts to be mounted by other animals without running away.

Steaming up: Provision of extra feeds of high nutritive value to an animal during the last weeks of pregnancy.

Steer: A male bovine castrated before reaching sexual maturity.

Sterilisation: The process of making sterile, the killing of all forms of life.

Sterility: A permanent inability of the animal to reproduce.

Stocking Rate: Number and classes of animals grazing a specific unit of land for a specific period of time.

Stolon: A horizontal stem which grows along the surface of the soil and roots at the nodes.

Strain: (1) A general term used for animals that have similar and distinct characteristics within a breed. (2) To tighten the wire in a fence, using a tool called a strainer.

Strip Cropping: Growing crops in narrow strips across a slope along the contours, alternating usually grass or cover crop with crop strips.

Strip grazing: Refers to temporarily subdividing a grazing area into subunits with temporary fences so grazing for short periods, often 4 hours or less, can be achieved.

Stripping out: The complete removal of the last milk from the udder using mainly hand milking.

Stubble: The basal portion of plants remaining after the top portion has been harvested.

Stunning: Any mechanical, electrical or other procedure that causes immediate loss of consciousness before slaughter of the animals.

Subsoiling: The breaking of compact subsoils, without inverting them.

Suckers: Sprouts arising from buds below the ground level.

Supplement: Nutritional additive (salt, protein, phosphorus) intended to remedy deficiencies of the range diet.

Supplemental feeding: Supplying concentrates, additives or extra nutritious feed to correct deficiencies of the range diet. Often erroneously used to mean emergency feeding.

Swamp: An area saturated with water throughout much of the year, but with the surface of the soil usually not deeply submerged.

Symbiosis: The living together in intimate association of two different organisms, so that the cohabitation is mutually beneficial.



Tag: A label attached to the animal for identification.

Tattooing: Identification of animals, by means of a tattooed mark.

Teaser: a male used to determine if a female is in heat and ready to be served.

Temperament: The individual character of an animal, shown by the way it behaves.

Thatch: Dried straws, grass or papyrus used in roofing farm buildings.

Thinning: The process of removing excess plants from a garden to reduce competition for water, nutrients and sunlight.

Threshing: A harvesting operation in which the seed is mechanically removed from the head and stalk of the plant.

Tillage: The act of digging and breaking up the soil.

Tiller: (1) An erect shoot that arises from the crown of a grass. (2) Asexual development of a new plant from a meristematic region of the parent plant.

Tilth: texture of the soil clods after cultivation.

Tipping: Cutting actively growing shoots to the desired height of the plucking table.

Tissue culture: The growing of cells in a special growth solution or medium. It is used for the rapid asexual multiplication of plants.

Tom: A male turkey.

Top-dress: To apply a material such as fertilizer or manure to the surface of the soil, or after the plants have germinated

Topsoil: The layer of soil used for cultivation, which usually contains more organic matter than underlying materials.

Total digestible nutrients: The total digested energy in a feedstuff expressed in units of weight or (TDN) percent.

Trace element: An element essential for normal growth and development of an organism, but required only in minute quantities.

Training: Modification of the natural growth habit of the crop to help the plant grow in the right shape and direction.

Trait: A characteristic or character of an animal.

Translocated herbicide A herbicide that is absorbed by the plant into its transport system and delivered to the entire plant structure.

Transpiration: Movement of water from the soil, into the roots, up the plant stems, and finally out of the plant leaves into the air as vapour.

Transplant: To remove plants from one place (usually the nursery) and replant them in another (usually the main garden).

Trellis: A support system for plants allowing the positioning of shoots, foliage and fruits for

maximum interception of sunlight, air and spray circulation and ease of management.

Tupping: A term to describe joining or mating of animals.

Turbidity: A measure of the clarity of water. The amount of suspended matter in a water sample or water supply

Turgid: State of a plant cell when the cell wall is rigid due to the hydrostatic pressure of liquid in the cell.

U

Udder: Mammary gland.

Undersown: A pasture sown with a crop. The crop is harvested and the pasture encouraged to take over the paddock.

Uterine horns: The site of implantation and development of the embryo

Uterus: Also site of implantation and a place where the embryo grows and develops from.

V

Vaccine: Biological preparation that provides active acquired immunity to particular diseases

Vaccination: the administration of a vaccine, in accordance with the manufacturer's instructions.

Variable cost: Expenses that change with the level of production on the farm. Examples of variable costs include supplemental feed, pesticides, veterinary services and supplies, and labor.

Variety: A group of individuals within a species that differs from the rest of the species.

Vector: an insect or any living carrier that transmits an infectious agent from an infected individual to a susceptible individual or its food or immediate surroundings.

Volatilization: The conversion of a chemical from a liquid or solid to a gas or vapour.

Volunteer plants: Plants not purposely planted germinating from seed laid down residues of crops that were growing on the site at some previous time.

W

Wasteland: Land not suitable for, or capable of, producing materials or services of value.

Water cycle: The path that water follows as it moves around earth in different states.

Water holding capacity: Total amount of water the soil can hold.

Water logging: the saturation of soil with water. All soil pores being filled with water.

Water retention: Soil's ability to hold water inside its pores.

Waterlogged: Soil that is saturated with water.

Watershed: (1) A total area of land above a given point on a waterway that contributes runoff water to the flow at that point. (2) A major subdivision of a drainage basin.

Water suckers: Sprouts arising from buds above the ground level.

Wean: To separate young animal from its mother, stopping it from suckling and allowing the mother to recover its body condition and allow for specialized feeding of the young.

Weathering: The breakdown of rocks to form soil particles.

Weed: A plant growing in the "wrong place."

Wether: A castrated male sheep or goat.

Wetland: Area characterised by soil that is usually saturated and supports mostly water-loving plants; i.e., hydrophytic plants.

Windbreak: A strip of trees or shrubs serving to reduce the force of wind; any protective shelter from the wind.

Z

Zero grazing: Keeping animals in stalls all the time and feeds and water are brought to them.

Zoning: A means by which governmental authority is used to promote a specific use of land under certain circumstances.

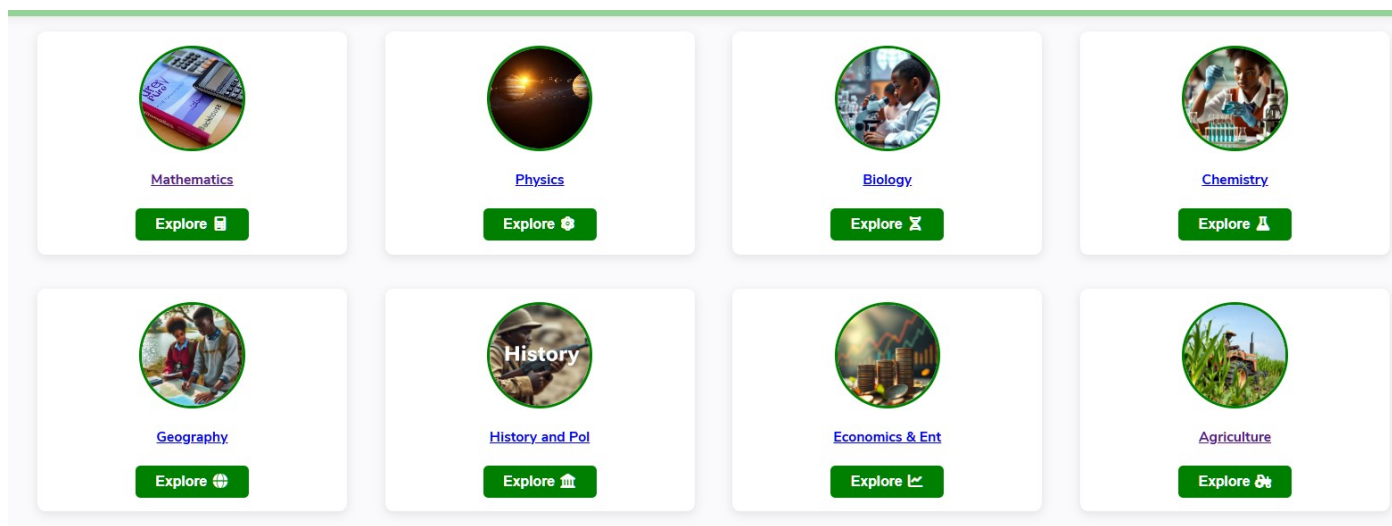
Zoonoses: disease which can be transmitted from animal to man or from man to animals e.g., tuberculosis, anthrax, brucellosis.



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